UNIVERSITY OF CAMBRIDGE

Core statistics

Cambridge Centre for Research Informatics Training

Housekeeping

- Introductions
- Attendance
- Questions
- Participation! ☺

What this course *isn't*

A mathematical statistics course:

$$SS_x = \sum (x_i - \bar{x})^2 = \sum (x_i^2 - 2x_i\bar{x} + \bar{x}^2) =$$
$$= \left(\sum x_i^2\right) - 2\left(\sum x_i\bar{x}\right) + \left(\sum \bar{x}^2\right) = \left(\sum x_i^2\right) - 2\bar{x}\sum x_i + n\bar{x}^2 =$$
$$= \left(\sum x_i^2\right) - 2\bar{x}n\frac{\sum x_i}{n} + n\bar{x}^2 = \left(\sum x_i^2\right) - 2\bar{x}n\bar{x} + n\bar{x}^2 =$$
$$= \left(\sum x_i^2\right) - 2n\bar{x}^2 + n\bar{x}^2 = \left(\sum x_i^2\right) - n\bar{x}^2$$
$$SS_y = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \left(\sum_{i=1}^{n} y_i^2\right) - n\bar{y}^2$$
$$SS_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) = \left(\sum_{i=1}^{n} x_i y_i\right) - n\bar{x}\,\bar{y}$$

Statistical Table 2: Critical values for the t statistic with v degrees of freedom

df	α (Two tails)						
v	.20	.10	.05	.02	.01	.005	.001
1	3.08	6.31	12.71	31.82	63.66	127.32	636.62
2	1.89	2.92	4.30	6.96	9.92	14.09	31.60
3	1.64	2.35	3.18	4.54	5.84	7.45	12.92
4	1.53	2.13	2.78	3.75	4.60	5.60	8.61
5	1.48	2.02	2.57	3.36	4.03	4.77	6.87
6	1.44	1.94	2.45	3.14	3.71	4.32	5.96
7	1.41	1.89	2.36	3.00	3.50	4.03	5.41
8	1.40	1.86	2.31	2.90	3.36	3.83	5.04
9	1.38	1.83	2.26	2.82	3.25	3.69	4.78
10	1.37	1.81	2.23	2.76	3.17	3.58	4.59
12	1.36	1.78	2.18	2.68	3.05	3.43	4.32
15	1.34	1.75	2.13	2.60	2.95	3.29	4.07
20	1.33	1.72	2.09	2.53	2.85	3.15	3.85
25	1.32	1.71	2.06	2.49	2.79	3.08	3.73
30	1.31	1.70	2.04	2.46	2.75	3.03	3.65
40	1.30	1.68	2.02	2.42	2.70	2.97	3.55
60	1.30	1.67	2.00	2.39	2.66	2.91	3.46
120	1.29	1.66	1.98	2.36	2.62	2.86	3.37
∞	1.28	1.64	1.96	2.33	2.58	2.81	3.29
	.10	.05	.025	.01	.005	.0025	.0005
	α (One tail)						

No mathematical derivations

No pen and paper calculations

What this course *isn't*

A "how to mindlessly use a stats program" course:

Analyze, graph and present scientific data easier than ever!
GraphPad PRISM 7

SPSS

Rcmdr

GenStat®

STATA®

What we’re aiming for

1. Know what statistical tests/analysis techniques are available

2. Know which ones are allowable

3. Be able to carry these out in R/Python and understand the results

Plan for the course

Six half-day sessions:

1. Statistical inference
2. Categorical predictors
3. Continuous predictors
4. Two predictors
5. Multiple predictors
6. Statistical uncertainty

Plan for the course

Lectures	~45-60 minutes Background/key concepts
Practicals	Implementation in R/Python Life sciences datasets Support from trainers

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Core statistics

Session 1: Statistical inference

Cambridge Centre for Research Informatics Training

Plan for the session

Session 1: **Statistical inference**

1. Statistical inference
2. Distributions & parameters
3. Hypothesis testing & p-values
4. Assumptions

1. Statistical inference

How taking samples can help us learn about the world

Populations & samples of stuff

Population

Populations & samples of stuff

Population

The complete set of stuff I want to know something about

Populations & samples of stuff

Population

The complete set of stuff I want to know something about

Sample

Populations & samples of stuff

Population
The complete set of stuff I want to know something about

Sample
The (much smaller) set of stuff I can actually see

Populations & samples of stuff

Inference

Population

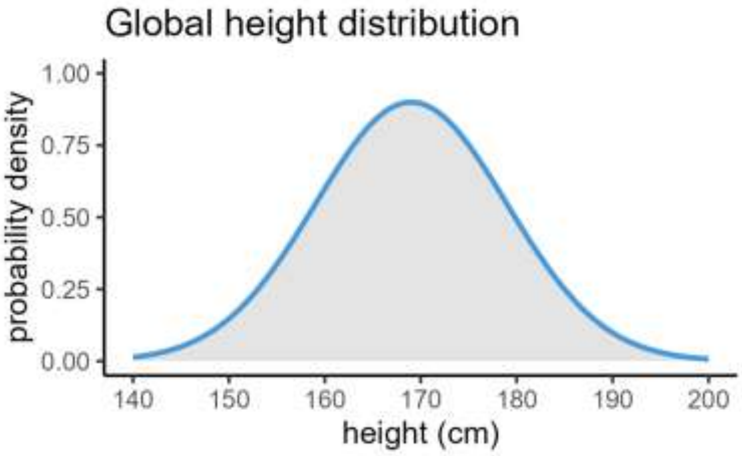
The complete set of stuff I want to know something about

Sample

The (much smaller) set of stuff I can actually see

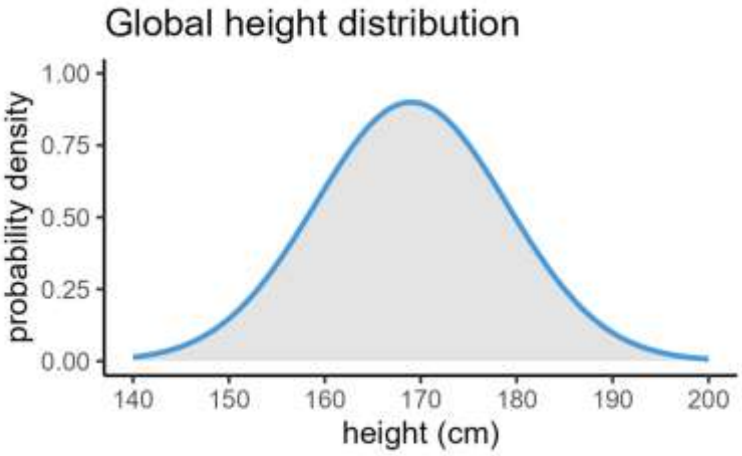
Statistical inference: simplifying framework

Statistical inference: simplifying framework



Abstraction

Statistical inference: simplifying framework

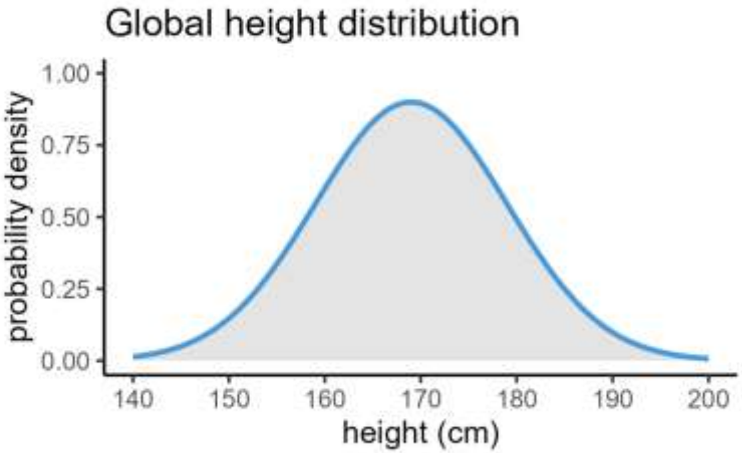


Abstraction

Distribution = function that shows:

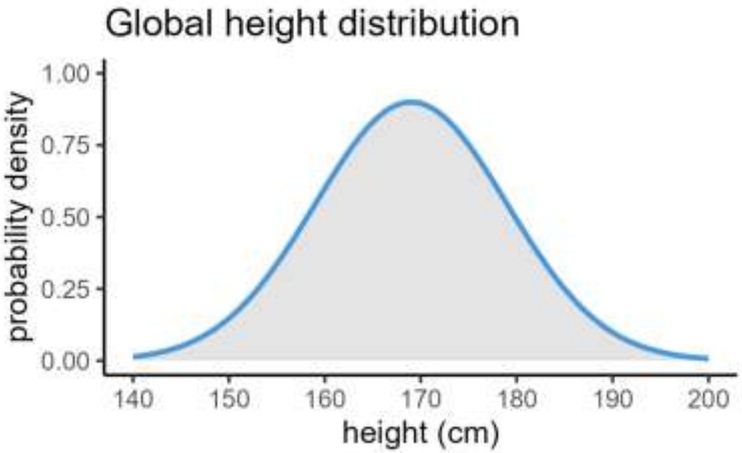
- The possible values a variable can take (x axis)
- And the likelihood of those values occurring (y axis)

Statistical inference: simplifying framework

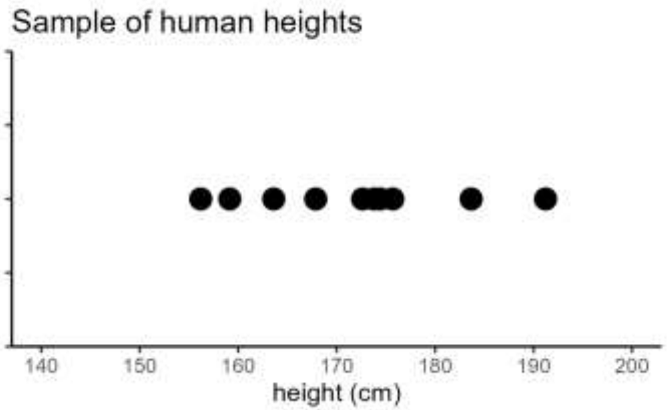


Abstraction

Statistical inference: simplifying framework

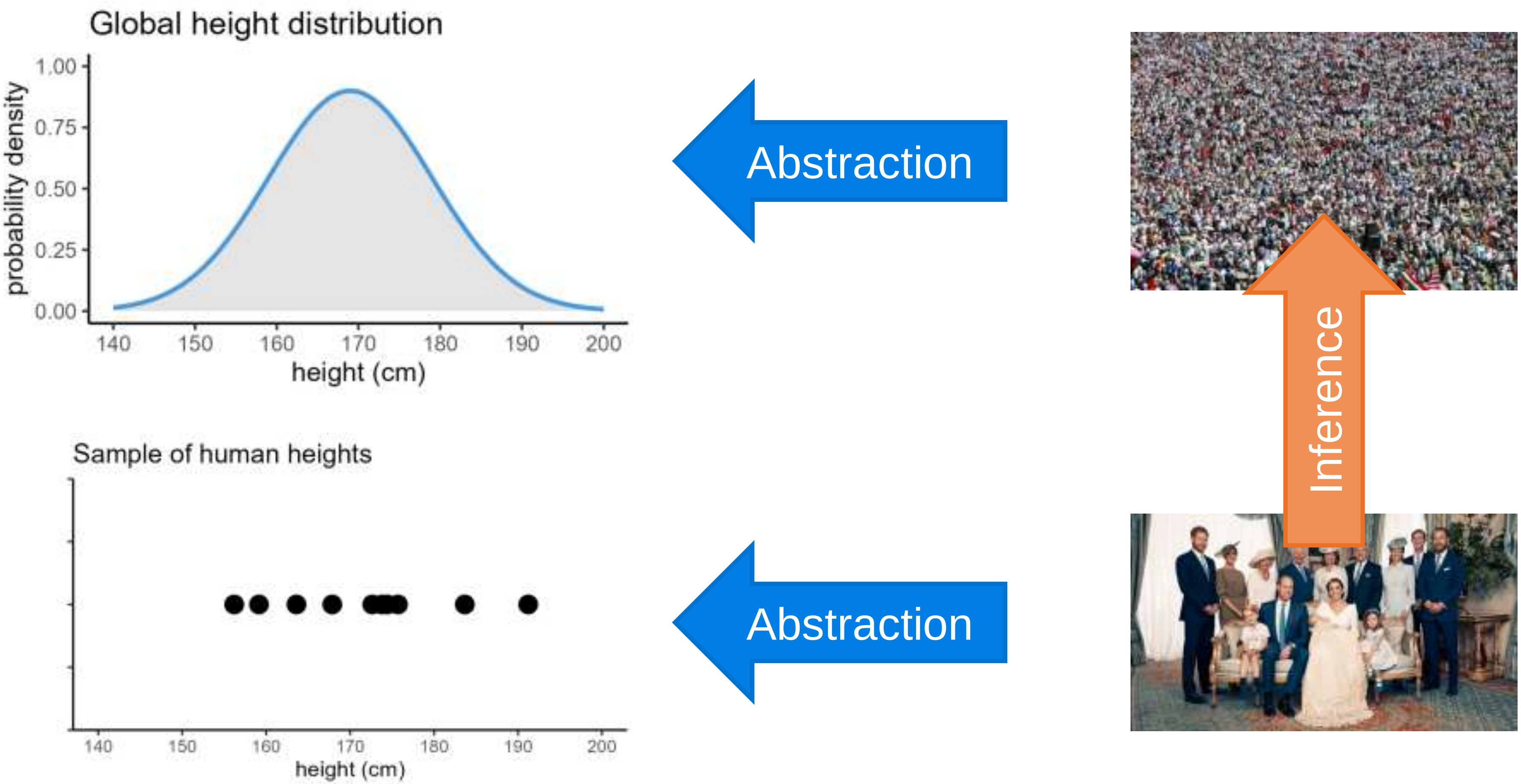


Abstraction

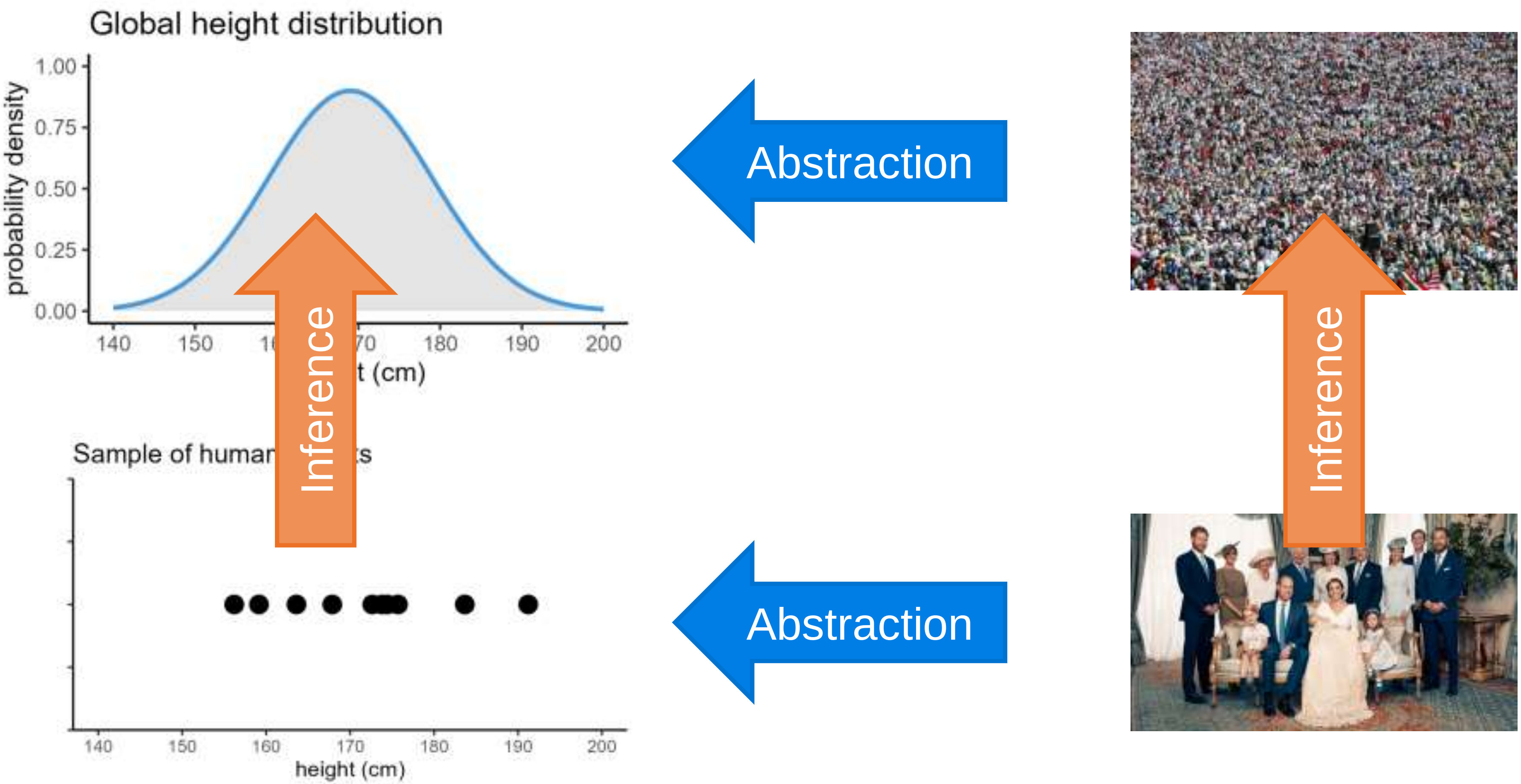


Abstraction

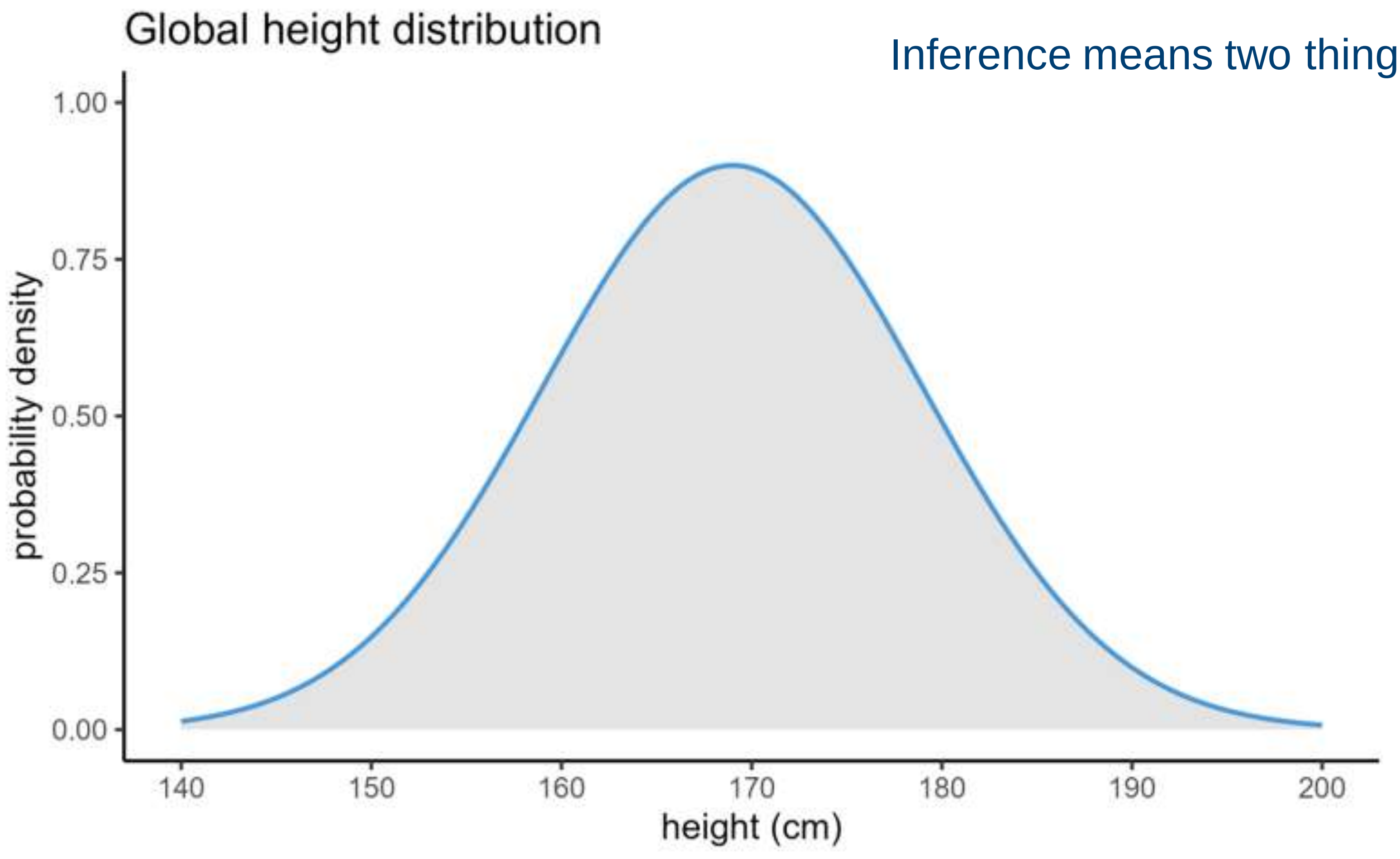
Statistical inference: simplifying framework



Statistical inference: simplifying framework

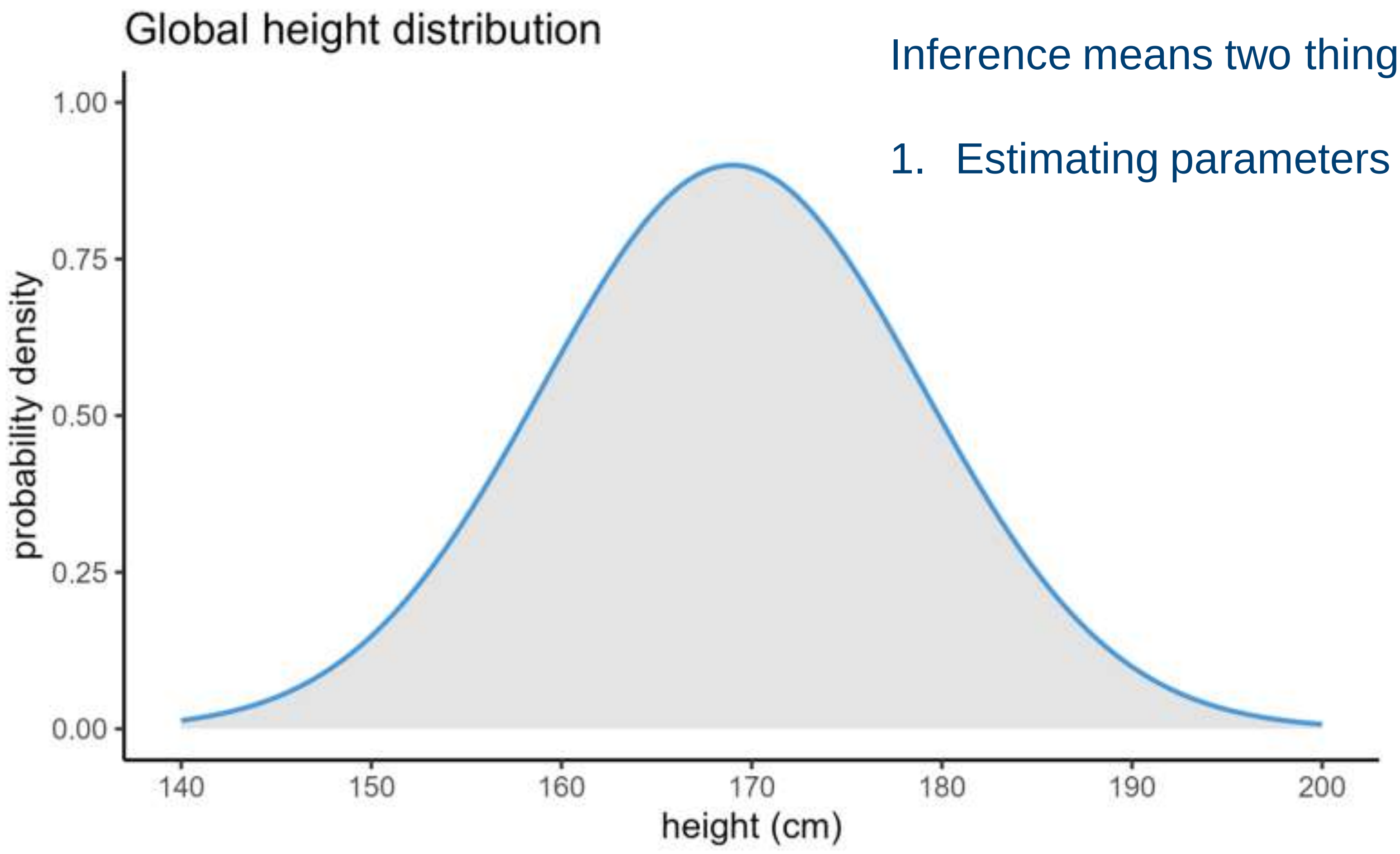


Distributions



Inference means two things:

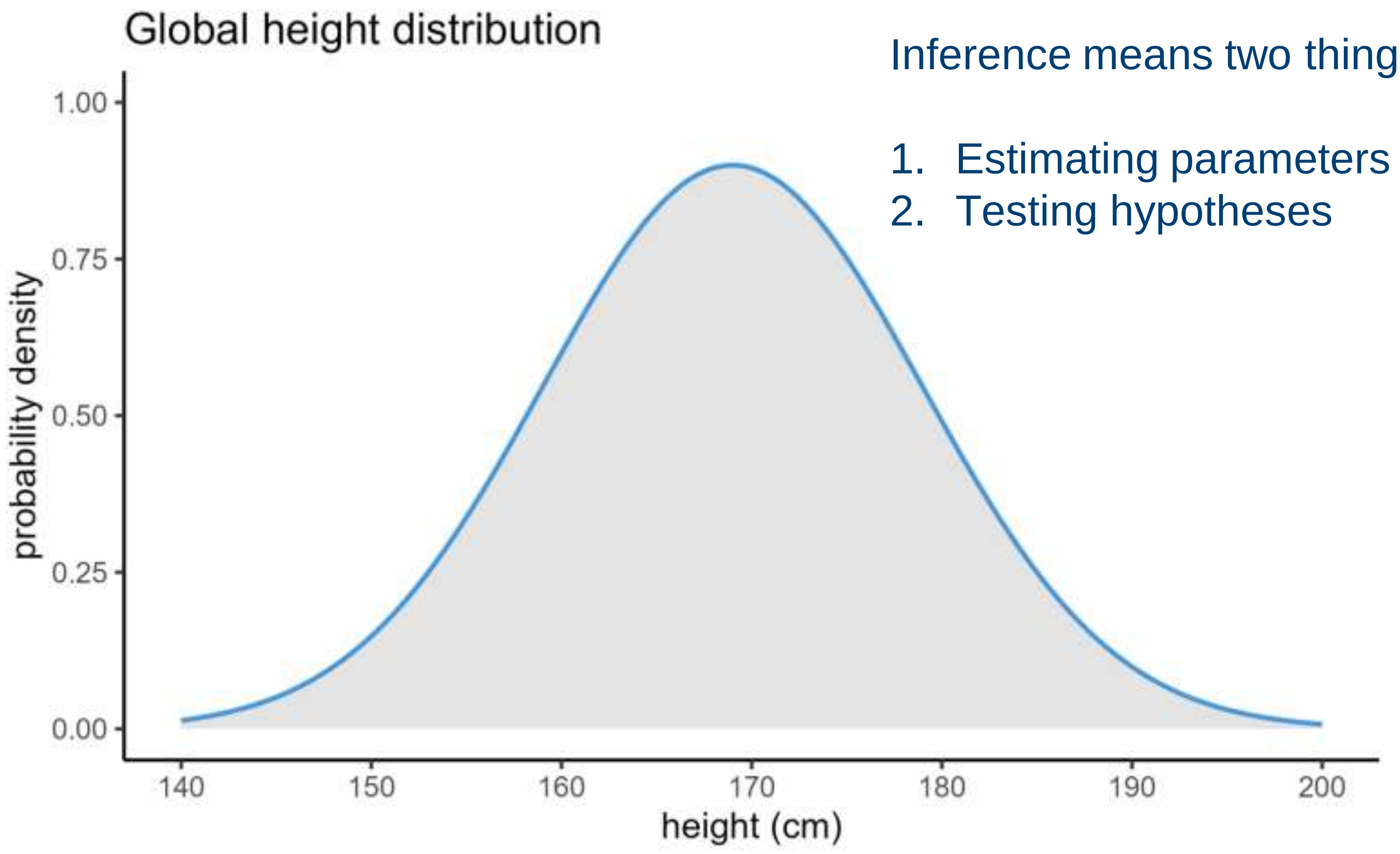
Distributions



Inference means two things:

1. Estimating parameters

Distributions



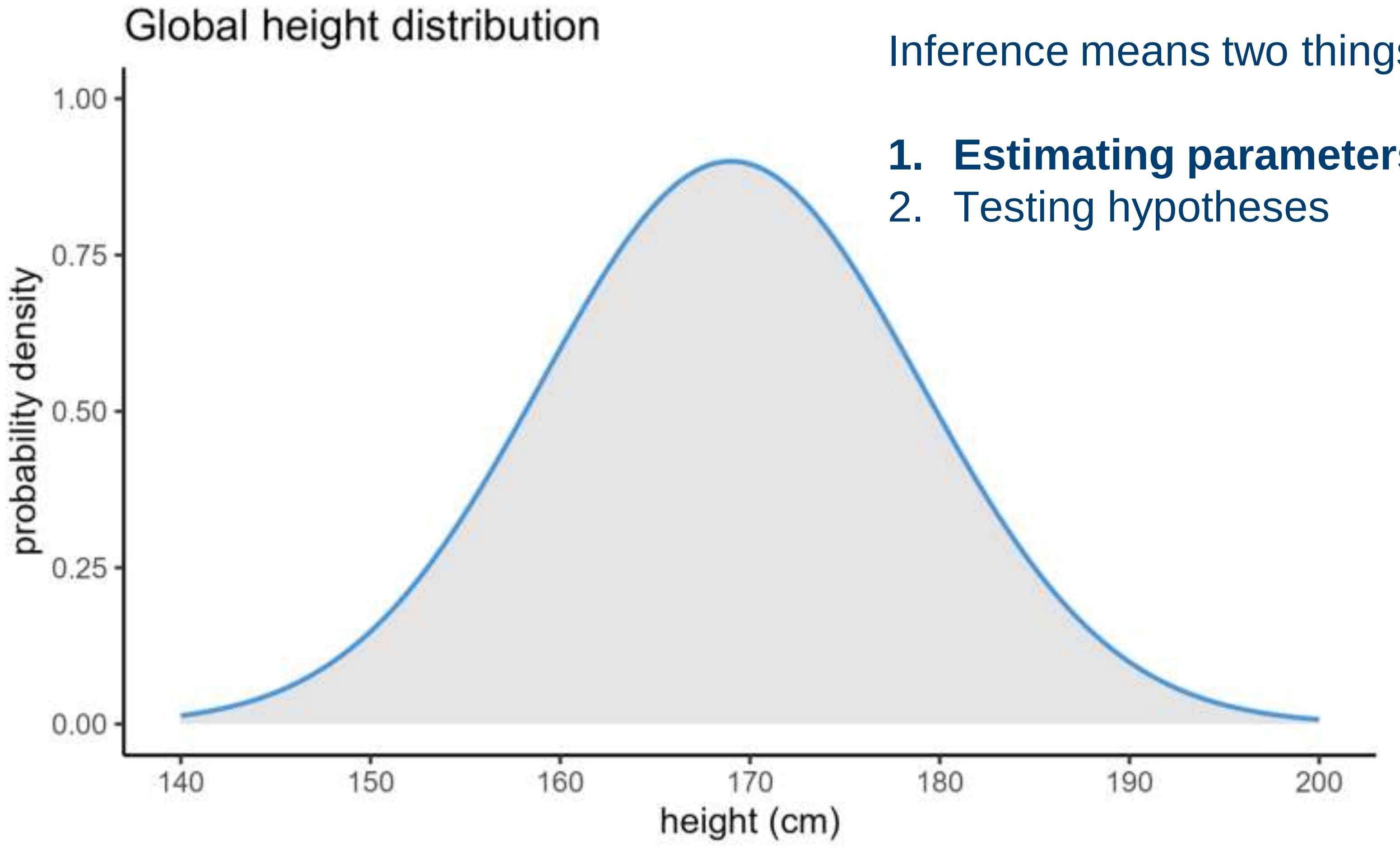
Inference means two things:

1. Estimating parameters
2. Testing hypotheses

2. Distributions & parameters

Describing a variable

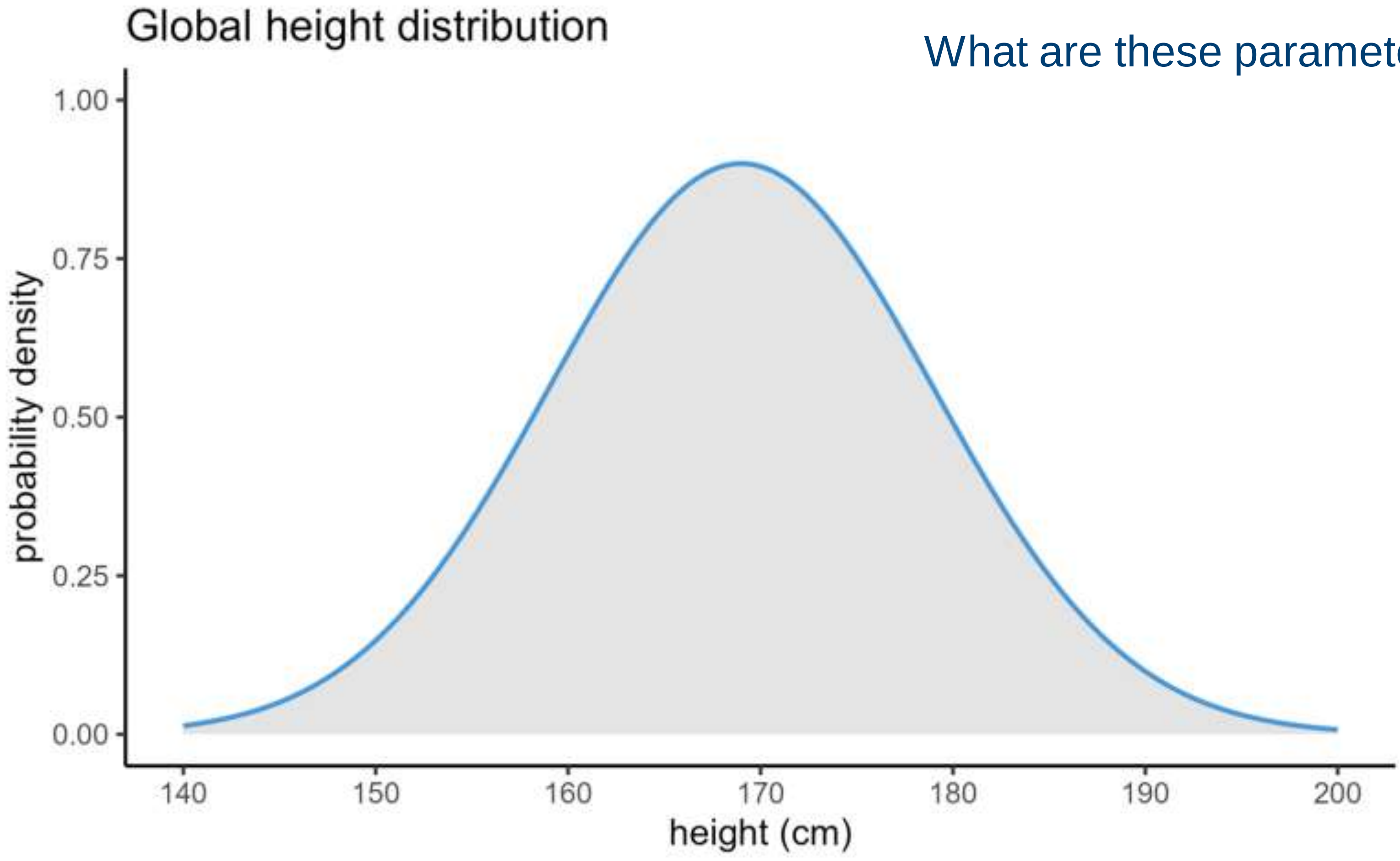
What parameters can we test?



Inference means two things:

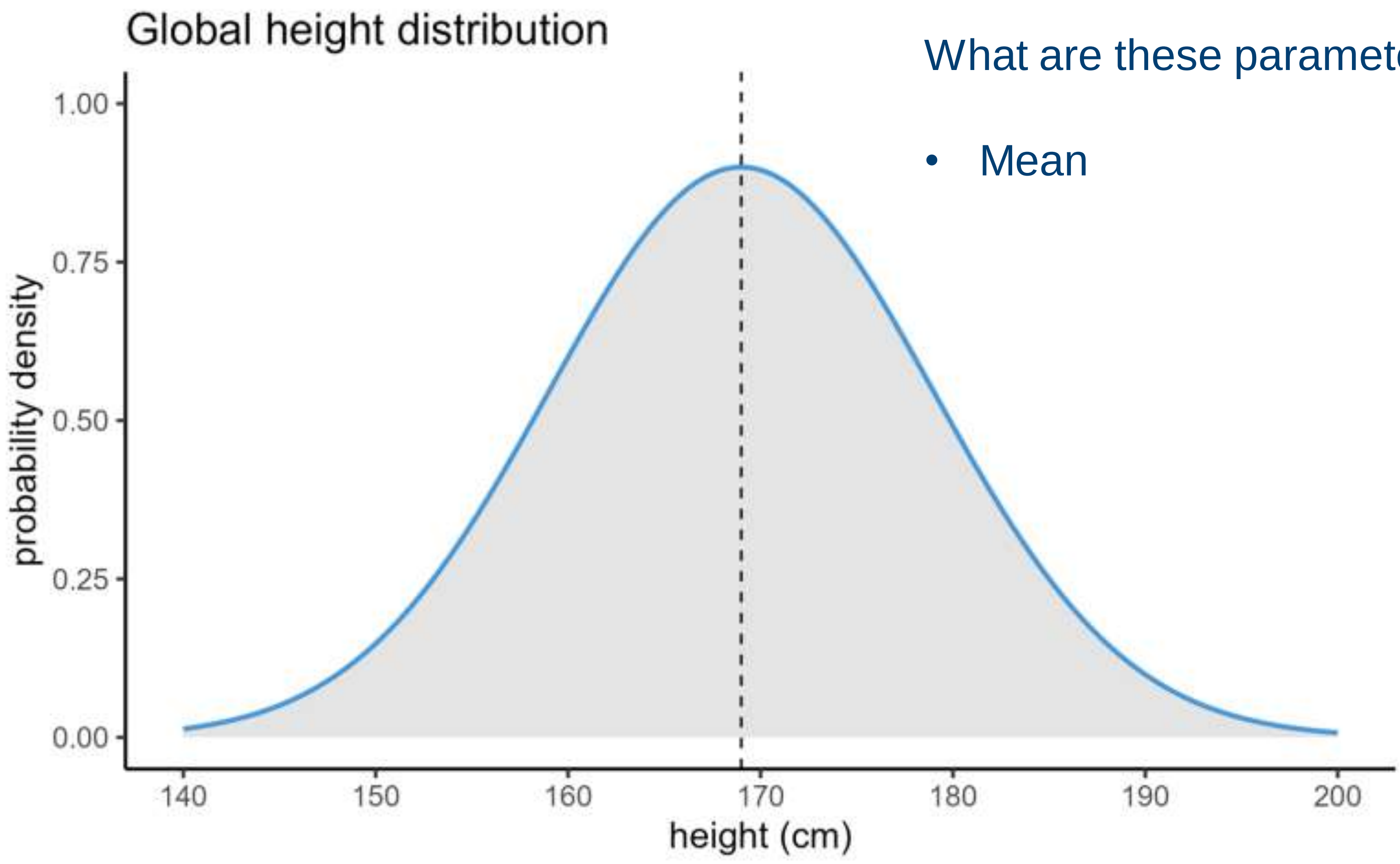
1. **Estimating parameters**
2. Testing hypotheses

What parameters can we test?



What are these parameters?

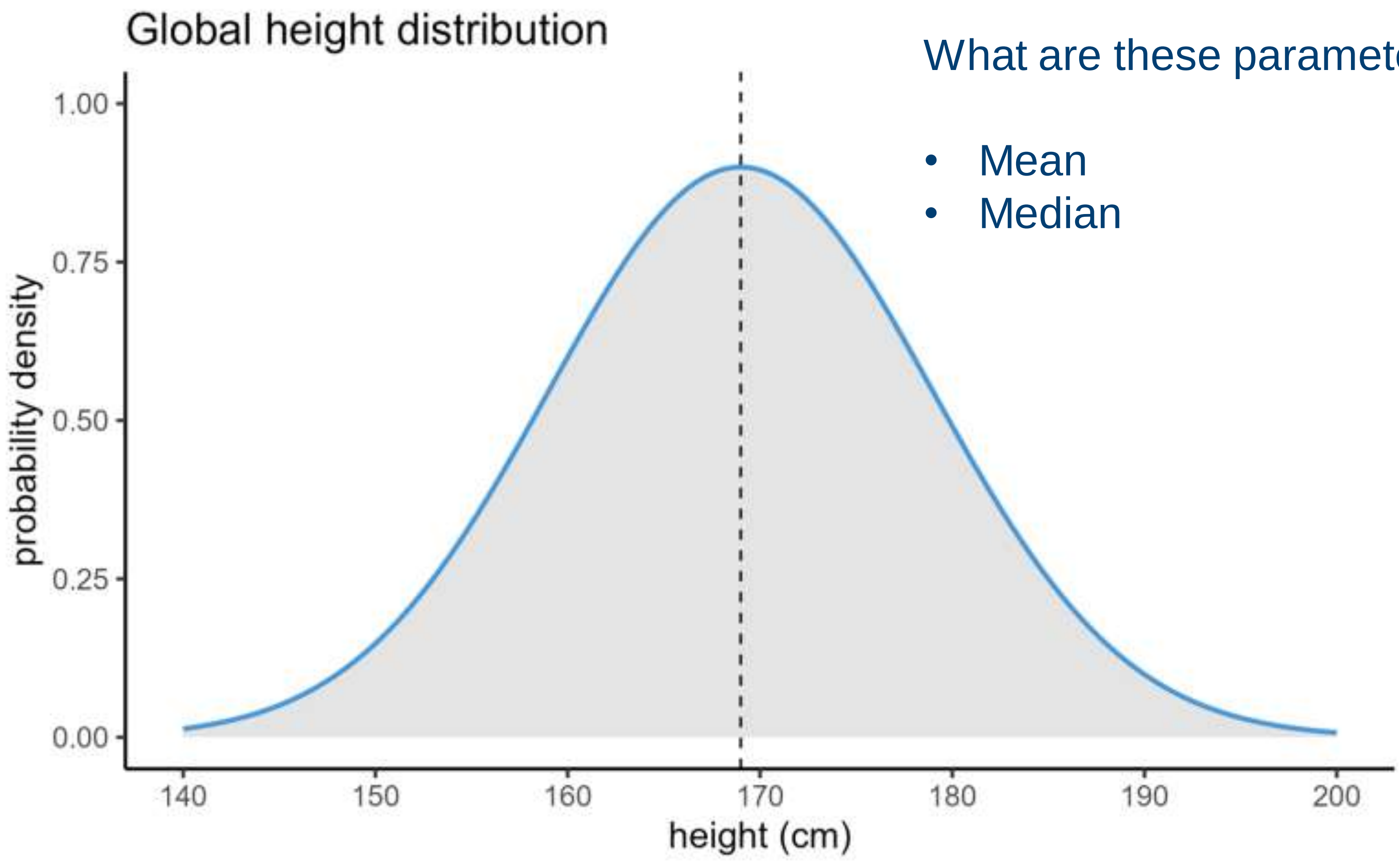
What parameters can we test?



What are these parameters?

- Mean

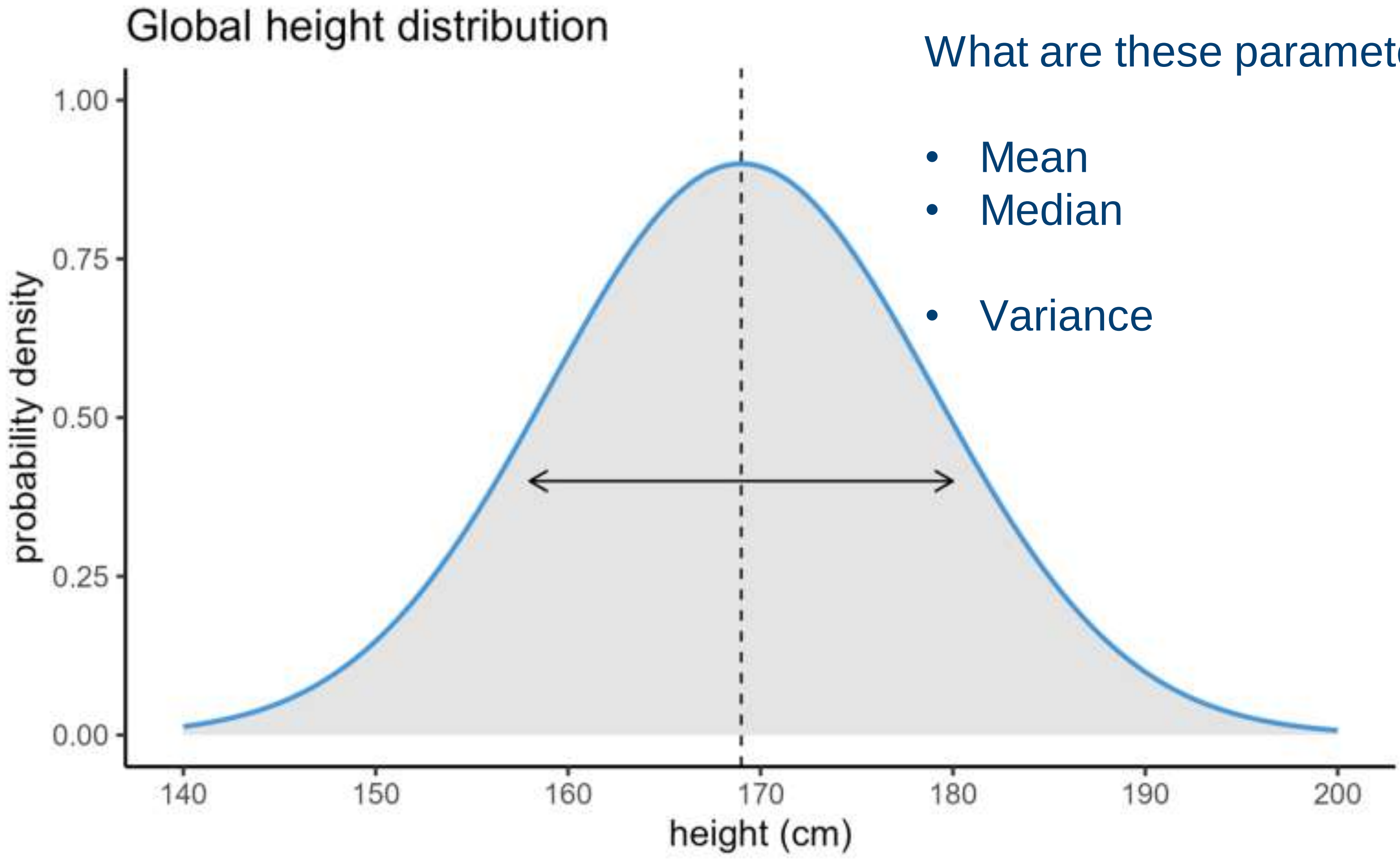
What parameters can we test?



What are these parameters?

- Mean
- Median

What parameters can we test?

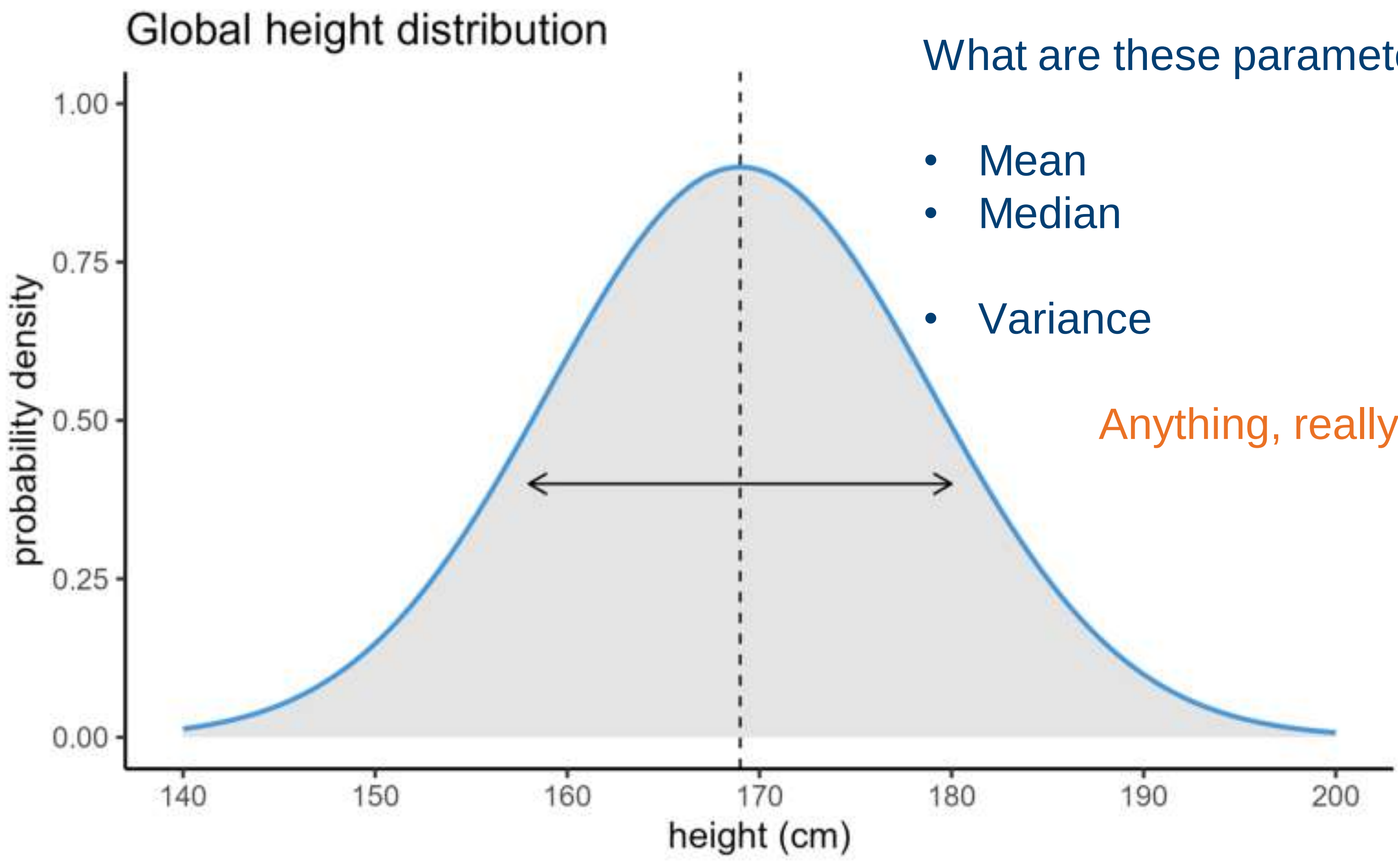


What are these parameters?

- Mean
- Median

- Variance

What parameters can we test?

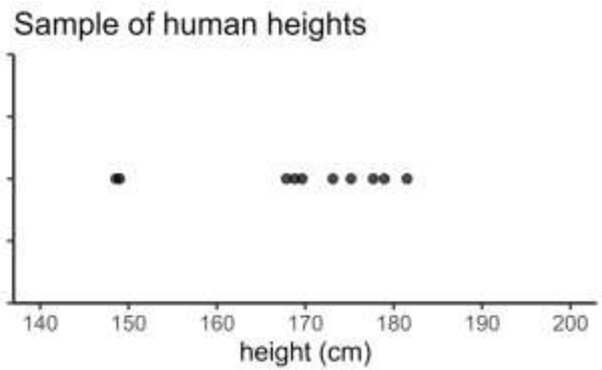


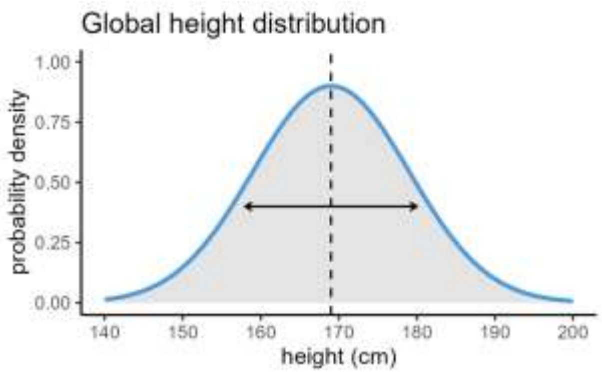
What are these parameters?

- Mean
- Median
- Variance

Anything, really!

Parameters vs statistics





Statistic

- Mean ($\bar{x}$) of sample
- Median ($\tilde{x}$) of sample
- Sample variance (s^2)

Inference

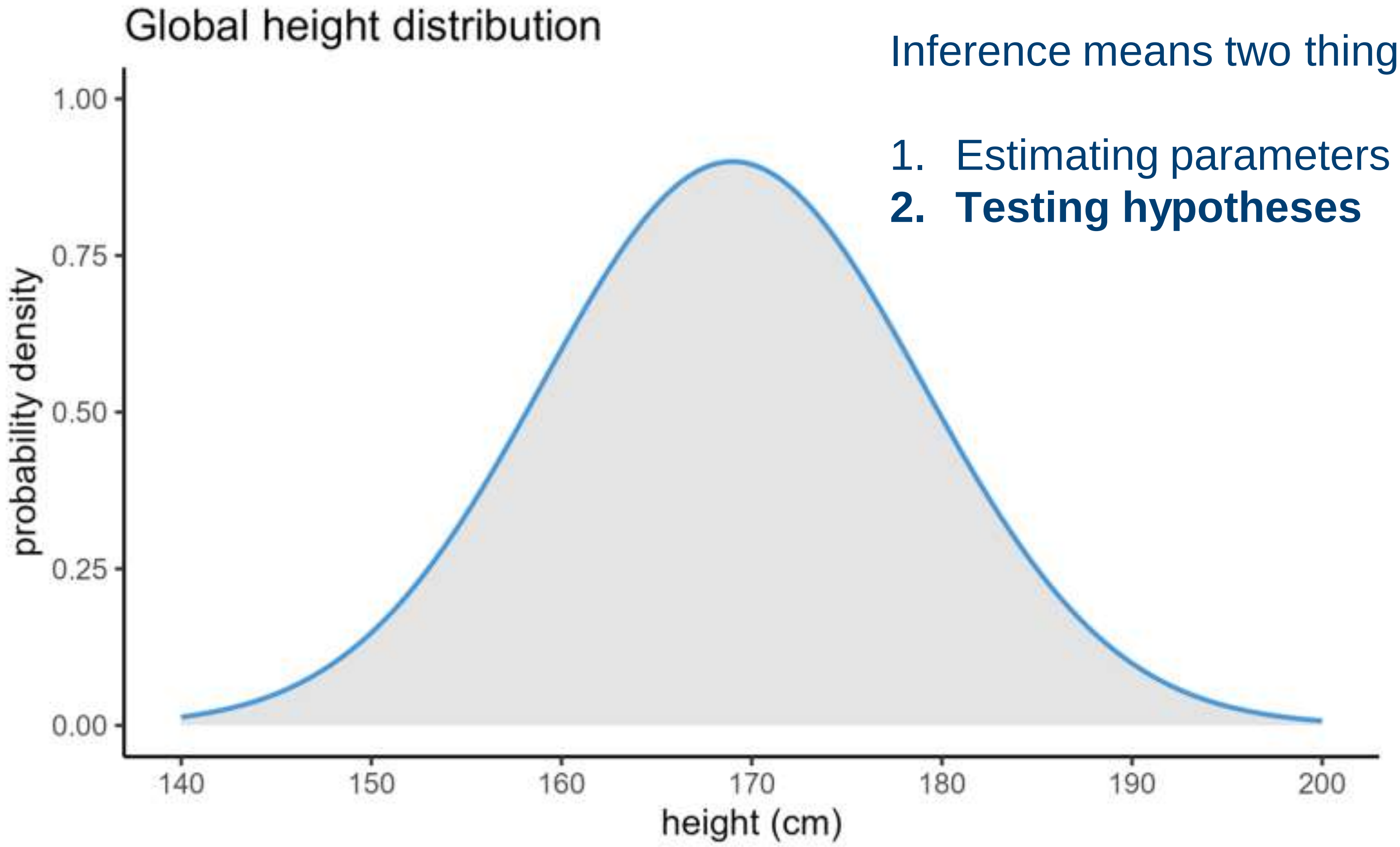
Parameter

- Mean (μ) of population
- Median (m) of population
- Variance (σ^2) of population

3. Hypothesis testing

Testing out theories about your variables

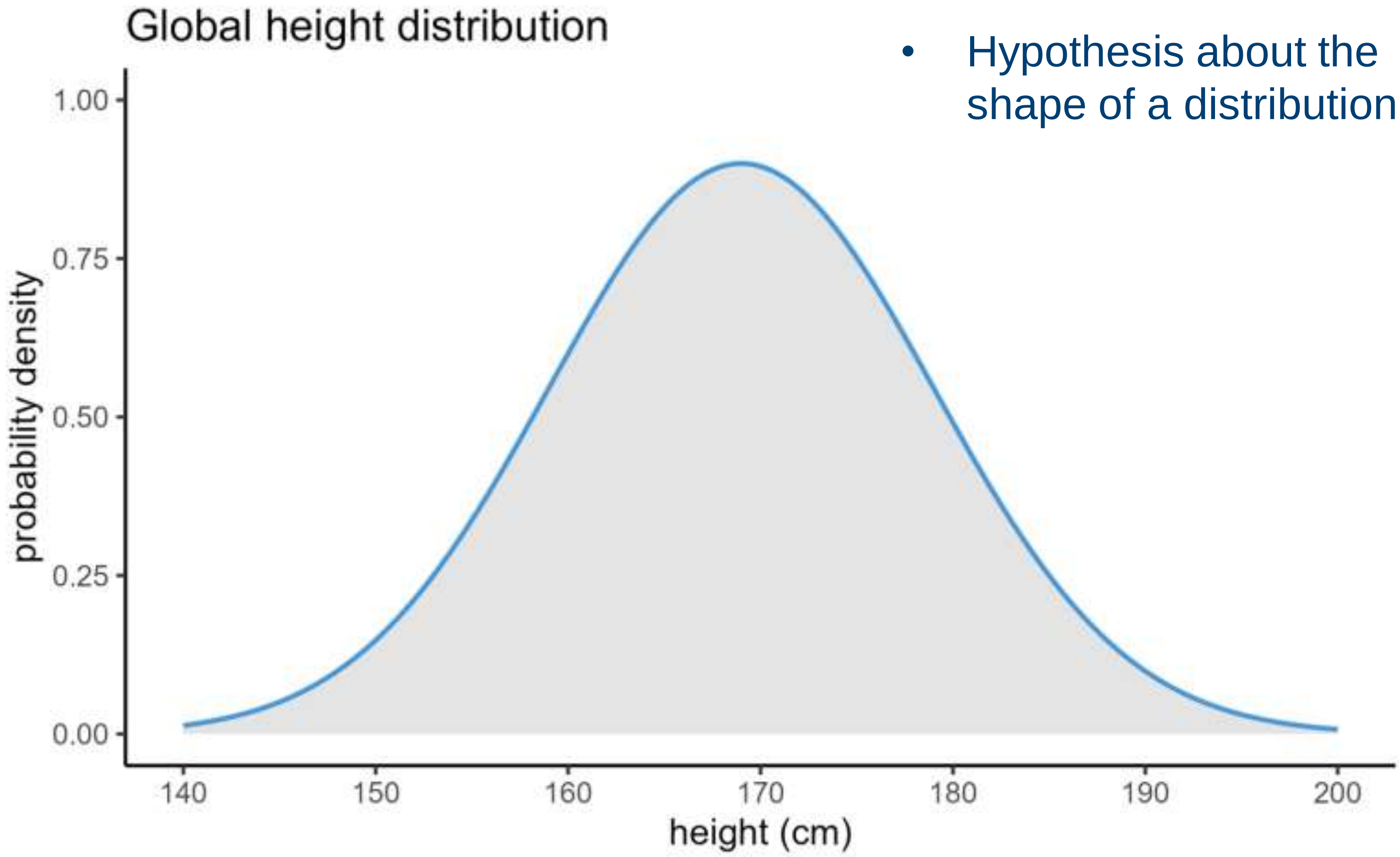
What hypotheses can we test?



Inference means two things:

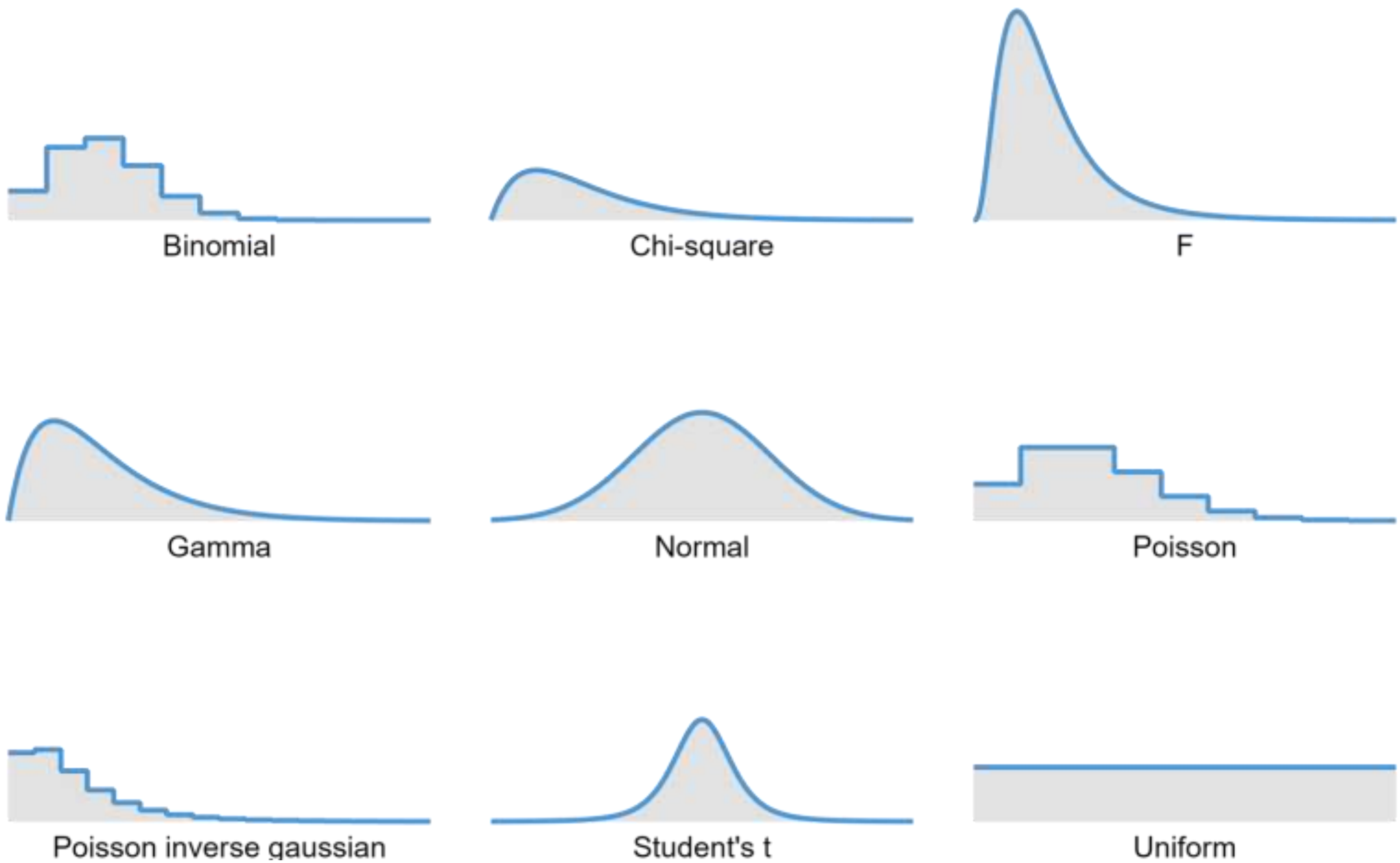
1. Estimating parameters
2. **Testing hypotheses**

What hypotheses can we test?

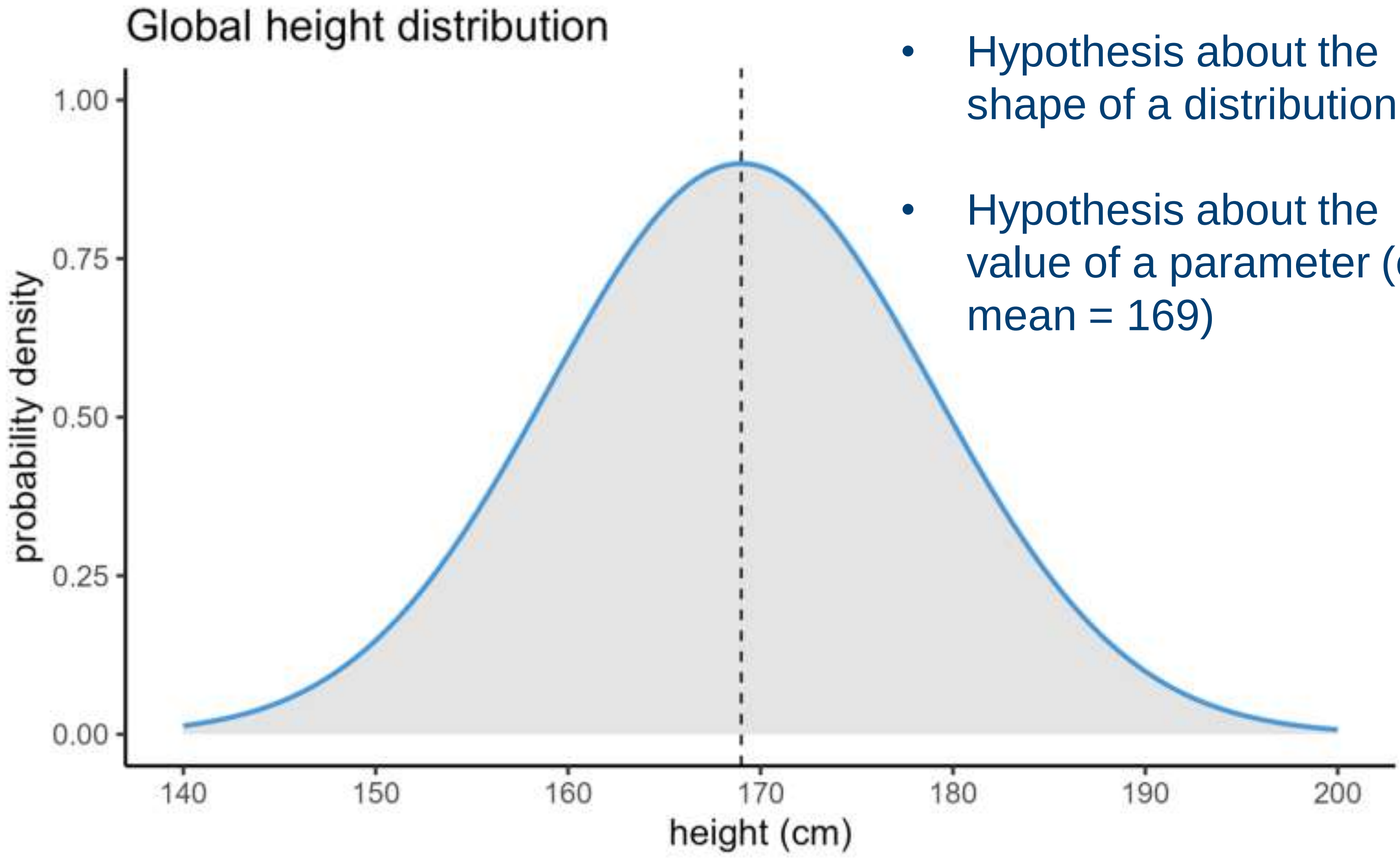


- Hypothesis about the shape of a distribution

What hypotheses can we test?

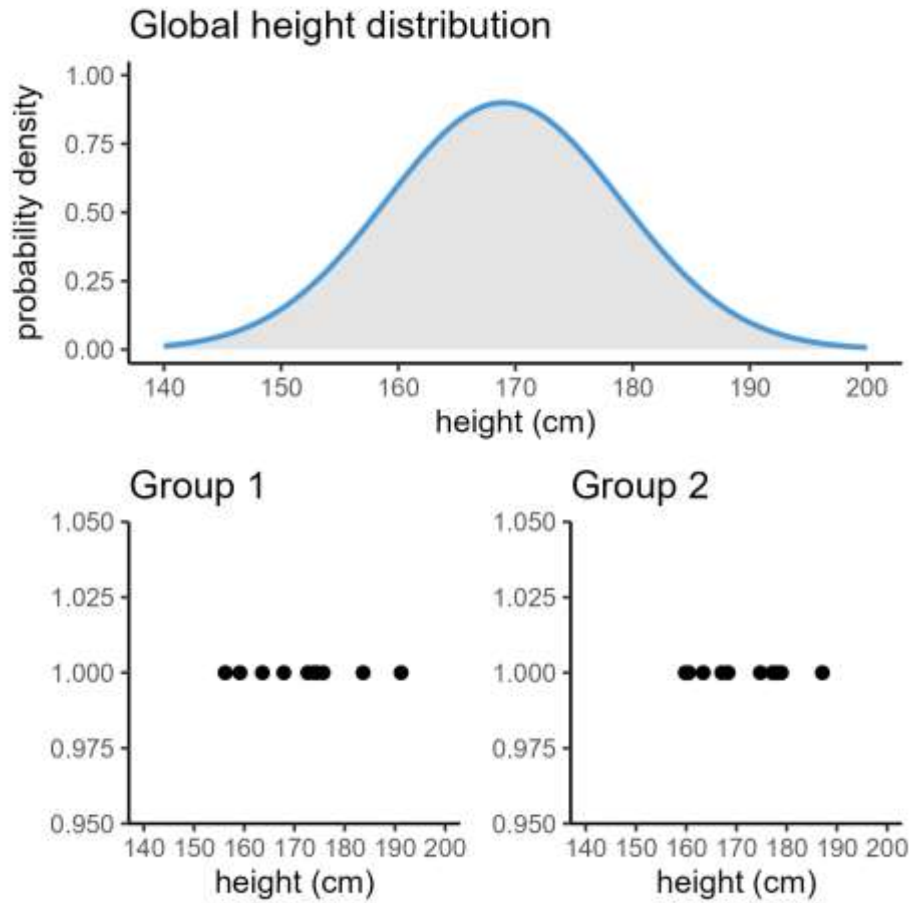


What hypotheses can we test?



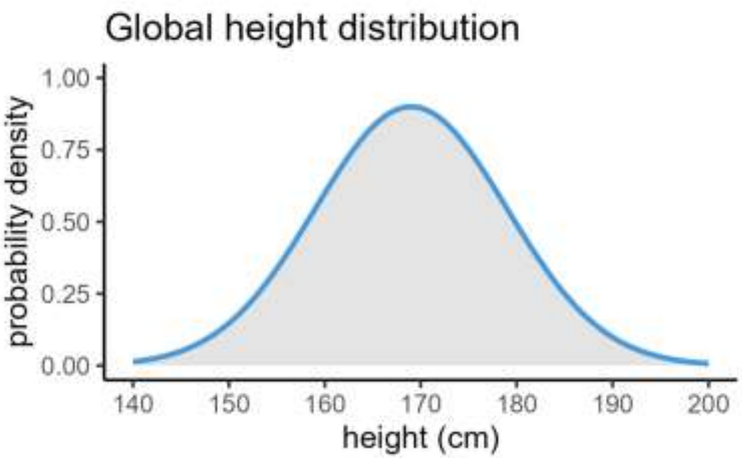
- Hypothesis about the shape of a distribution
- Hypothesis about the value of a parameter (e.g., mean = 169)

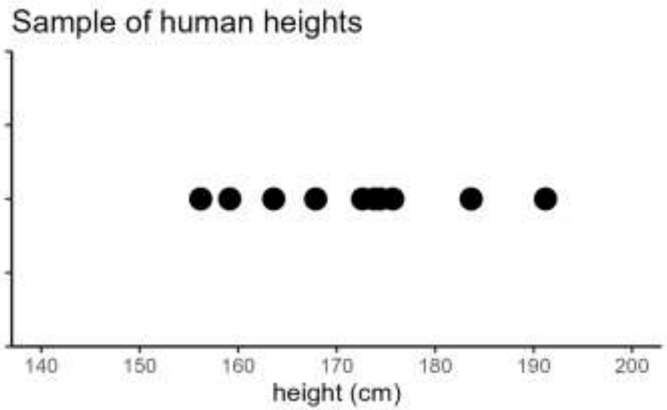
What hypotheses can we test?



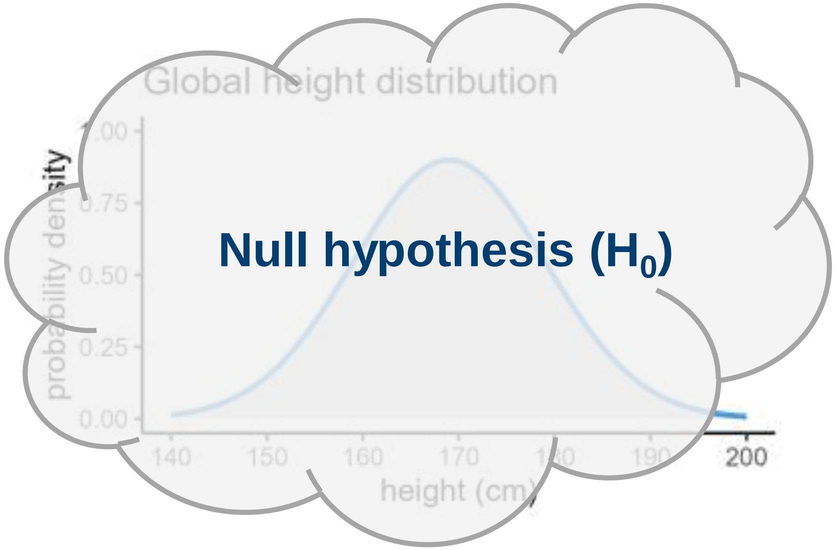
- Hypothesis about the shape of a distribution
- Hypothesis about the value of a parameter (e.g., mean = 169)
- Hypothesis that two groups were sampled from the same distribution

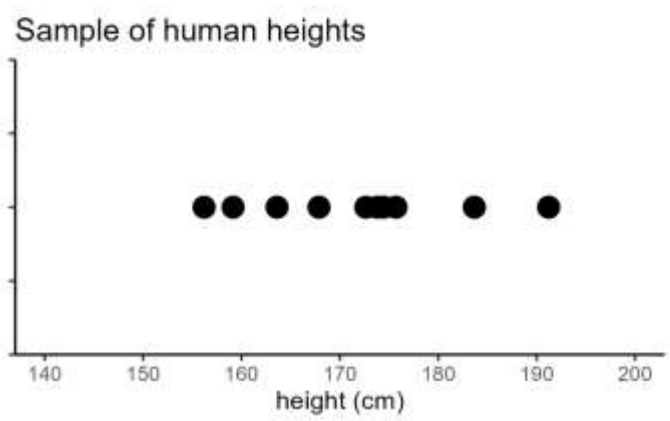
Classical hypothesis testing



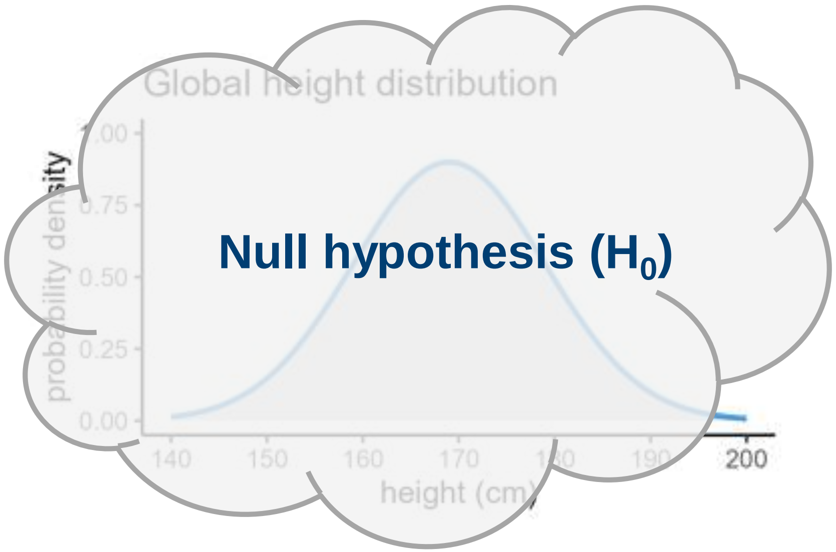


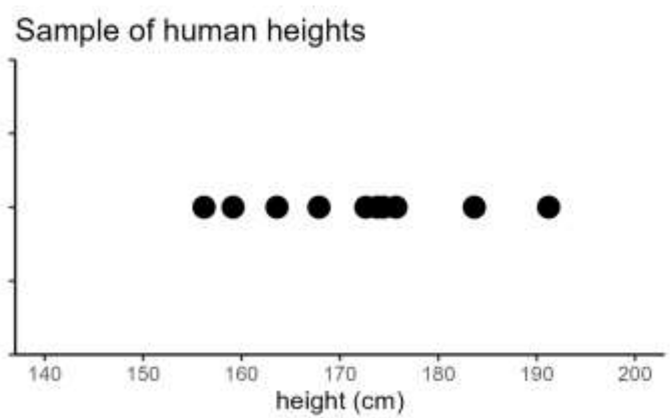
Classical hypothesis testing





Classical hypothesis testing

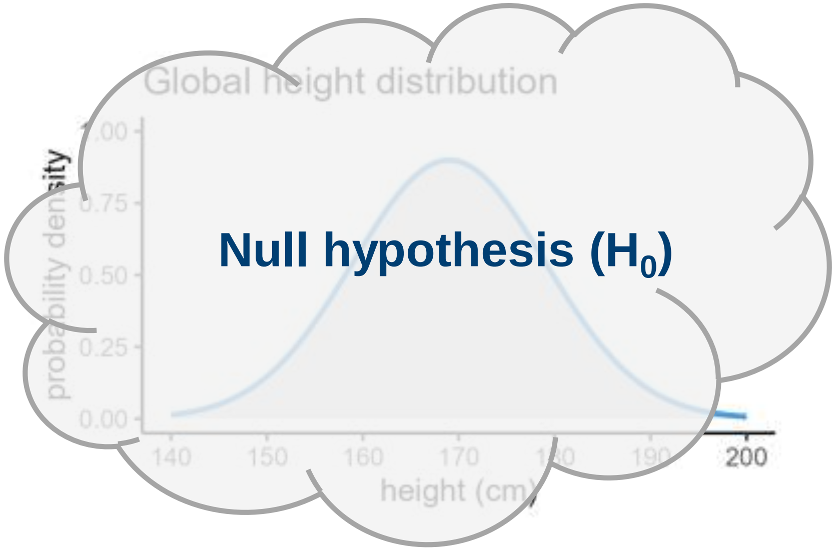


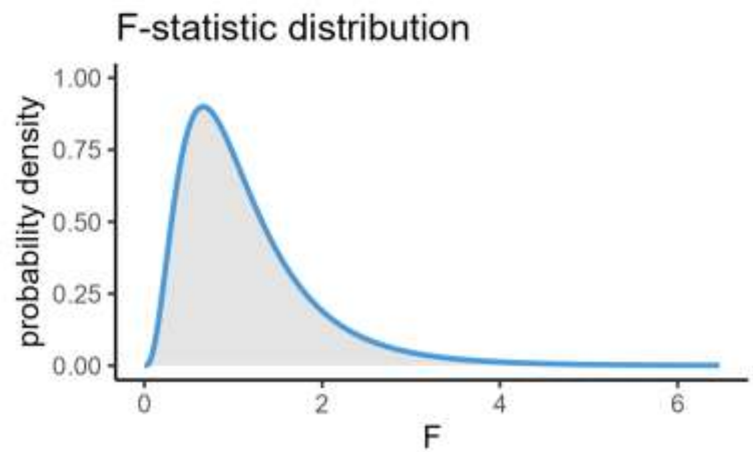


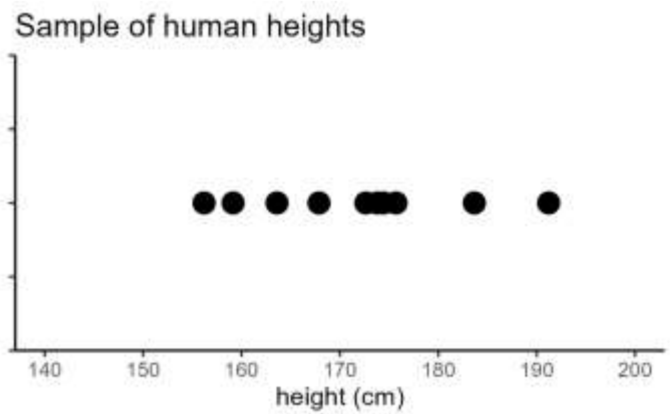
1. Choose test & calculate a test statistic

- Formula that uses the dataset to create a single number
- This number is the **statistic**
- e.g., t-statistic, F-statistic

Classical hypothesis testing



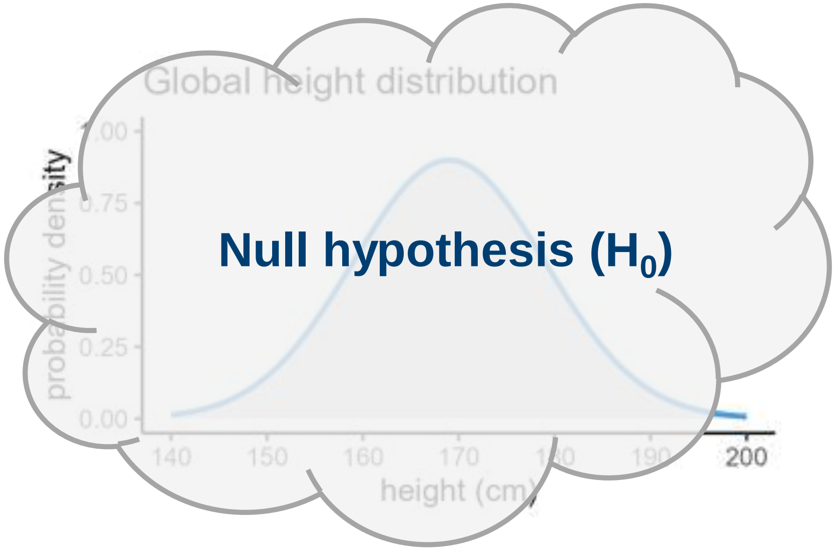


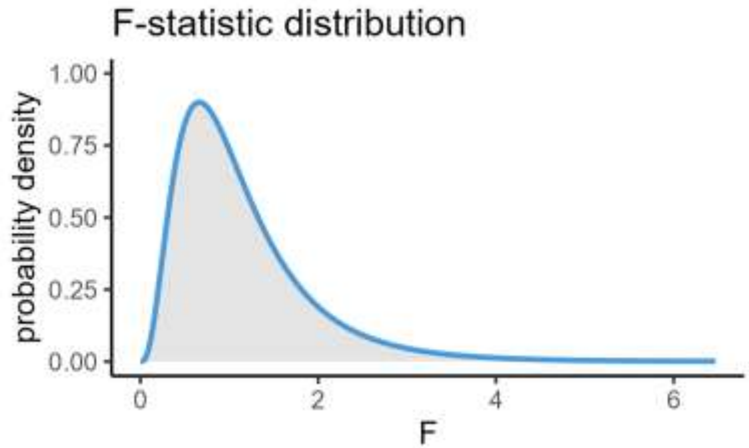


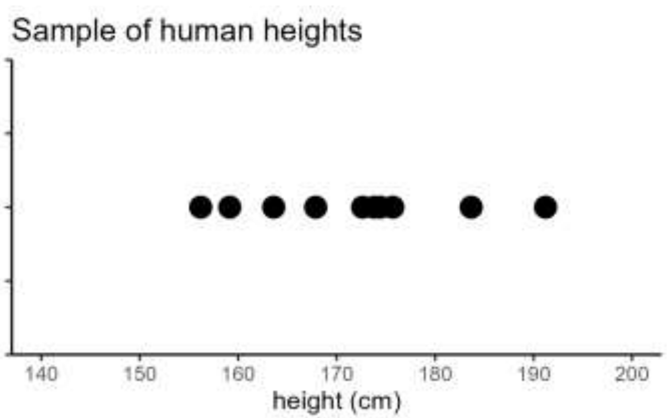
2. Assess probability of test statistic

- Using the distribution of the statistic
- Calculates a p-value

Classical hypothesis testing



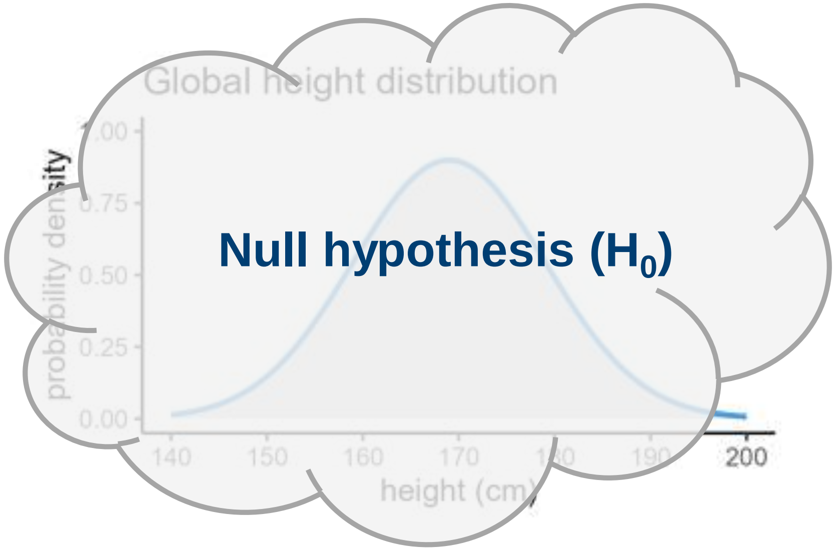


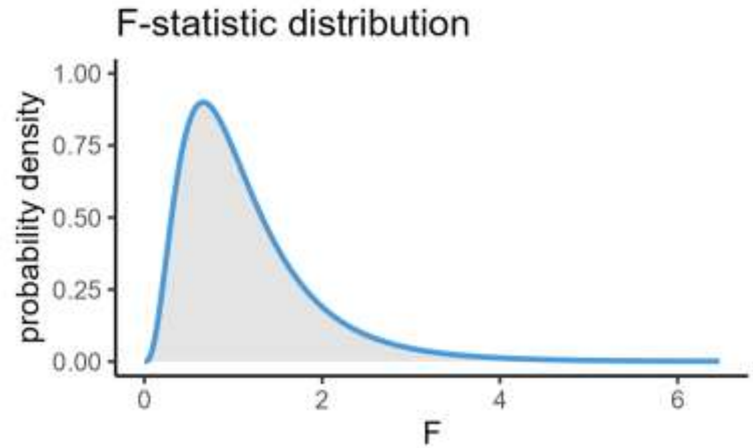


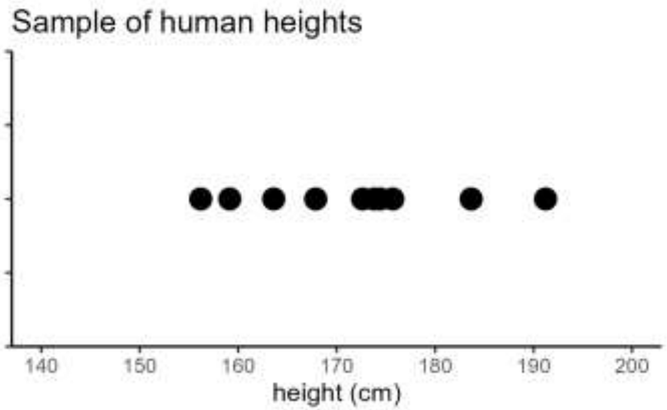
3. Reject or retain null hypothesis

- Based on the size of the p-value

Classical hypothesis testing





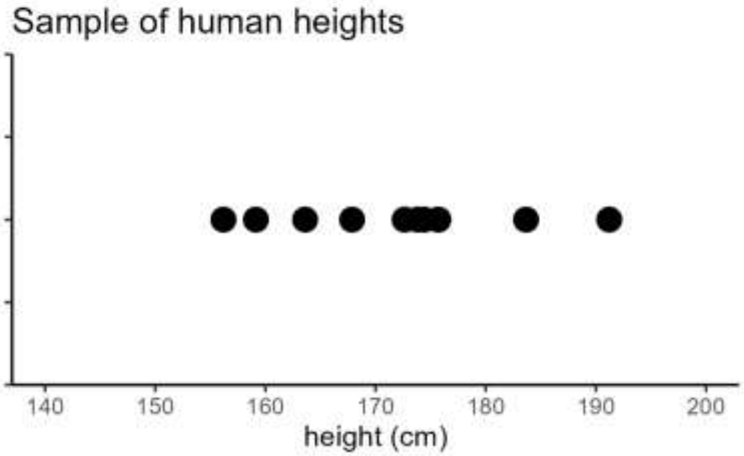


1. Choose test & calculate a test statistic

2. Assess probability of test statistic

3. Reject or retain null hypothesis

Hypothesis testing in practice



Hypothesis testing in practice

Hypothesis testing in practice

1. Determine your null hypothesis

Hypothesis testing in practice

1. Determine your null hypothesis

2. Choose appropriate statistical test

Hypothesis testing in practice

1. Determine your null hypothesis

2. Choose appropriate statistical test

3. Computer calculates test statistic and associated p-value

Hypothesis testing in practice

1. Determine your null hypothesis
2. Choose appropriate statistical test
3. Computer calculates test statistic and associated p-value
4. Interpret result based on research question

Hypothesis testing in practice

1. Determine your null hypothesis
2. Choose appropriate statistical test
3. Computer calculates test statistic and associated p-value
4. Interpret result based on research question

Interpreting results and p-values

p-value = probability of observing the data, if the null hypothesis is true

- Property of the dataset
- Not equal to the probability of the null hypothesis being true
- Cannot provide “proof”

This means that all statistical tests are probabilistic and can be wrong!

4. Assumptions

How do we choose a statistical test?

Choosing a statistical test

Research question

What do you want to find out?

Choosing a statistical test

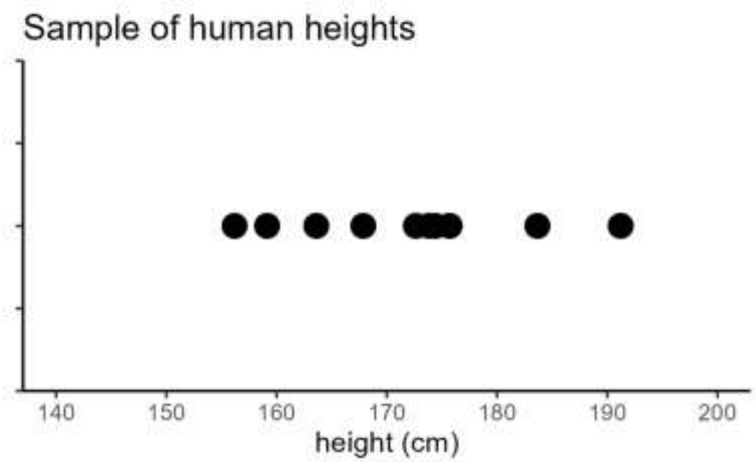
Research question

- Difference between groups?
- Predicting an outcome?
- Association between variables?

What do you want to find out?

Choosing a statistical test

Research question



What do you want to find out?

What are your data like?

What are your data like?

All statistical tests are based on expectations about your dataset, known as **assumptions**

What are your data like?

All statistical tests are based on expectations about your dataset, known as assumptions

Things you need to know

- Are the variables categorical or continuous?

- Are your groups paired or independent?

…

What are your data like?

All statistical tests are based on expectations about your dataset, known as assumptions

Things you need to know

- Are the variables categorical or continuous?

- Are your groups paired or independent?

…

Things you need to find out

- Does your variable follow a normal distribution?

- Do your groups have equal variance?

…

Assumptions

All statistical tests are based on expectations about your dataset, known as assumptions

What happens if your data don’t meet all the assumptions of a test?

- You **cannot trust** the results of that test
- You might be able to **choose an alternative** test

Session 1: Statistical inference

Lecture summary:

- We make inferences about a population (distribution) from a sample (dataset)

- We test hypotheses by calculating statistics and p-values
- The size of the p-value represents the probability of the dataset

- All statistical tests have assumptions that you have to check before you can trust the results

Session 1: Statistical inference

Course materials:

Performing t-tests in R/Python

- One vs two samples
- Paired vs independent samples

Testing assumptions

- Normality
- Equality of variance

Core statistics

Session 1: Statistical inference

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Core statistics

Session 2: Categorical predictors

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Plan for the session

Session 2: **Categorical predictors**

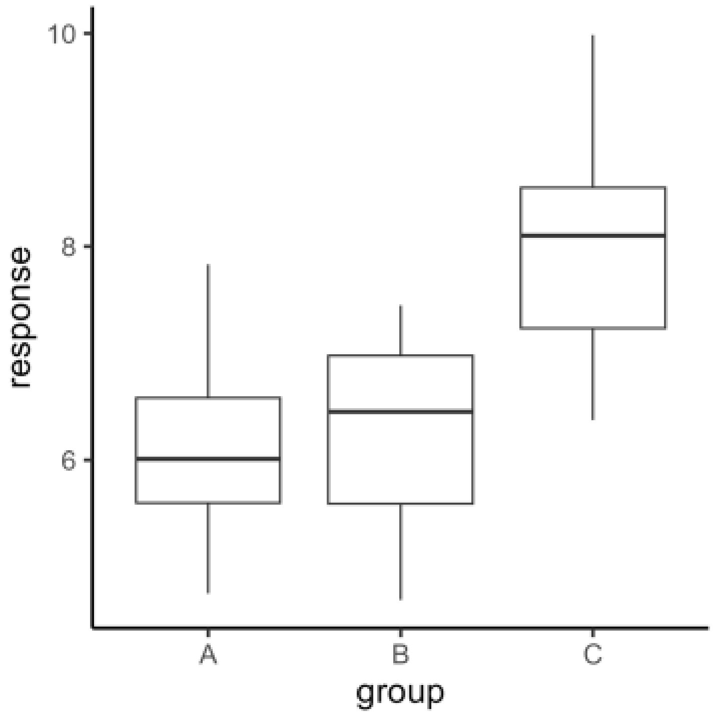
1. Comparing multiple groups
2. Analysis of variance
3. Post hoc tests

1. Comparing multiple groups

Expanding beyond t-tests

Problem

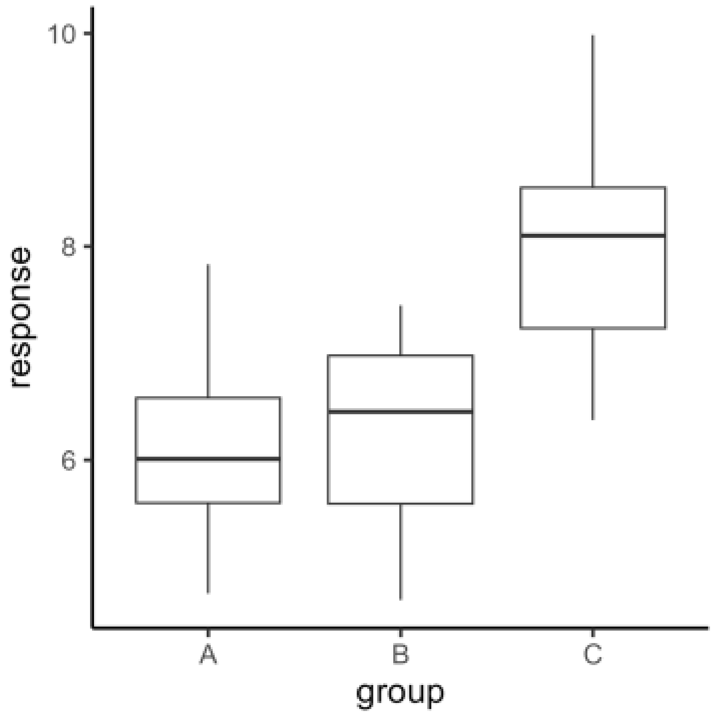
Were these groups sampled from the same distribution?



- Do all the groups have the same mean?
- If not, which groups are different?

Approach

Technique called one-way ANOVA



Assumptions

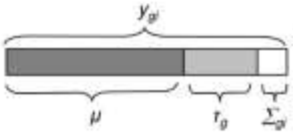
- Normally distributed
- Equal variance
- Independent groups

Classical ANOVA Explanation

Partitioning of individual values

Each weight can be partitioned into three components:

$$y_{gi} = \mu + \tau_g + \varepsilon_{gi}$$



where,

μ = overall mean
τ_g = the group effect
ε_{gi} = the random error component

Partitioning variation

You saw in the previous lectures how variation can be best described by using the Sum of Squared Devations (*SS*) from the mean.

$$s_y^2 = \frac{\sum_{i=1}^{n}(y_i - \bar{y})^2}{n-1}$$

We now need to construct *SS* for each component, in order to obtain

$$SS_{Tot} = SS_G + SS_E$$

Let us visualise each of this components:

Additive SS

SST = SSG + SSE

$$SST = \sum_{g=1}^{k}\sum_{i=1}^{n}(y_{gi} - \bar{y})^2 = \sum_{g=1}^{k}\sum_{i=1}^{n}\left[(y_{gi} - \bar{y}_g) + (\bar{y}_g - \bar{y})\right]^2 =$$

$$= \sum_{g=1}^{k}\sum_{i=1}^{n}\left[(y_{gi} - \bar{y}_g)^2 + (\bar{y}_g - \bar{y})^2 + 2(y_{gi} - \bar{y}_g)\cdot(\bar{y}_g - \bar{y})\right] =$$

$$= \sum_{g=1}^{k}\sum_{i=1}^{n}(y_{gi} - \bar{y}_g)^2 + \sum_{g=1}^{k}\sum_{i=1}^{n}(\bar{y}_g - \bar{y})^2 +$$

$$+ 2\sum_{g=1}^{k}\sum_{i=1}^{n}(y_{gi} - \bar{y}_g)\cdot(\bar{y}_g - \bar{y})$$

$$SST = \underbrace{\sum_{g=1}^{k}\sum_{i=1}^{n}(y_{gi} - \bar{y}_g)^2}_{\text{SSE}} + \underbrace{\sum_{g=1}^{k}\sum_{i=1}^{n}(\bar{y}_g - \bar{y})^2}_{\text{SSG}} +$$

$$+ 2\sum_{g=1}^{k}\sum_{i=1}^{n}(y_{gi} - \bar{y}_g)\cdot(\bar{y}_g - \bar{y}) \quad = 0$$

$$\sum_{g=1}^{k}\left[(\bar{y}_g - \bar{y})\cdot\sum_{i=1}^{n}(y_{gi} - \bar{y}_g)\right] = \qquad \boxed{\bar{y}_g = \sum_{i=1}^{n} y_{gi} / n}$$

$$\sum_{g=1}^{k}\left[(\bar{y}_g - \bar{y})\cdot\left(\sum_{i=1}^{n} y_{gi} - n\bar{y}_g\right)\right] = \sum_{g=1}^{k}\left[(\bar{y}_g - \bar{y})\cdot\left(\cancel{n\bar{y}_g - n\bar{y}_g}\right)\right]$$

Total Sum of Squares (SS_{Tot})

$$SS_{Tot} = \sum_{g=1}^{k}\sum_{i=1}^{n_g}(y_{gi} - \bar{y})^2$$

In the meerkat example, the overall mean is 572.6. So, the total sum of squares is:

$$SSTot = (514 - 572.6)^2 + (519 - 572.6)^2 +$$

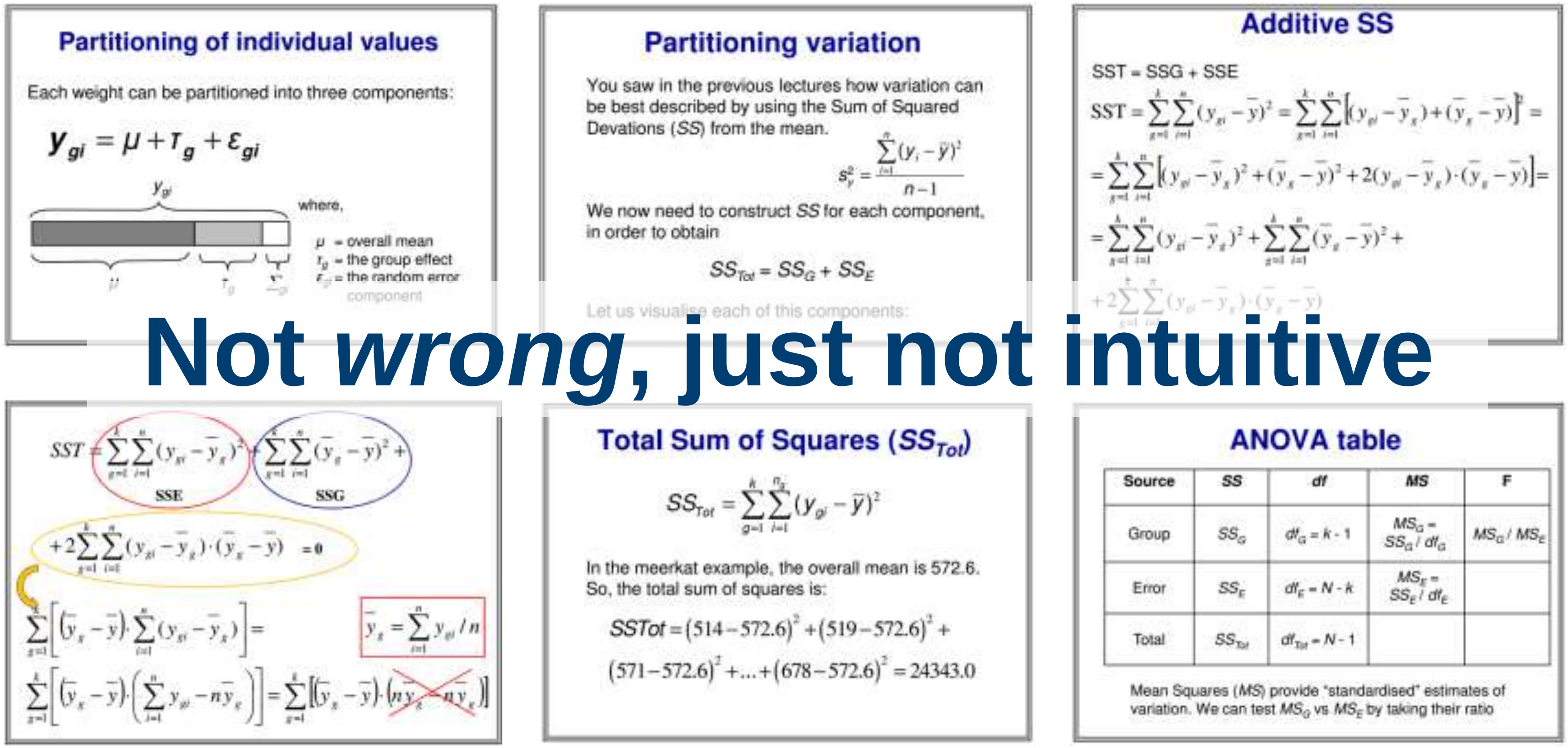
$$(571 - 572.6)^2 + \ldots + (678 - 572.6)^2 = 24343.0$$

ANOVA table

Source	*SS*	*df*	*MS*	F
Group	SS_G	$df_G = k - 1$	$MS_G = SS_G / df_G$	MS_G / MS_E
Error	SS_E	$df_E = N - k$	$MS_E = SS_E / df_E$	
Total	SS_{Tot}	$df_{Tot} = N - 1$		

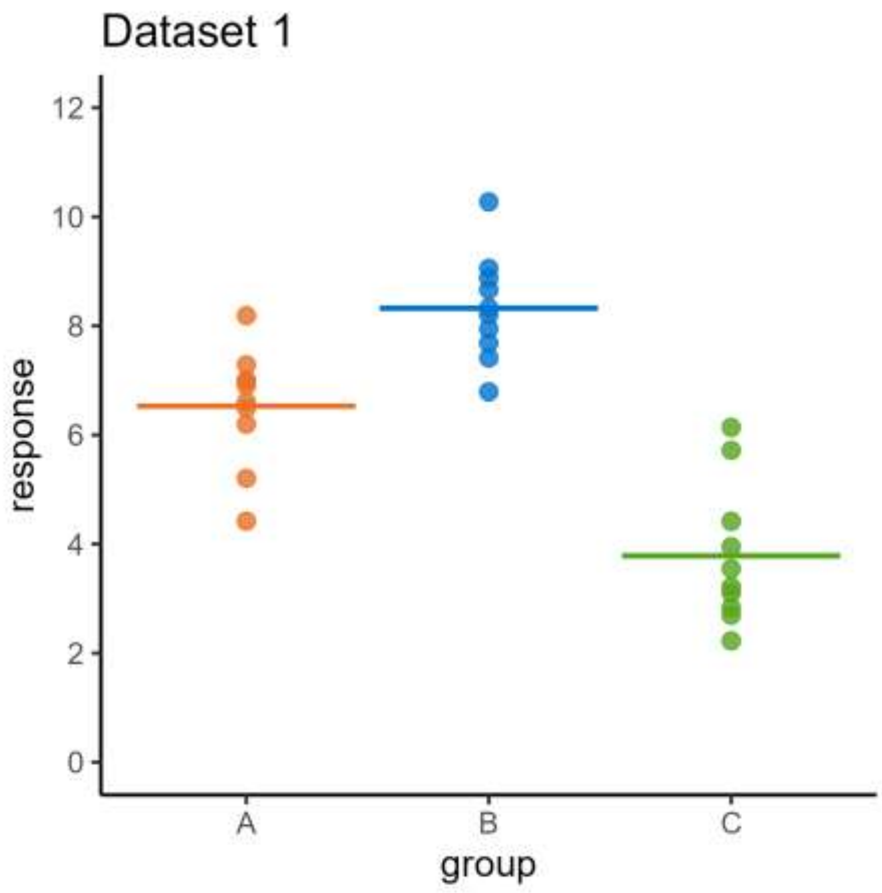
Mean Squares (*MS*) provide "standardised" estimates of variation. We can test MS_G vs MS_E by taking their ratio

Classical ANOVA Explanation



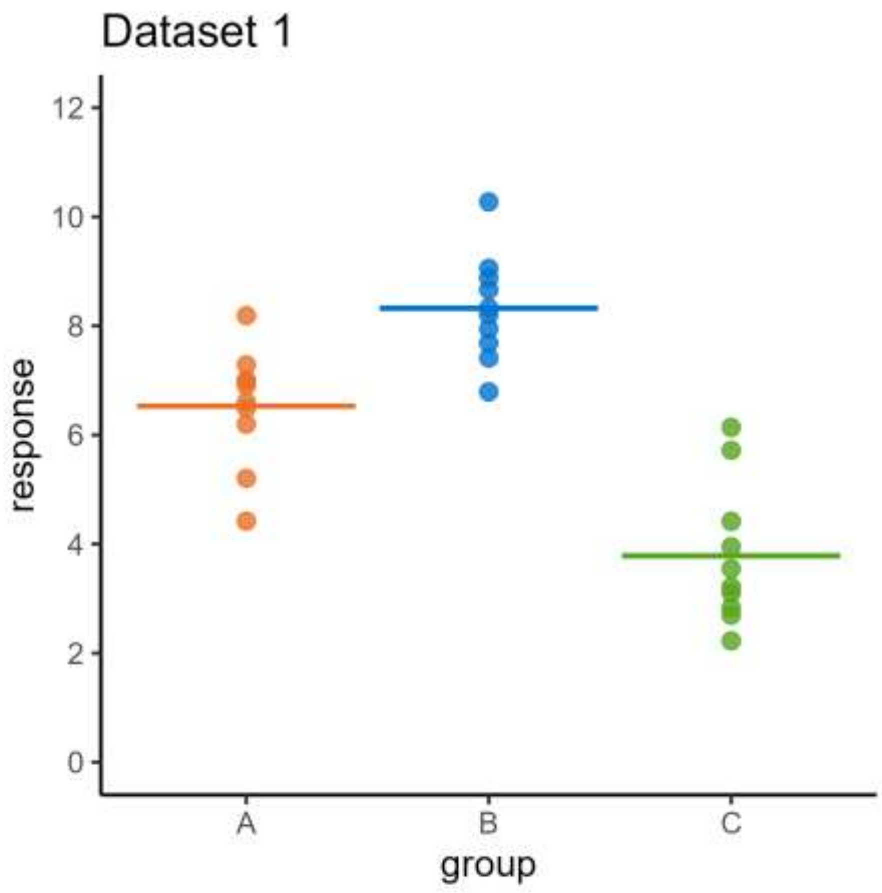
Not *wrong*, just not intuitive

Intuitive ANOVA explanation



- 3 groups
- 10 points per group
- Equal variances
- Is there a difference between the groups?

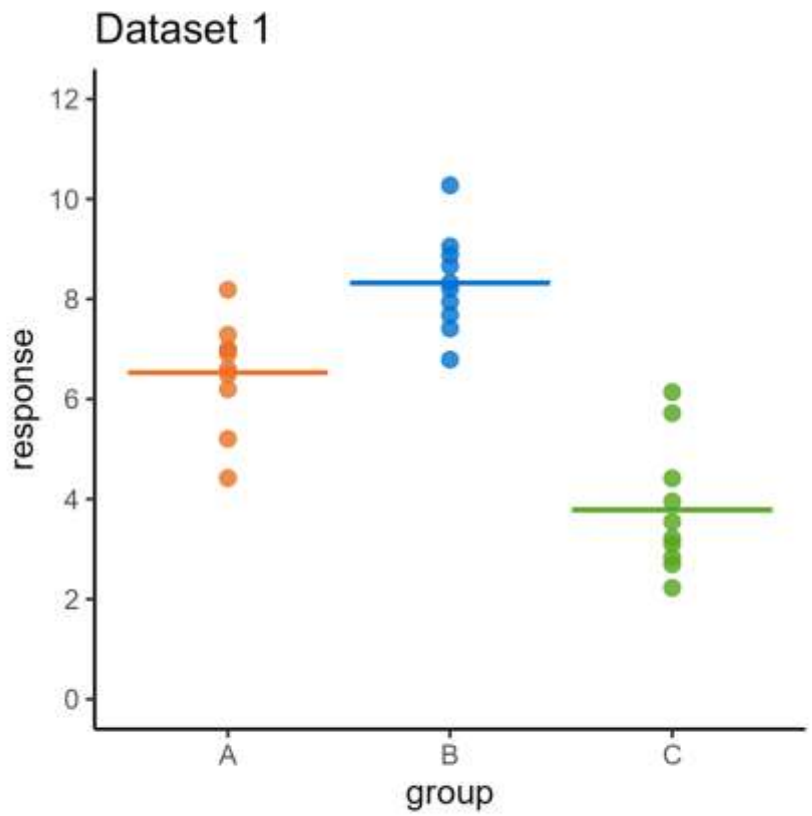
Intuitive ANOVA explanation

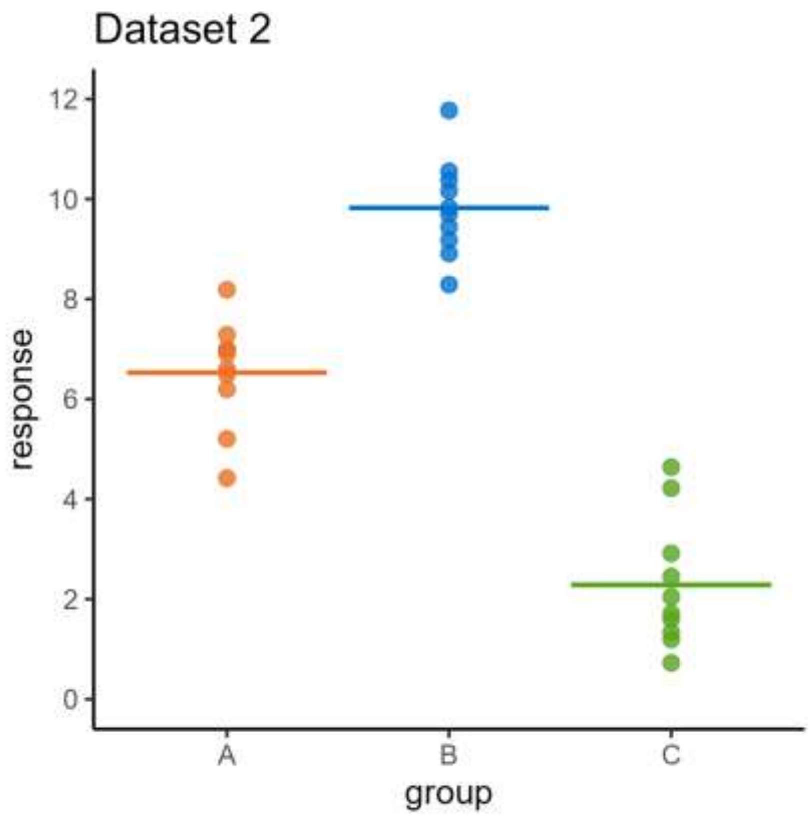


Two questions:

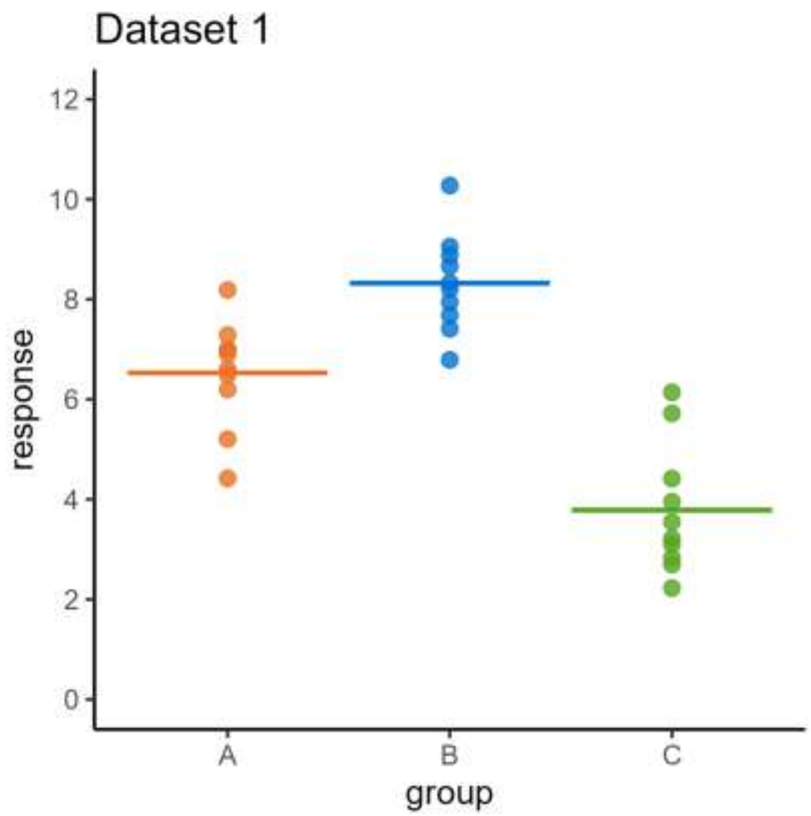
1. What affects our intuitive sense of **difference**?
2. How can we **measure** difference?

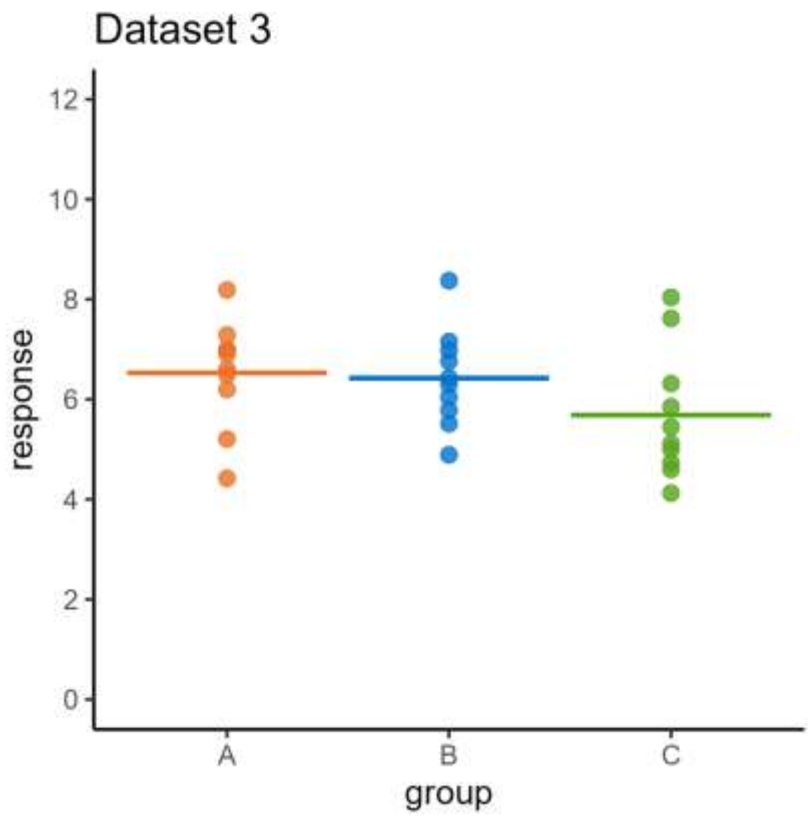
What affects our sense of difference?



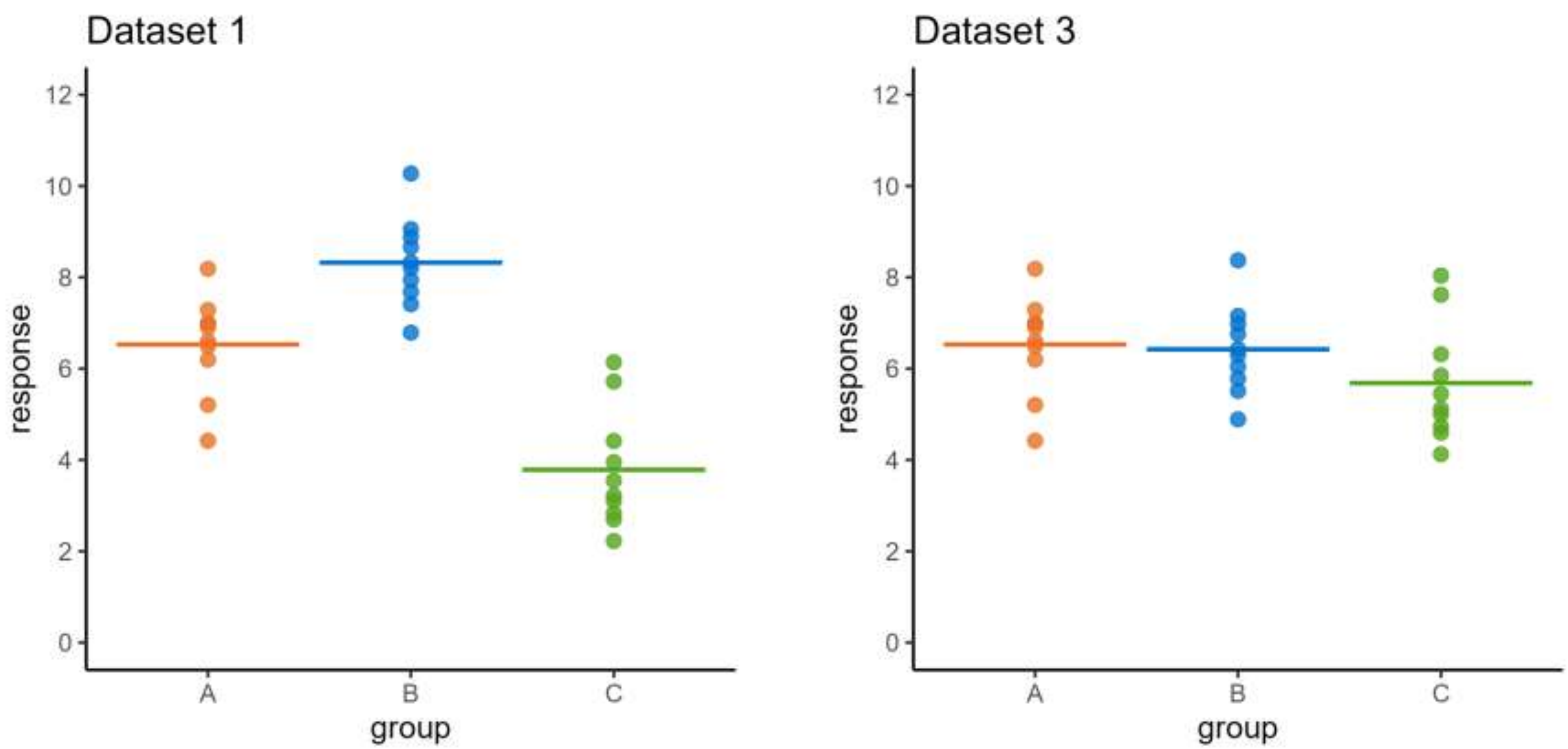


What affects our sense of difference?



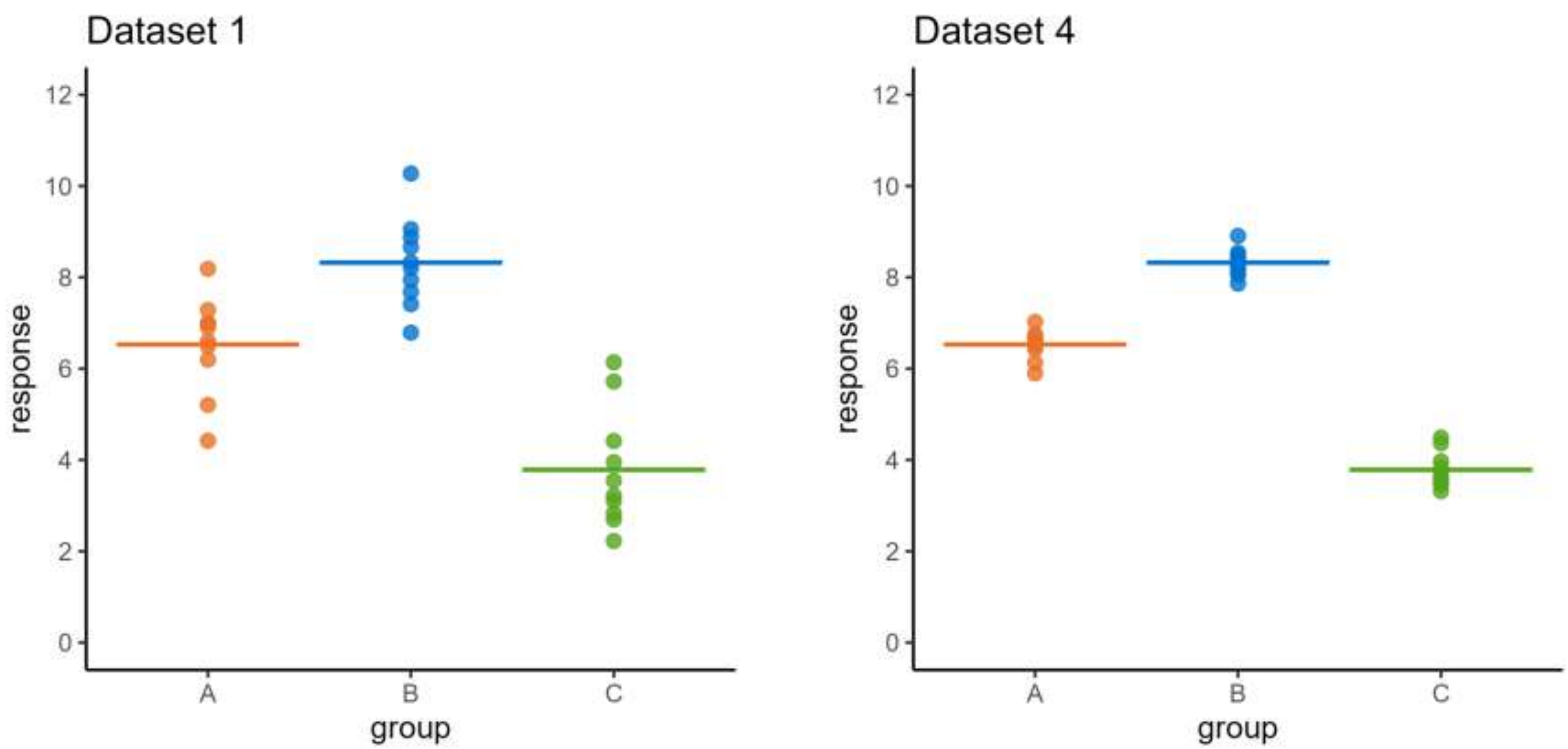


What affects our sense of difference?

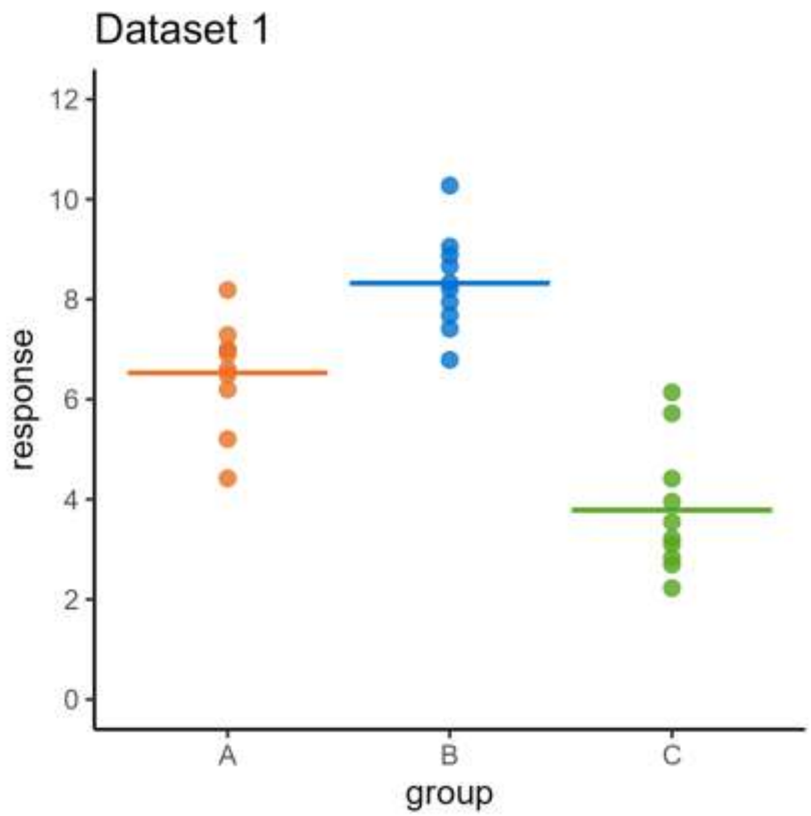


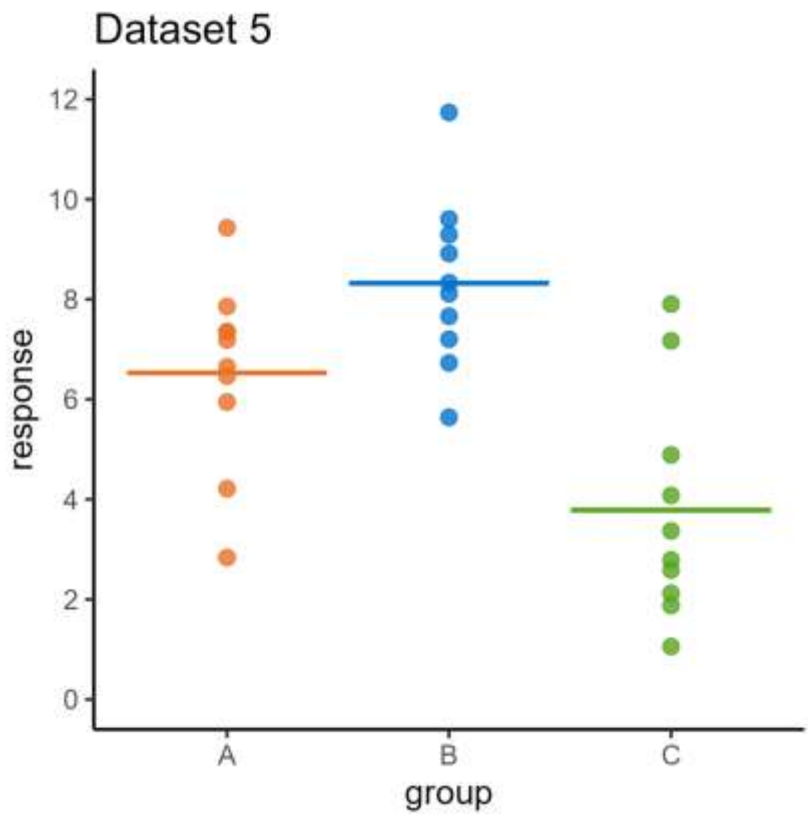
The spread of the means

What affects our sense of difference?



What affects our sense of difference?





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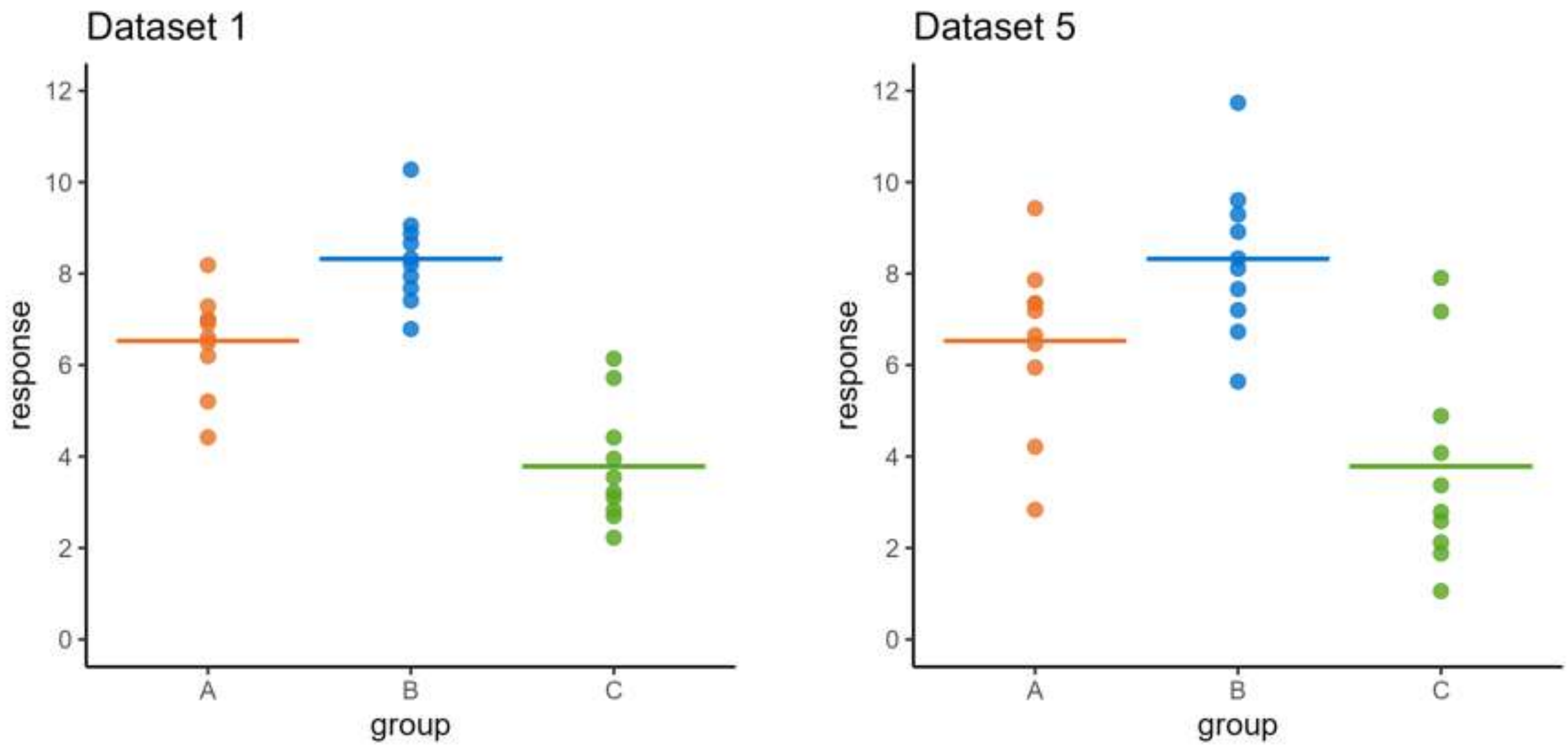
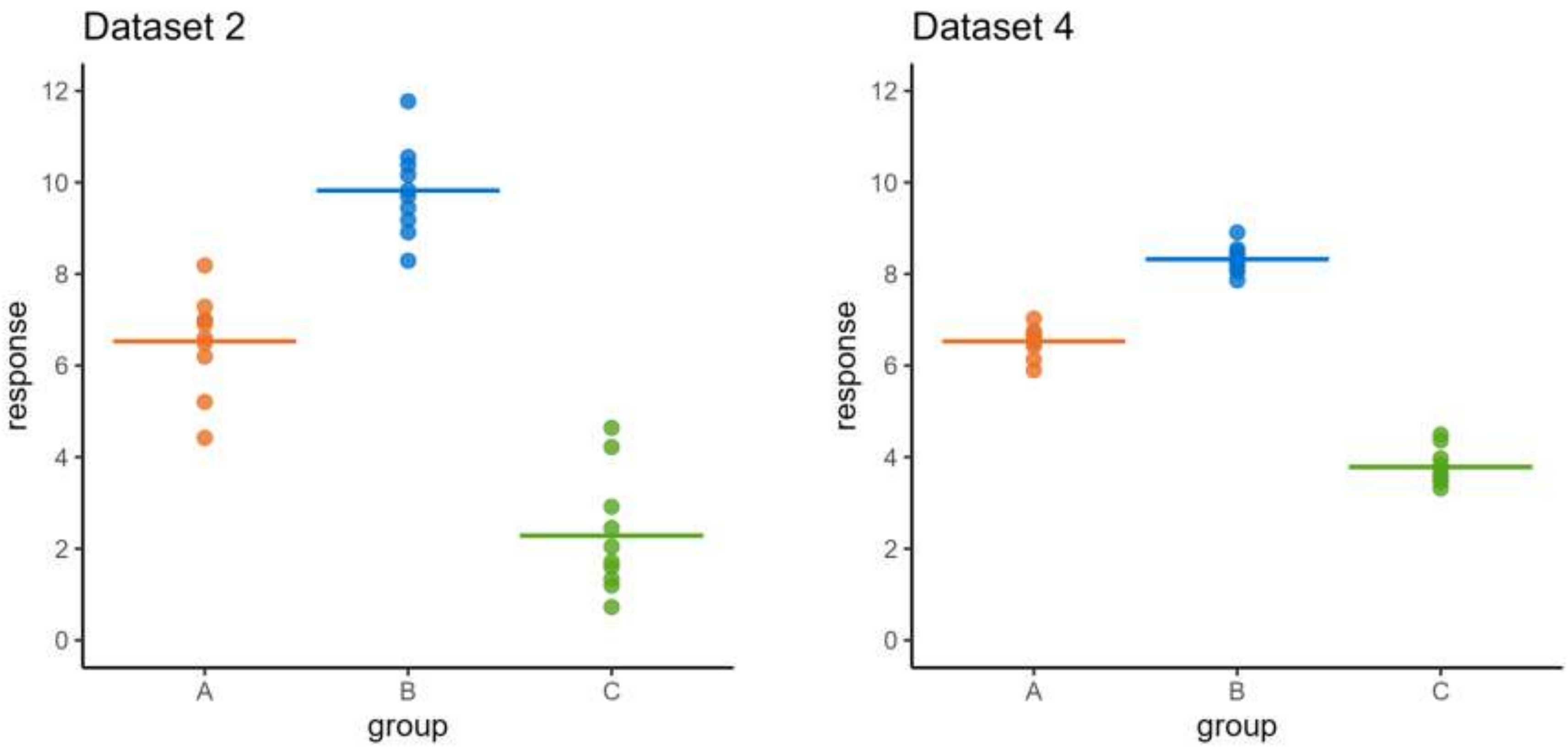


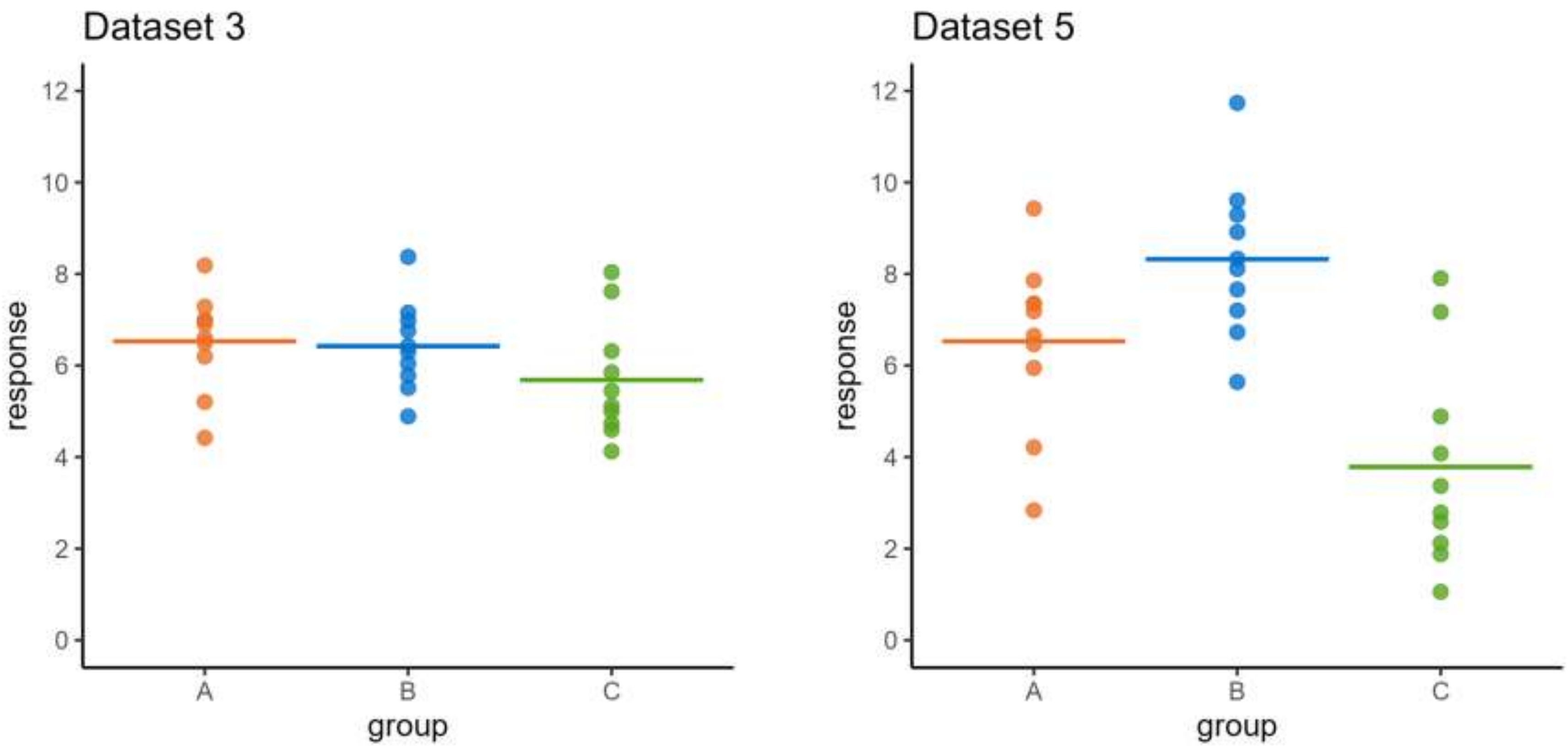
The spread of the data (within the groups)

What affects our sense of difference?



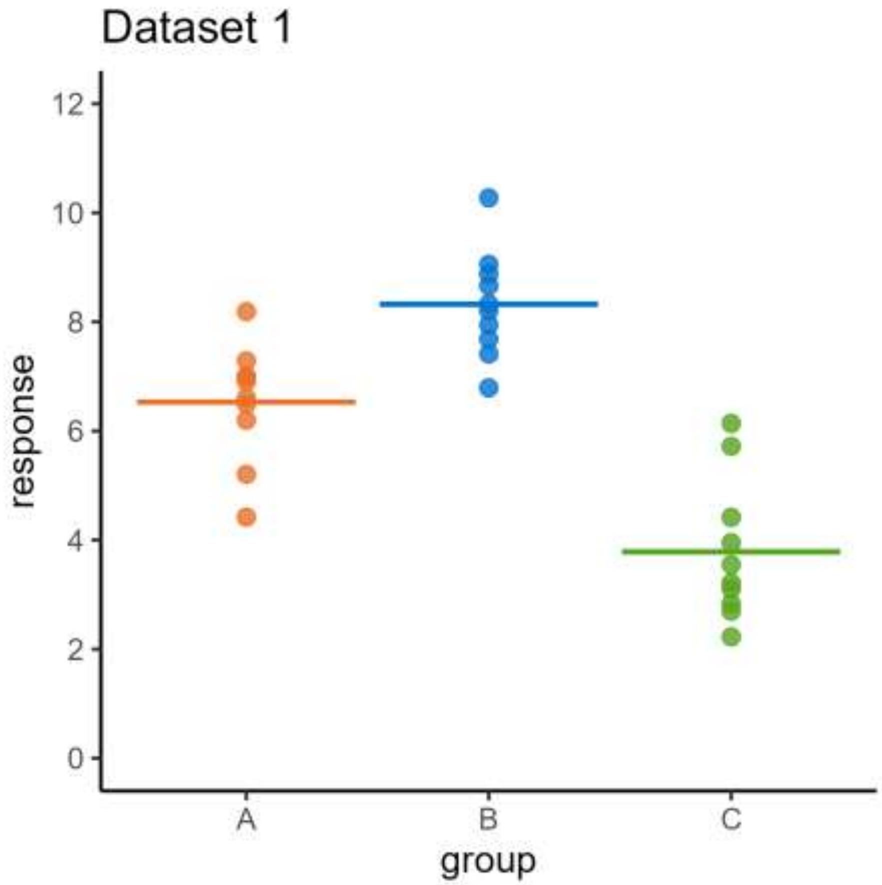
Groups different when the spread of means is big relative to spread of data

What affects our sense of difference?



Groups not different when the spread of means is small relative to spread of data

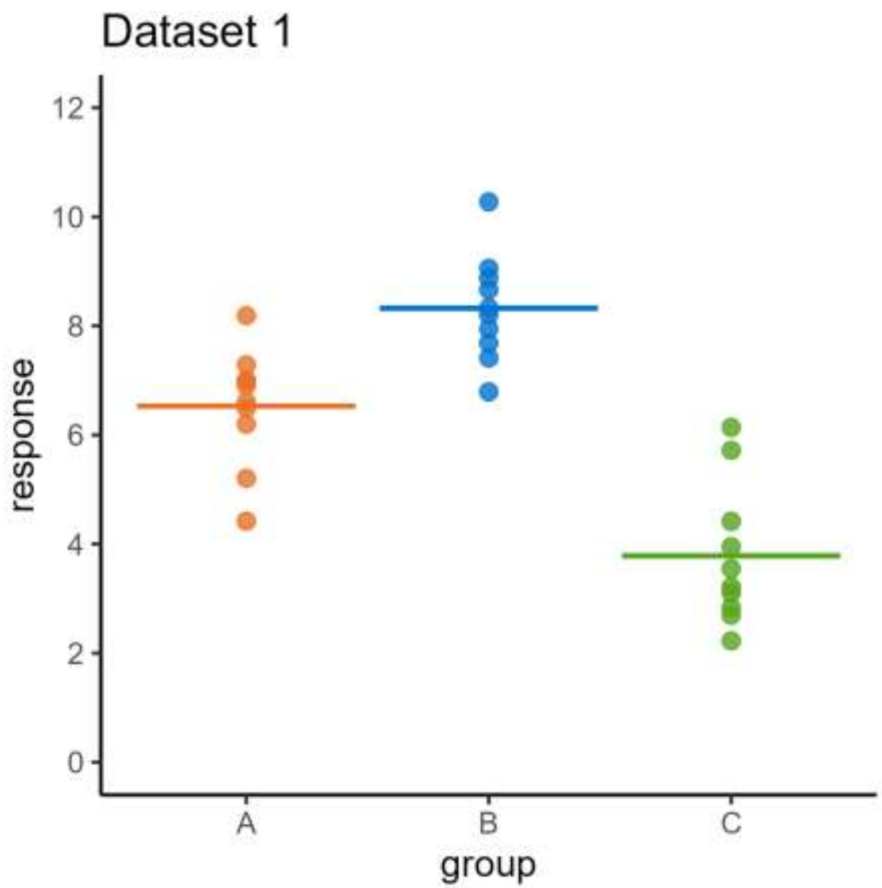
How do we measure difference?



We need two numbers:

- One to measure the spread of the means
- One to measure the spread of the data

How do we measure difference?



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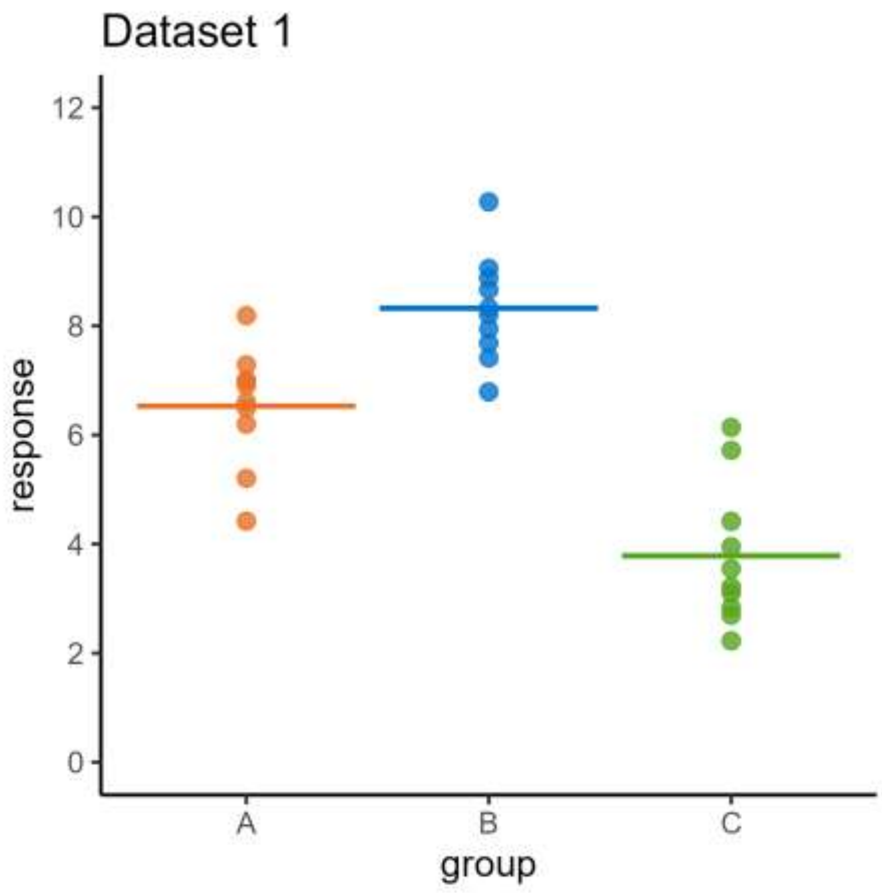
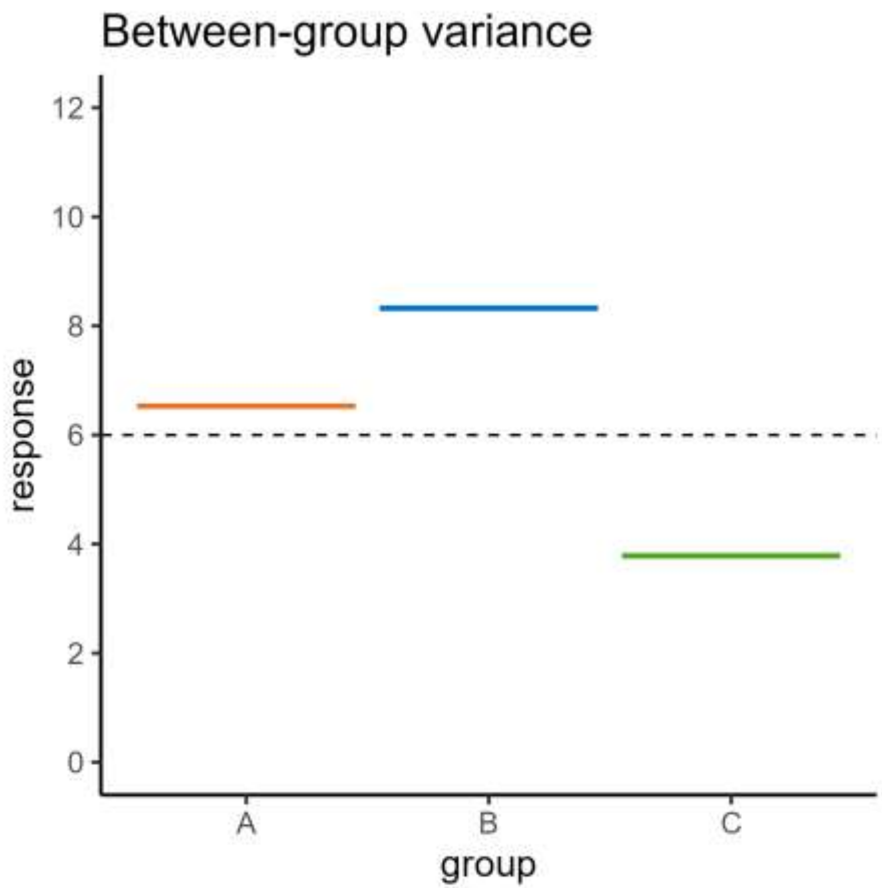


We need two numbers:

- One to measure the spread of the means
- One to measure the spread of the data

Use the **variance** of the means

How do we measure difference?



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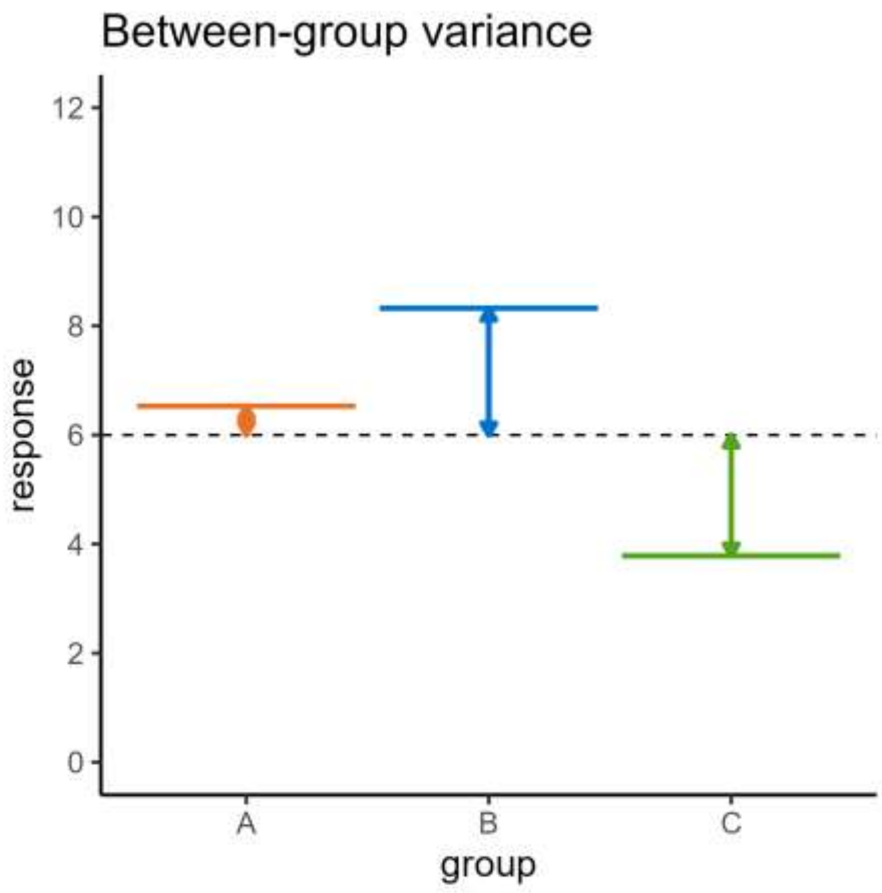

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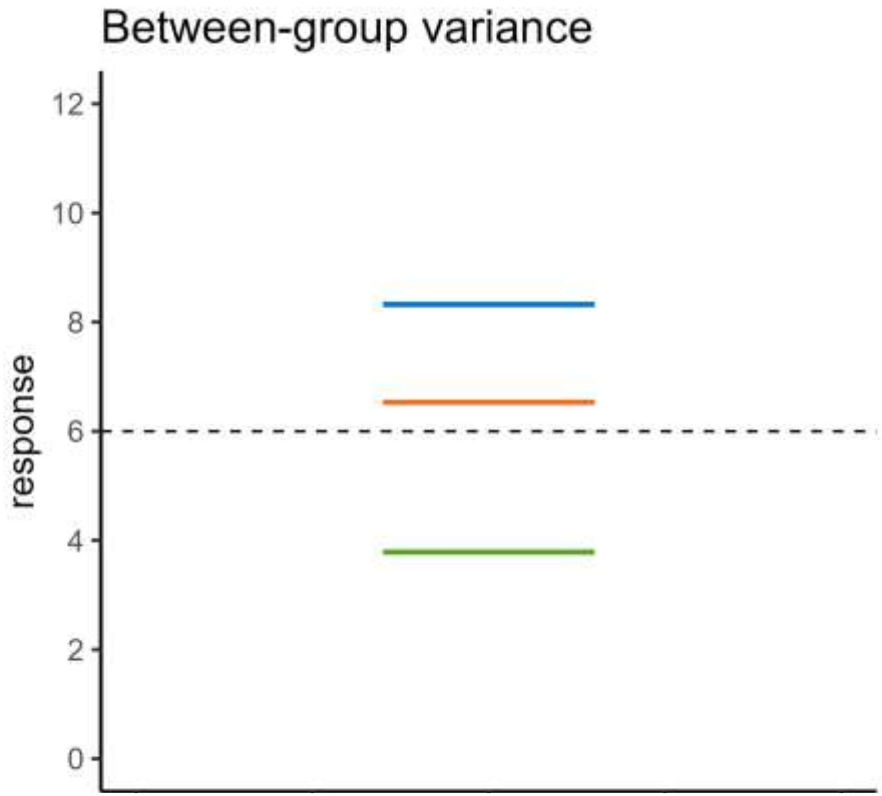

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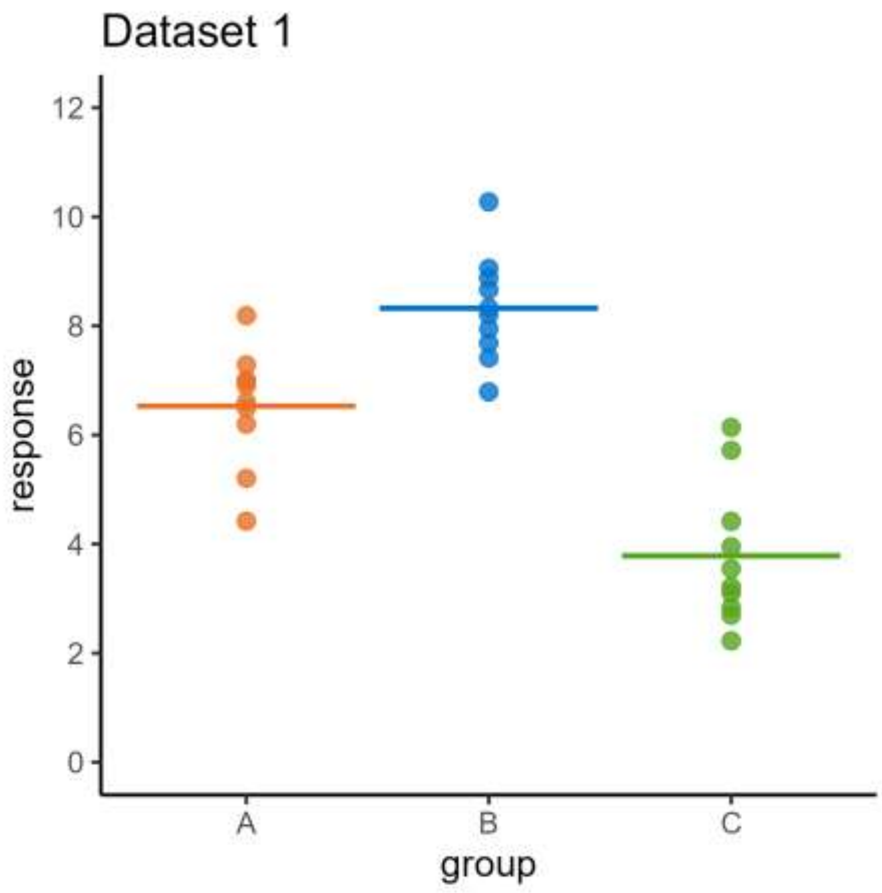

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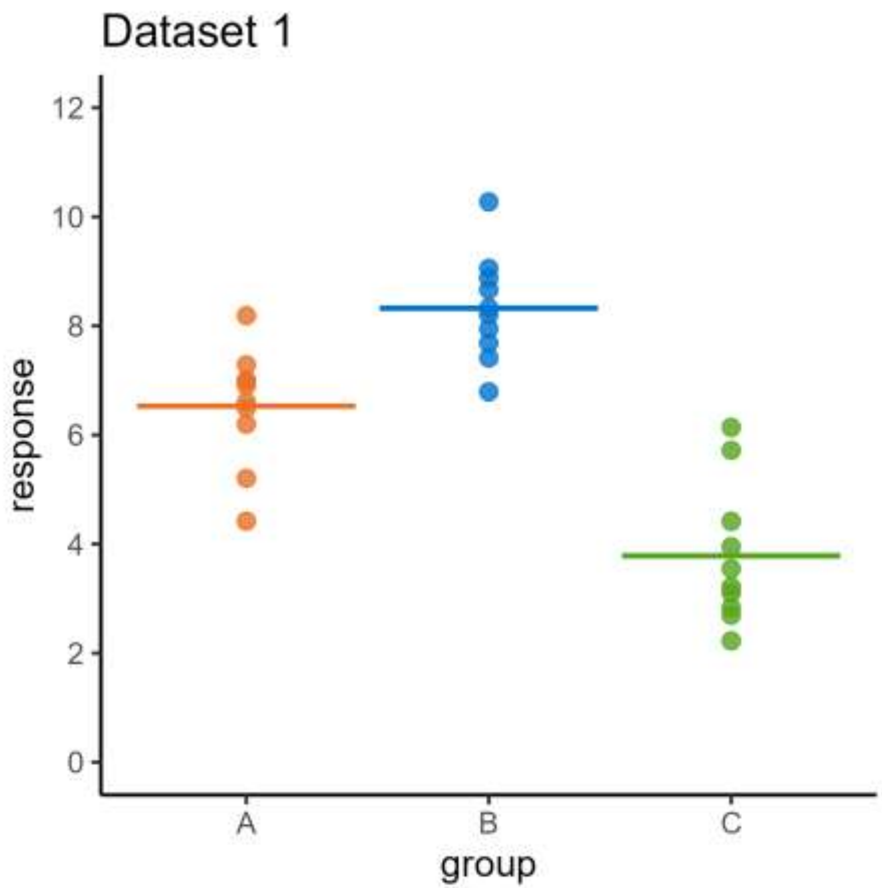

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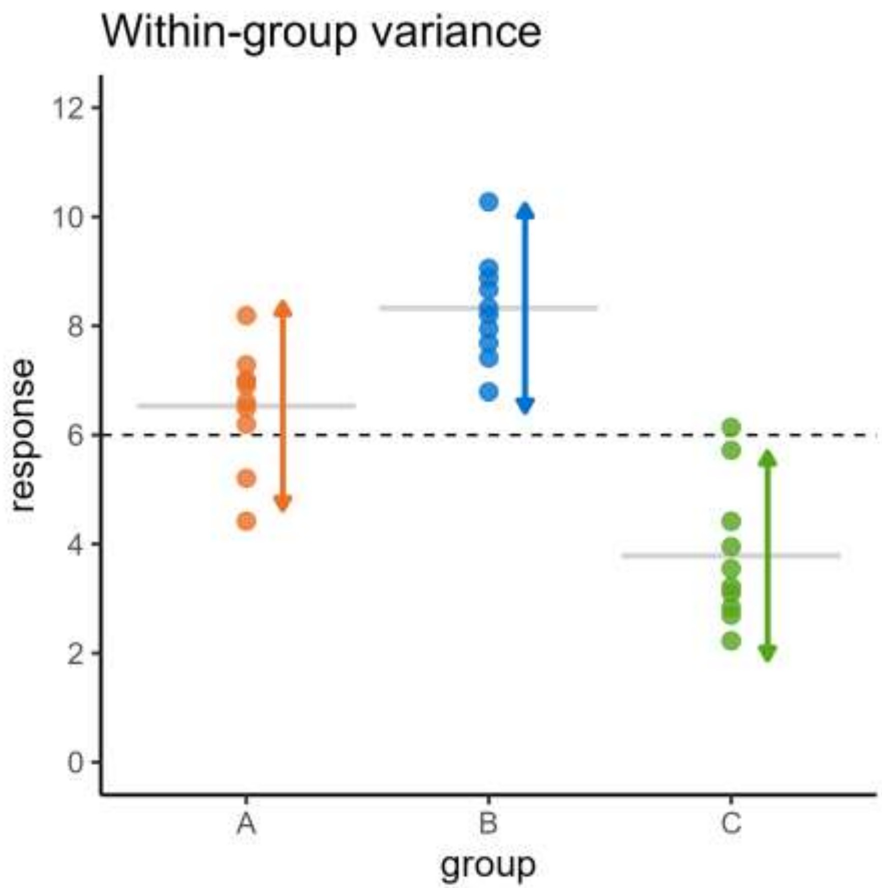

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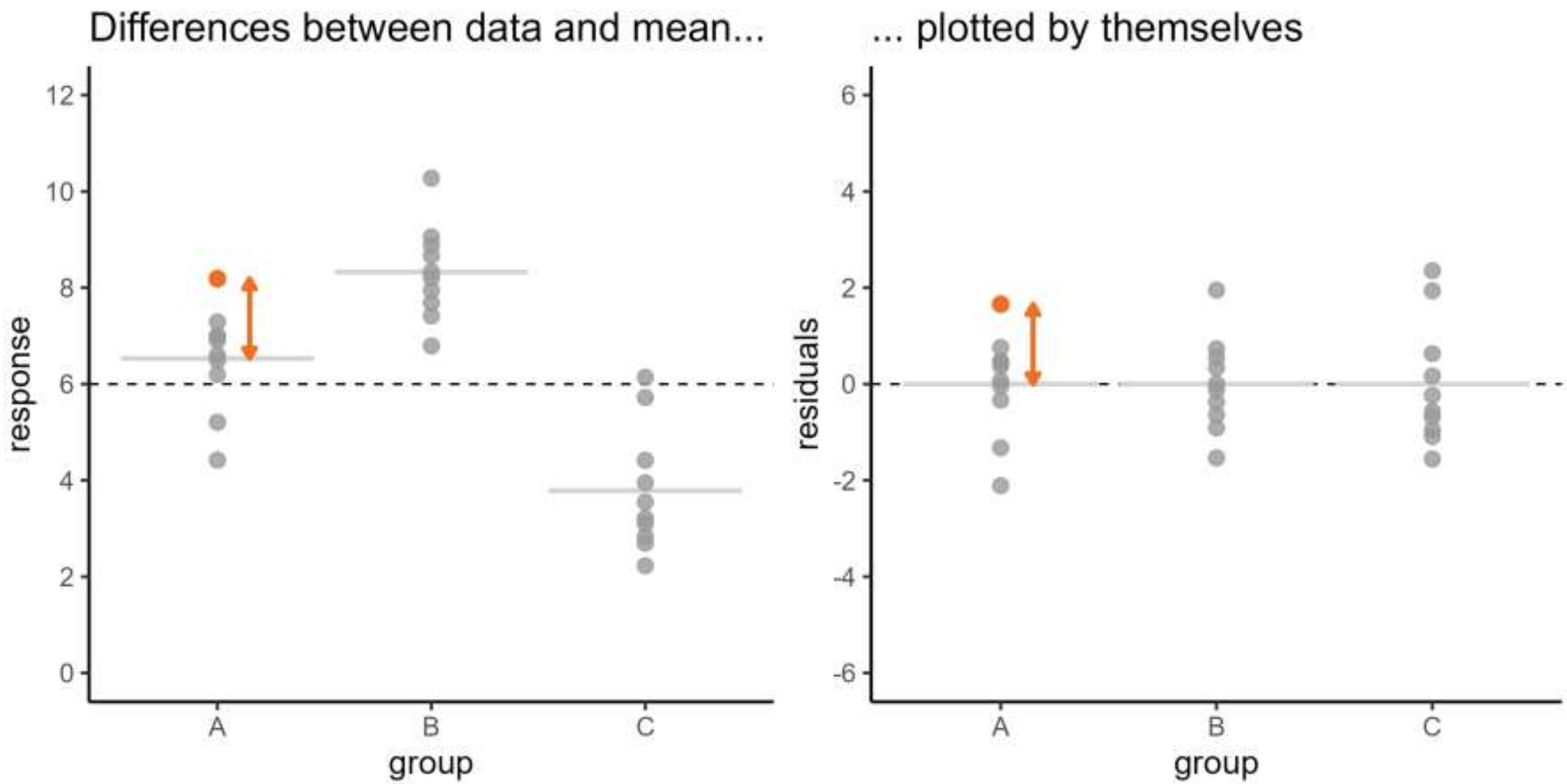

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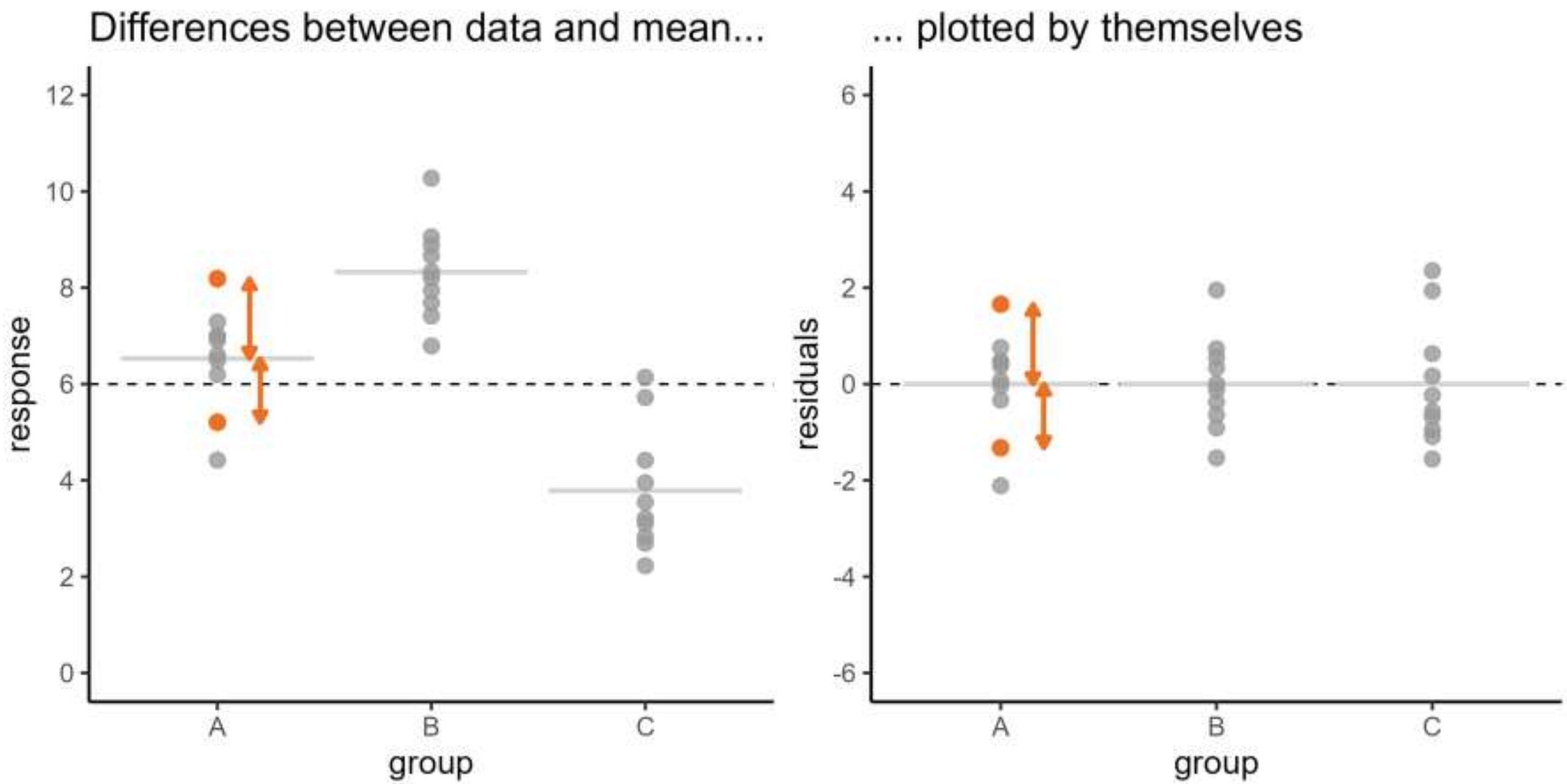

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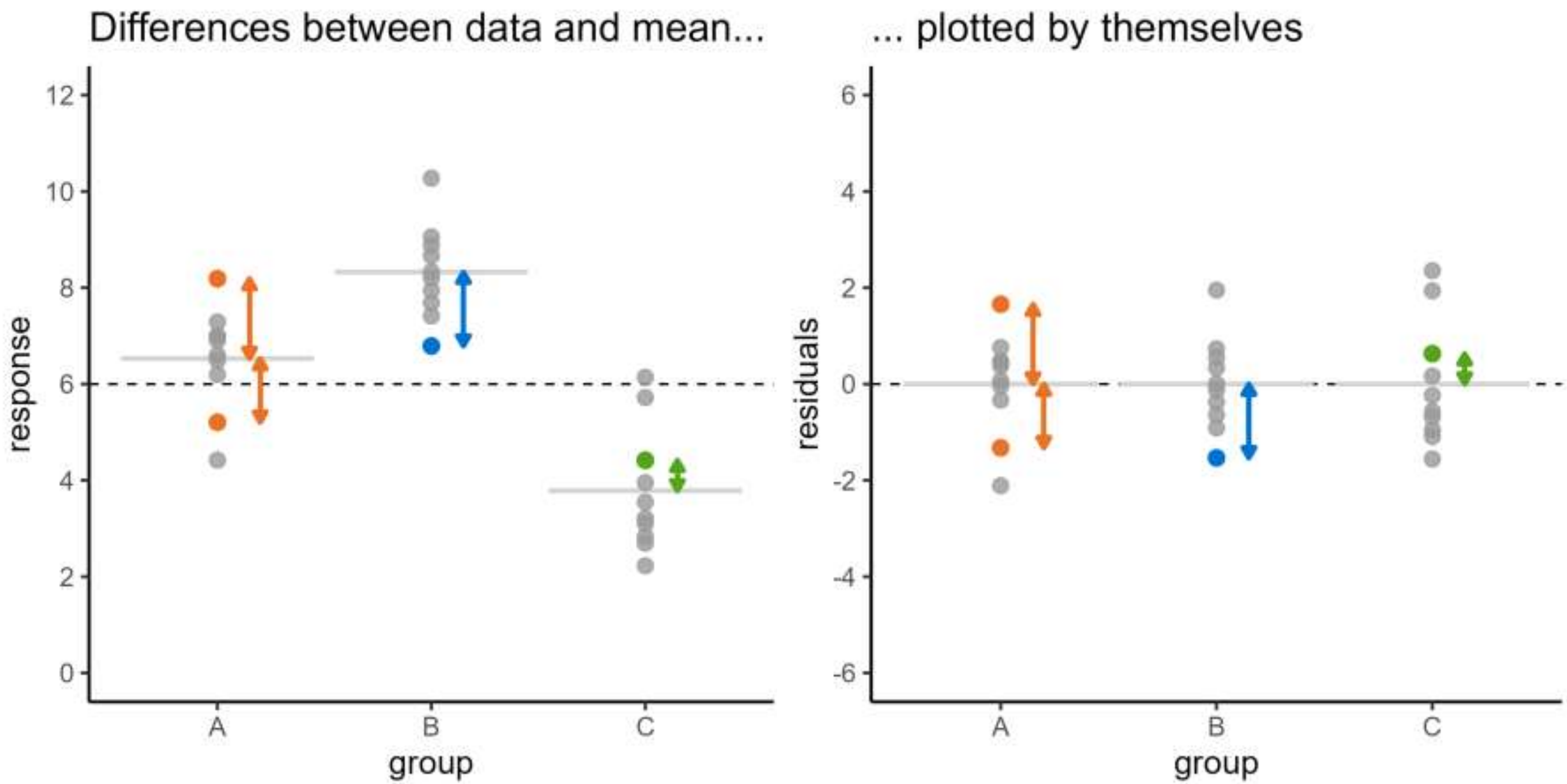

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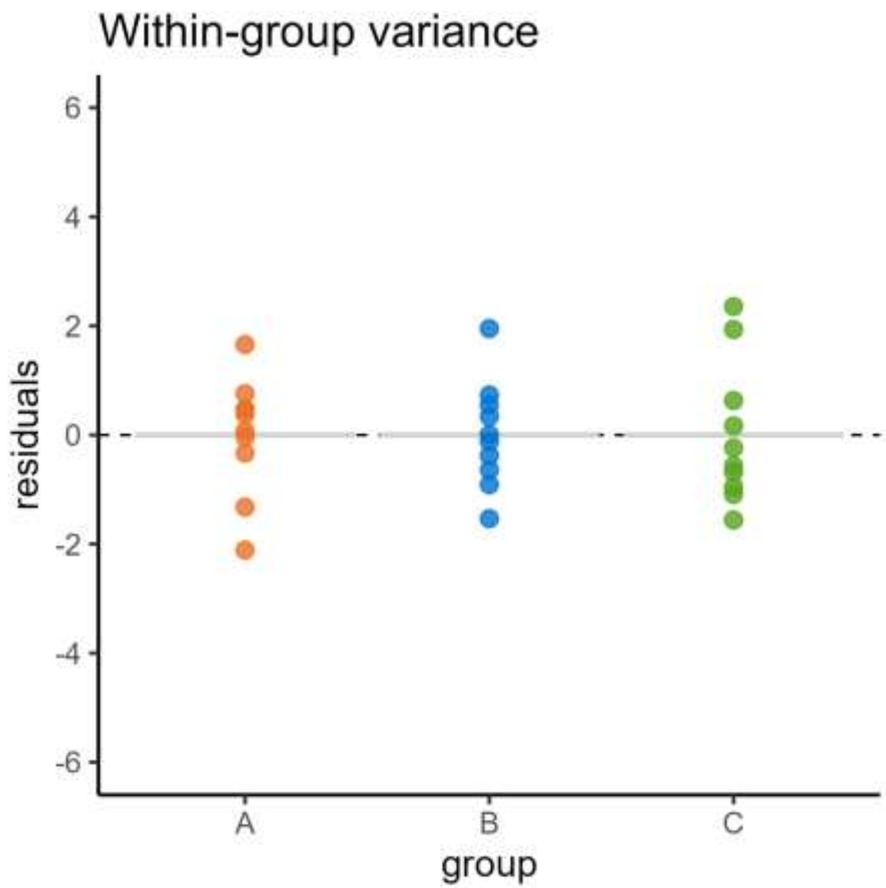

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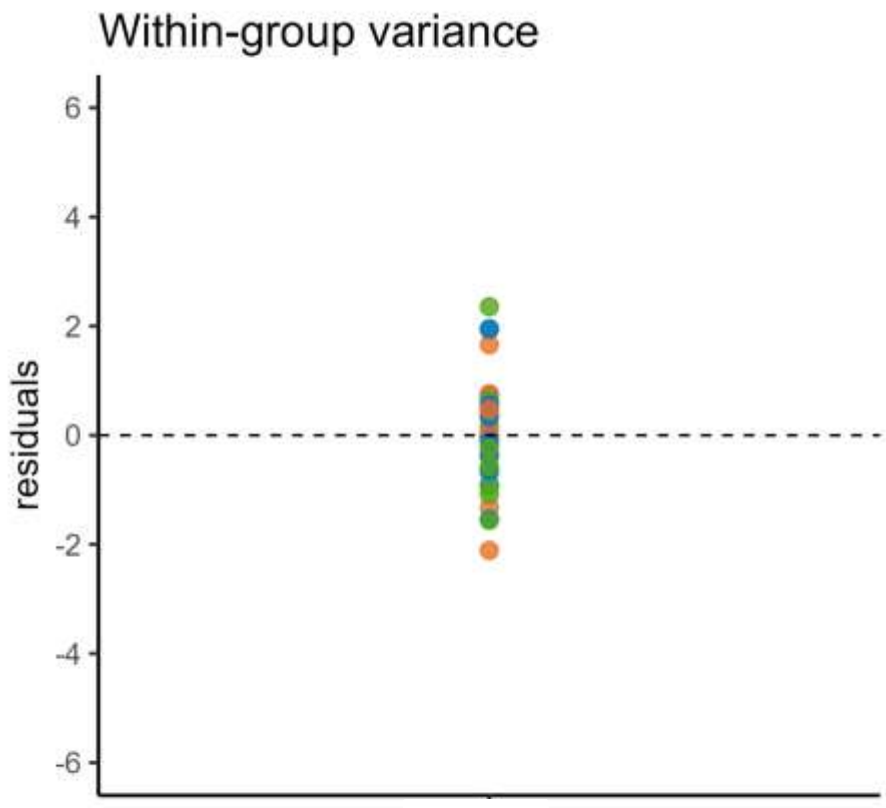
Use the **variance** of the data

How do we measure difference?



We need two numbers:

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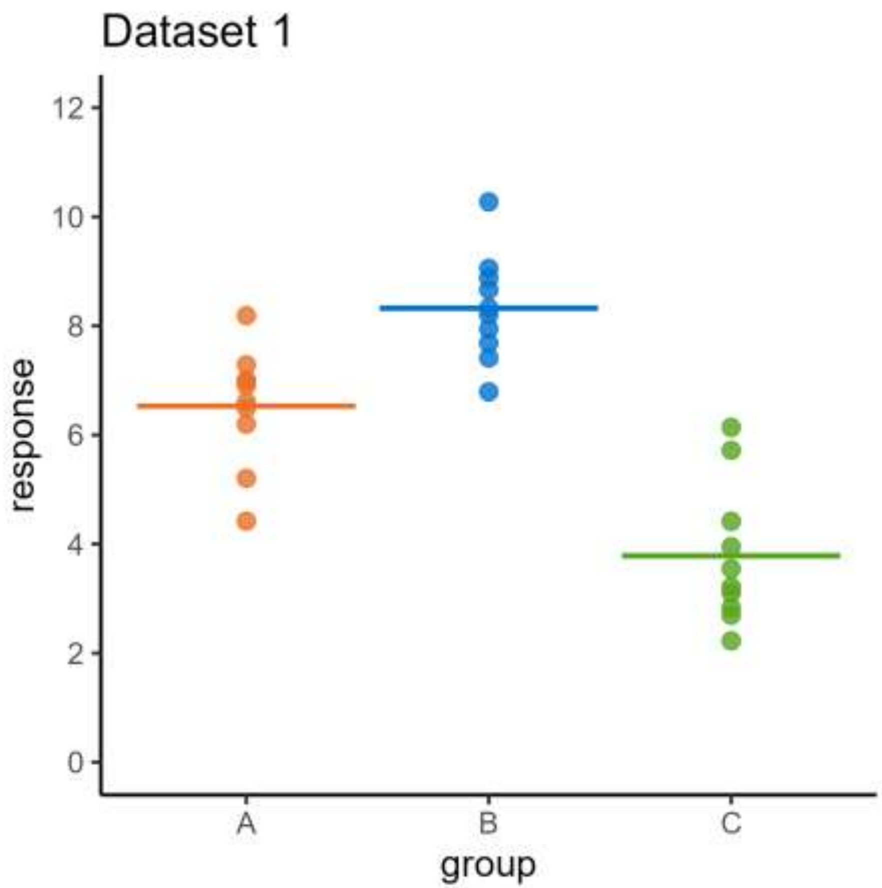
- One to measure the spread of the data

Use the **variance** of the data

2. Analysis of variance

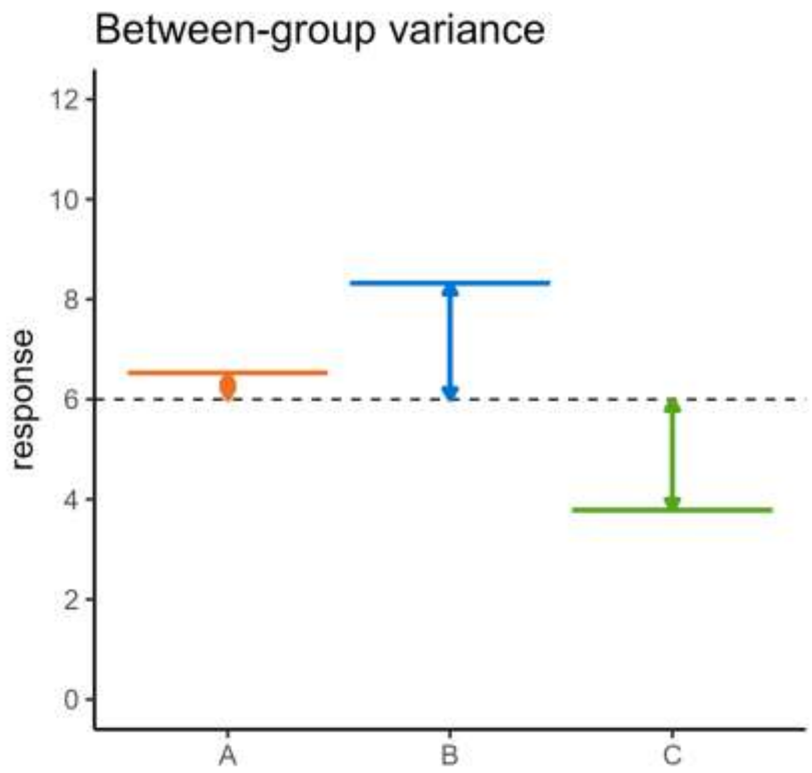
Is there a significant difference between group means?

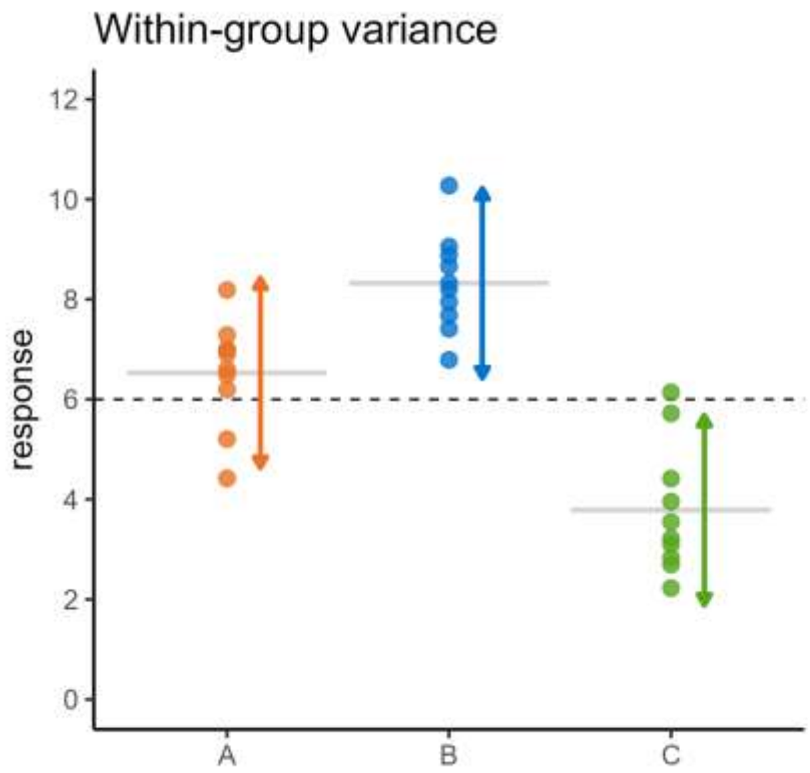
ANalysis Of VAriance



- 3 groups
- 10 points per group
- Equal variances
- Is there a difference between the **means**?

ANalysis Of VAriance

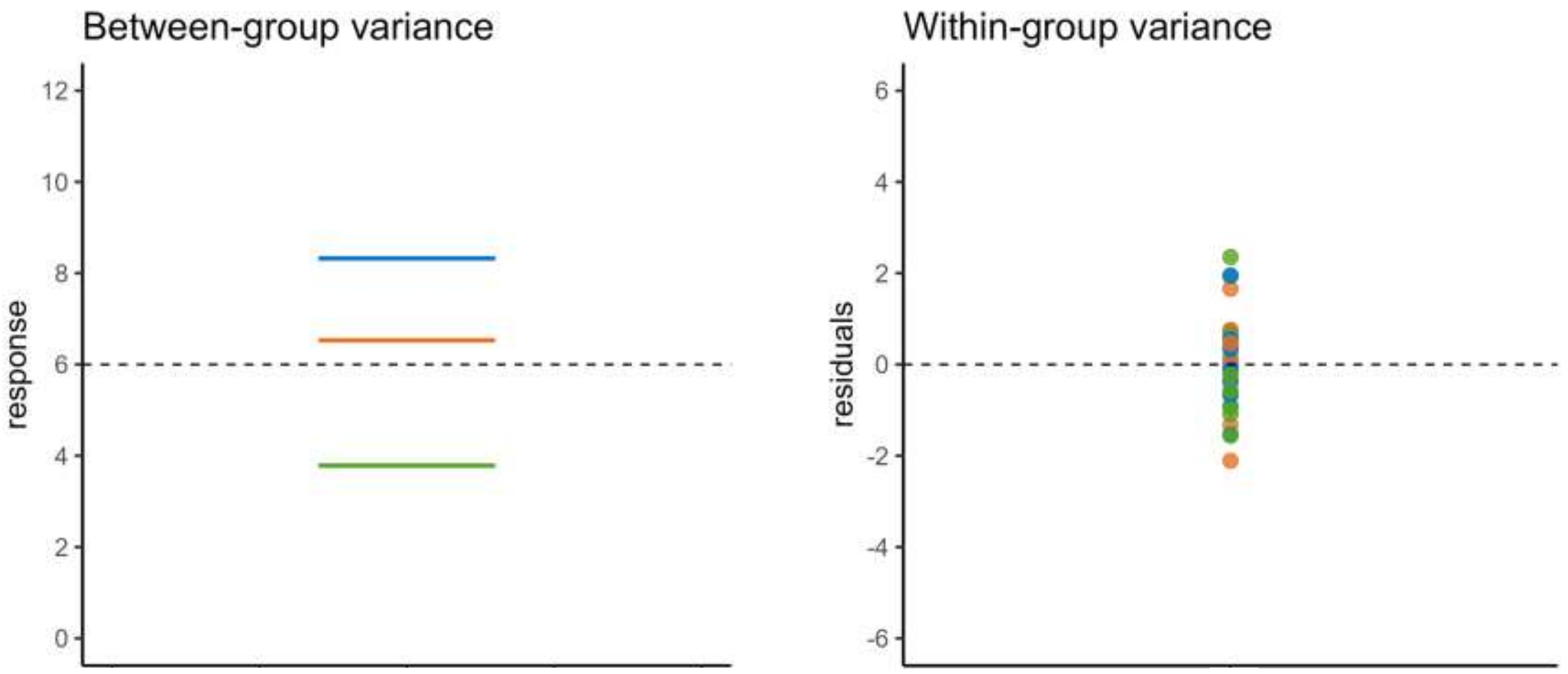




How much do the means vary **between** groups…

… compared with the data **within** each group?

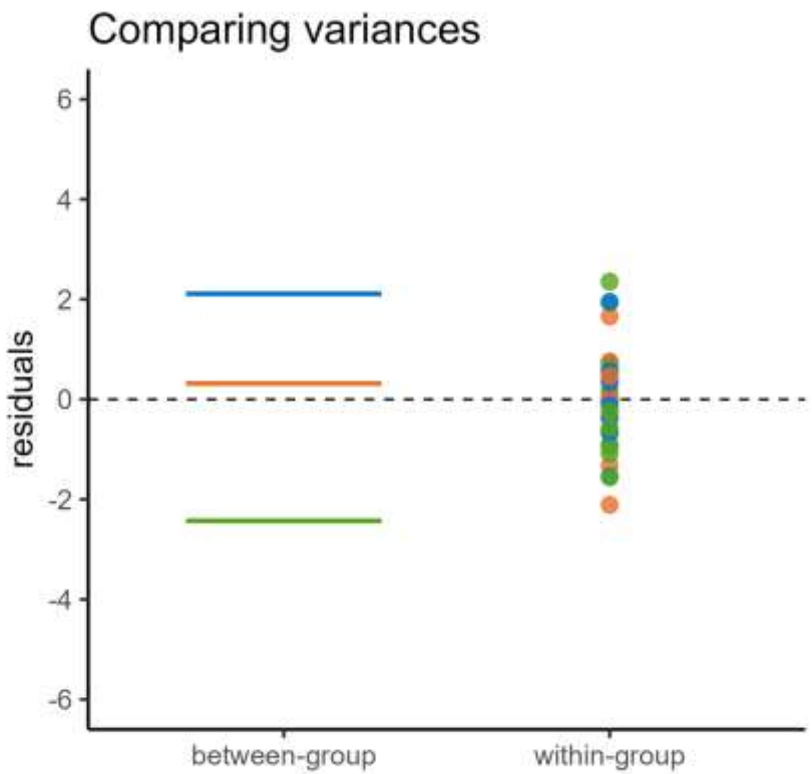
ANalysis Of VAriance



Group means relative to global mean

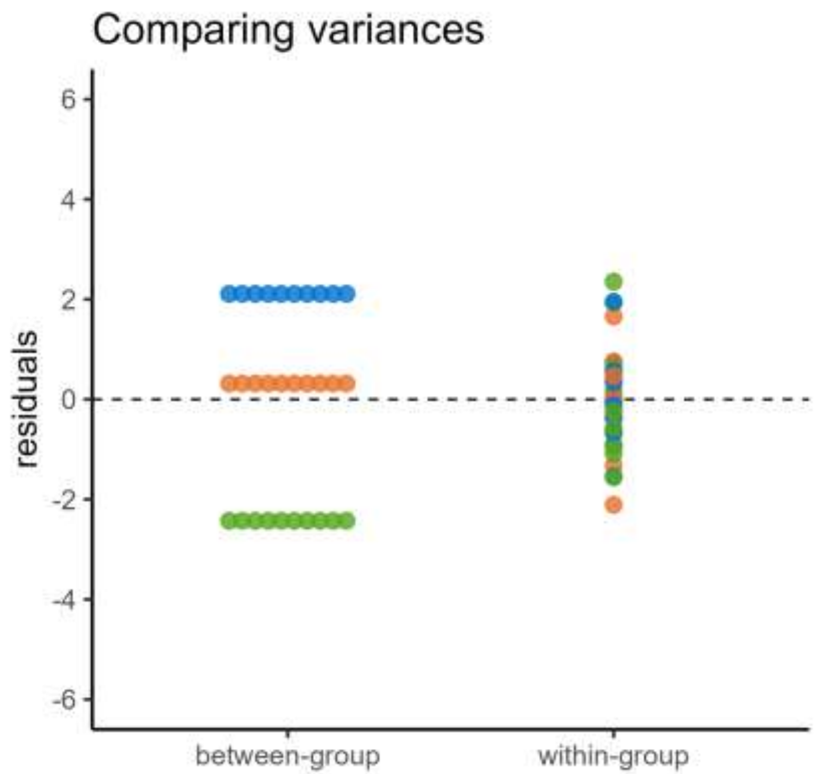
Group data relative to group means

ANalysis Of VAriance



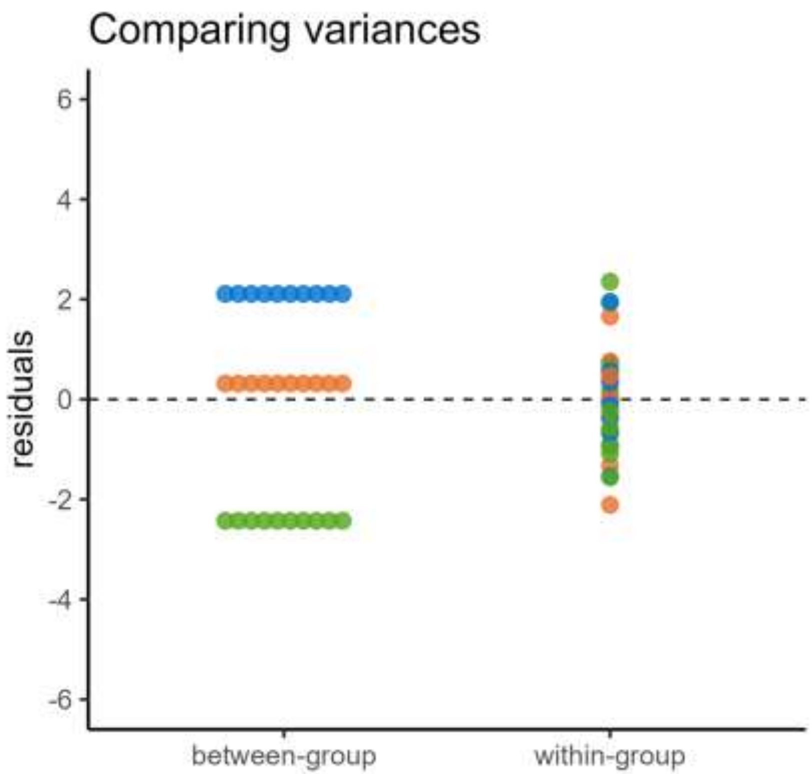
Can’t quite just compare the variance of these two sets of numbers…

ANalysis Of VAriance



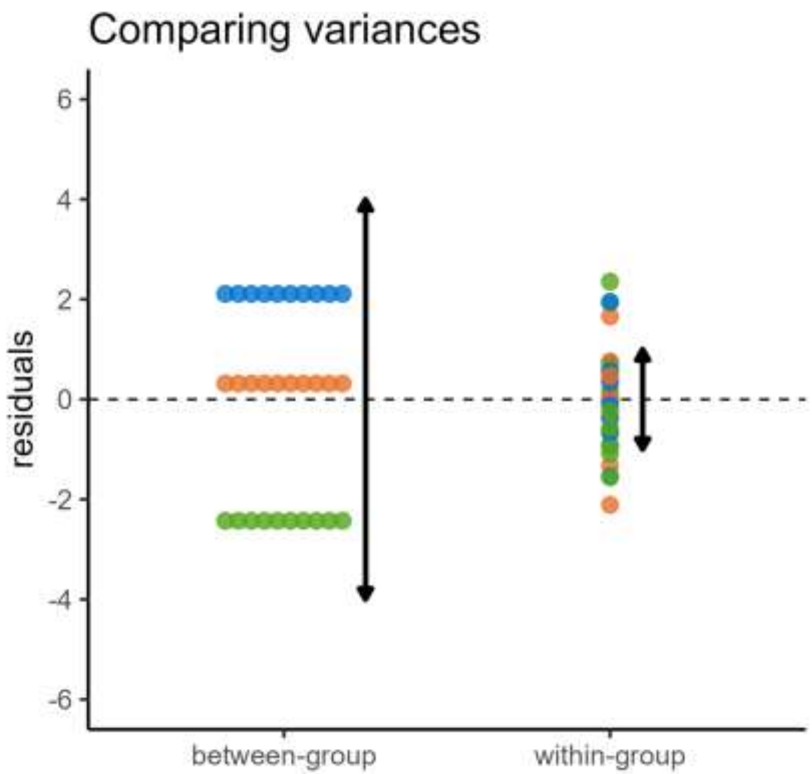
Each group mean is actually represented by multiple data points

ANalysis Of VAriance



var = 52.26   var = 1.25

ANalysis Of VAriance

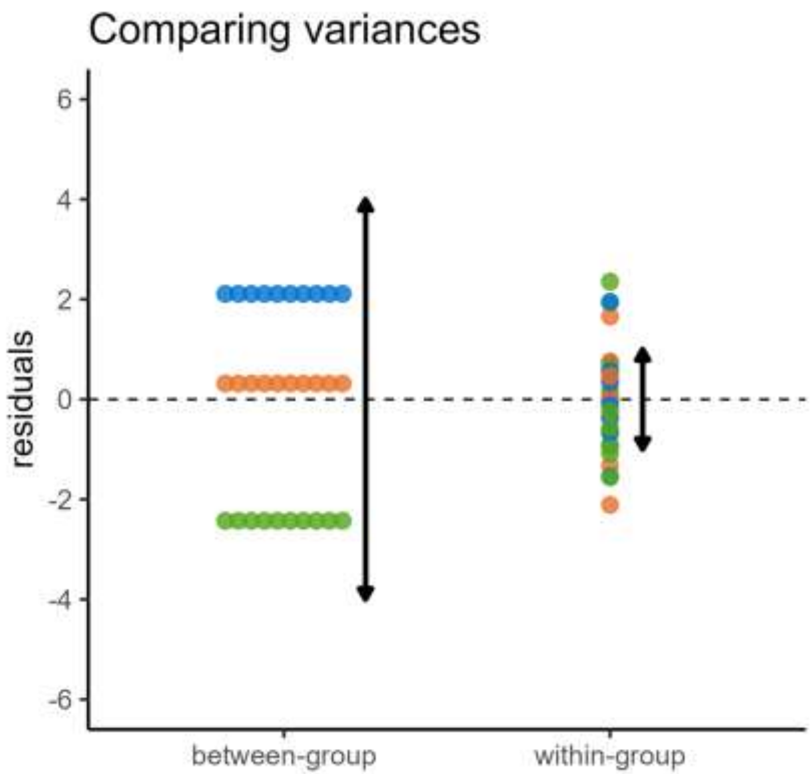


var = 52.26 var = 1.25

We then compare these variances as a fraction:

$$\frac{Between\ group\ variance}{Within\ group\ variance} = \frac{52.26}{1.25} = 41.7$$

ANalysis Of VAriance



var = 52.26 var = 1.25

We then compare these variances as a fraction:

$$\frac{Between\ group\ variance}{Within\ group\ variance} = \frac{52.26}{1.25} = 41.7$$

Can be considered as a **signal-to-noise** ratio

If this ratio is big, we have a significant effect

ANalysis Of VAriance

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Group	2	104.51	52.26	41.71	5.52e-09
Residuals	27	33.83	1.25		

ANalysis Of VAriance

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Signal-to-noise ratio
(Ratio of variances)

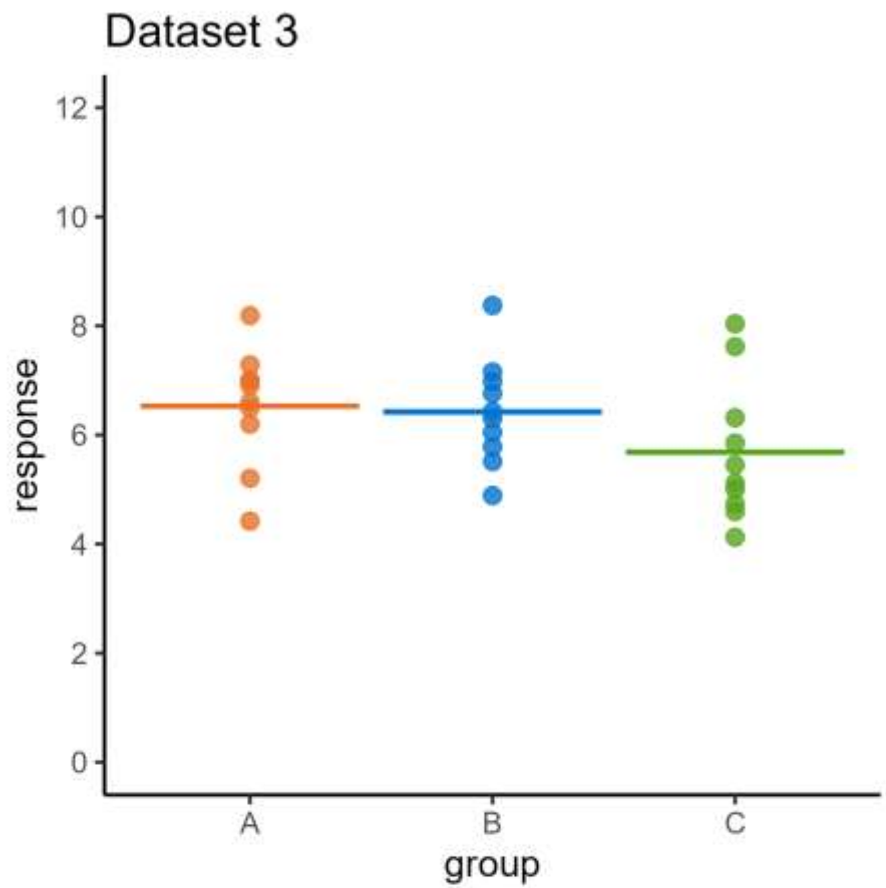
ANalysis Of VAriance

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Group	2	104.51	52.26	41.71	5.52e-09
Residuals	27	33.83	1.25		

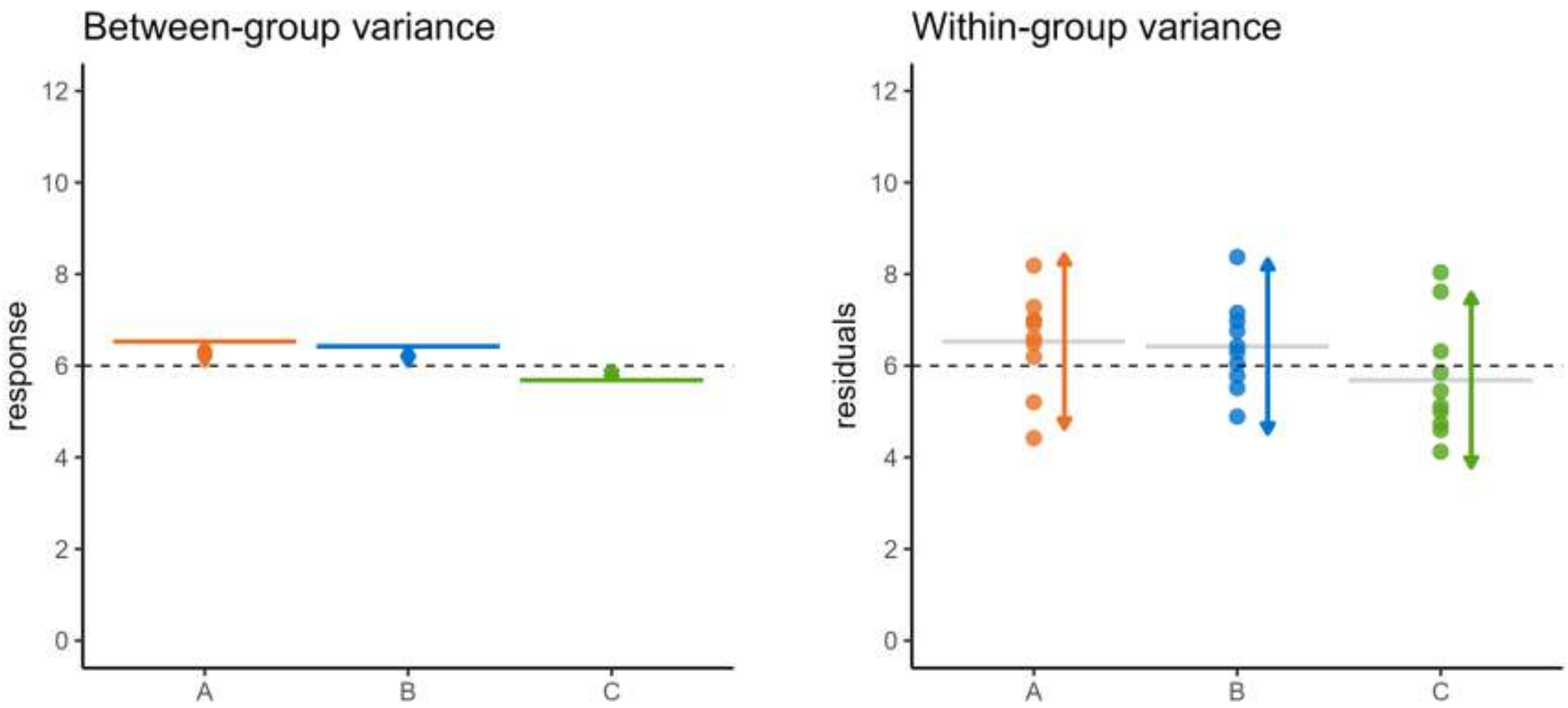
Signal-to-noise ratio
(Ratio of variances)

Measure of whether this
is a big/unlikely ratio

ANalysis Of VAriance



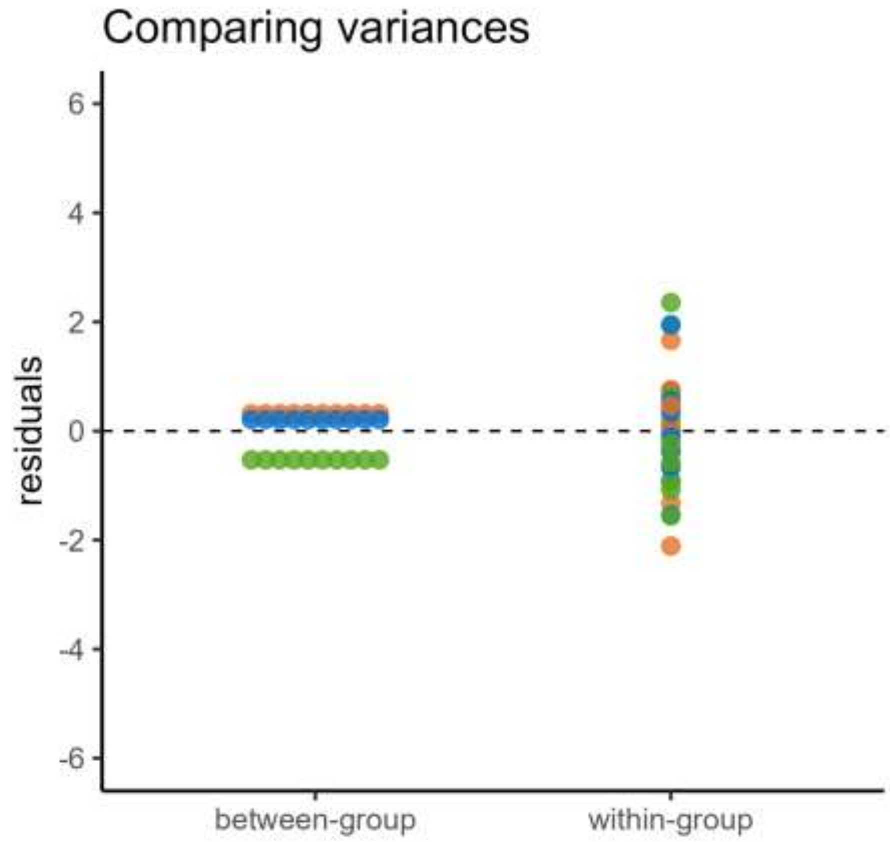
ANalysis Of VAriance



How much do the means vary **between** groups…

… compared with the data **within** each group?

ANalysis Of VAriance



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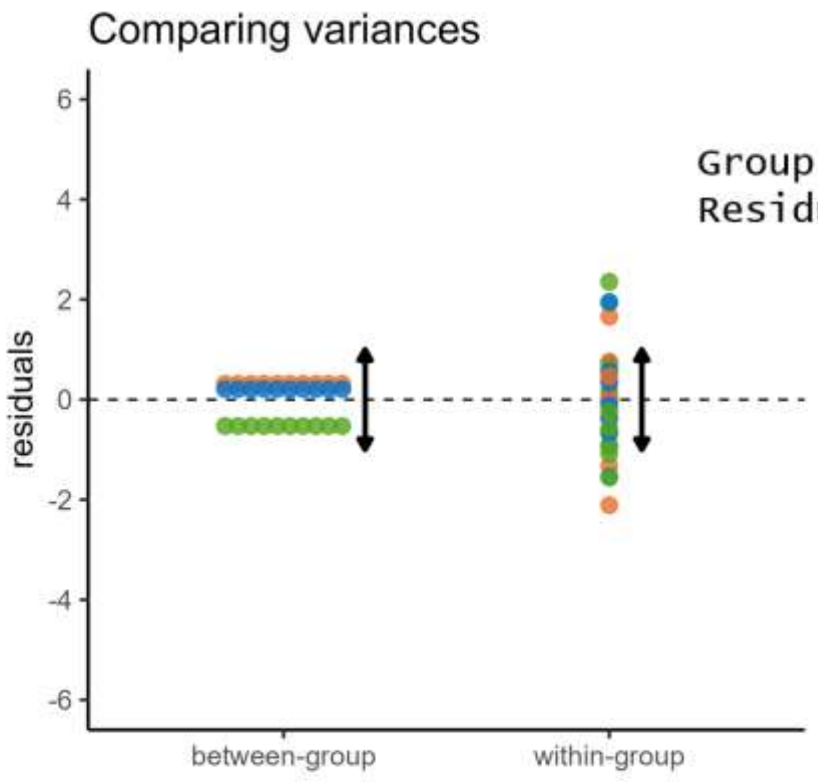


	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Group	2	4.24	2.119	1.69	0.203
Residuals	27	33.83	1.253		

Signal-to-noise ratio is small

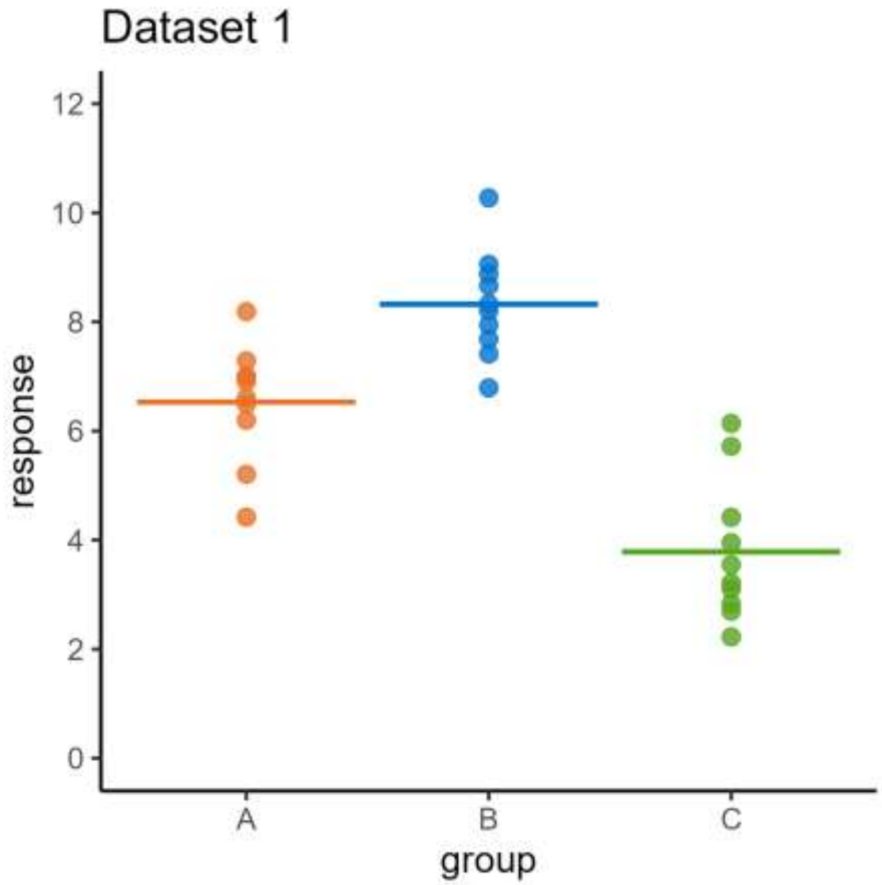
Means are **not** significantly different

var = 2.12 var = 1.25

3. Post hoc tests

Which groups have significantly different means?

Post hoc tests

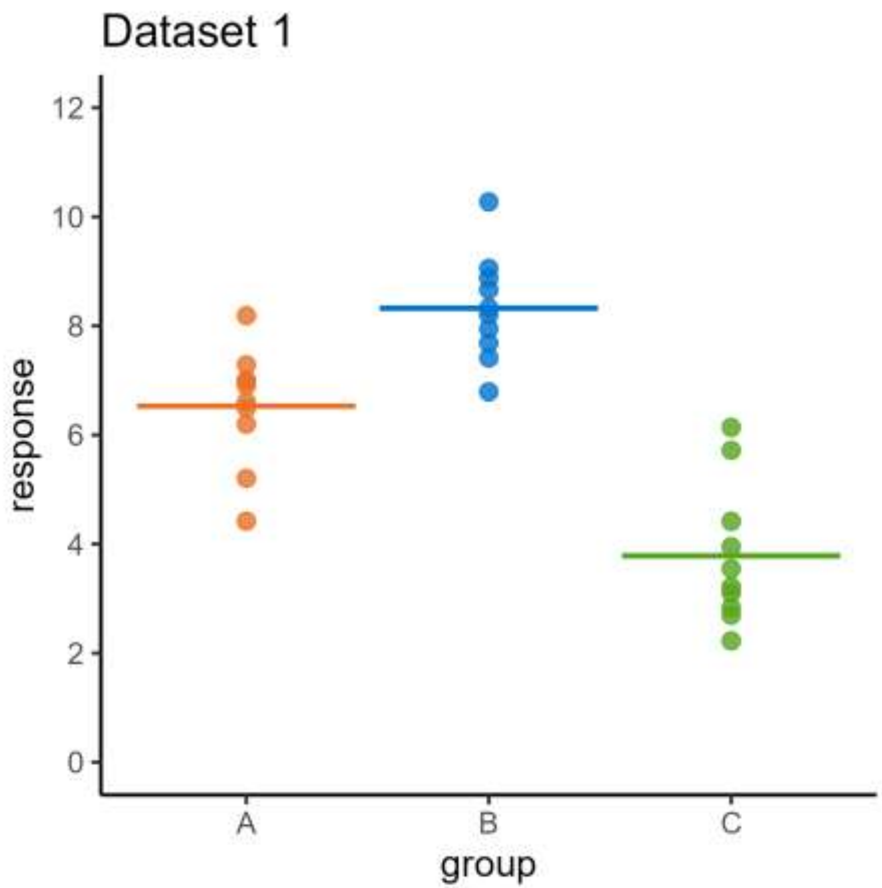


Were these groups sampled from the same distribution?

Two stage investigation:

- Do all the groups have the same mean?
- If not, which groups are different?

Post hoc tests



Were these groups sampled from the same distribution?

p-value = 5.52e-09

- Groups do not have the same mean
- But **where** is the difference(s)?

Post hoc tests

Post hoc tests:

- Compare the groups in a pairwise manner
- Tells us **where** the differences are
- While controlling for false positives (adjusting p-values)

Tukey's HSD

```
      Tukey multiple comparisons of means
        95% family-wise confidence level

Fit: aov(formula = lm_anova)

$Group
         diff        lwr       upr     p adj
B-A  1.793566  0.5524446  3.034688 0.0036529
C-A -2.745240 -3.9863613 -1.504118 0.0000242
C-B -4.538806 -5.7799276 -3.297684 0.0000000
```

Session 2: Categorical predictors

Lecture summary:

- One-way ANOVA test is performed by comparing between-group and within-group variance
- This can be thought of as a signal-to-noise ratio
- Post hoc tests can be used following a significant ANOVA to determine which groups are significantly different

Session 2: Categorical predictors

Course materials:

Performing ANOVA (and Kruskal-Wallis) in R/Python

Testing assumptions

- Normality (Q-Q plots)
- Equality of variance

Core statistics

Session 2: Categorical predictors

Cambridge Centre for Research Informatics Training

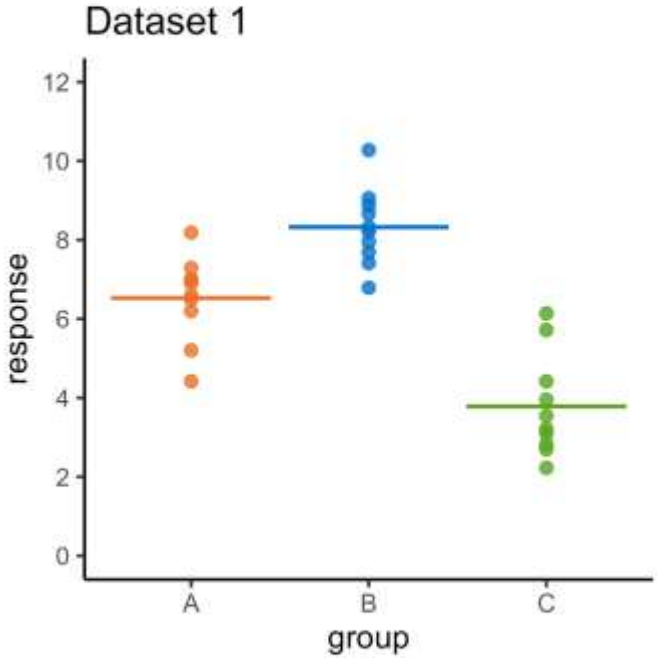
Core statistics

Session 3: Continuous predictors

Cambridge Centre for Research Informatics Training

Recap of tests so far

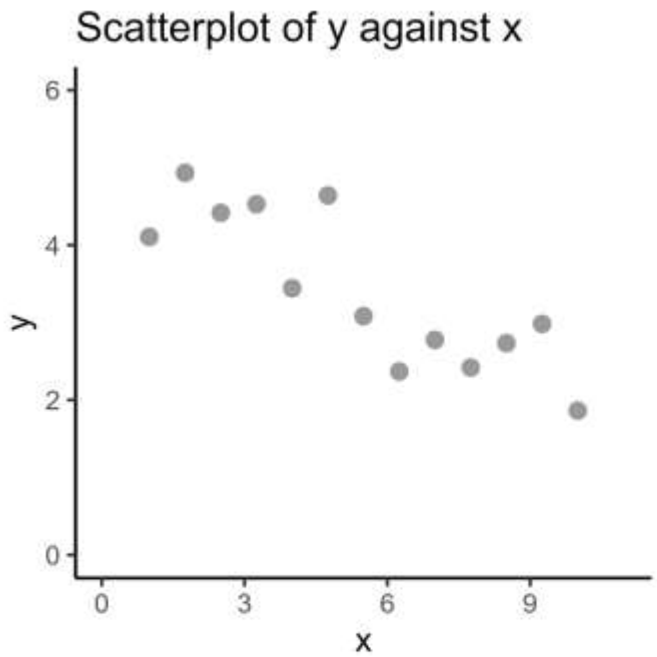
So far, all tests have had a **categorical** predictor



- Is there a difference between the **means**?
- Technique called ANOVA

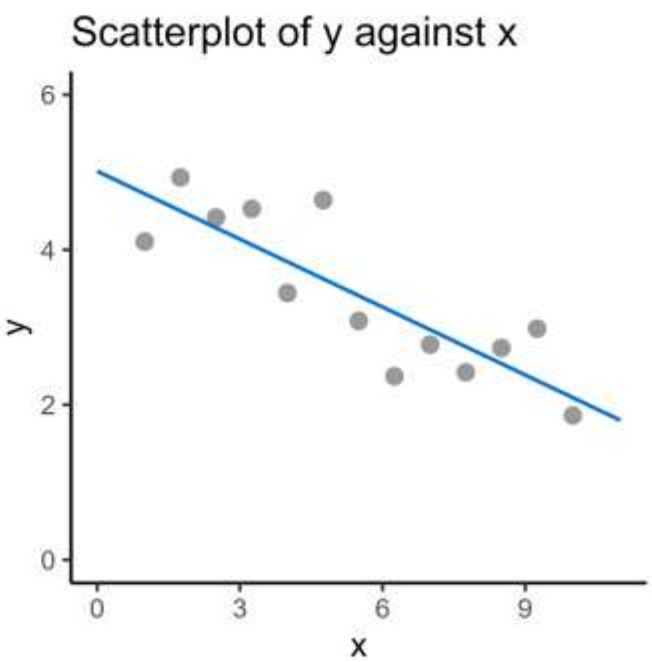
Recap of tests so far

What happens when the predictor variable is **continuous?**



Plan for the session

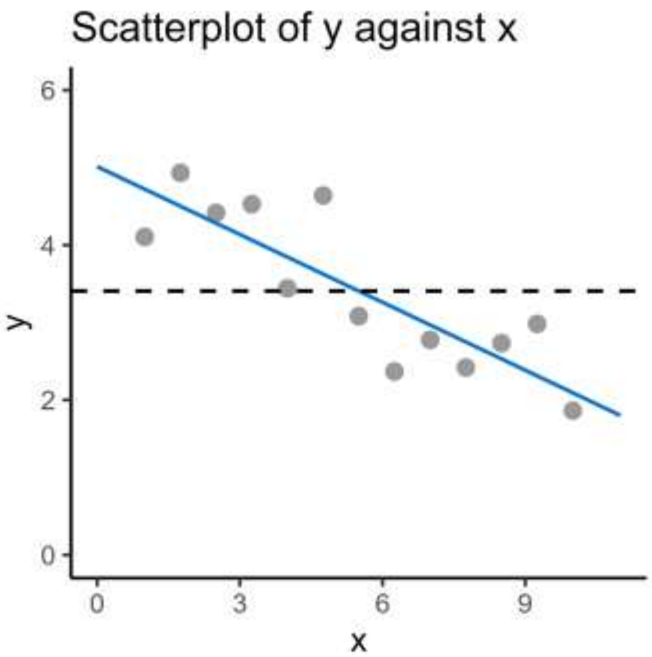
Session 3: **Continuous predictors**



1. Line of best fit

Plan for the session

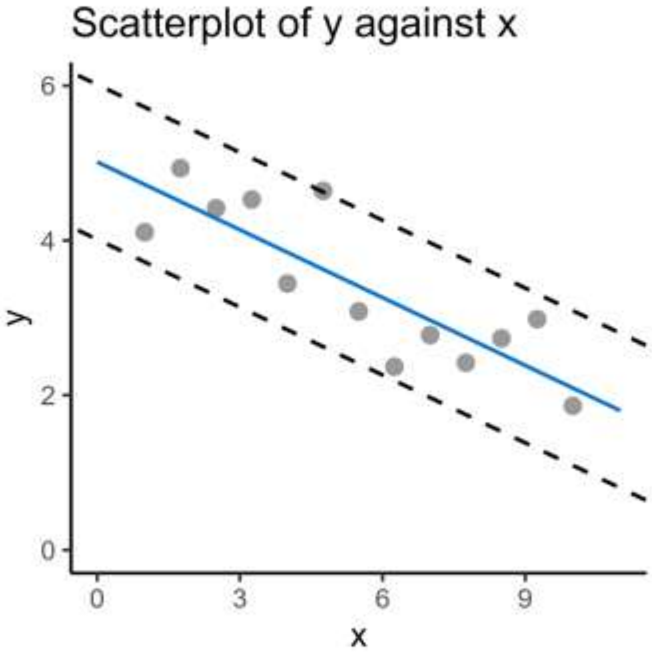
Session 3: **Continuous predictors**



1. Line of best fit
2. Regression analysis

Plan for the session

Session 3: **Continuous predictors**

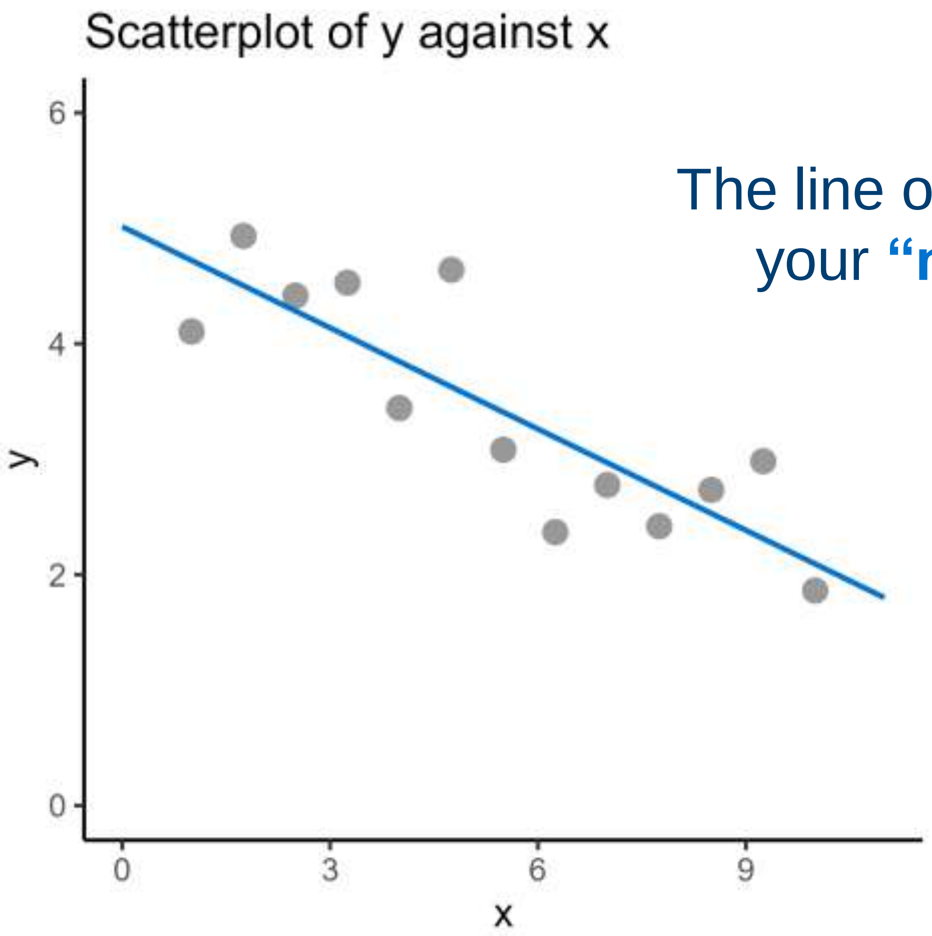


1. Line of best fit
2. Regression analysis
3. Correlation

1. Line of best fit

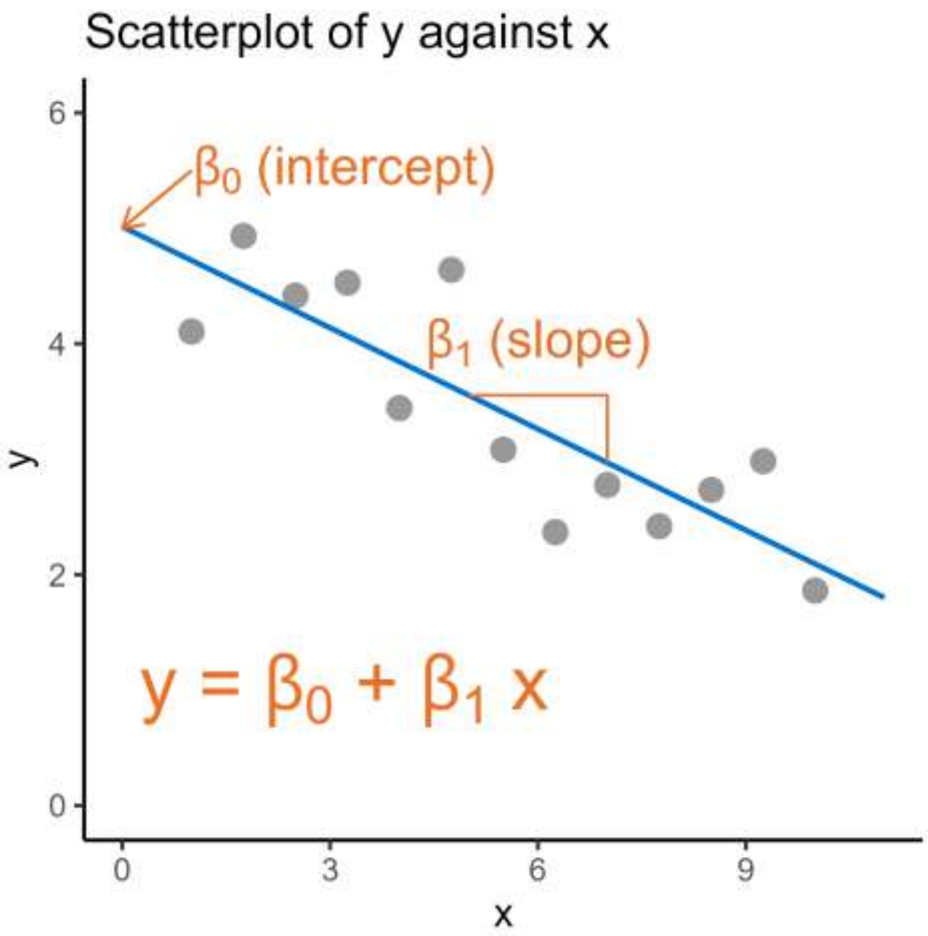
Describing the trend in a dataset

Line of best fit

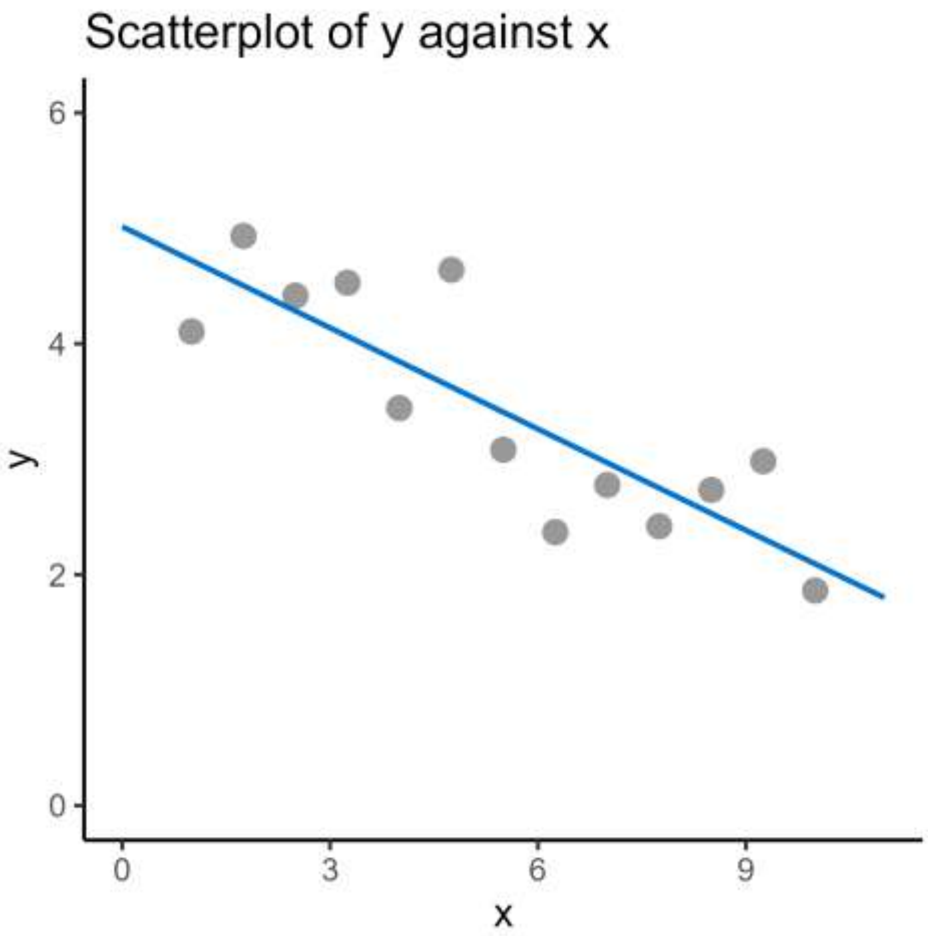


The line of best fit is your **"model"**

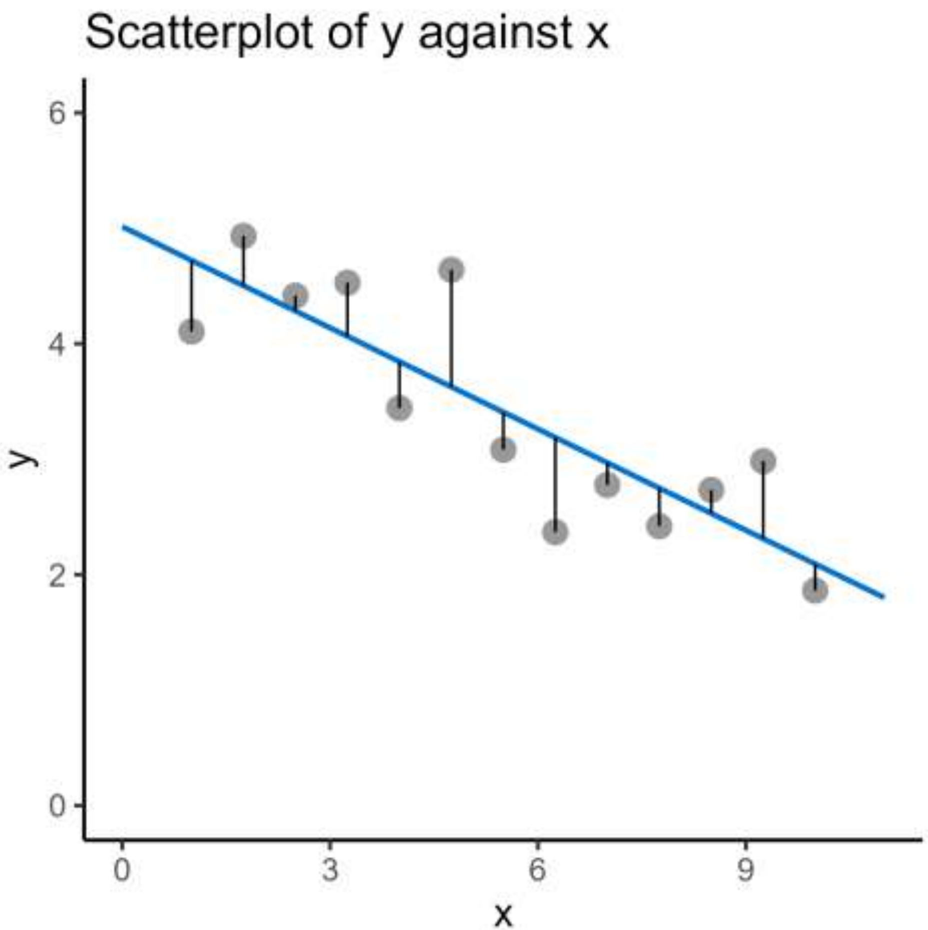
Line of best fit



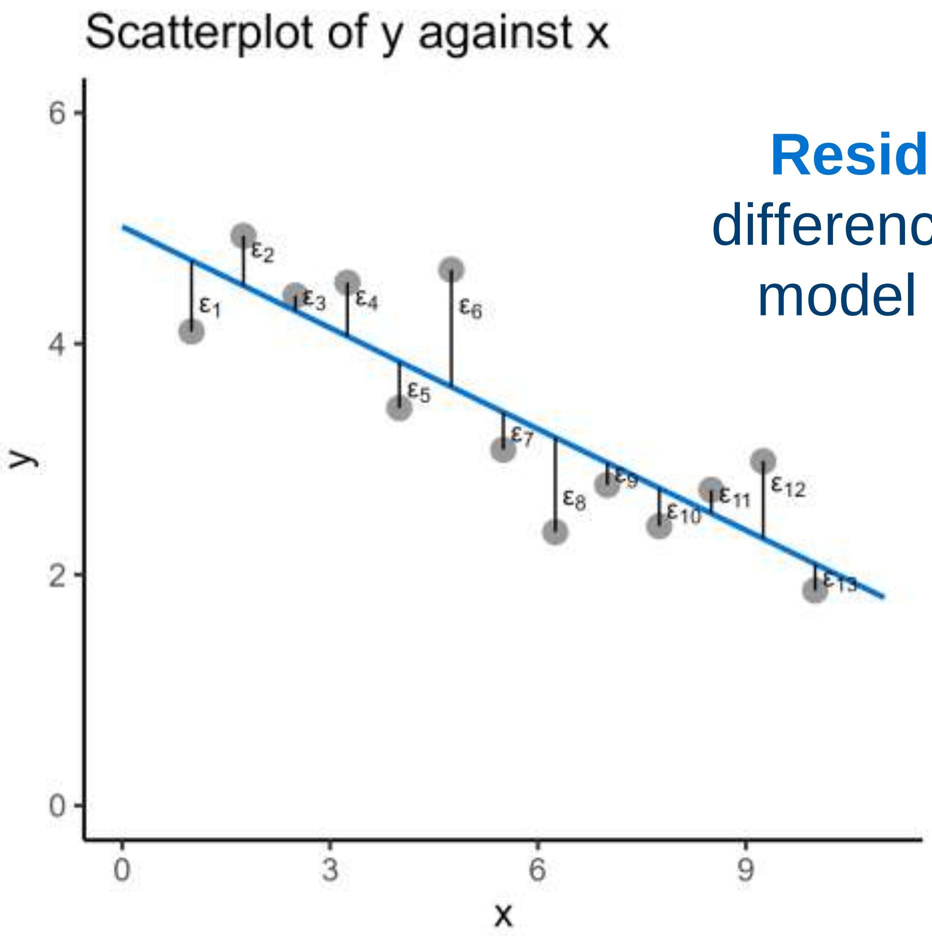
Line of best fit



Line of best fit

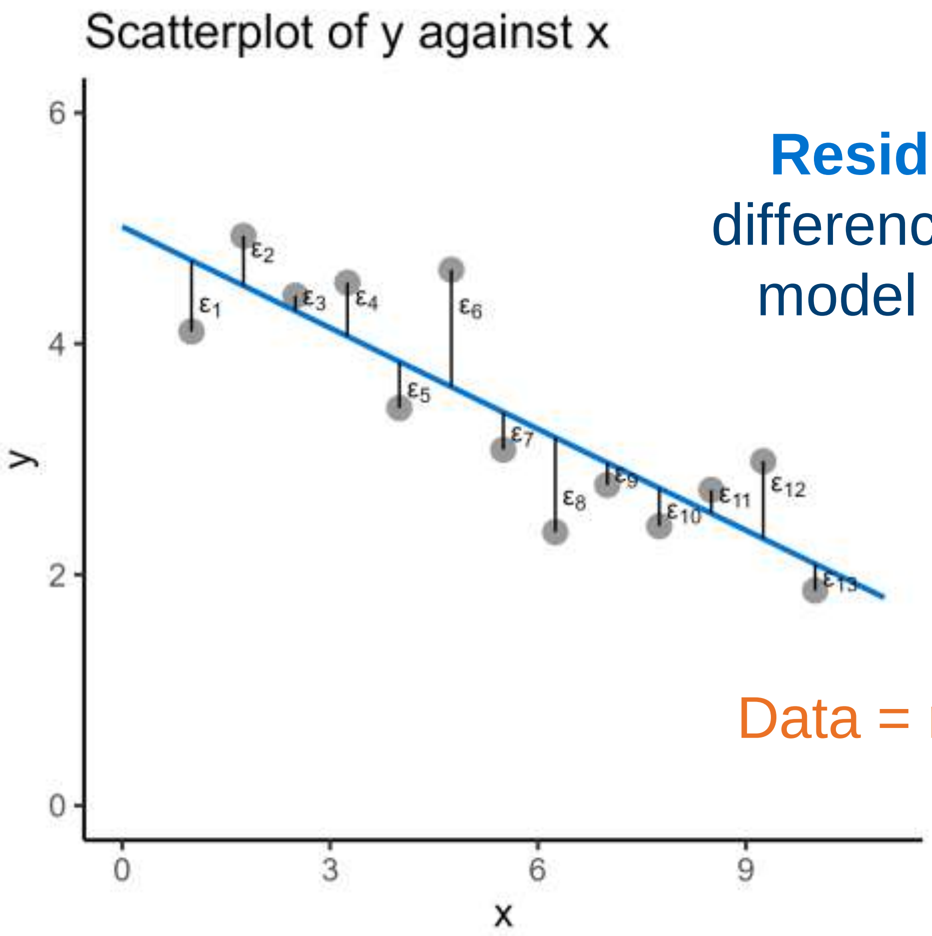


Line of best fit



Residuals are the difference between the model and the data

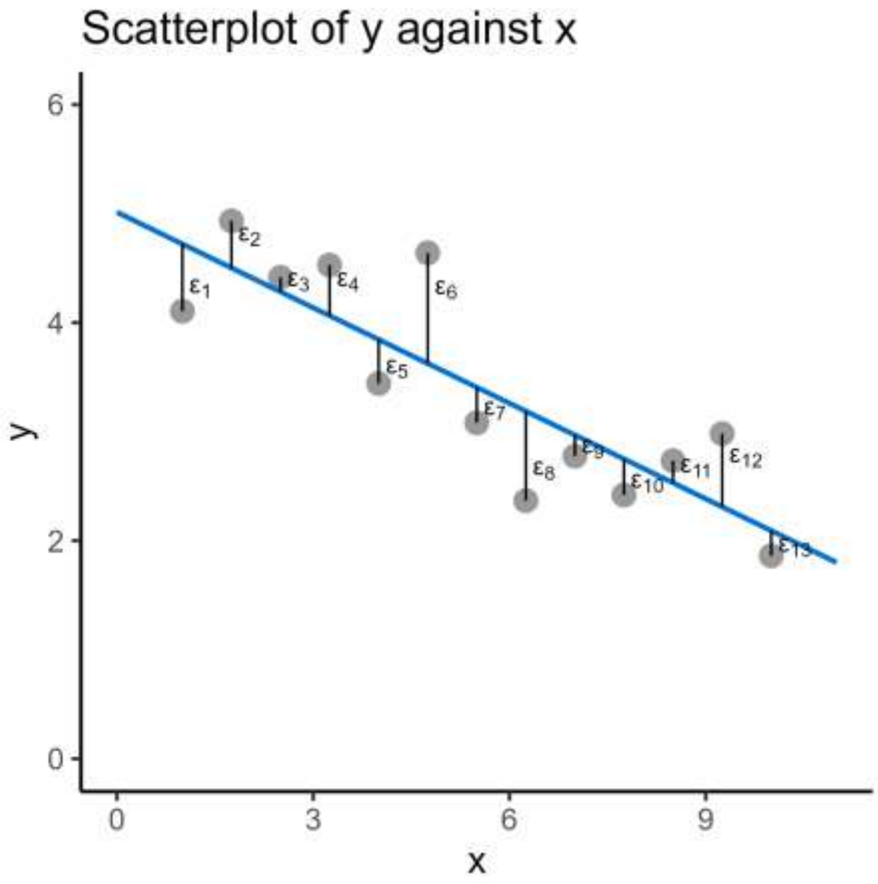
Line of best fit

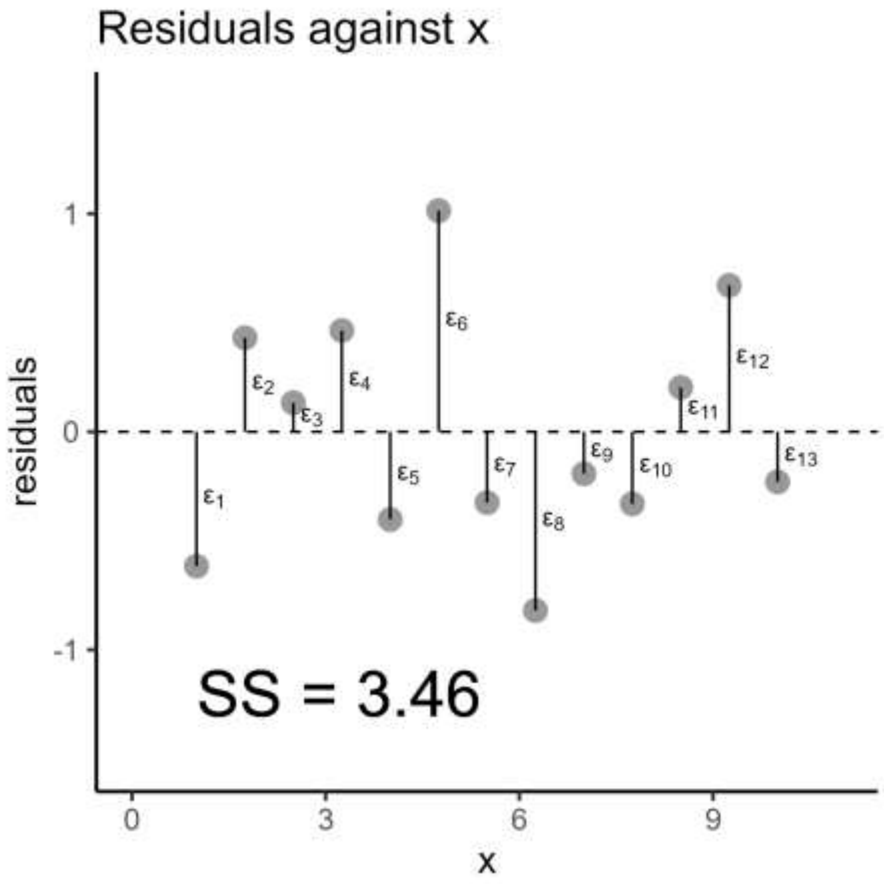


Residuals are the difference between the model and the data

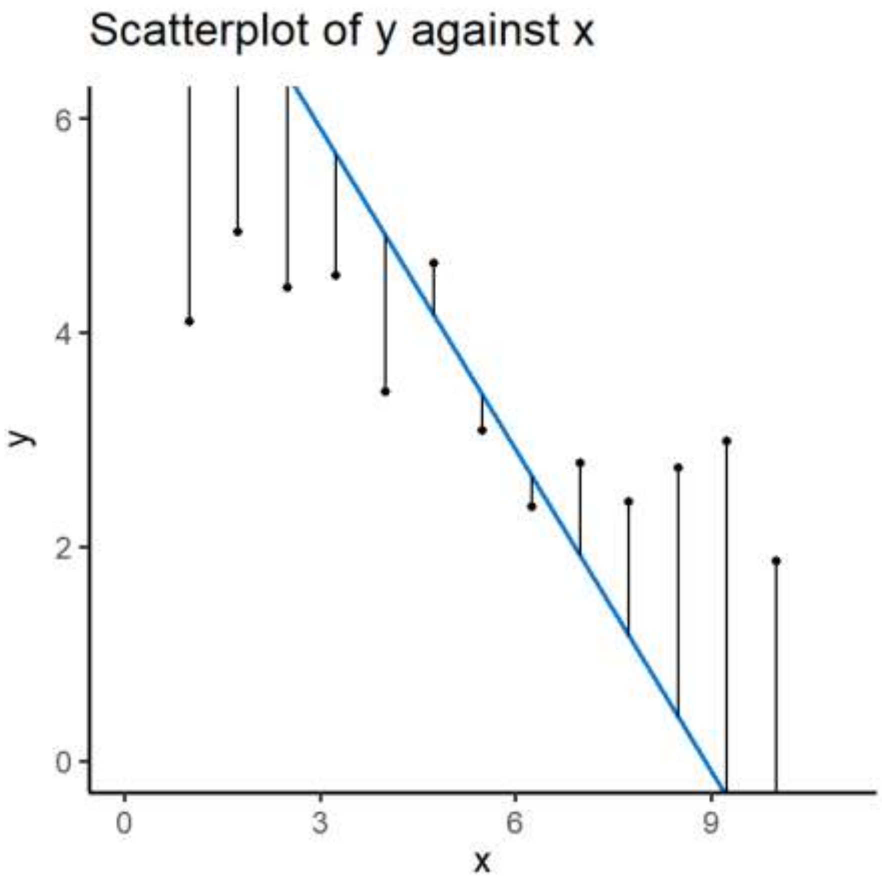
Data = model + error

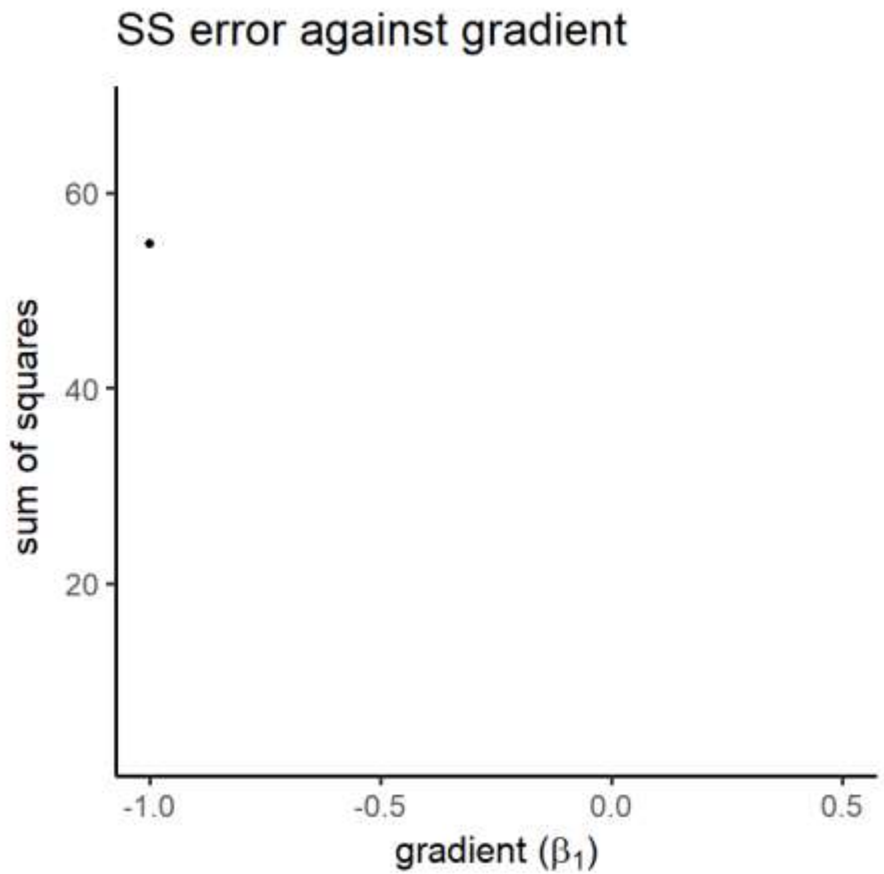
Line of best fit



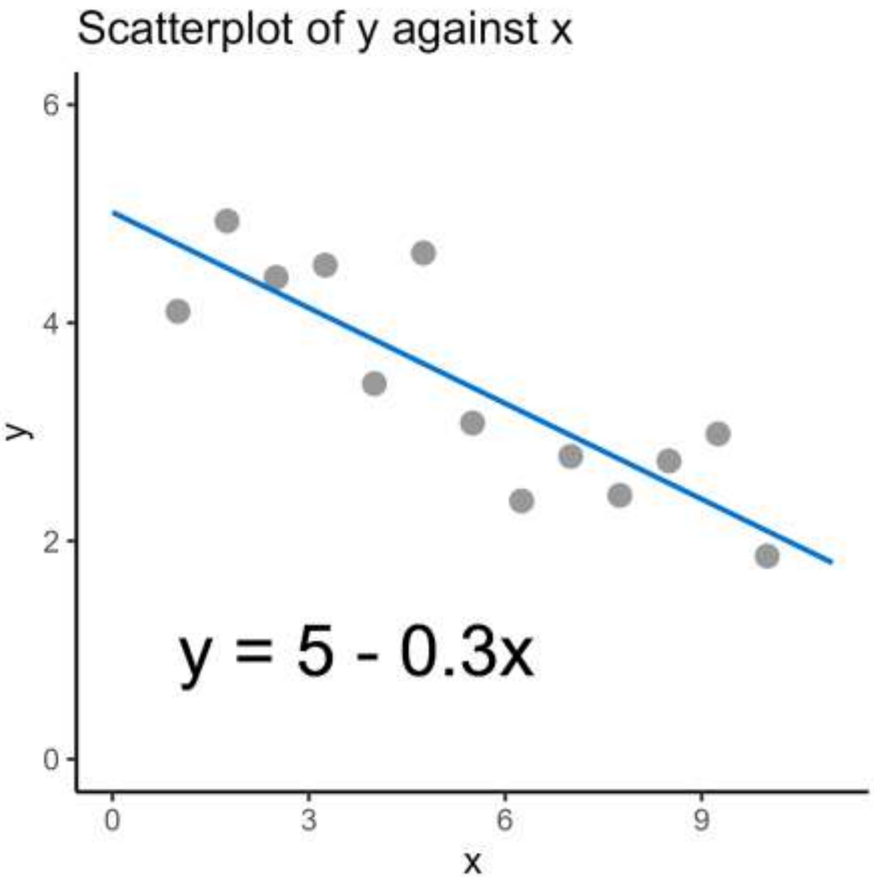


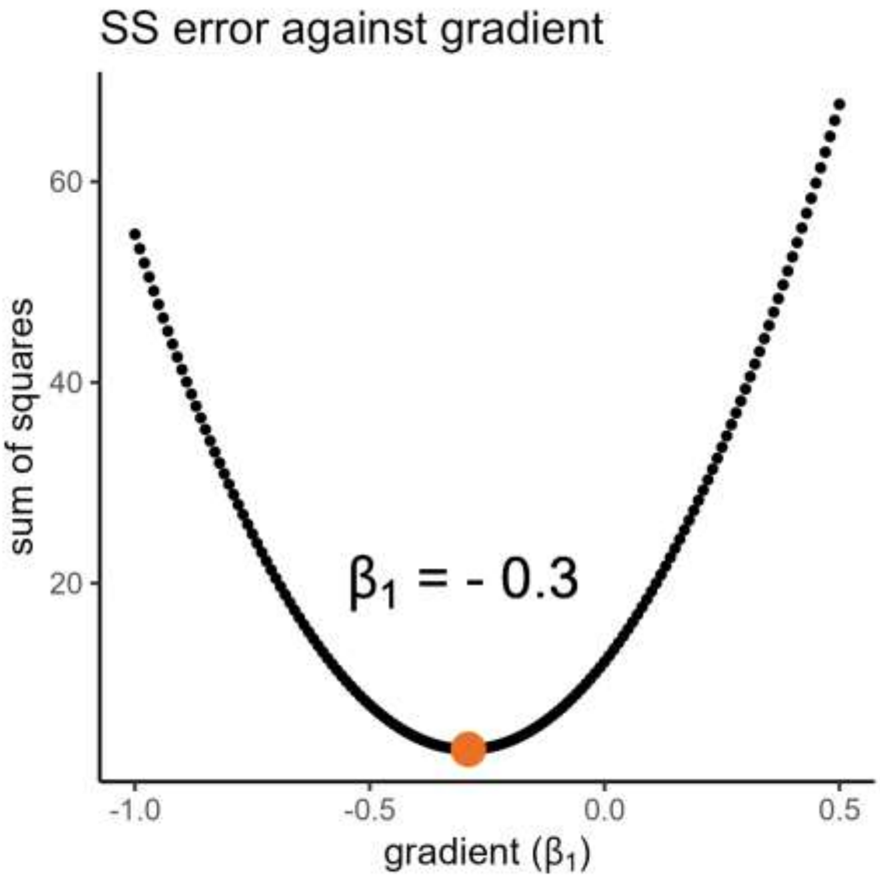
Line of best fit



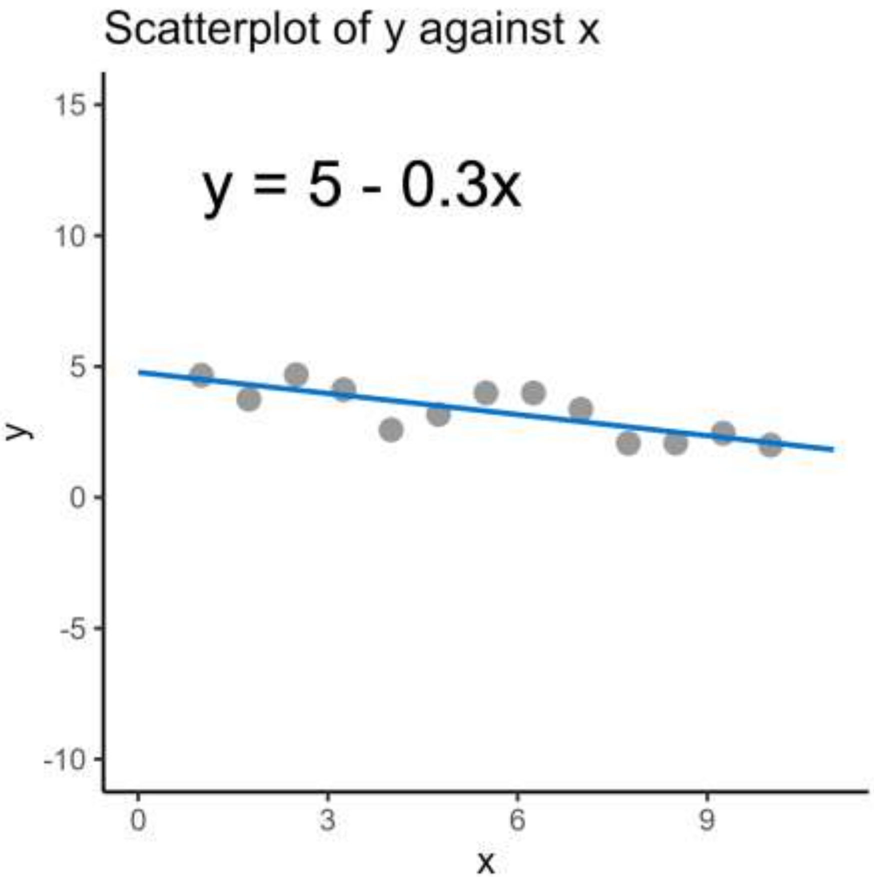


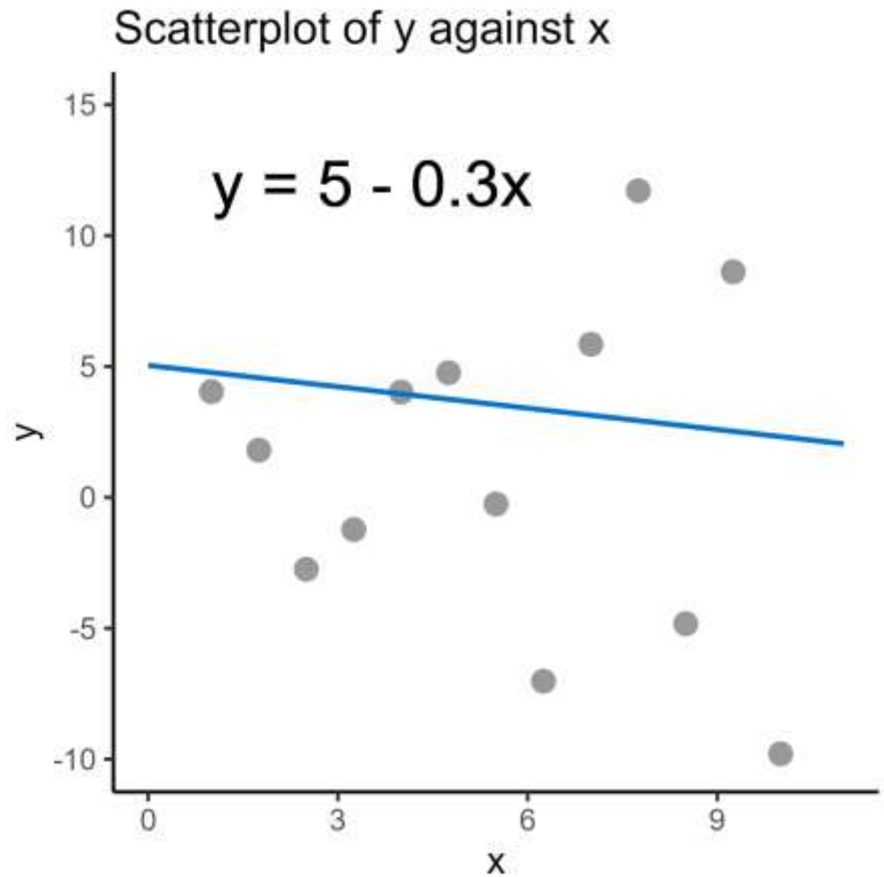
Line of best fit





Line of best fit

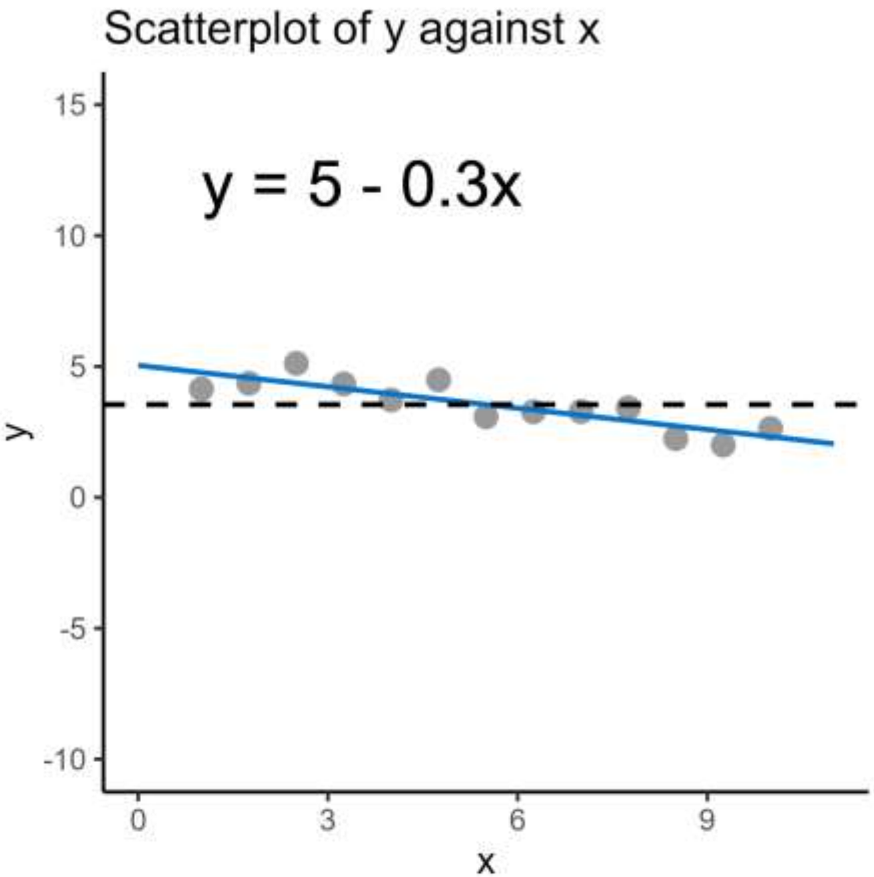


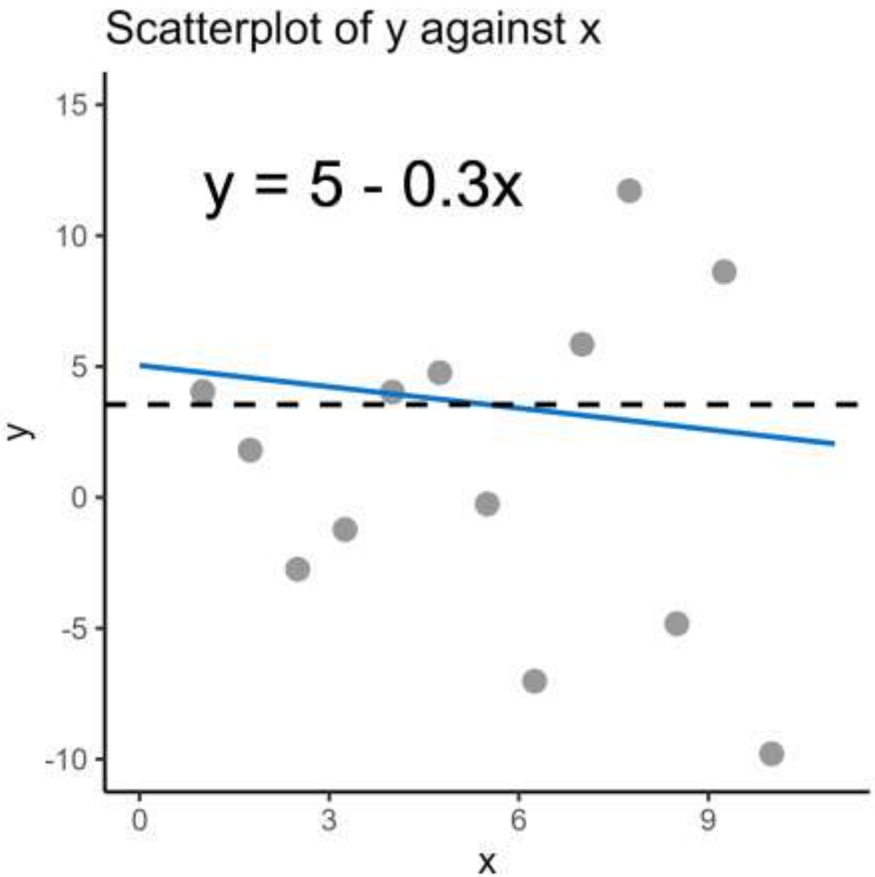


2. Regression analysis

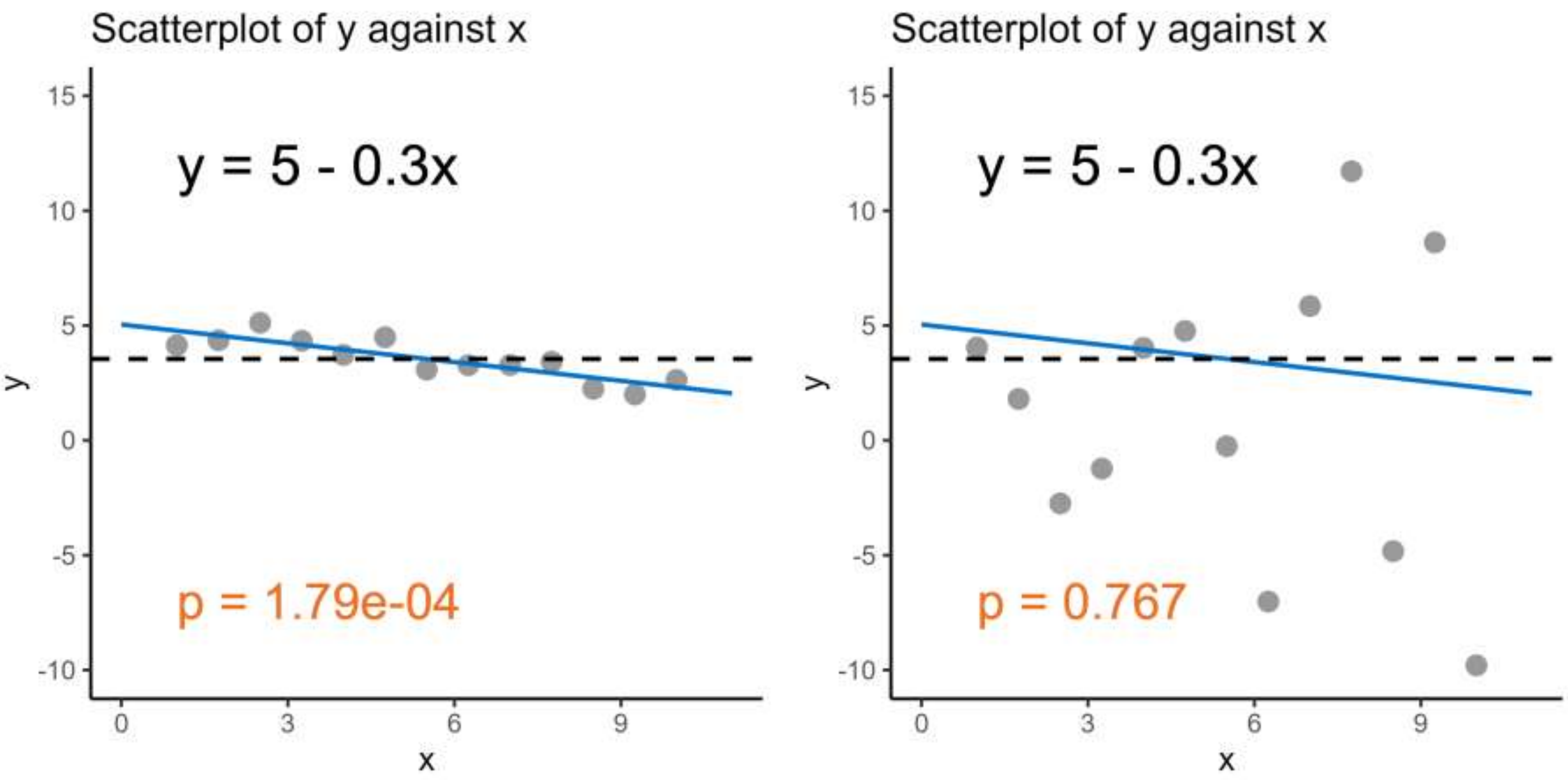
Significance testing for a line of best fit

Significance

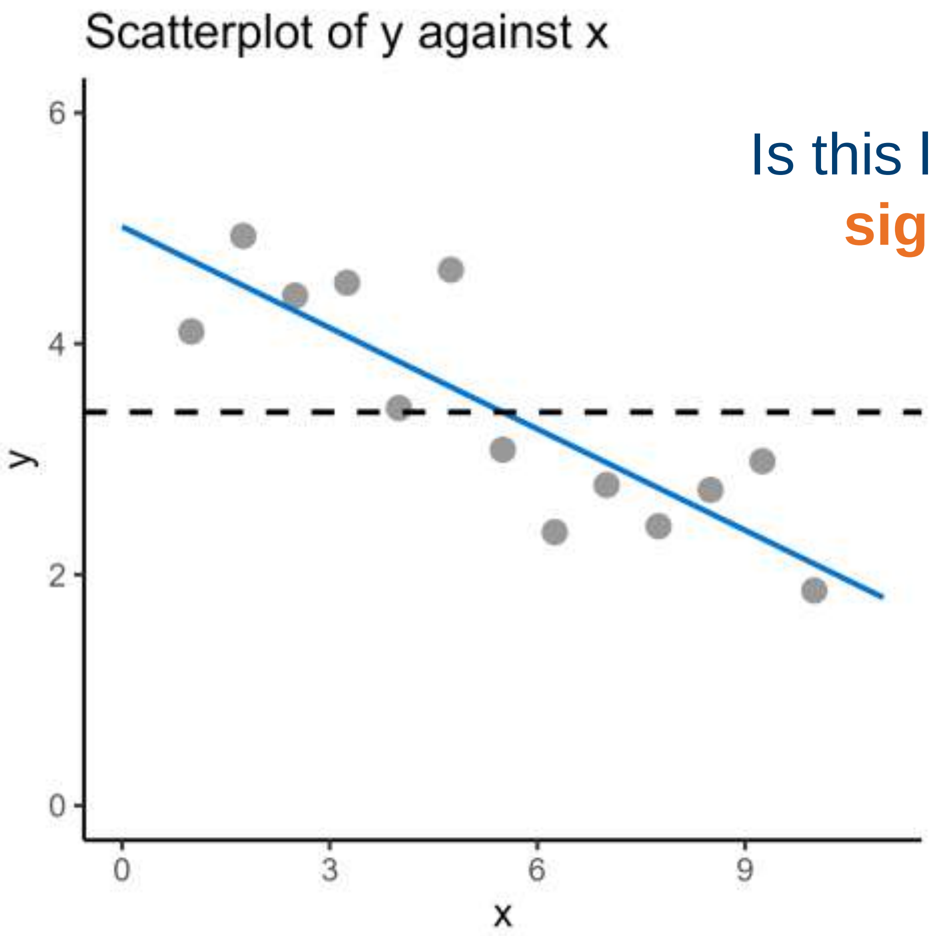




Significance

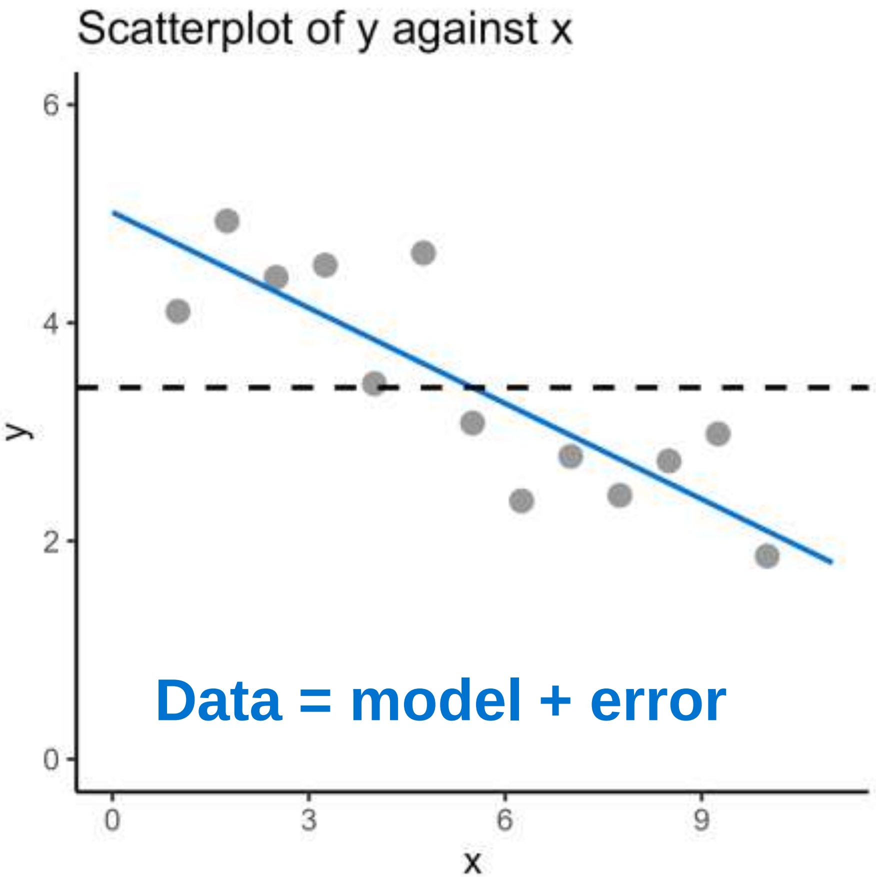


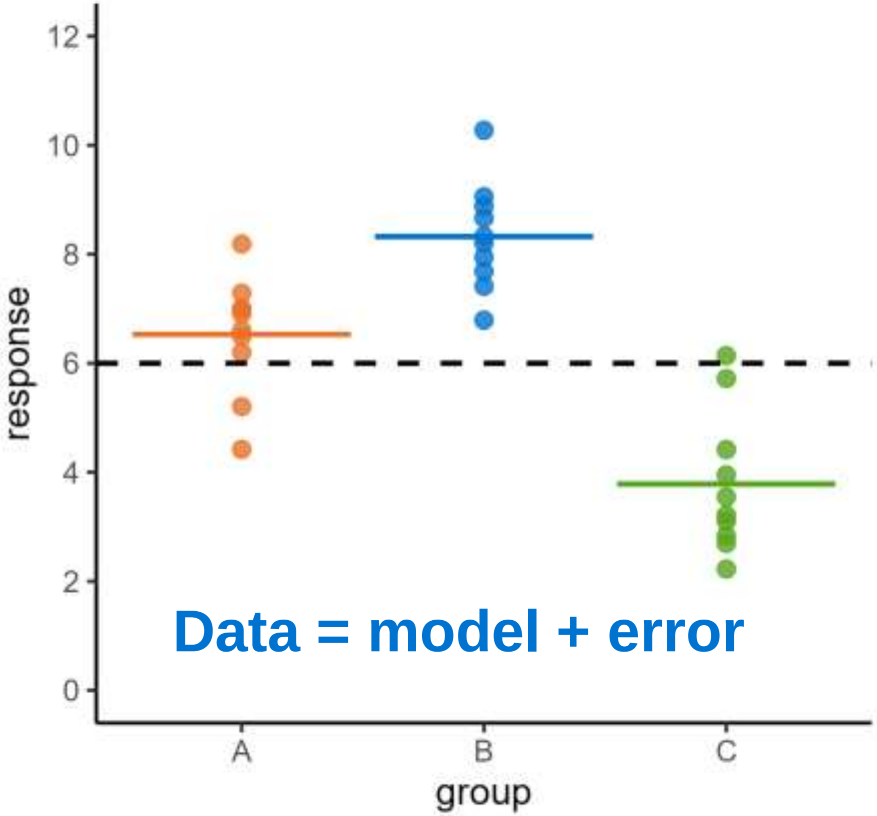
Regression analysis



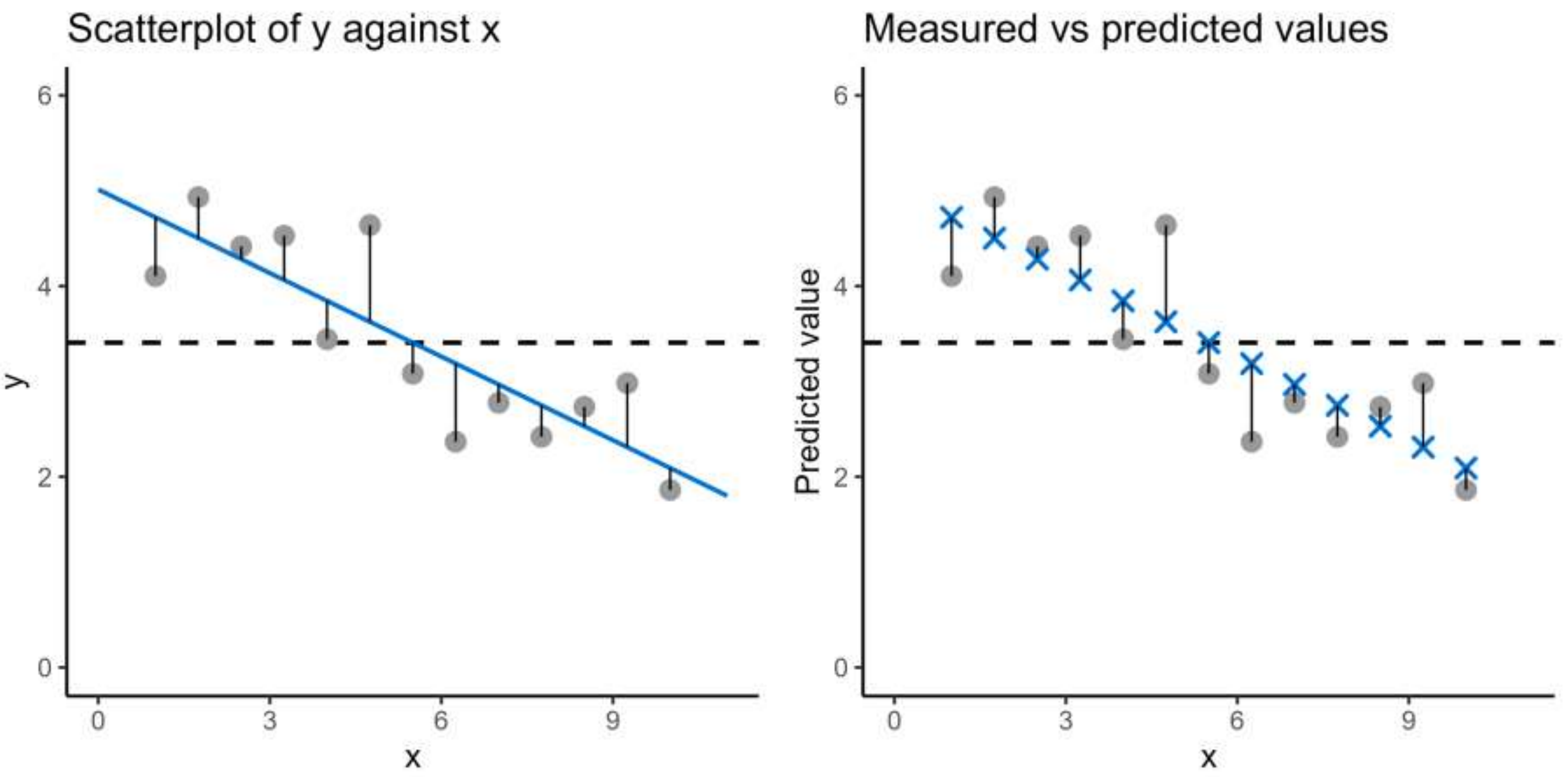
Is this line of best fit **significant**?

Regression analysis

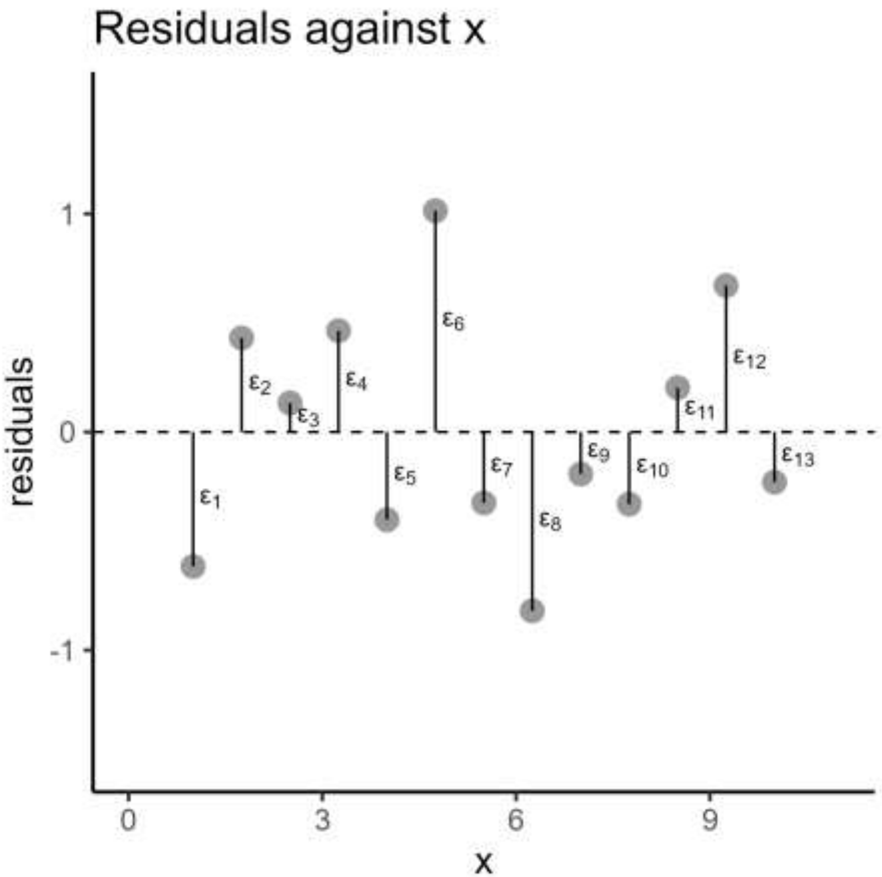


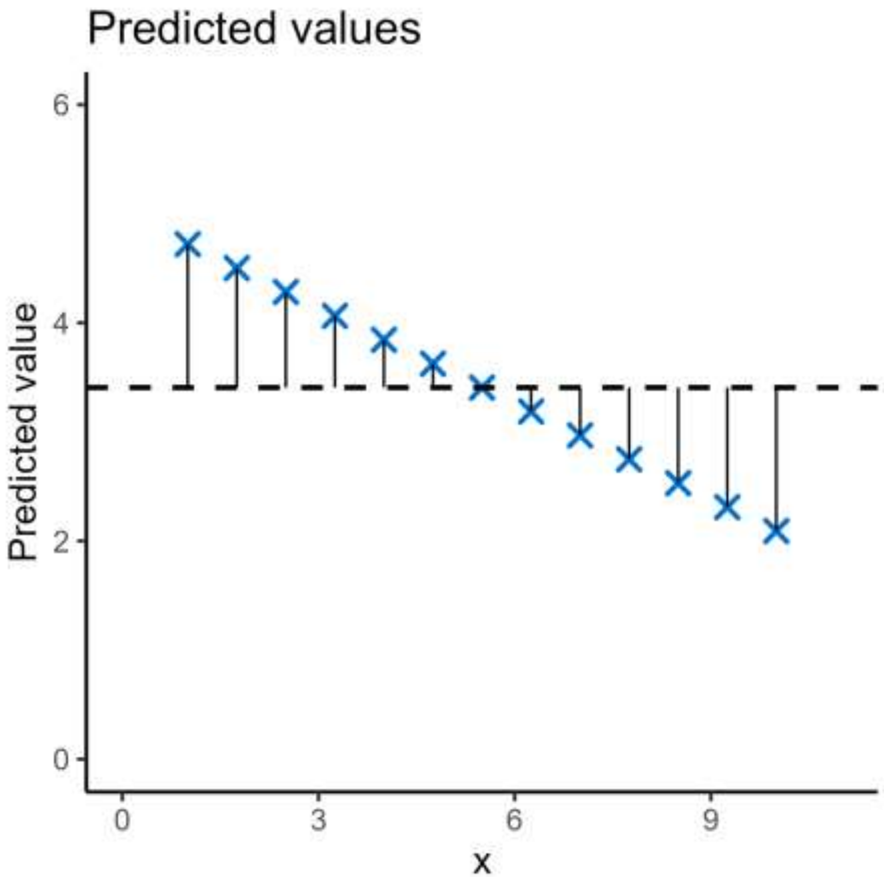


Regression analysis

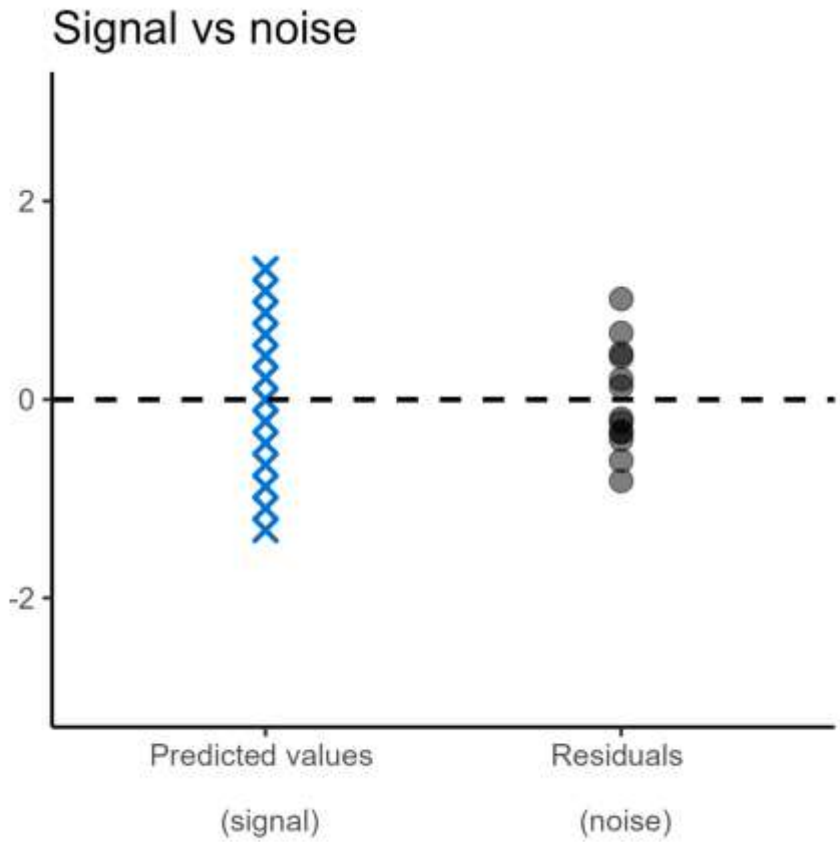


Regression analysis

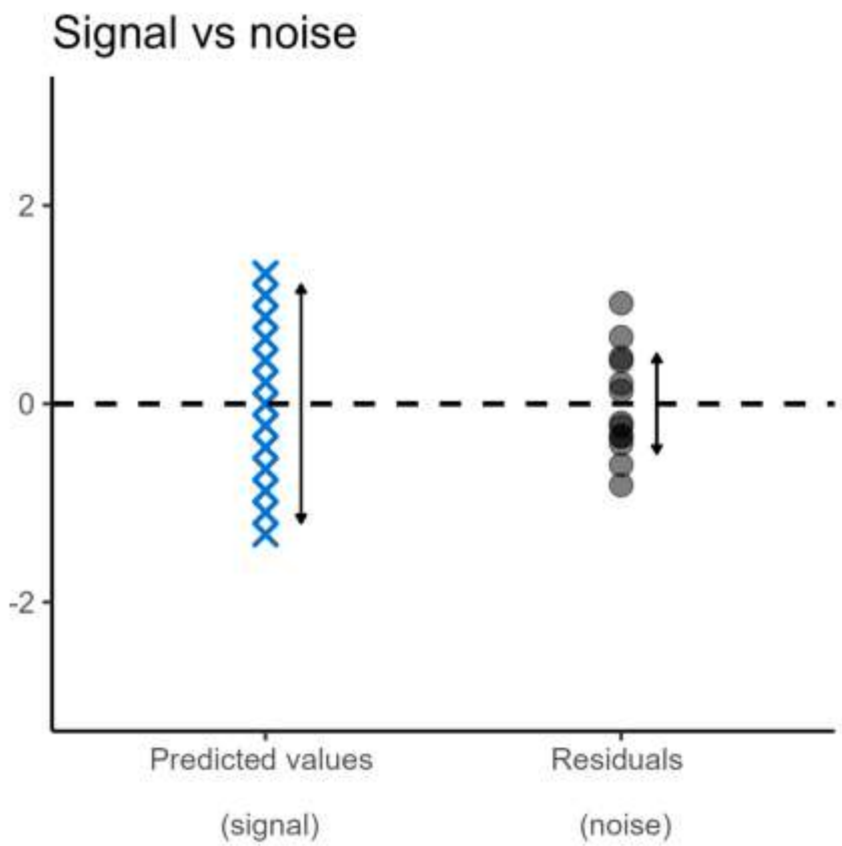




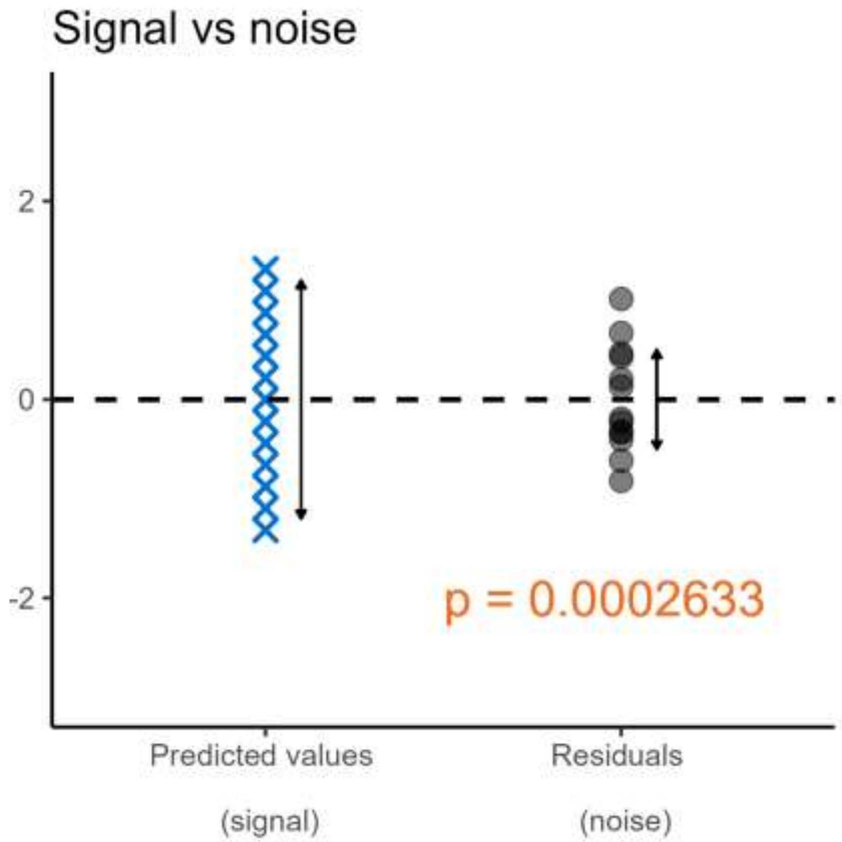
Regression analysis



Regression analysis

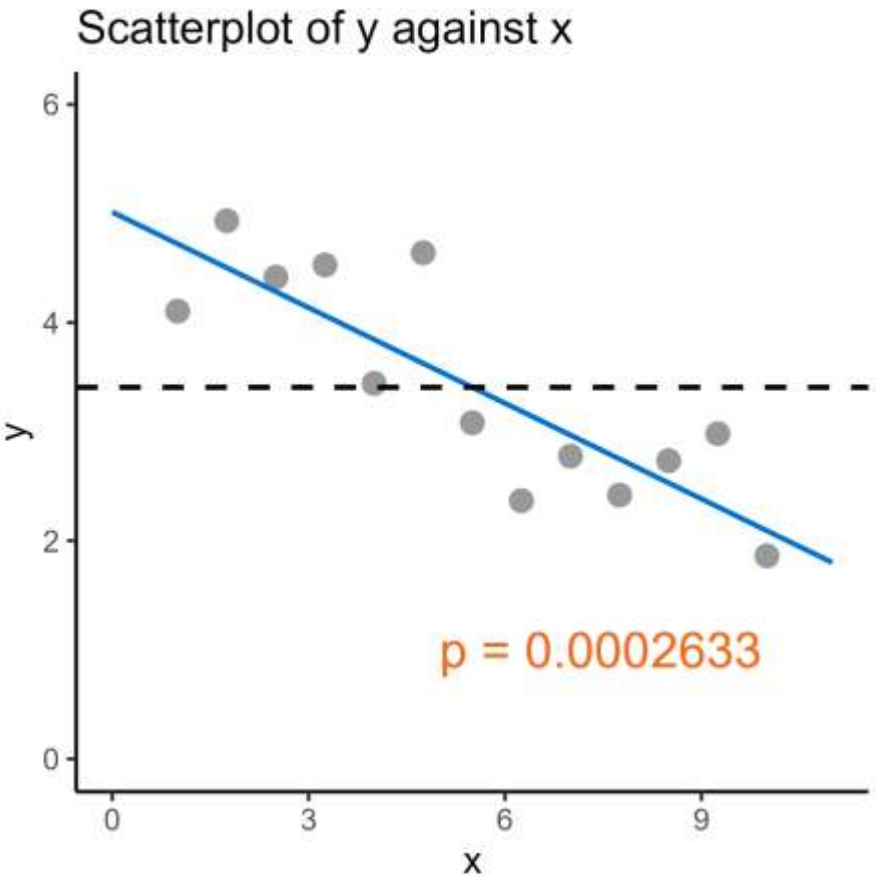


Regression analysis



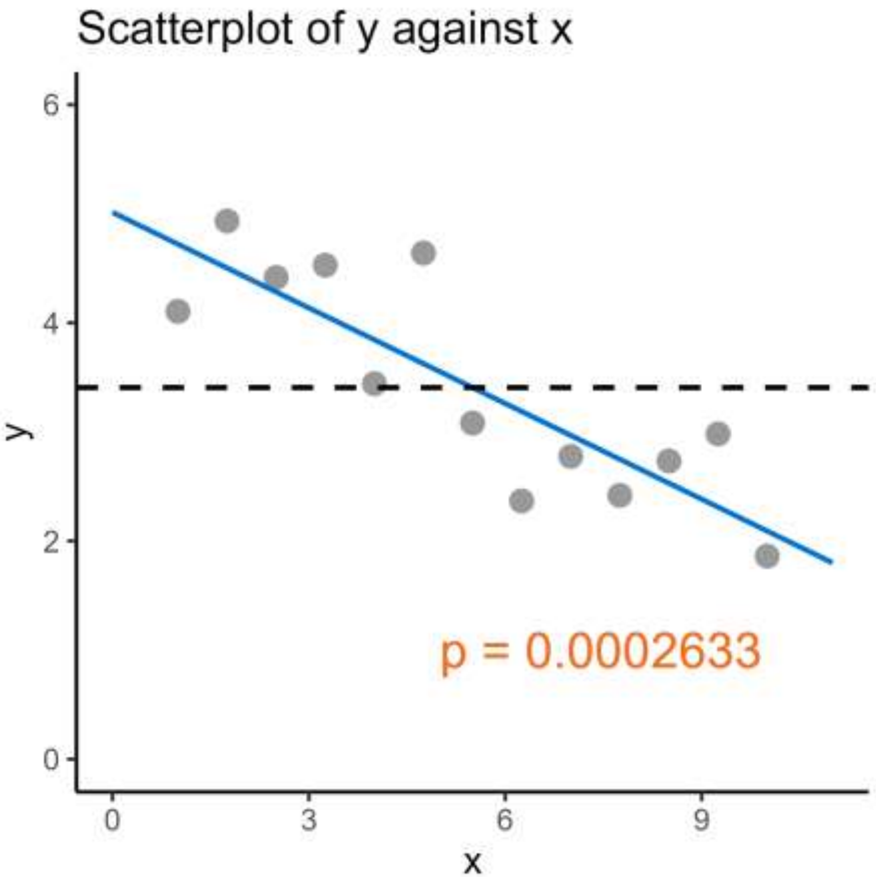
- Signal-to-noise ratio is significantly big

Regression analysis



- Signal-to-noise ratio is significantly big

Regression analysis



- Signal-to-noise ratio is significantly big
- Line of best fit is significantly better than the null
- The slope of the line is significantly different from 0
- As x changes, y also changes

Assumptions of simple linear regression

Assumptions:

Assumptions of simple linear regression

Assumptions:

- Continuous response variable
- Independent observations

Assumptions of simple linear regression

Assumptions:

- Continuous response variable
- Independent observations

Information you need to know about the dataset

Assumptions of simple linear regression

Assumptions:

- Continuous response variable
- Independent observations

Information you need to know about the dataset

- Linear relationship (coefficients)
- Homogeneity of variance in residuals
- Normally distributed residuals

- *Lack of points with high leverage*

Assumptions of simple linear regression

Assumptions:

- Continuous response variable
- Independent observations

Information you need to know about the dataset

- Linear relationship (coefficients)
- Homogeneity of variance in residuals
- Normally distributed residuals

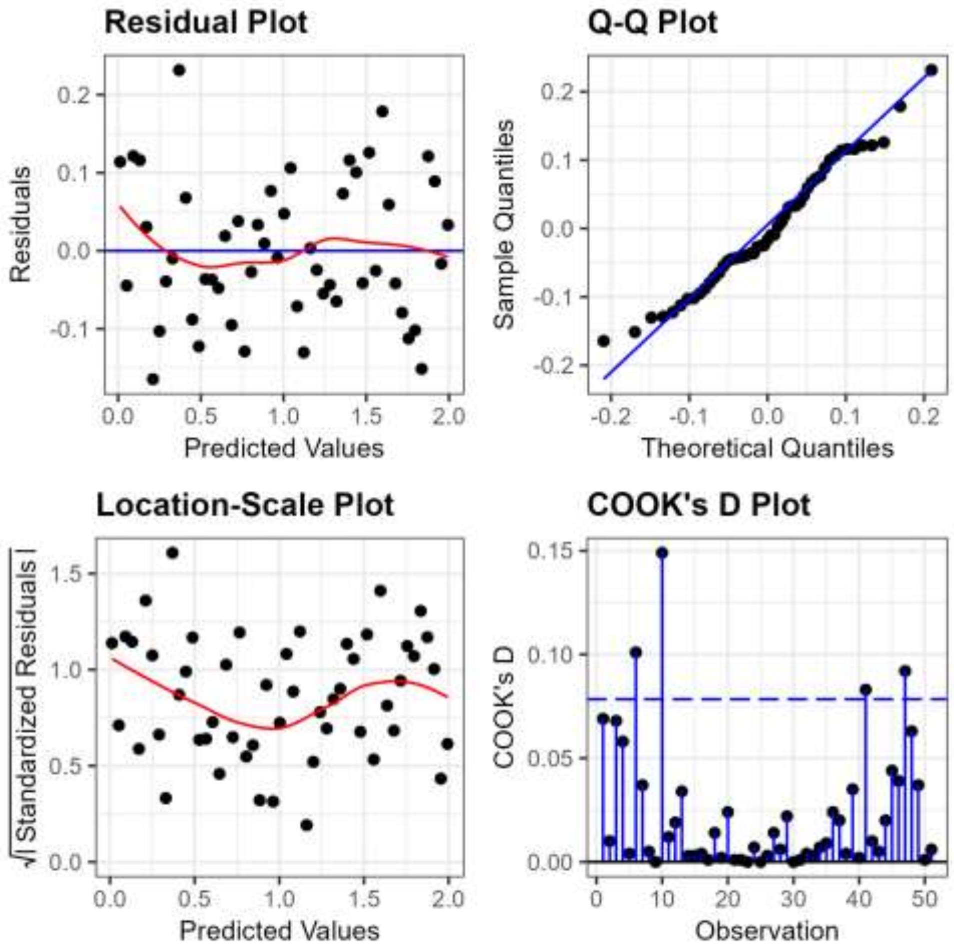
- *Lack of points with high leverage*

Information you need to find out

Diagnostic plots

Linearity

Normality



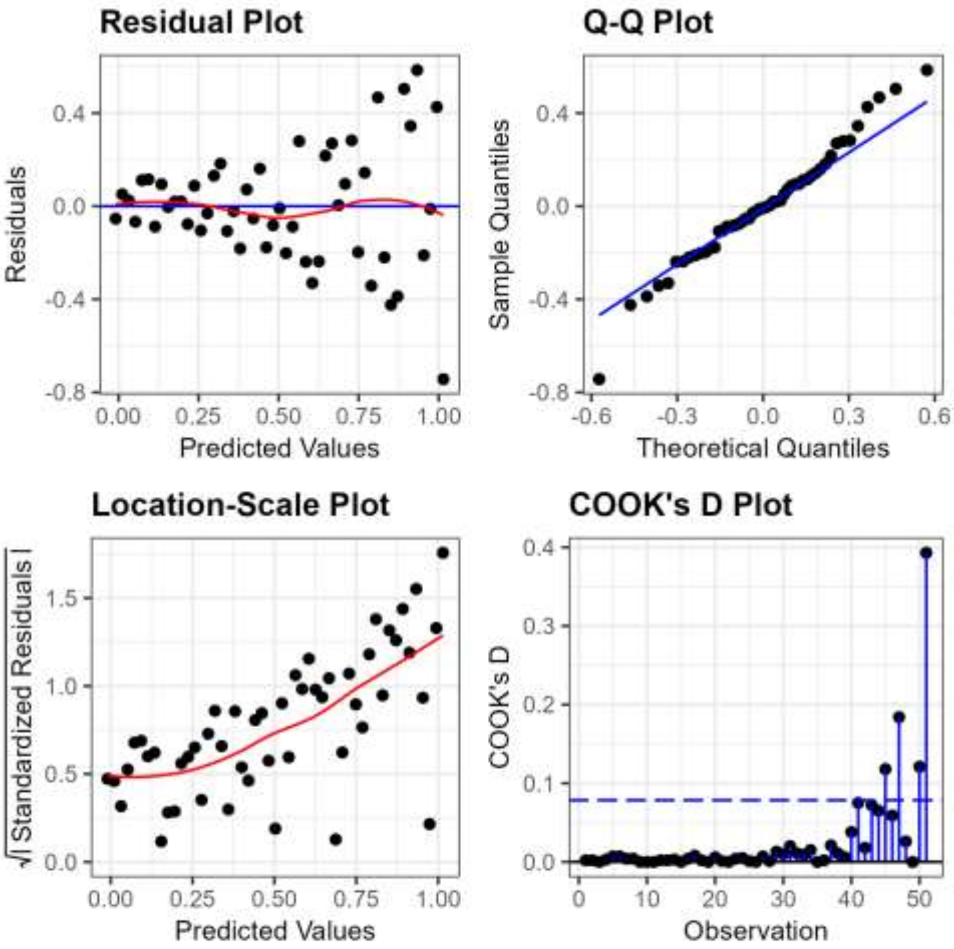
Equality of Variance

Influential Points

Diagnostic plots

Linearity

Normality

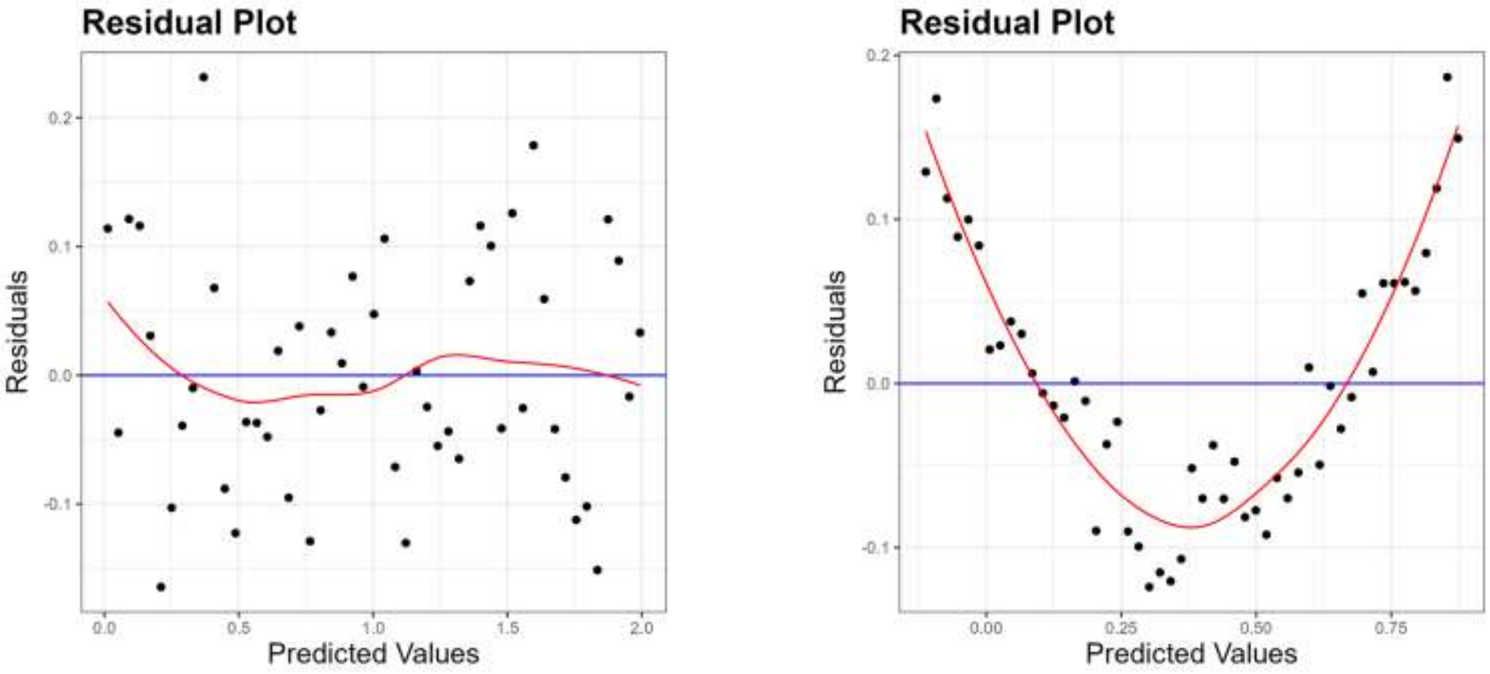


Equality of Variance

Influential Points

Residuals (vs fitted) plot

- This checks for **linearity** in the relationship between variables
- The red line should have no distinct pattern



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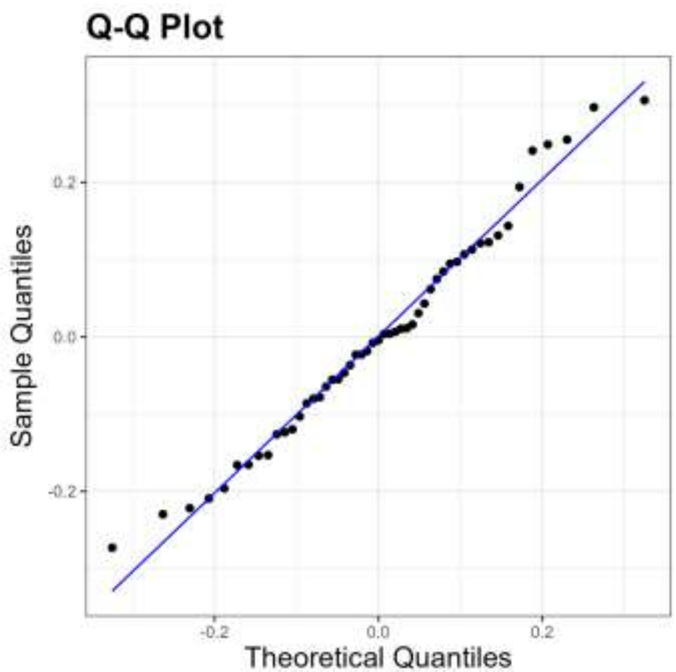
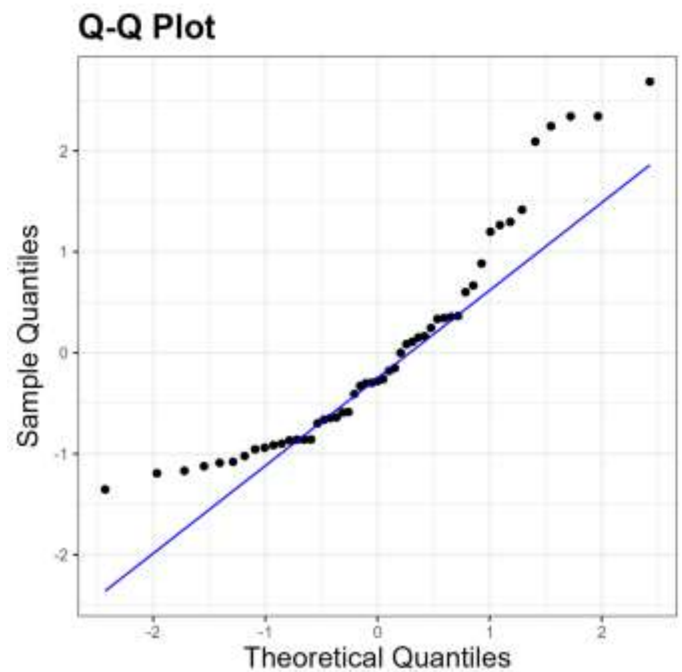
Normal Q-Q plot

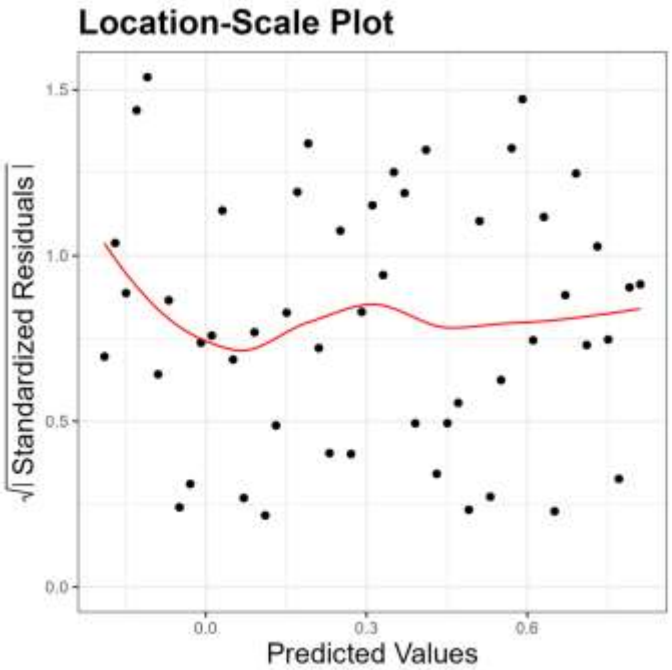
- This checks if the residuals are **normally distributed**
- Residuals should not deviate massively from the straight line

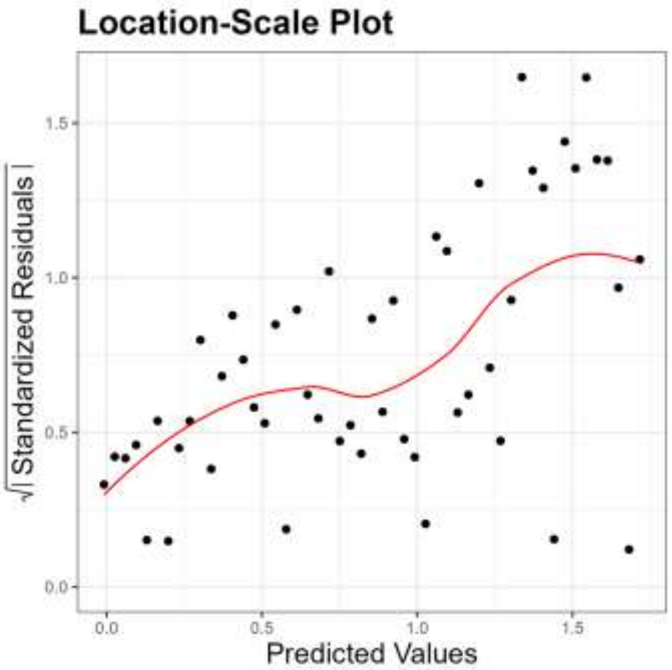




Location-Scale Plot

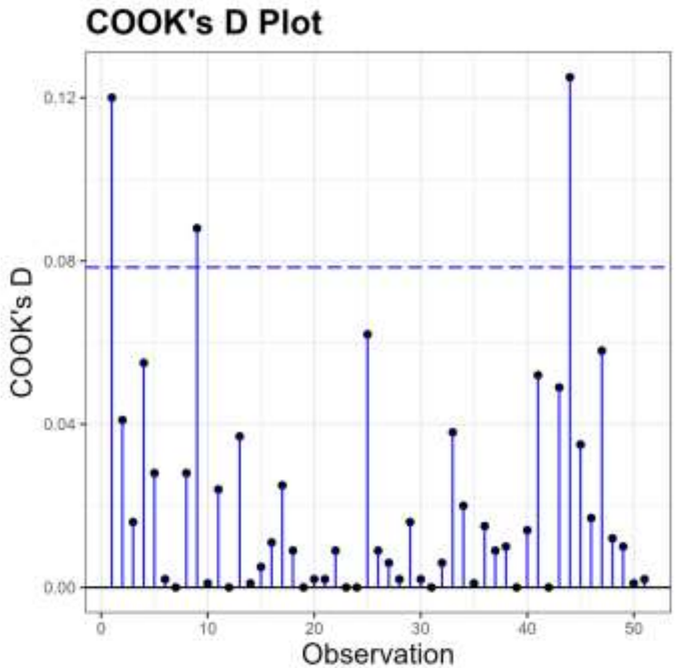
- This shows **equality of variance**
- You should see a horizontal line with equally spread points

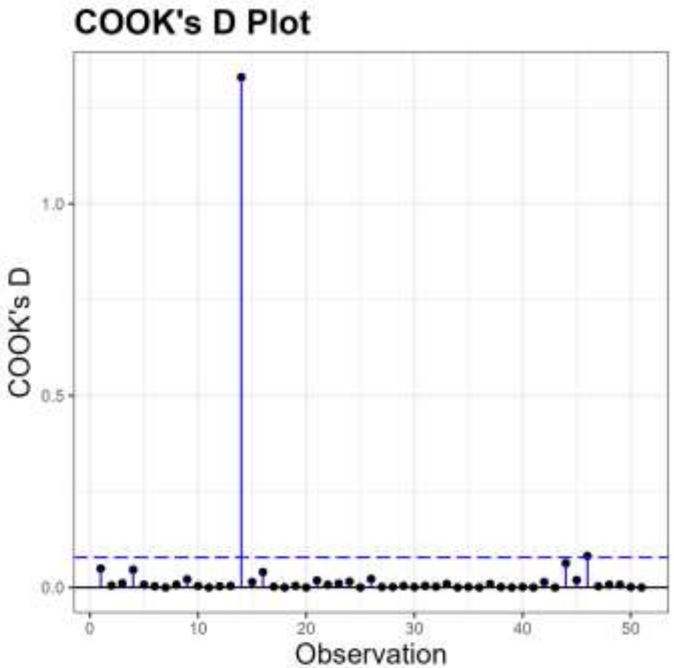




Cook's D plot

- This ensures no observations are overly influential
- You should perform checks on data points >1 Cook's distance



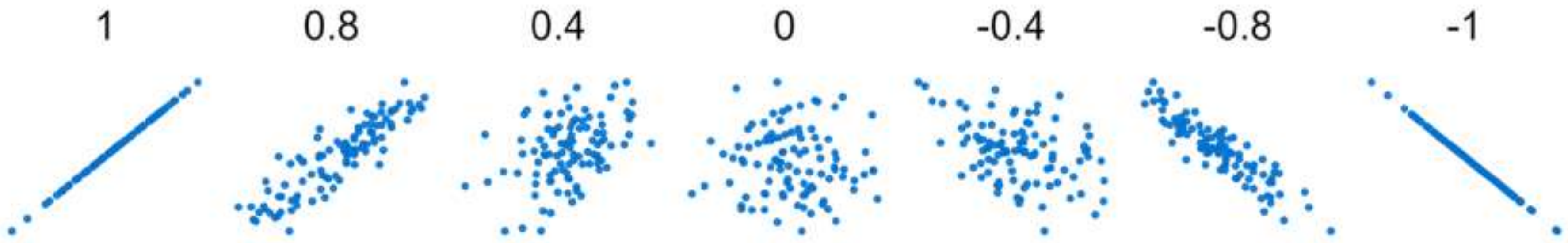


3. Correlation

Measuring the strength of association between variables

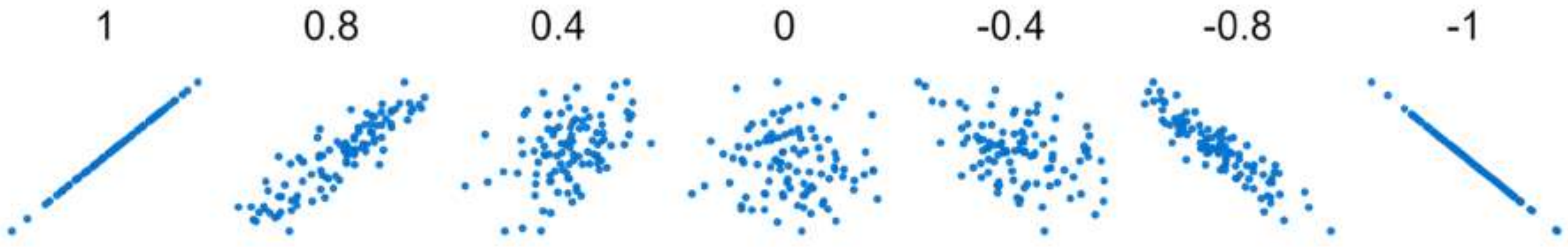
Correlation

Correlation = strength of association between two variables



Correlation

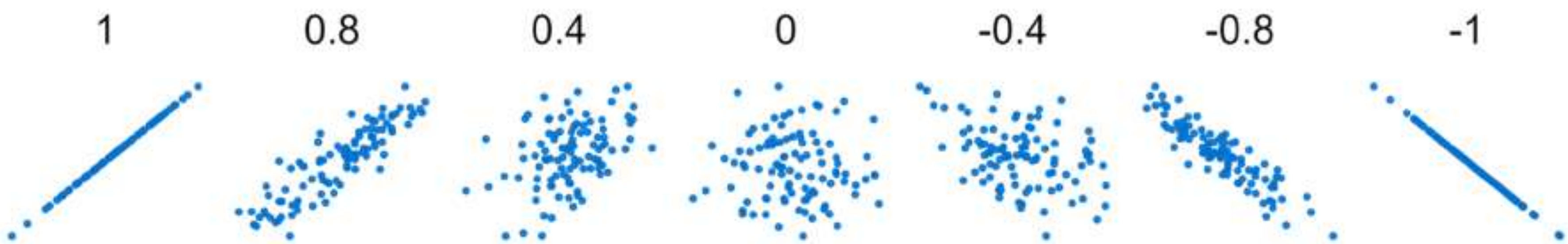
Correlation = strength of association between two variables



Requires a linear relationship between the variables

Correlation

Correlation = strength of association between two variables



The gradient of the slope does not matter

Pearson product moment correlation coefficient

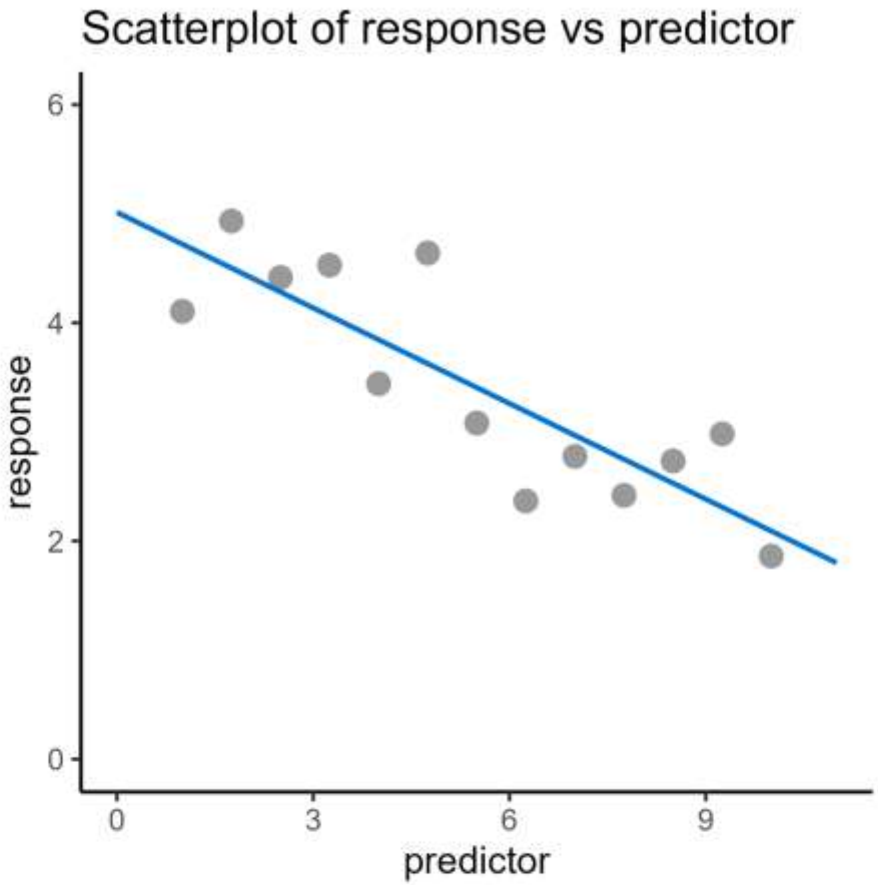
Pearson product moment correlation coefficient

- Denoted by r
- Estimation (statistic) of the true population correlation
- Varies between -1 and 1

R^2 has a special interpretation

- Represents amount of variance explained
- Varies between 0 and 1

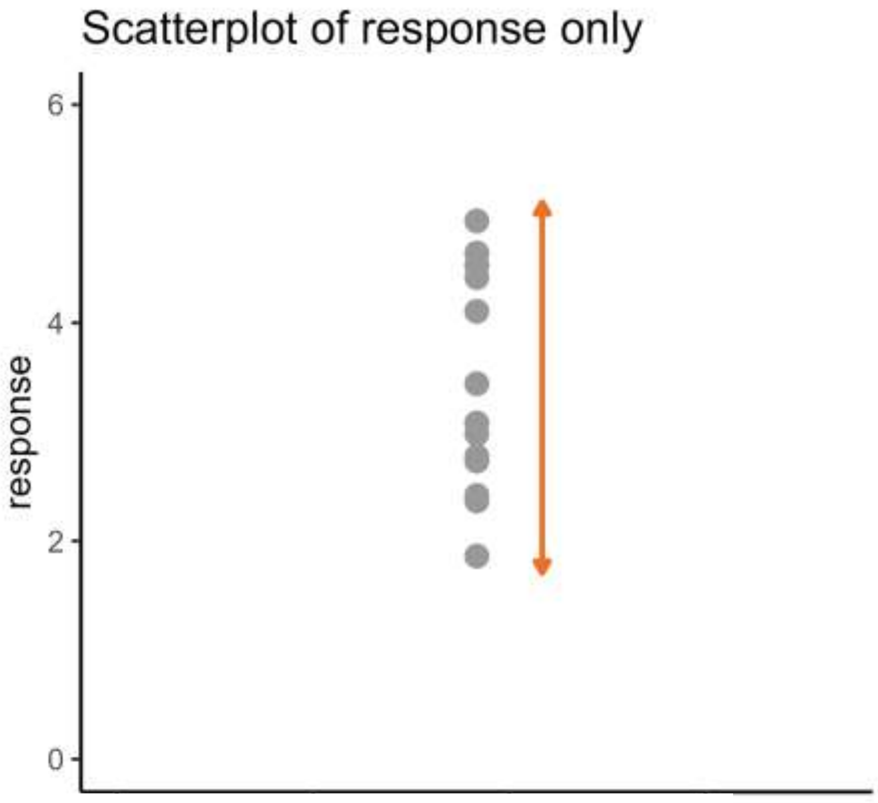
Pearson product moment correlation coefficient



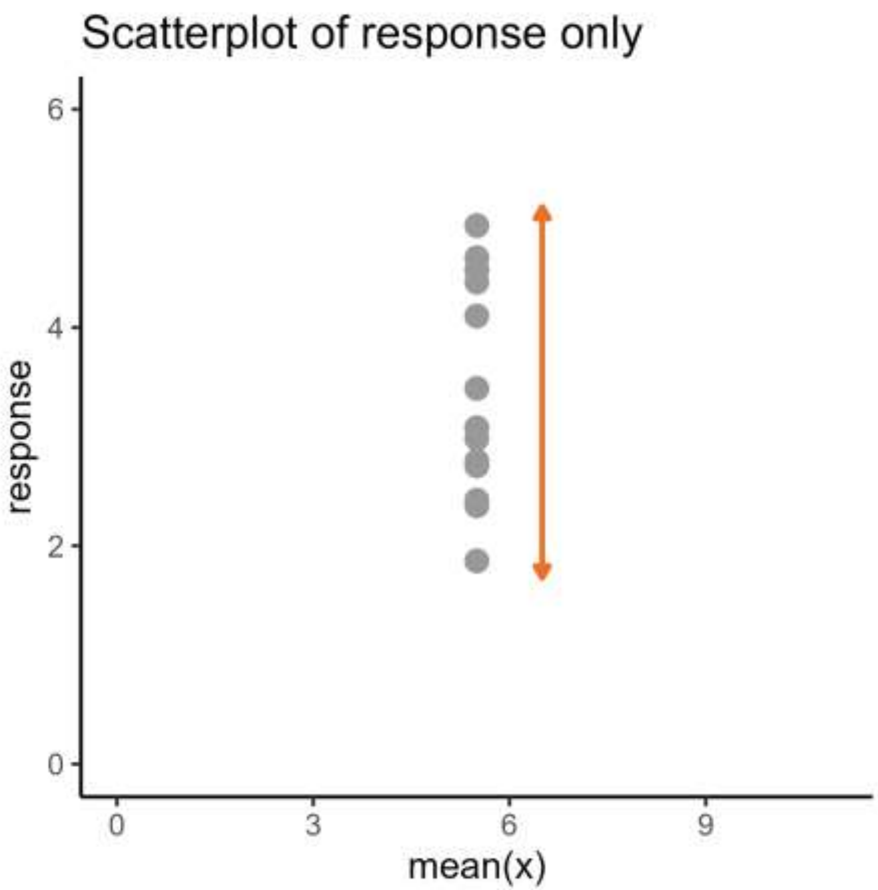
$R^2 = 0.72$

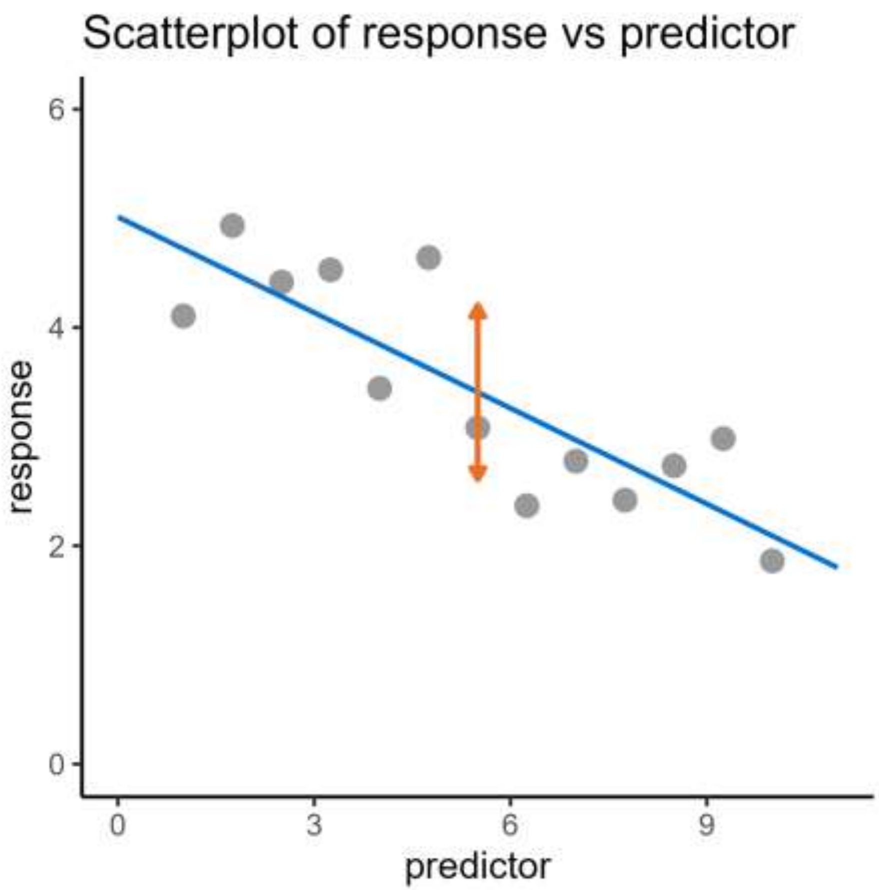
72% of the variance in y is explained by the model

Pearson product moment correlation coefficient

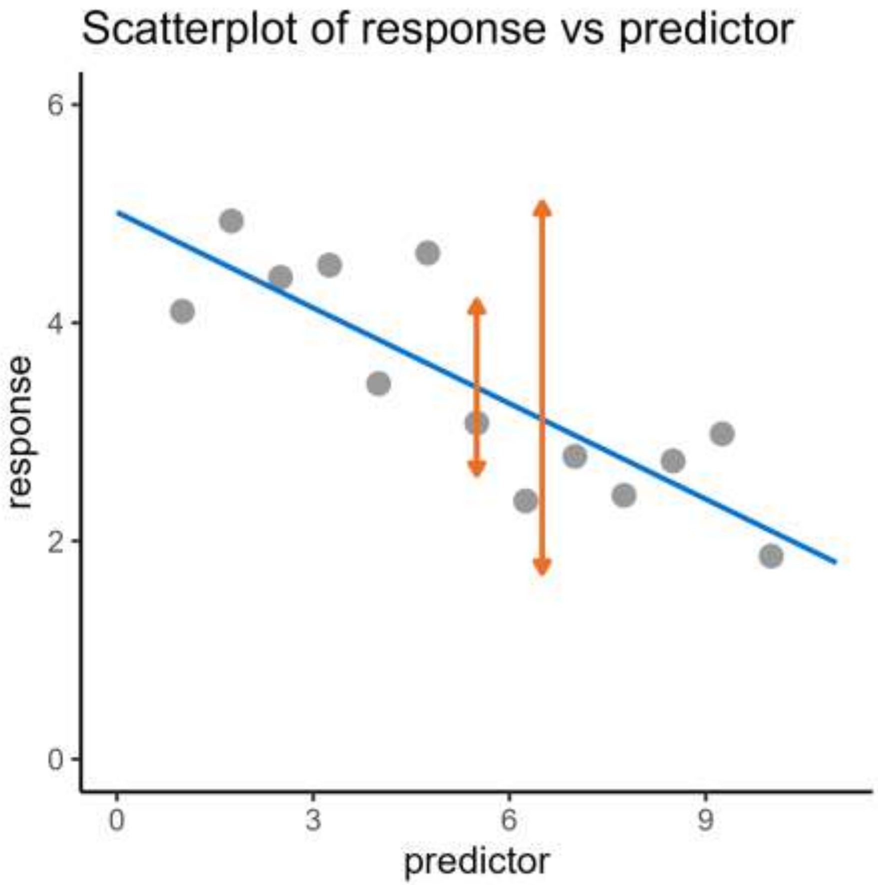


Significance

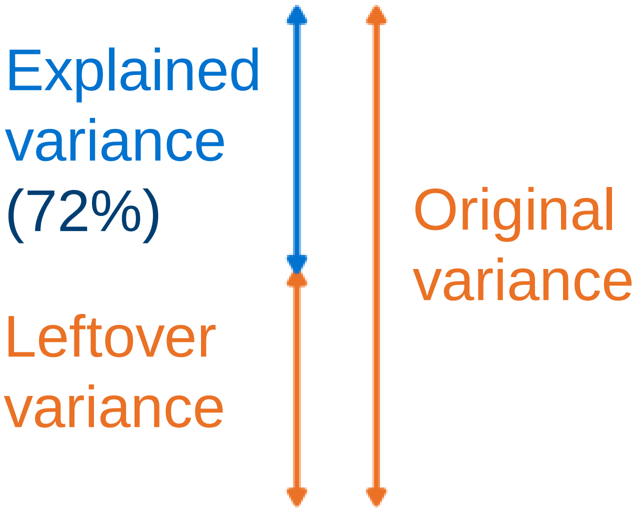




Pearson product moment correlation coefficient



$R^2 = 0.72$



Session 3: Continuous predictors

Lecture summary:

Line of best fit: **the signal**

- Describes the trend in the dataset
- How much does y (response) change as x (predictor) changes?

Correlation: **the noise**

- Strength of association between variables
- How much error around the line of best fit?

Regression analysis: **the signal-to-noise ratio**

- Compares the line of best fit to the null
- Is there sufficient evidence of a relationship between x and y?

Session 3: Continuous predictors

Course materials:

Regression analysis

- Diagnostic plots

Correlation

- Pearson's r
- Spearman's ρ (non-parametric)

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Core statistics

Session 3: Continuous predictors

Cambridge Centre for Research Informatics Training

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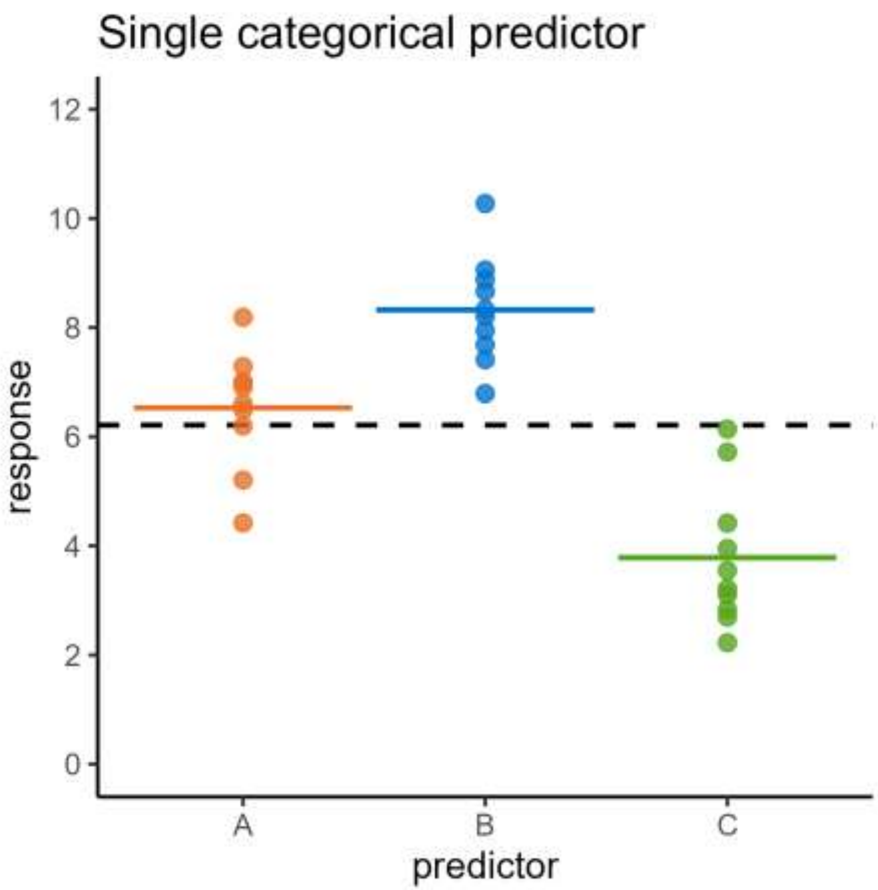
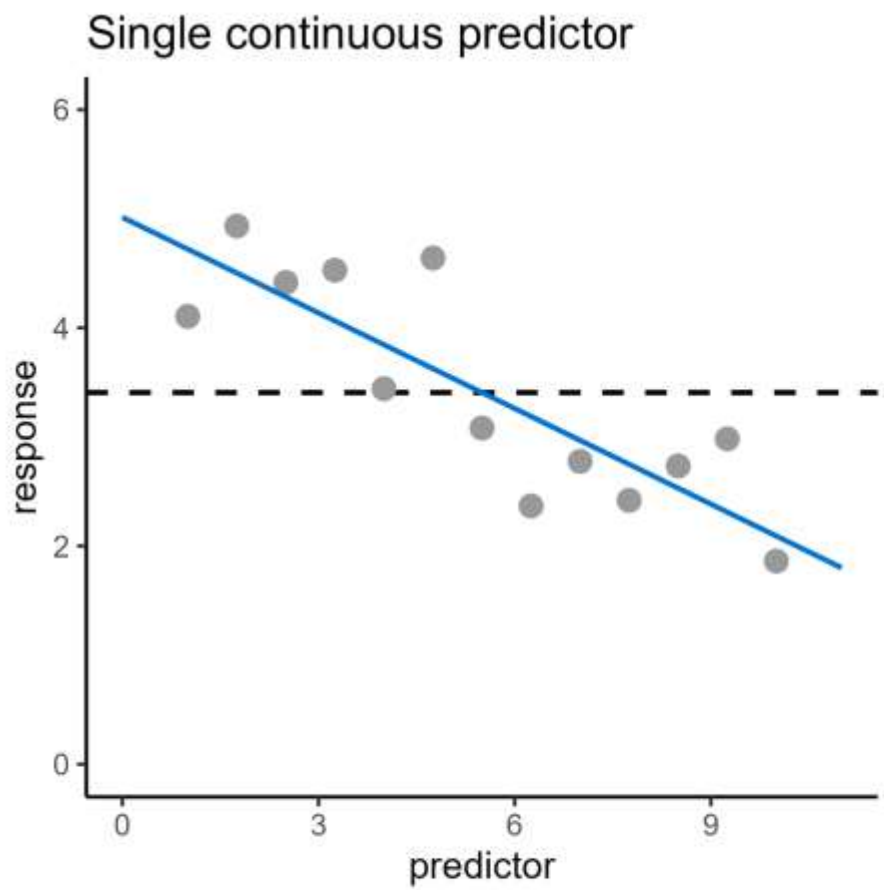
Core statistics

Session 4: Two predictors

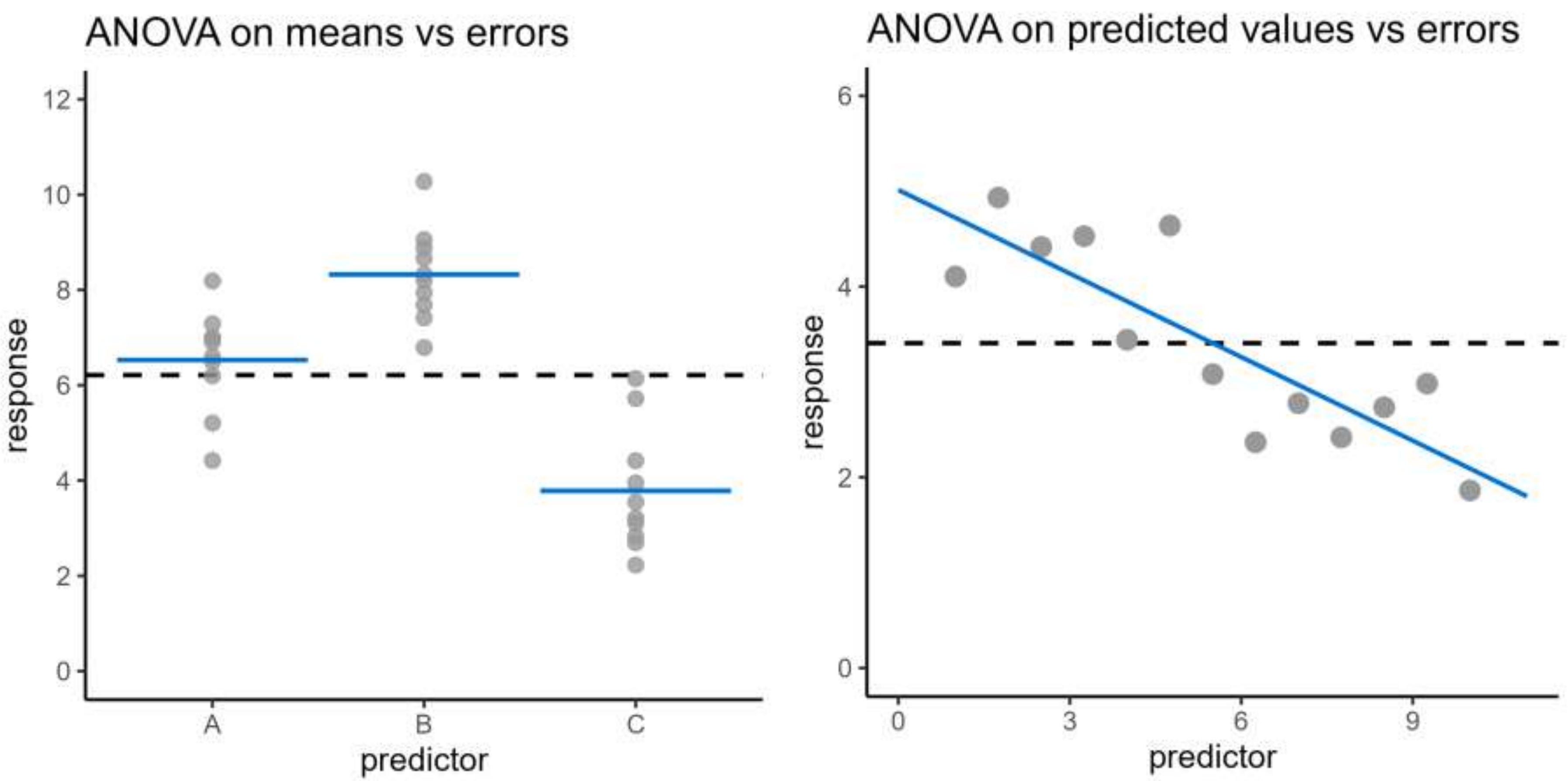
Cambridge Centre for Research Informatics Training

Recap of regression analysis

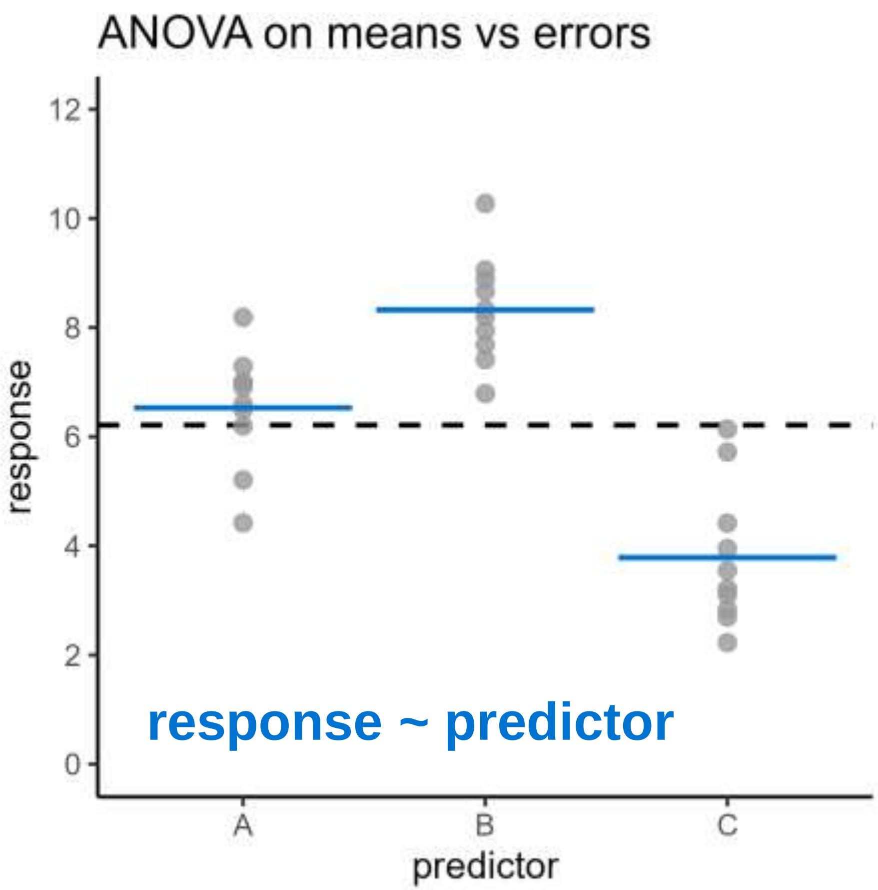


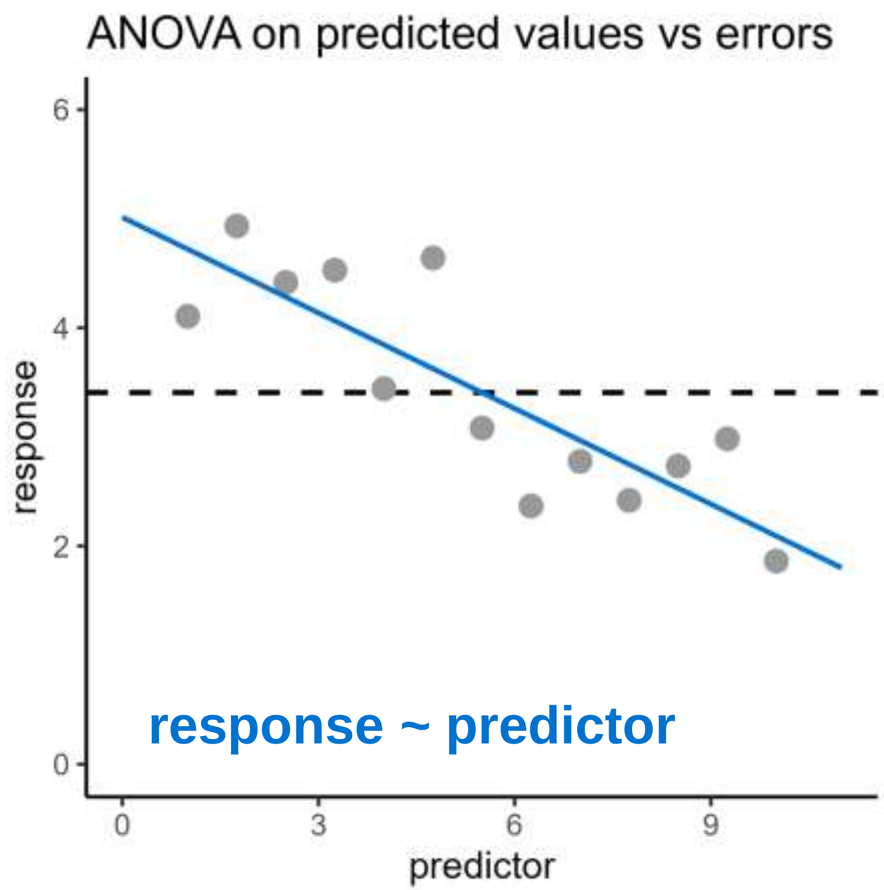


Recap of regression analysis



Recap of regression analysis





Plan for the session

Session 4: **Two predictors**

1. Two categorical predictors
2. Interpreting interactions
3. Grouped linear regression

1. Two categorical predictors

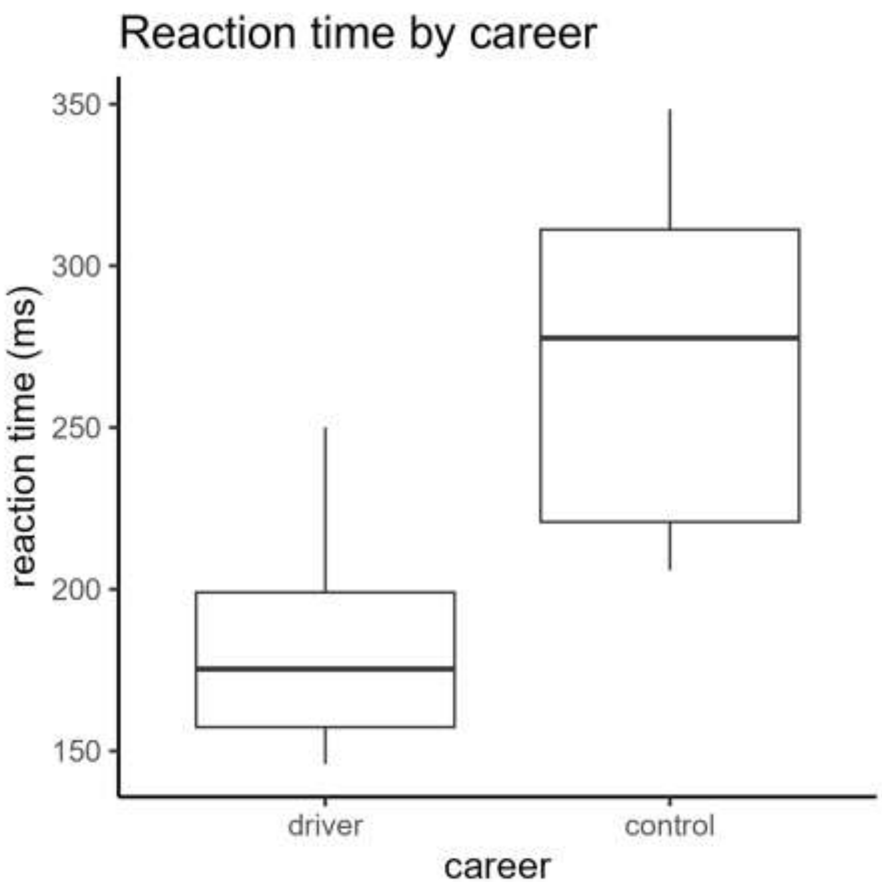
Running a two-way ANOVA test

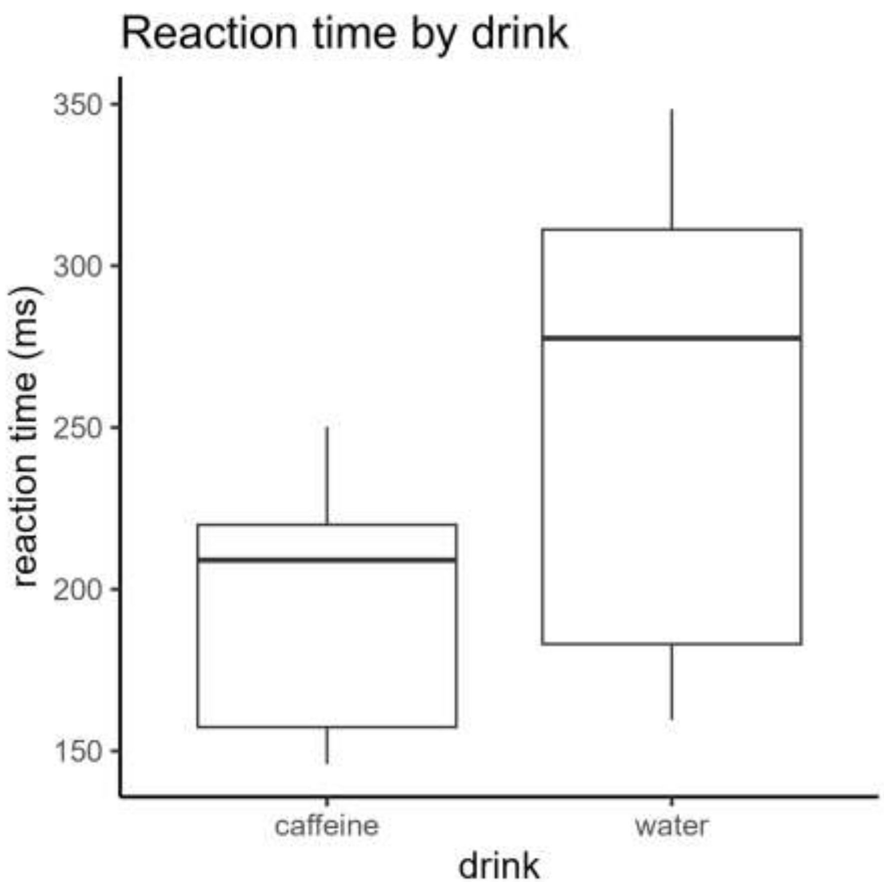
Two categorical predictors

reaction time (ms)		career	
		driver	control
drink	caffeine	141	219
		151	108
		158	205
	water	168	323
		153	336
		191	291

reaction time ~ **drink** AND **career**?

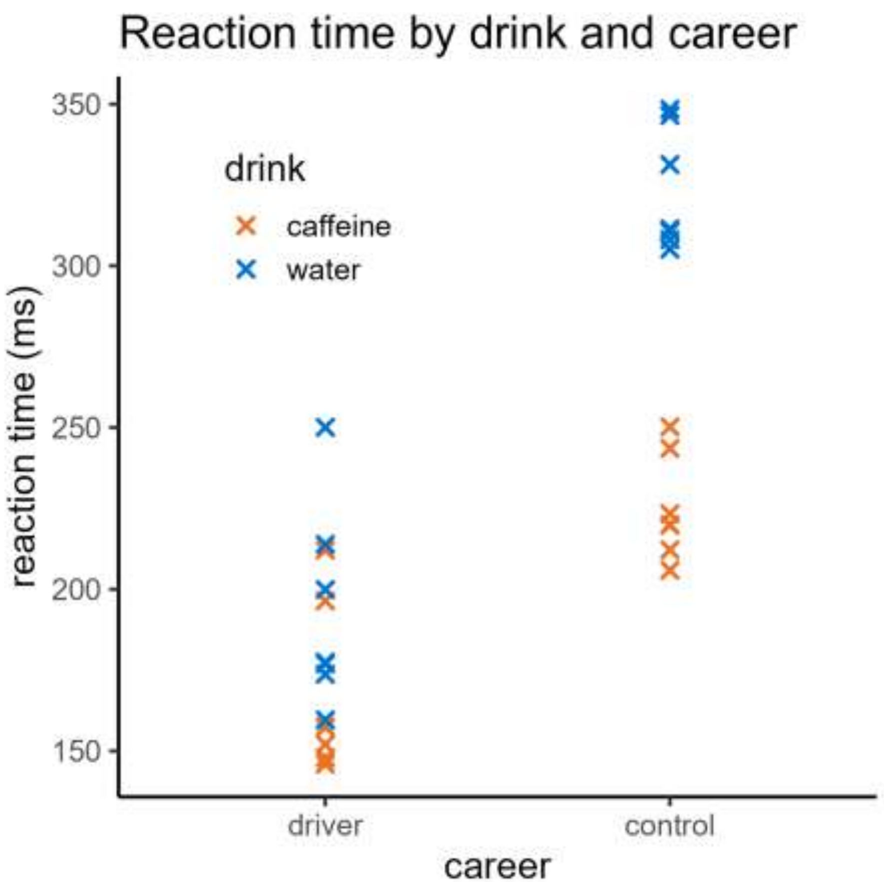
Two categorical predictors

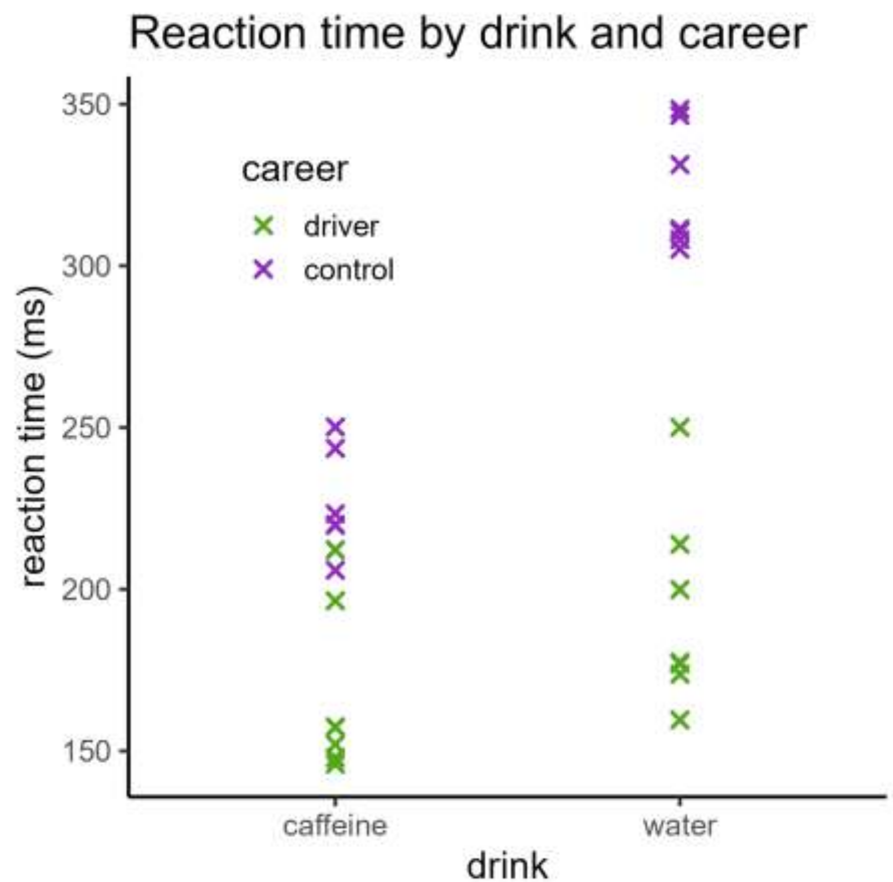




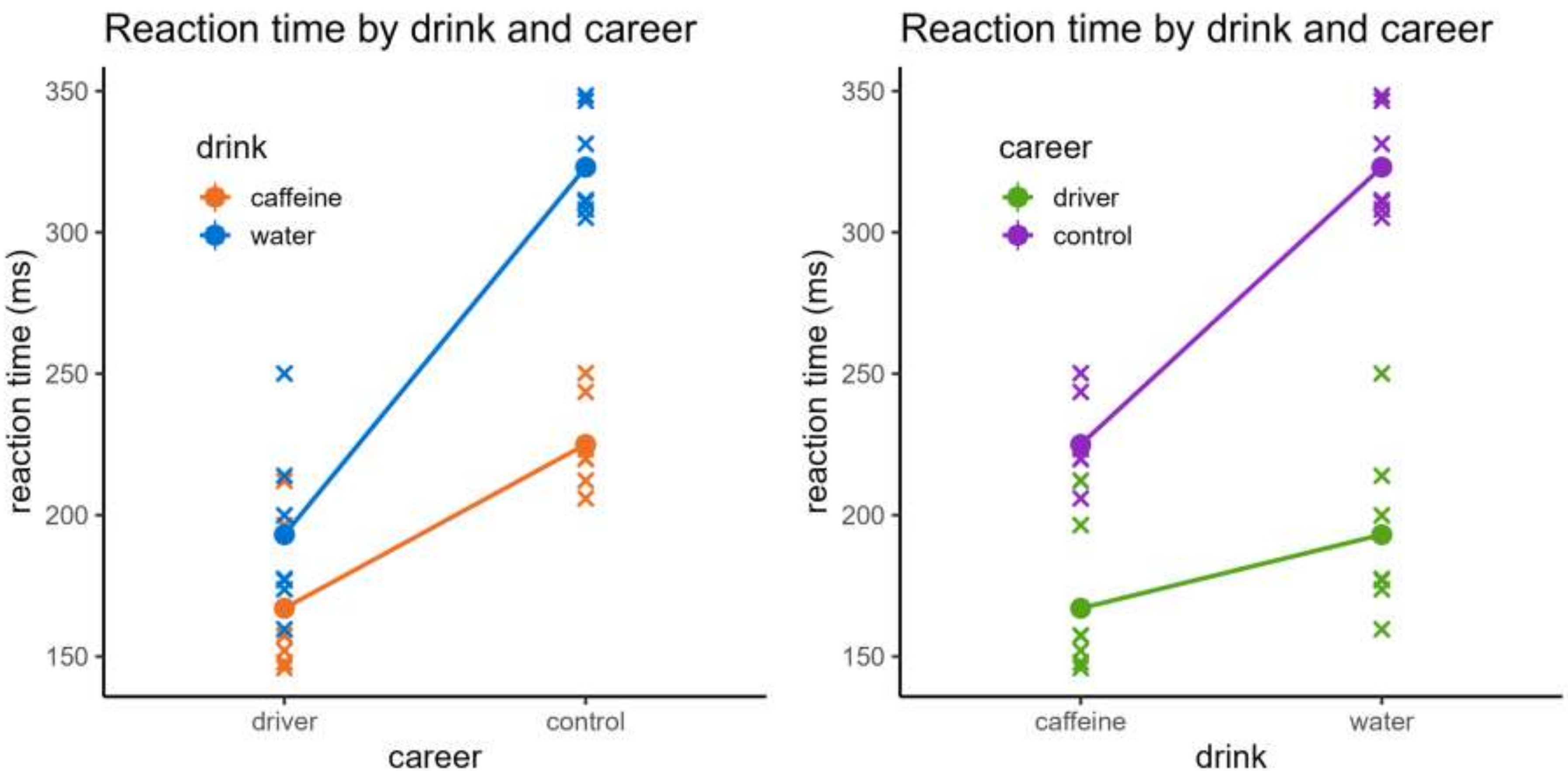
Two categorical predictors

Two categorical predictors

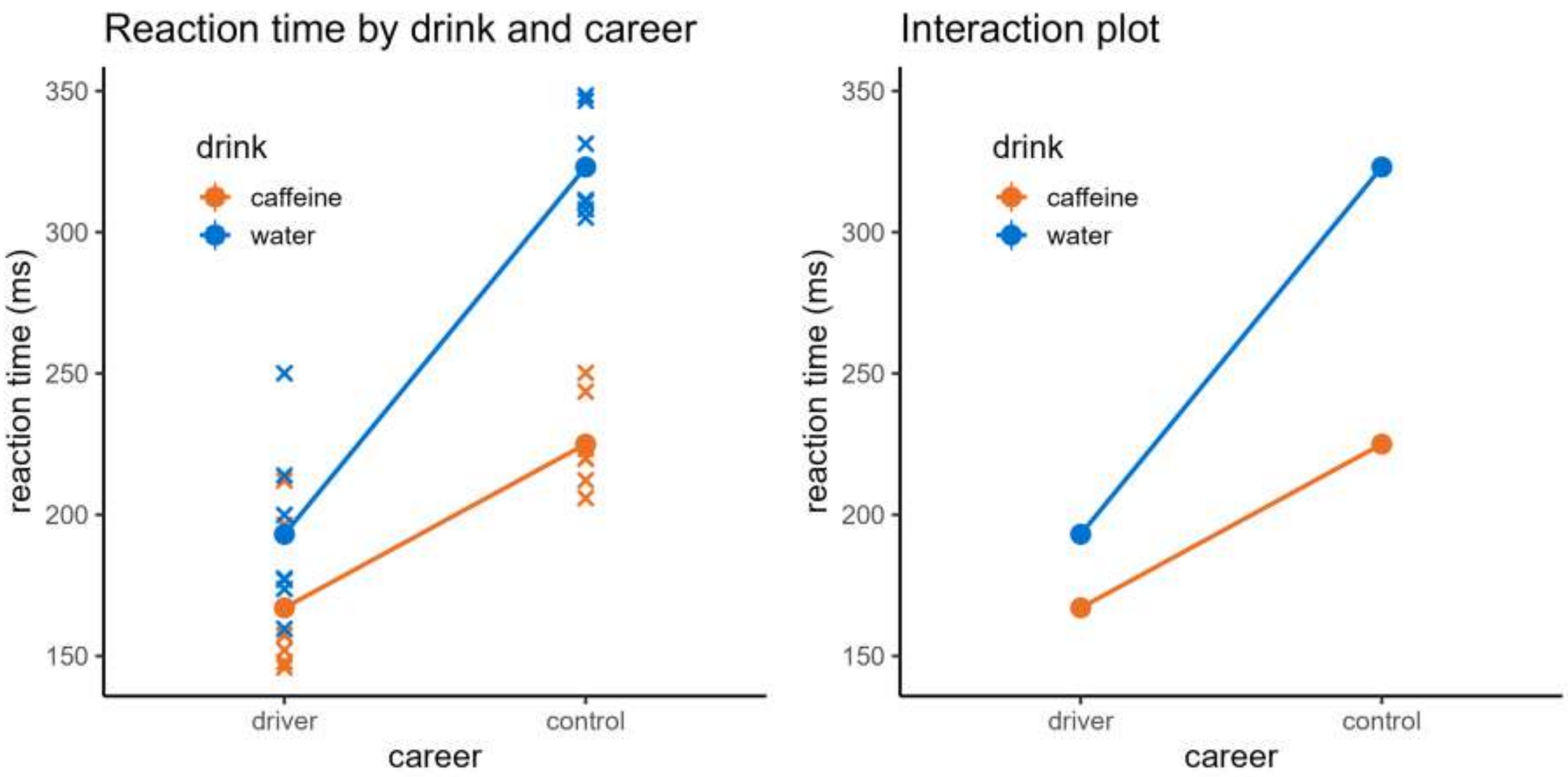




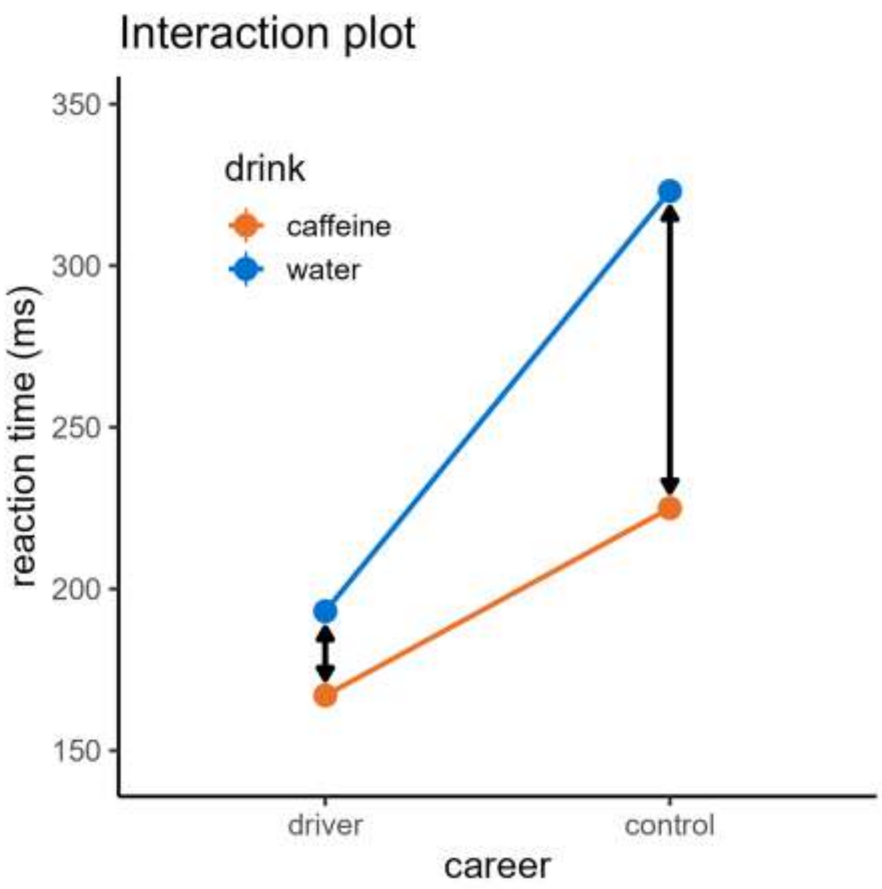
Two categorical predictors



Two categorical predictors



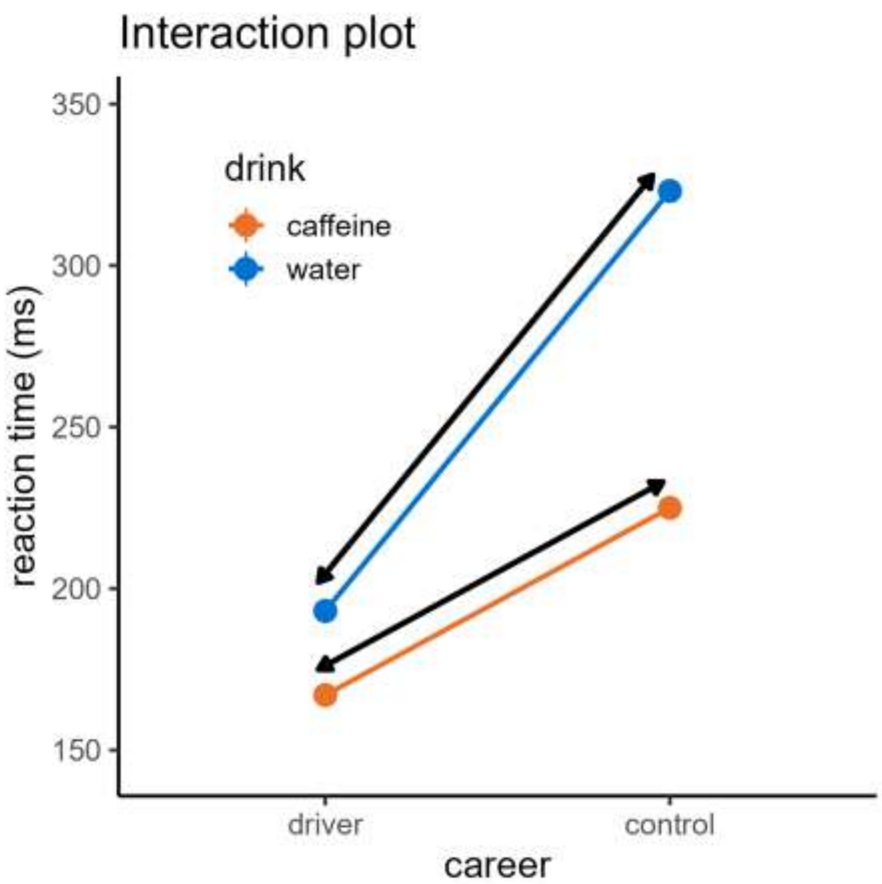
Visualising interactions



Different effect of Drink

For each Career

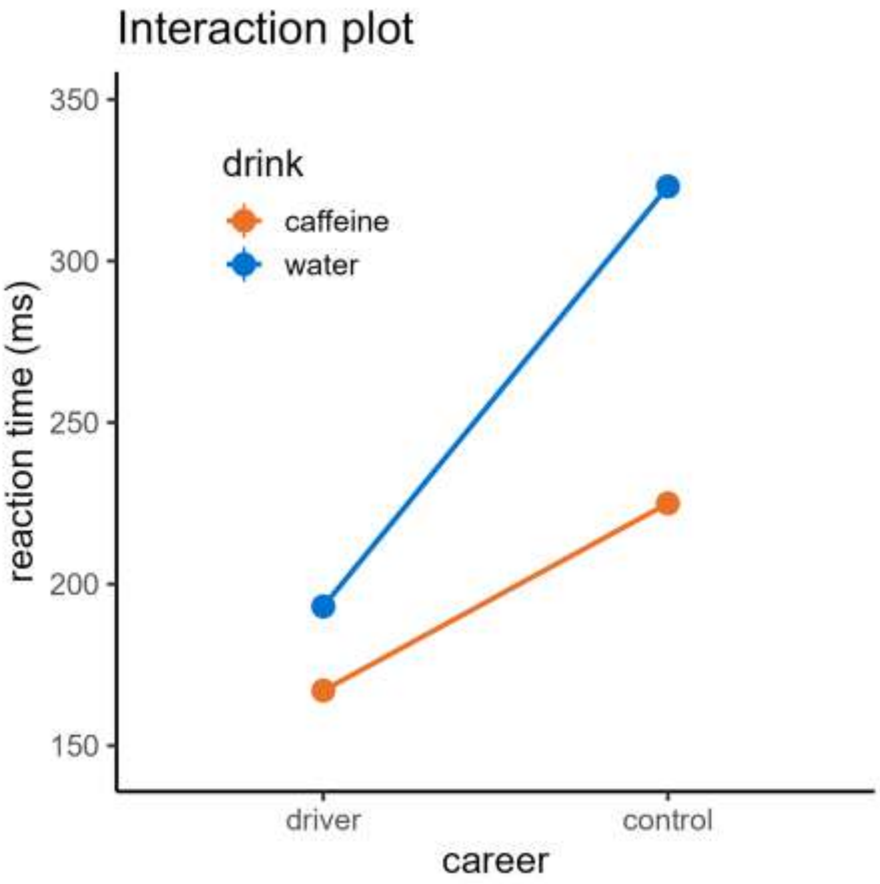
Visualising interactions



Different effect of Career

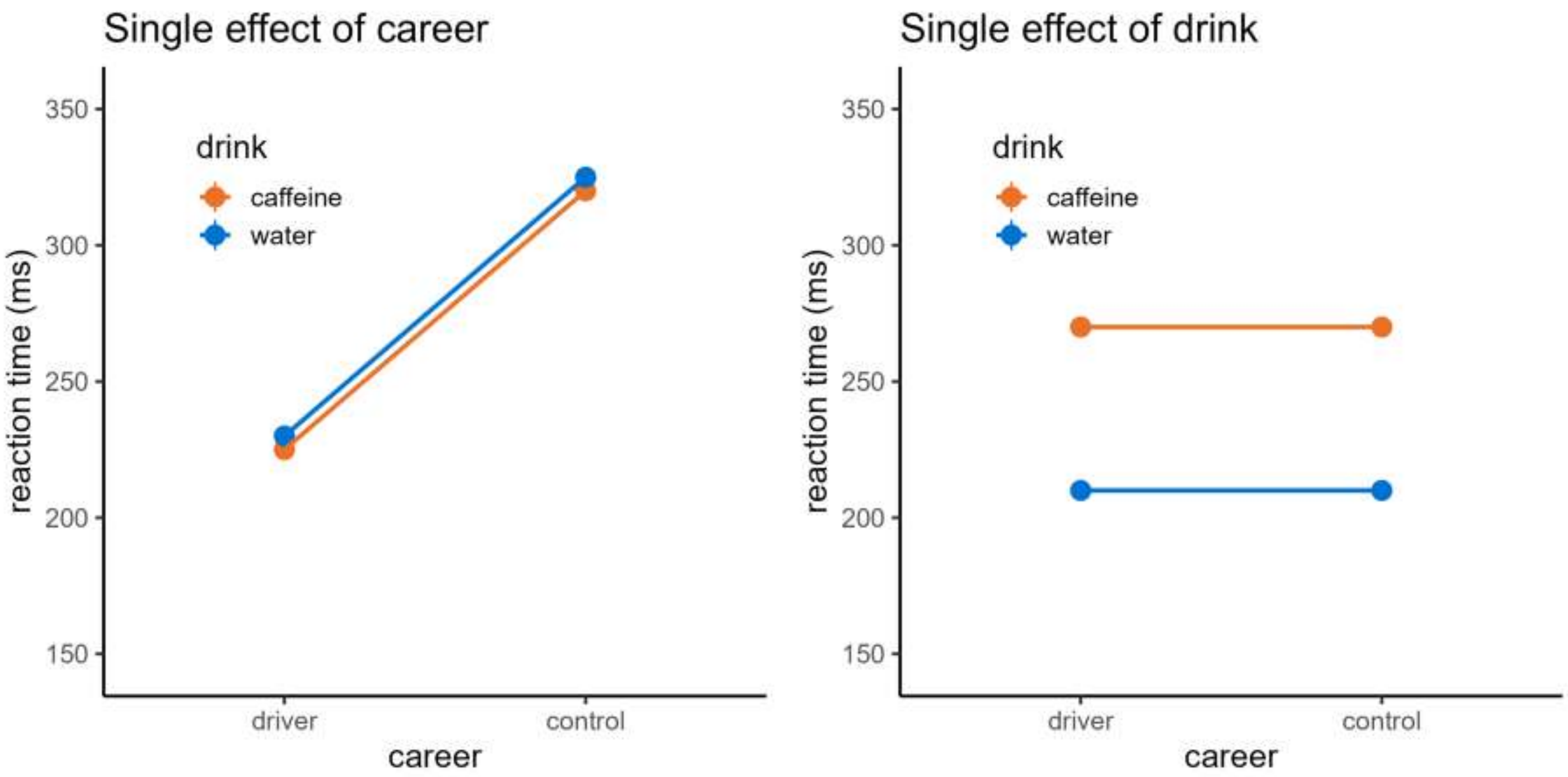
For each Drink

Other situations?

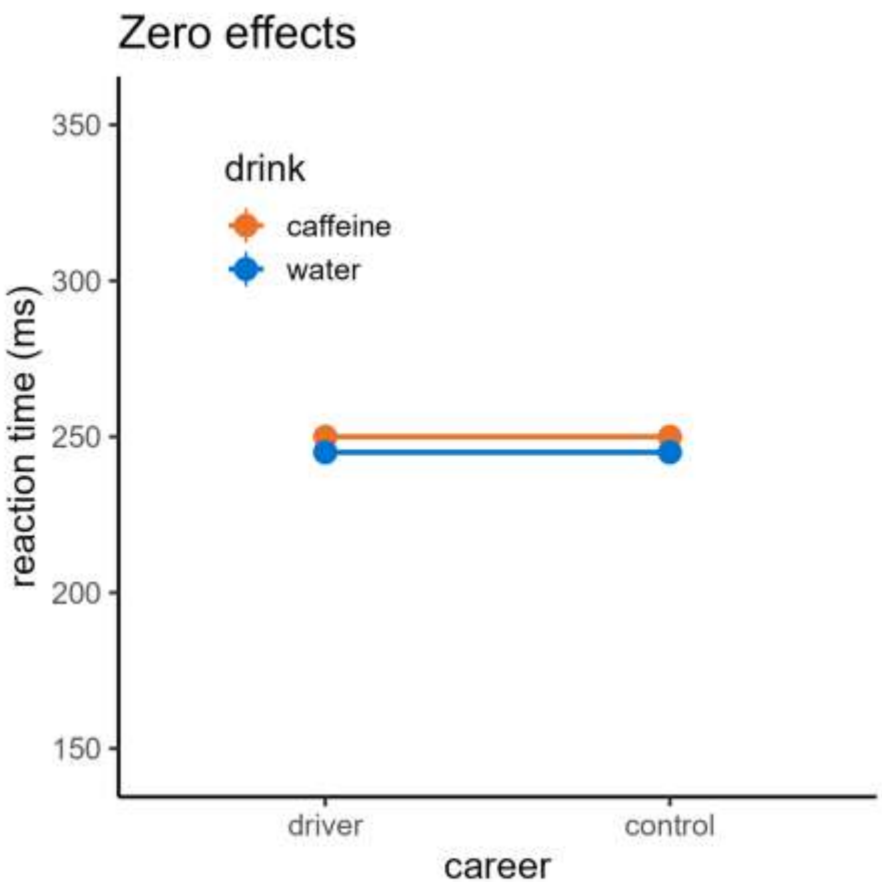


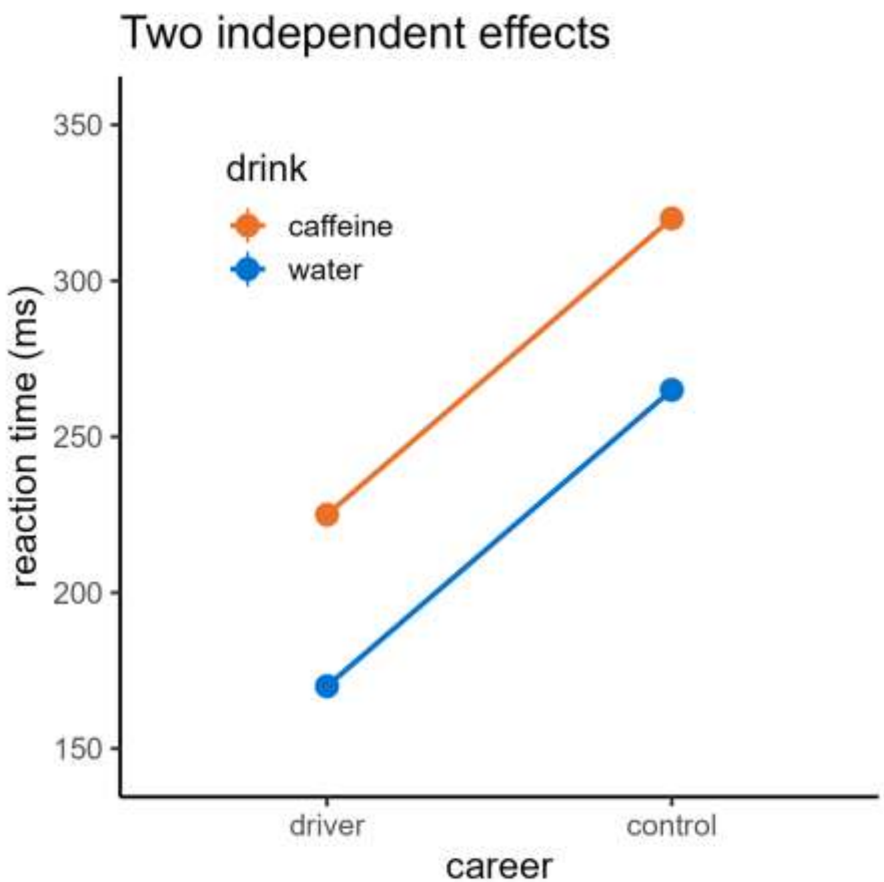
What other scenarios could exist with two predictors?

Two categorical predictors

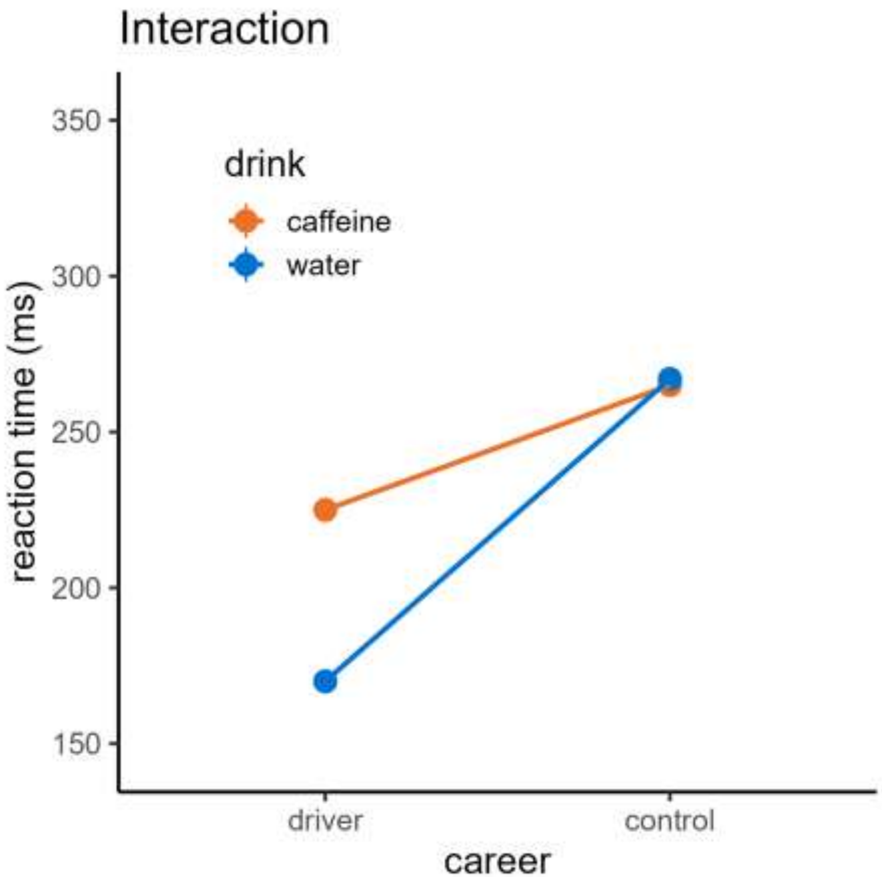


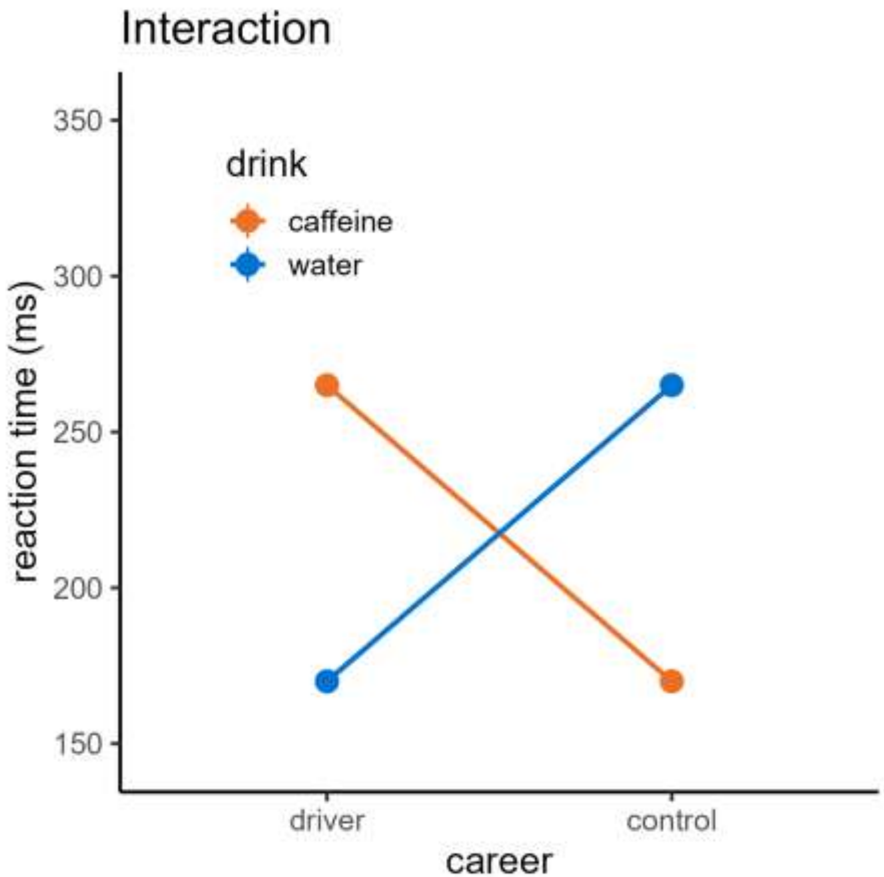
Two categorical predictors



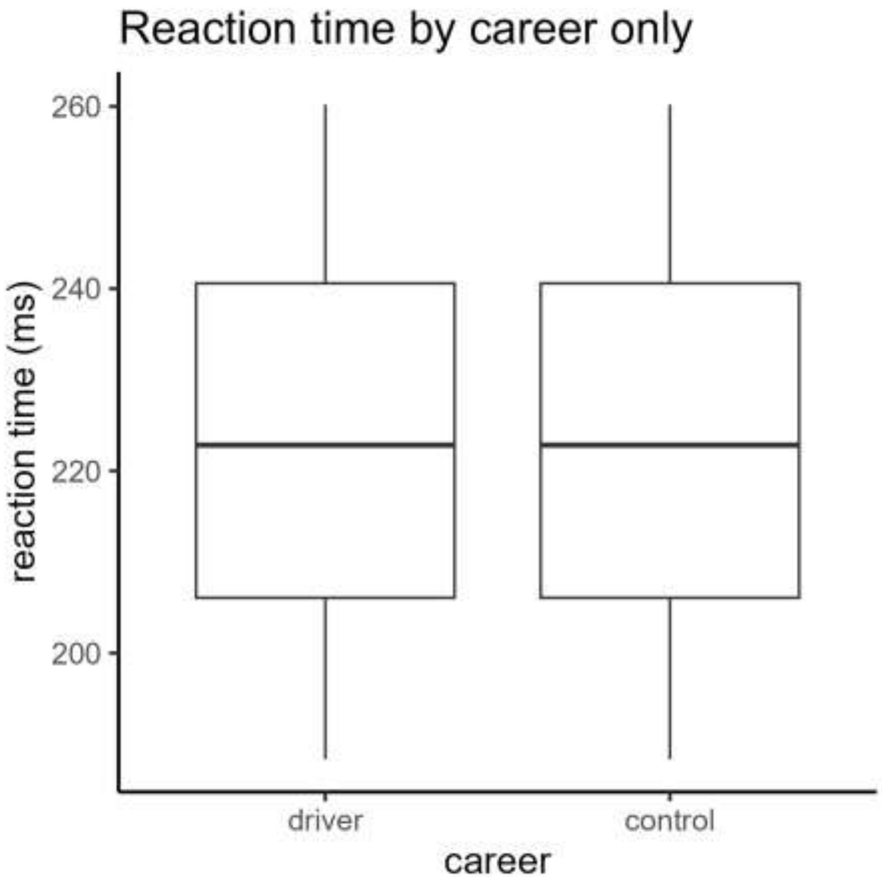


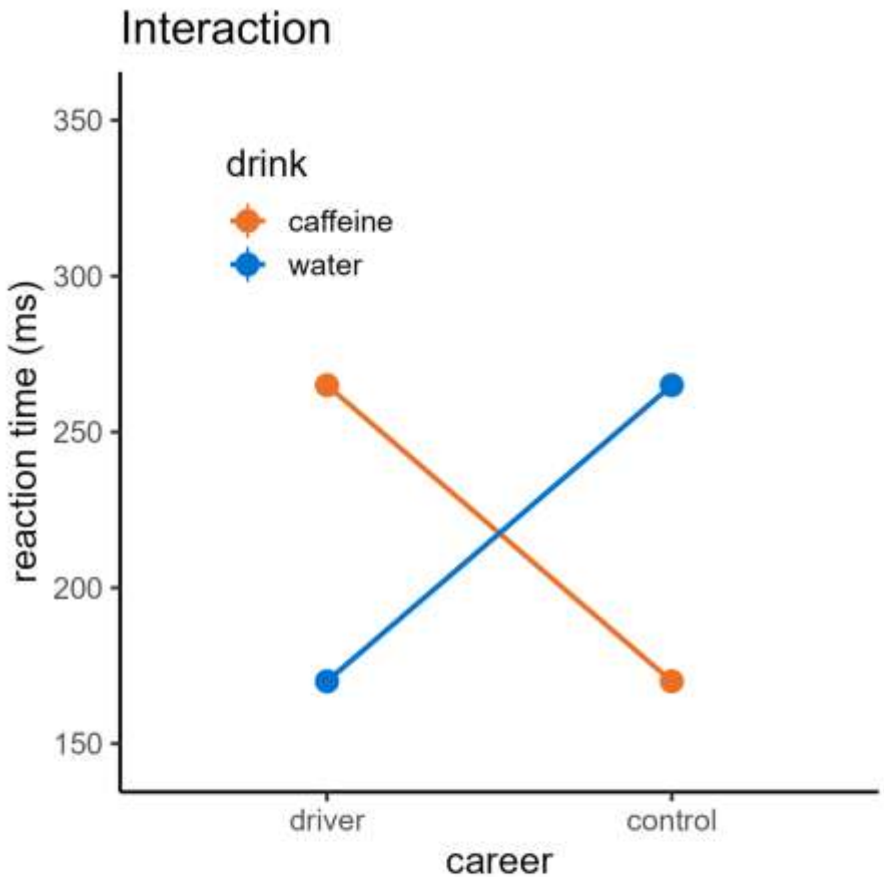
Two categorical predictors





Two categorical predictors

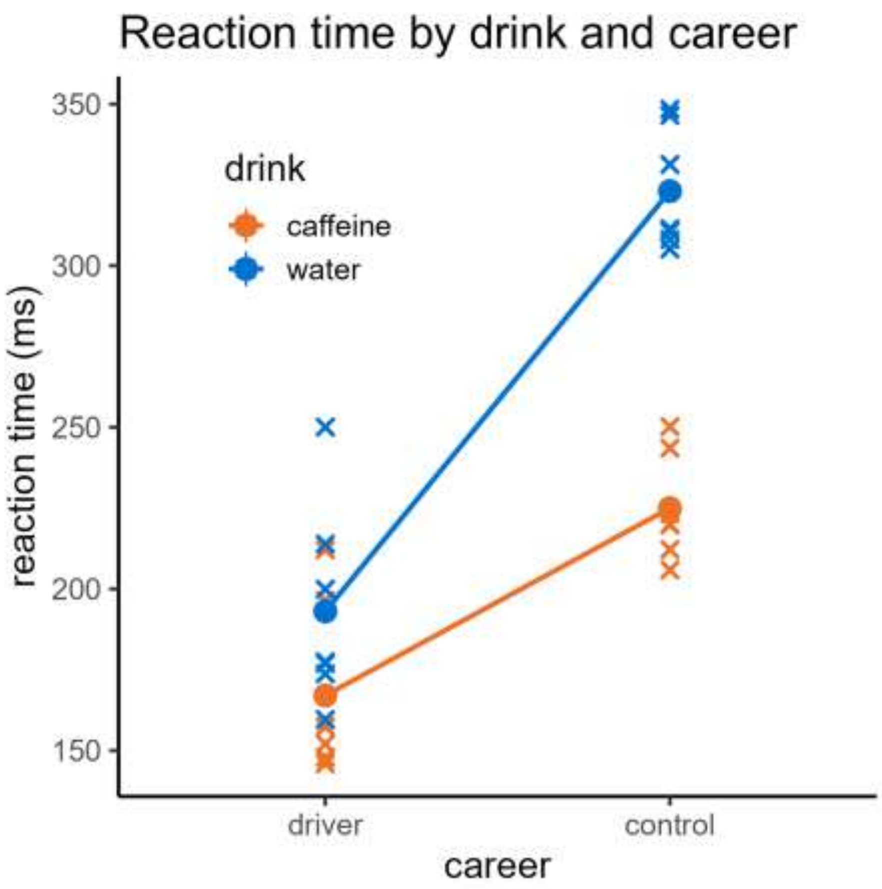




2. Interpreting interactions

How do we know if an interaction is meaningful?

Testing significance



- Effect of Career?
- Effect of Drink?
- Effect of Career:Drink interaction?

Testing significance

Two stage investigation:

1. Is there a significant interaction?

2. If not, do either of the predictor variables have a significant independent effect?

ANOVA again

```
Analysis of Variance Table

Response: Reaction
             Df Sum Sq Mean Sq F value    Pr(>F)
Drink         1  26966   26966  47.919 3.697e-07 ***
Career        1  61821   61821 109.858 1.947e-10 ***
Drink:Career  1   9080    9080  16.136 0.0005048 ***
Residuals    24  13506     563
```

ANOVA again

```
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Drink:Career  1   9080    9080  16.136 0.0005048 ***
Residuals    24  13506     563                      
```

Interaction is significant

ANOVA again

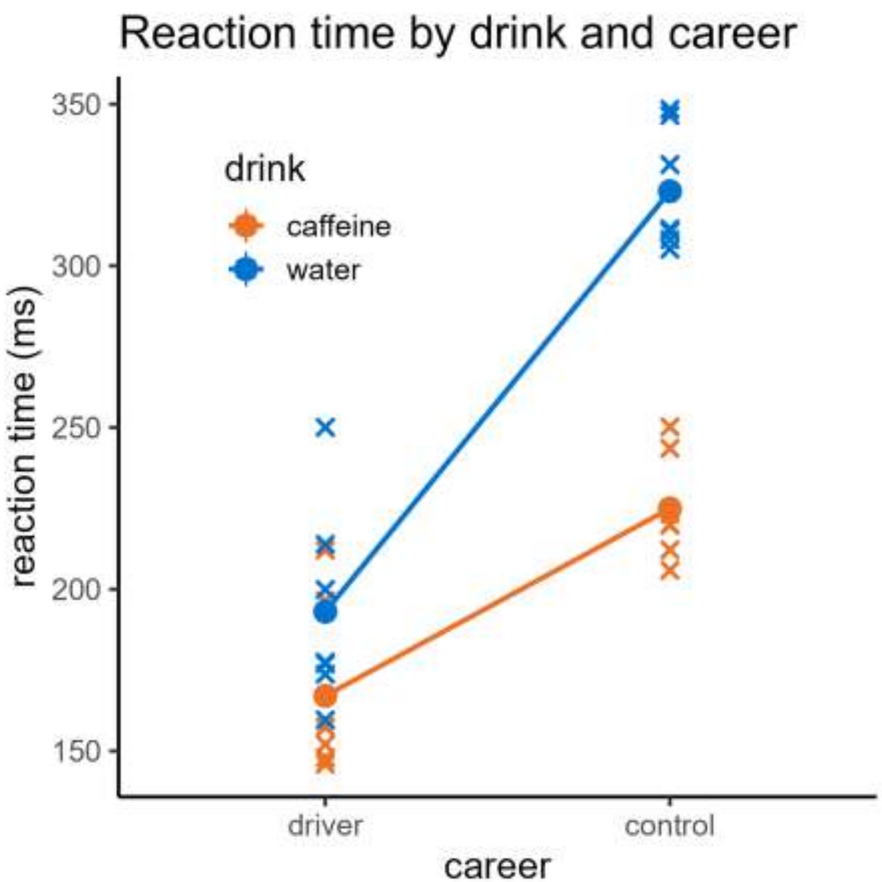
```
Analysis of Variance Table

Response: Reaction
             Df Sum Sq Mean Sq F value    Pr(>F)
Drink         1  26966   26966  47.919 3.697e-07 ***
Career        1  61821   61821 109.858 1.947e-10 ***
Drink:Career  1   9080    9080  16.136 0.0005048 ***
Residuals    24  13506     563
```

Ignore main effects

Interaction is significant

Interpretation



- There is a significant interaction between Career and Drink
- Therefore, we cannot comment on the main effects in isolation

3. Grouped linear regression

Combining categorical and continuous predictors

Two predictors

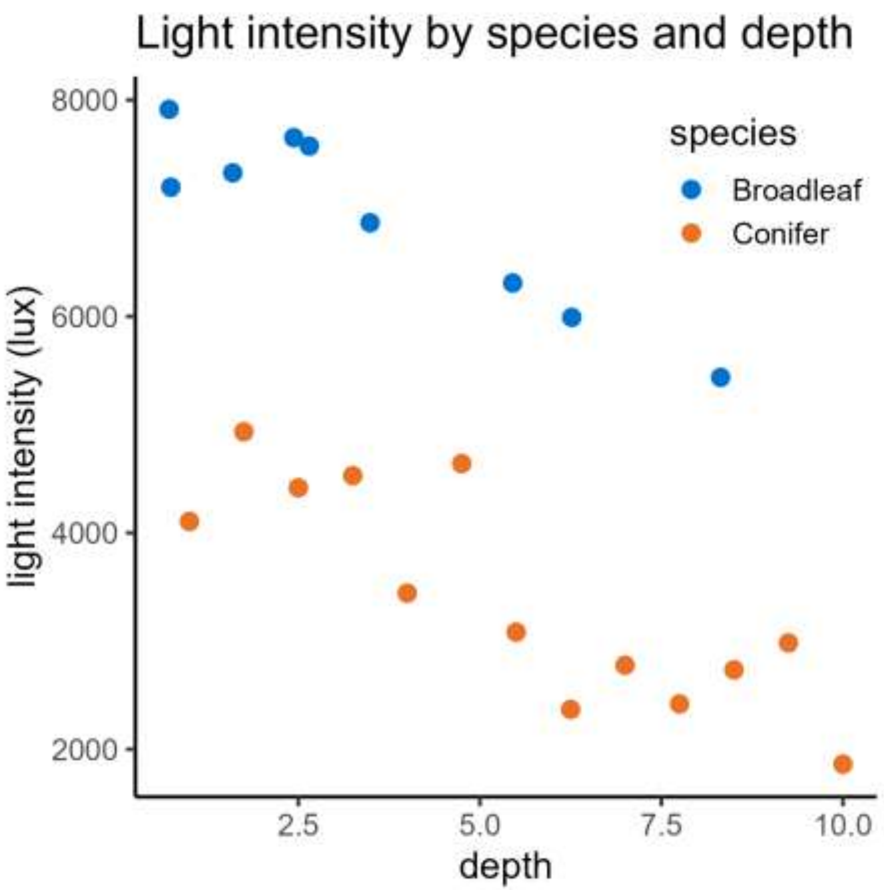
These ideas generalise to any dataset with two predictors

Two predictors

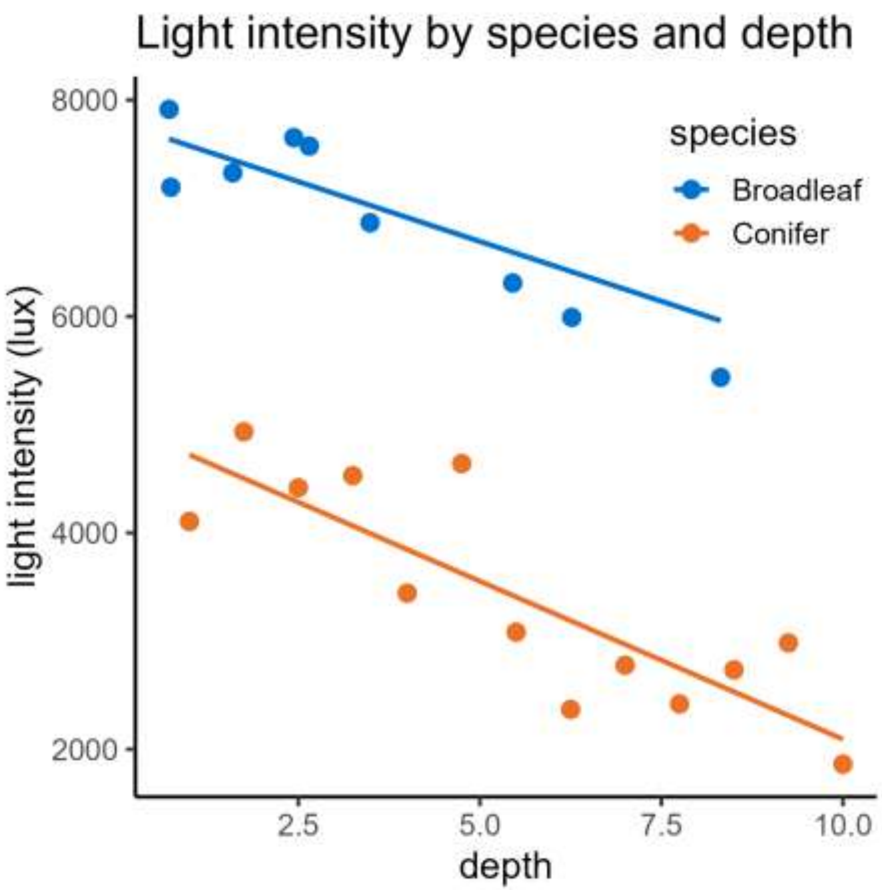
These ideas generalise to any dataset with two predictors

response	predictor (continuous)	predictor (categorical)
light (lux)	**depth (m)**	**species**
4105	1.0	Conifer
3081	5.5	Conifer
2734	8.5	Conifer
7652	2.4	Broadleaf
7914	0.7	Broadleaf
6309	5.5	Broadleaf

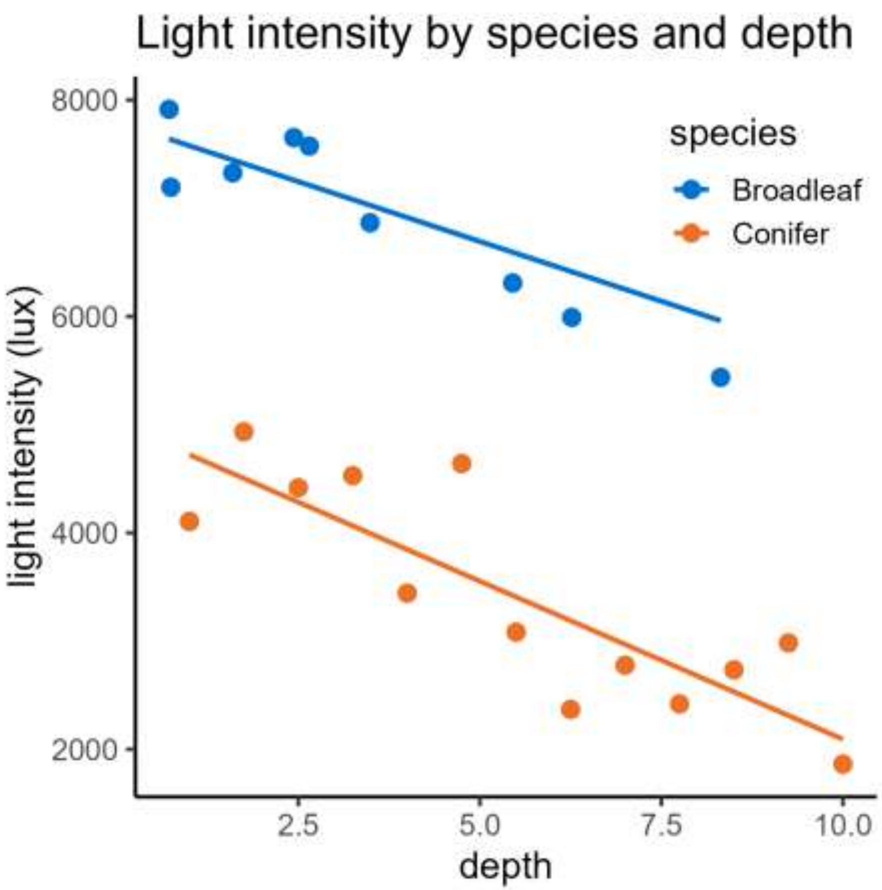
Grouped linear regression



Grouped linear regression



Grouped linear regression



Same two stage investigation:

1. Is there an interaction?

2. If not, do the predictors have independent effects?

Still ANOVA

```
Analysis of Variance Table

Response: Light
               Df   Sum Sq  Mean Sq  F value    Pr(>F)
Depth           1 30812910 30812910 107.8154 2.861e-09 ***
Species         1 51029543 51029543 178.5541 4.128e-11 ***
Depth:Species   1   218138   218138   0.7633    0.3932
Residuals      19  5430069   285793
```

Still ANOVA

```
Analysis of Variance Table

Response: Light
              Df   Sum Sq  Mean Sq  F value    Pr(>F)
Depth          1 30812910 30812910 107.8154 2.861e-09 ***
Species        1 51029543 51029543 178.5541 4.128e-11 ***
Depth:Species  1   218138   218138   0.7633    0.3932
Residuals     19  5430069   285793
```

Interaction is not significant

Still ANOVA

```
Analysis of Variance Table

Response: Light
               Df   Sum Sq  Mean Sq  F value    Pr(>F)
Depth           1 30812910 30812910 107.8154 2.861e-09 ***
Species         1 51029543 51029543 178.5541 4.128e-11 ***
Depth:Species   1   218138   218138   0.7633    0.3932
Residuals      19  5430069   285793
```

But both main effects are significant

Interaction is not significant

Session 4: Two predictors

Lecture summary:

- Two variables have an interaction if the effect of one variable depends on the other
- Interactions can be visualised with interaction plots
- We can perform analysis of variance on an interaction as well as the predictor variables individually
- If there is a significant interaction, we cannot comment on main/independent effects in isolation

Session 4: Two predictors

Course materials:

Fitting & testing interactions in R/Python:

- Two-way ANOVA
- Grouped linear regression
- Two continuous predictors

Visualising and interpreting interactions

UNIVERSITY OF CAMBRIDGE

Core statistics

Session 4: Two predictors

Cambridge Centre for Research Informatics Training

UNIVERSITY OF CAMBRIDGE

Core statistics

Session 5: Multiple predictors

Cambridge Centre for Research Informatics Training

Plan for the session

Session 5: **Multiple predictors**

1. Statistical models
2. Linearity
3. Model comparison

1. Statistical models

Revisiting the tests we’ve seen so far

Statistical tests so far

• One-sample, paired & two-sample t-tests	CS1
• One-way ANOVA	CS2
• Simple linear regression	CS3
• Two-way ANOVA	CS4
• Grouped linear regression	CS4

Statistical tests so far

• One-sample, paired & two-sample t-tests	CS1
• One-way ANOVA	CS2
• Simple linear regression	CS3
• Two-way ANOVA	CS4
• Grouped linear regression	CS4

All special cases of **linear models**!

What is a (linear) model?

Describing the relationship between:

- A **response variable**

- And any number of **predictor variables**

What is a (linear) model?

Describing the relationship between:

- A **response variable**
  e.g., height

- And any number of **predictor variables**

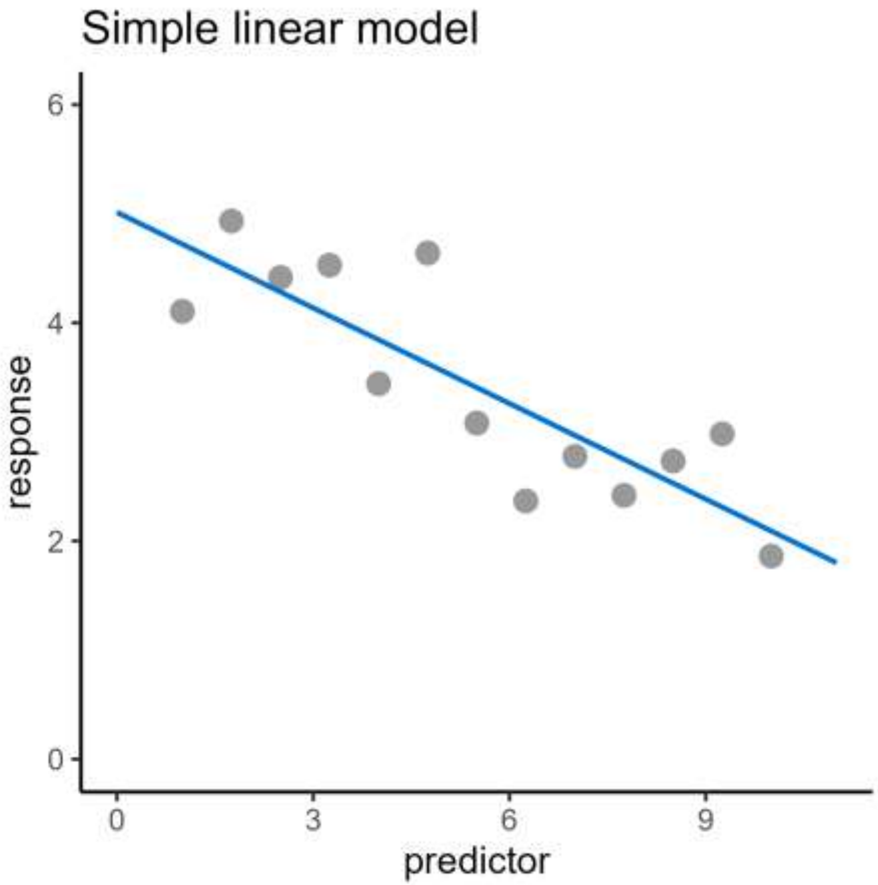
What is a (linear) model?

Describing the relationship between:

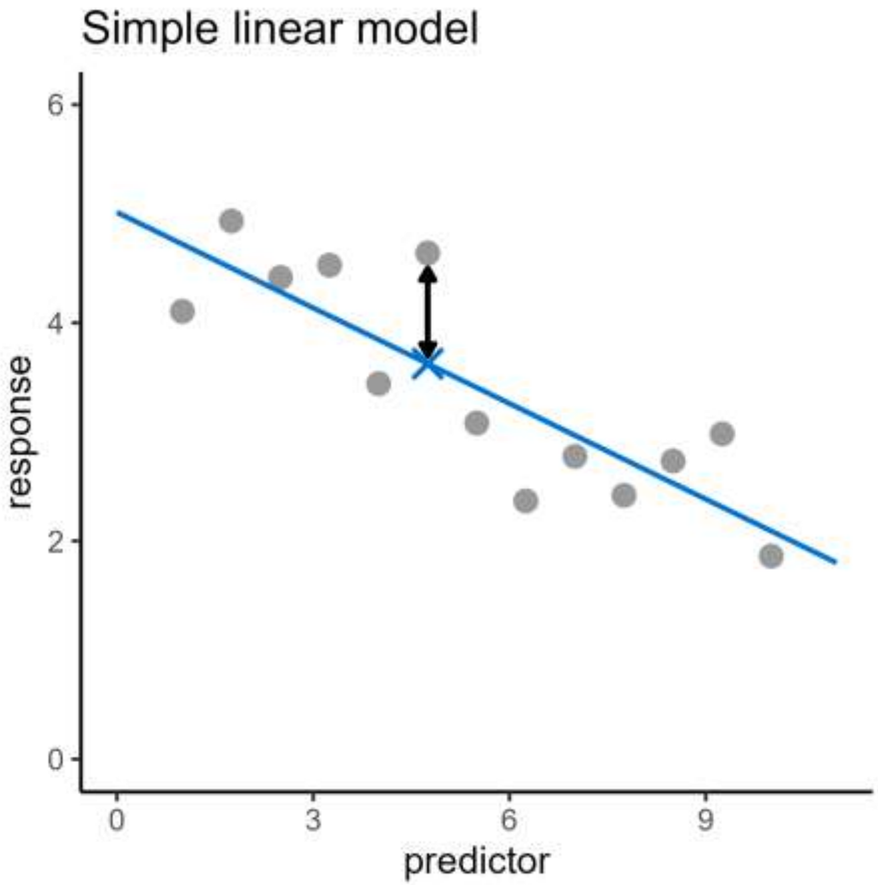
- A **response variable**
  e.g., height

- And any number of **predictor variables**
  e.g., age, diet, gender, mother's height, time of day, country of birth, star sign, type of pet they own, number of Instagram followers

Describing a trend/relationship

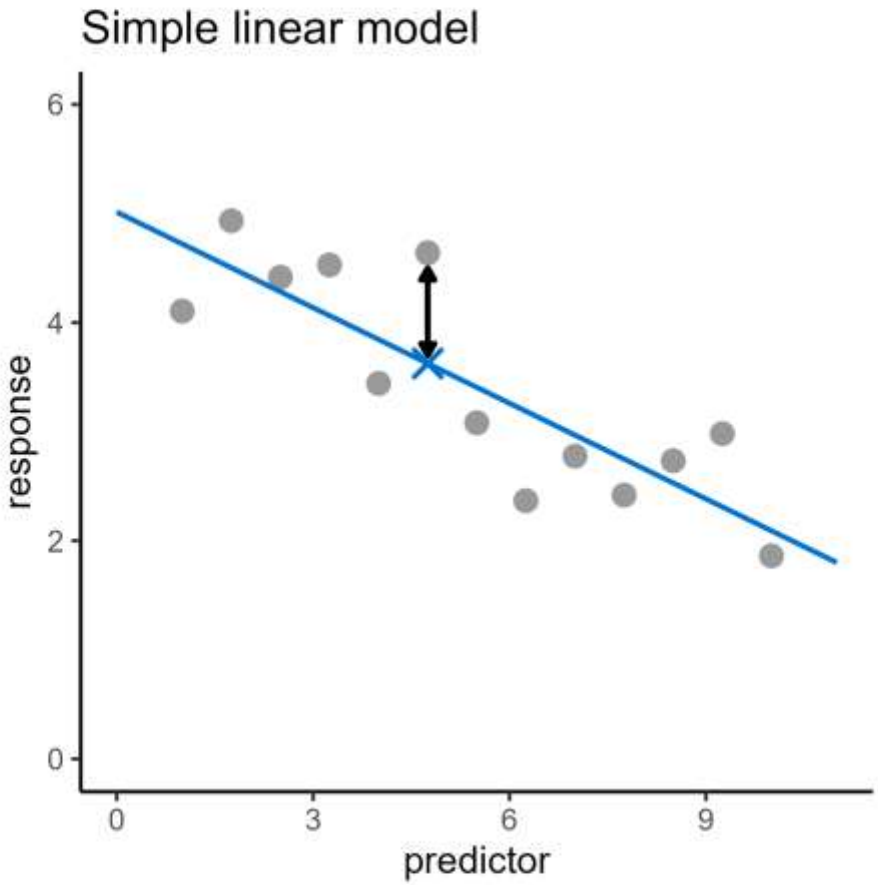


Describing a trend/relationship



Data = model + error
= prediction + residual

Describing a trend/relationship

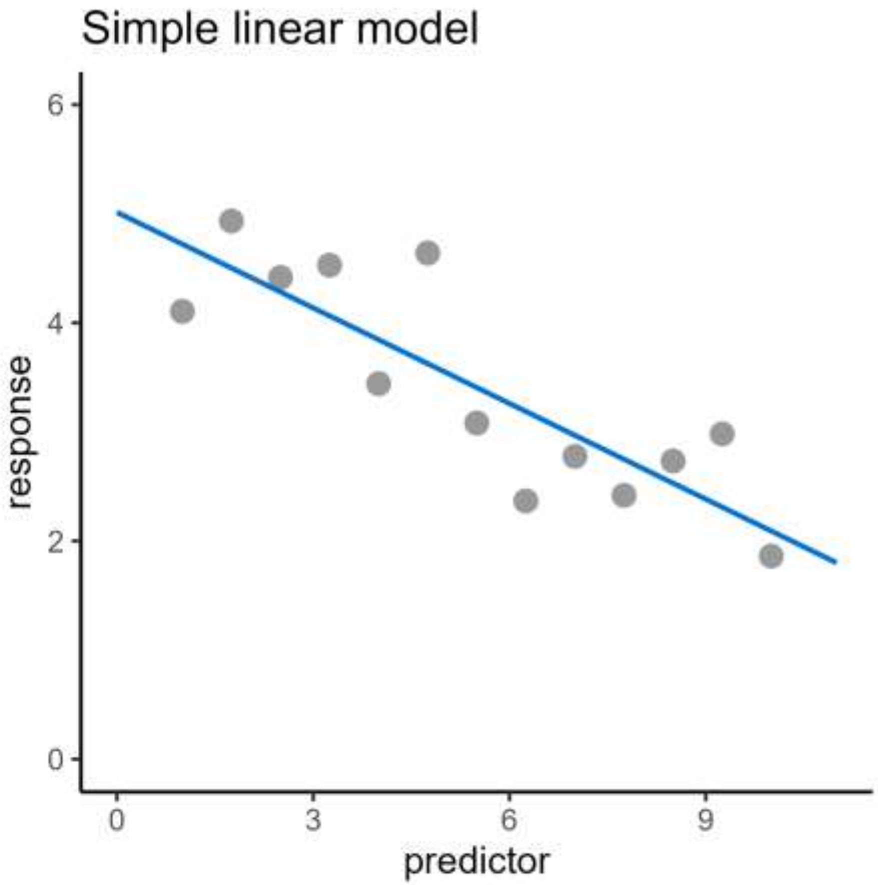


Data = model + error
= prediction + residual

Descriptive only

Model ≠ statistical test

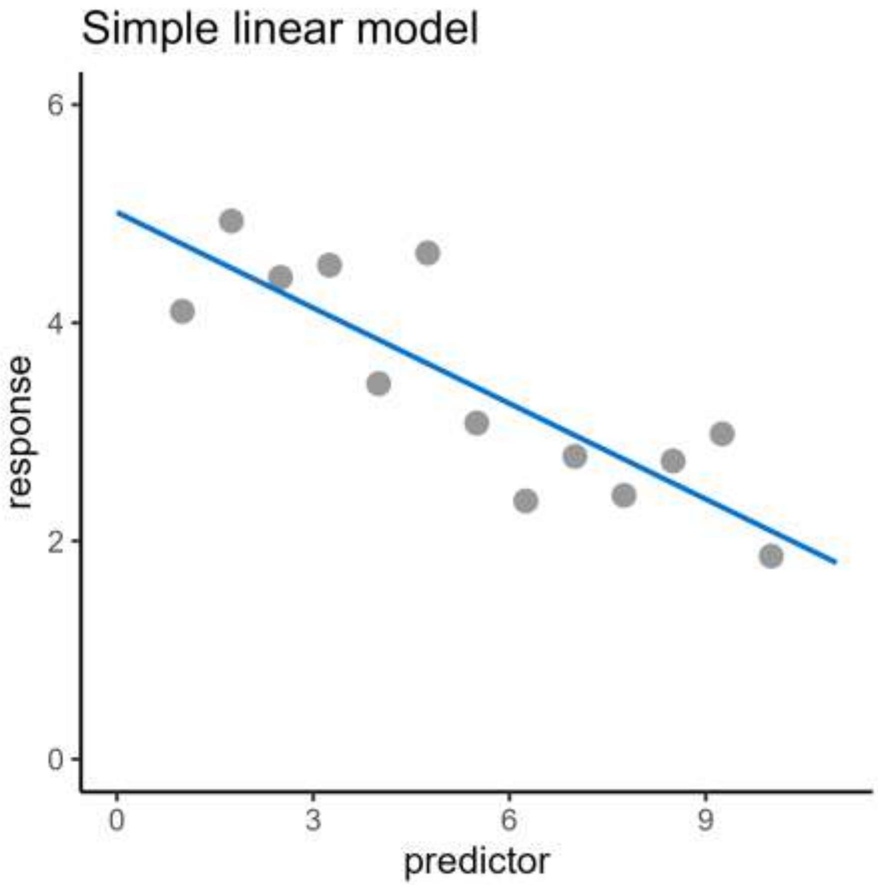
Revisiting simple linear regression



We describe the model using an equation:

$$y = \beta_0 + \beta_1 x$$

Revisiting simple linear regression



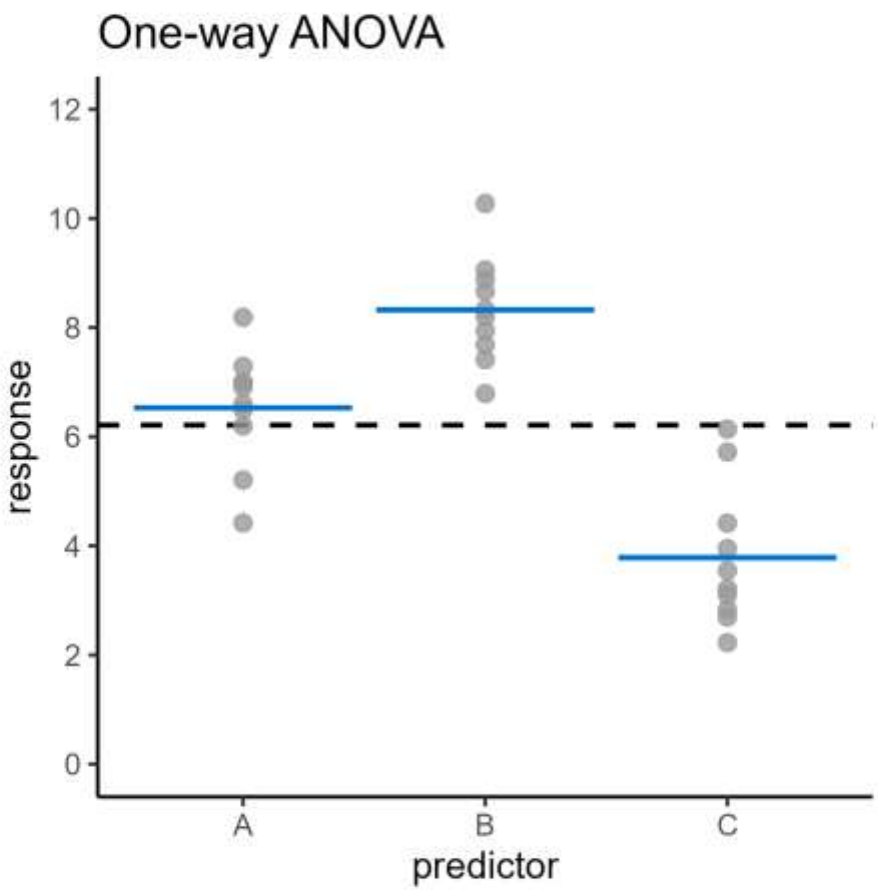
We describe the model using an equation:

$$y = \beta_0 + \beta_1 x$$

$$Response = 5 \; - 0.3 \; \times Predictor$$

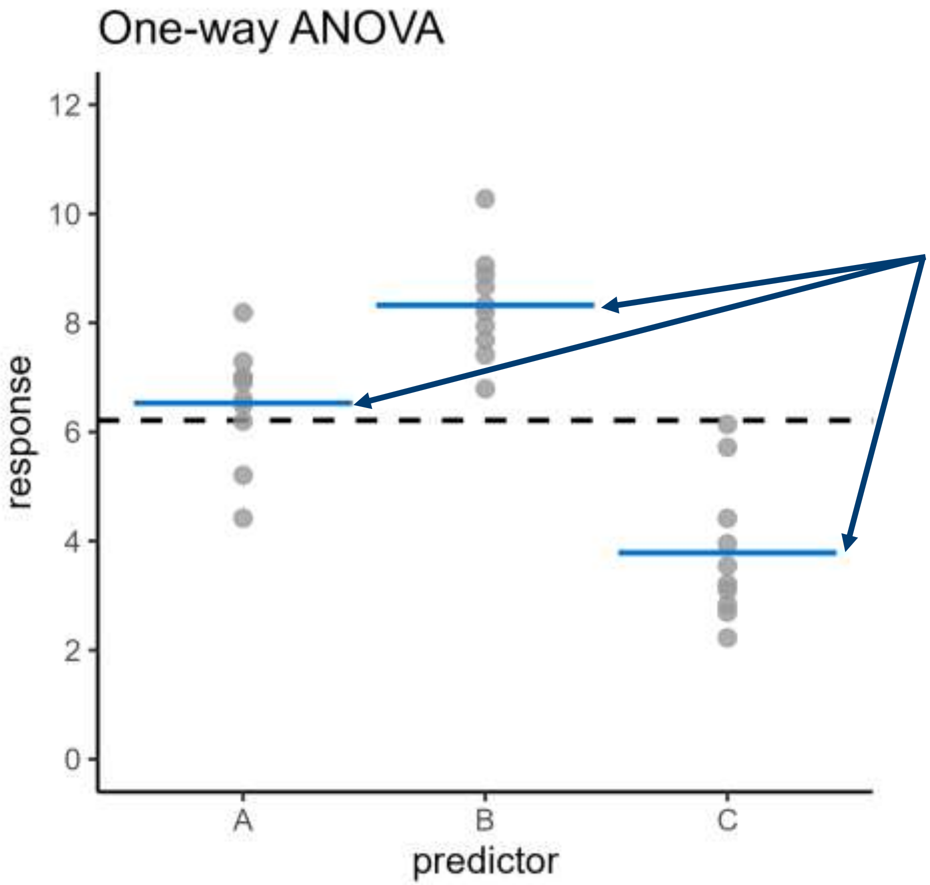
y = Response β_0 = 5
x = Predictor β_1 = -0.3

Revisiting one-way ANOVA



What is the model?

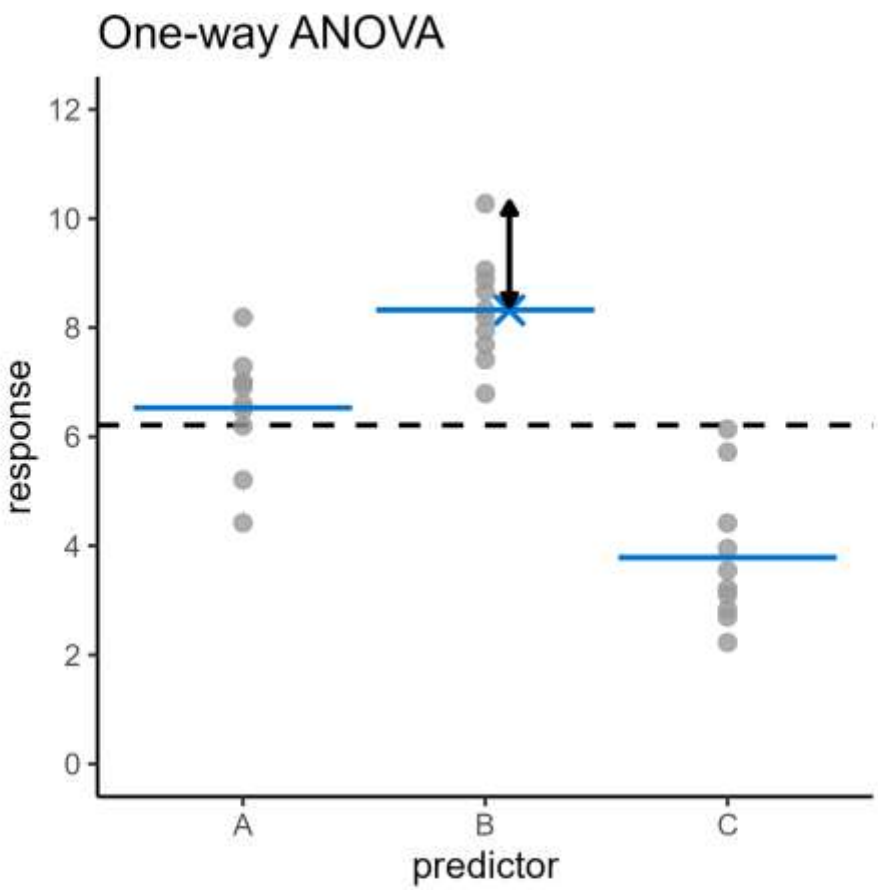
Revisiting one-way ANOVA



The model is the group means

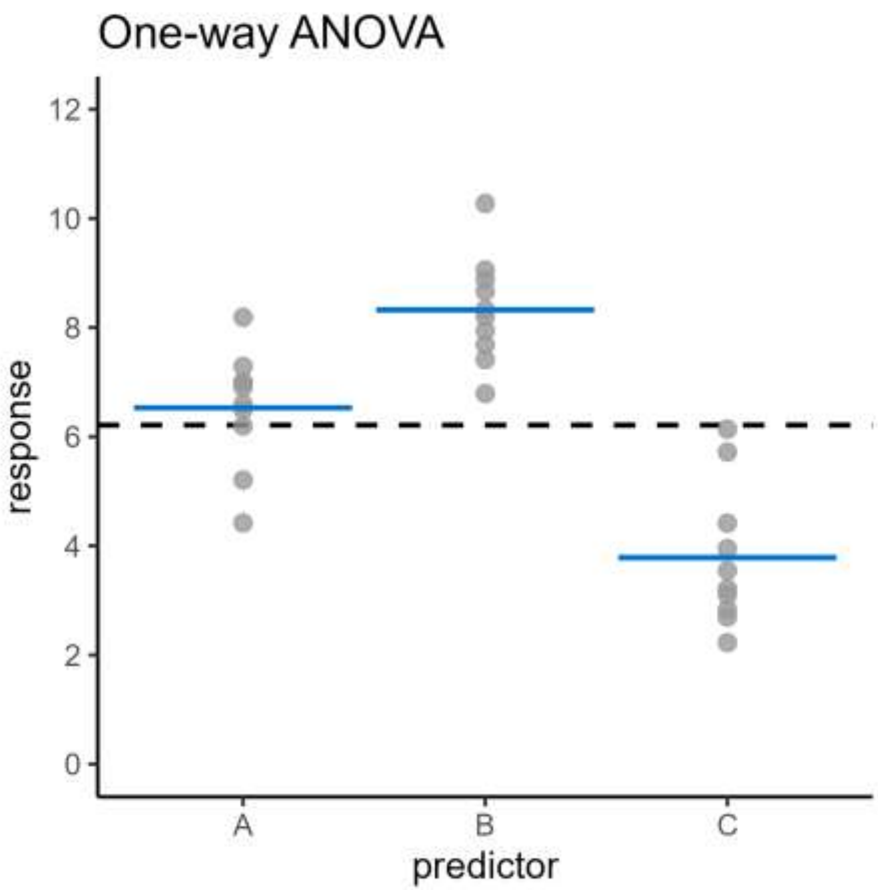
Model

Revisiting one-way ANOVA



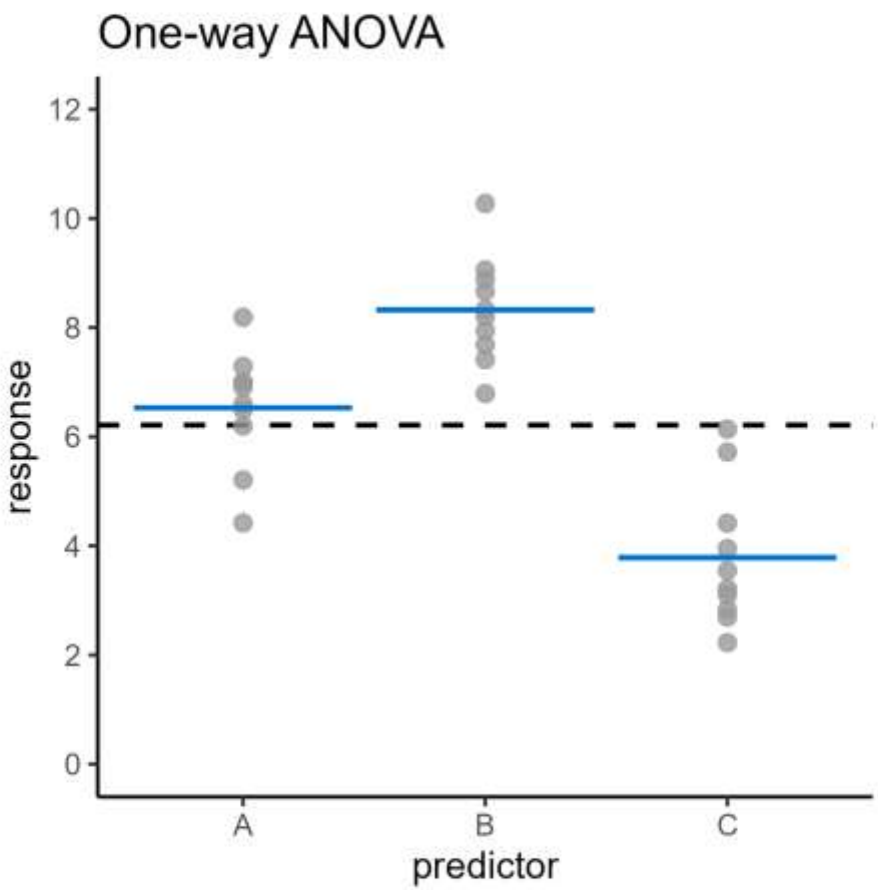
Data = model + error
= prediction + residual

Revisiting one-way ANOVA



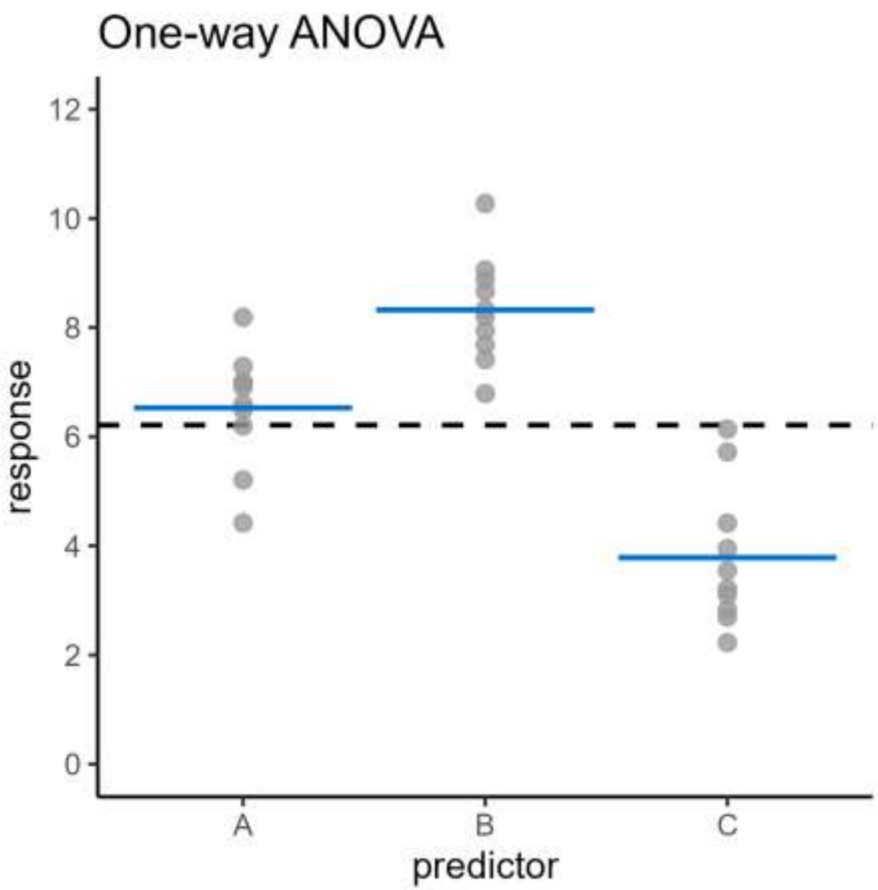
How do we describe this model in an equation?

Revisiting one-way ANOVA



$$response = \begin{pmatrix} 6.5 \\ 8.3 \\ 3.8 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \end{pmatrix}$$

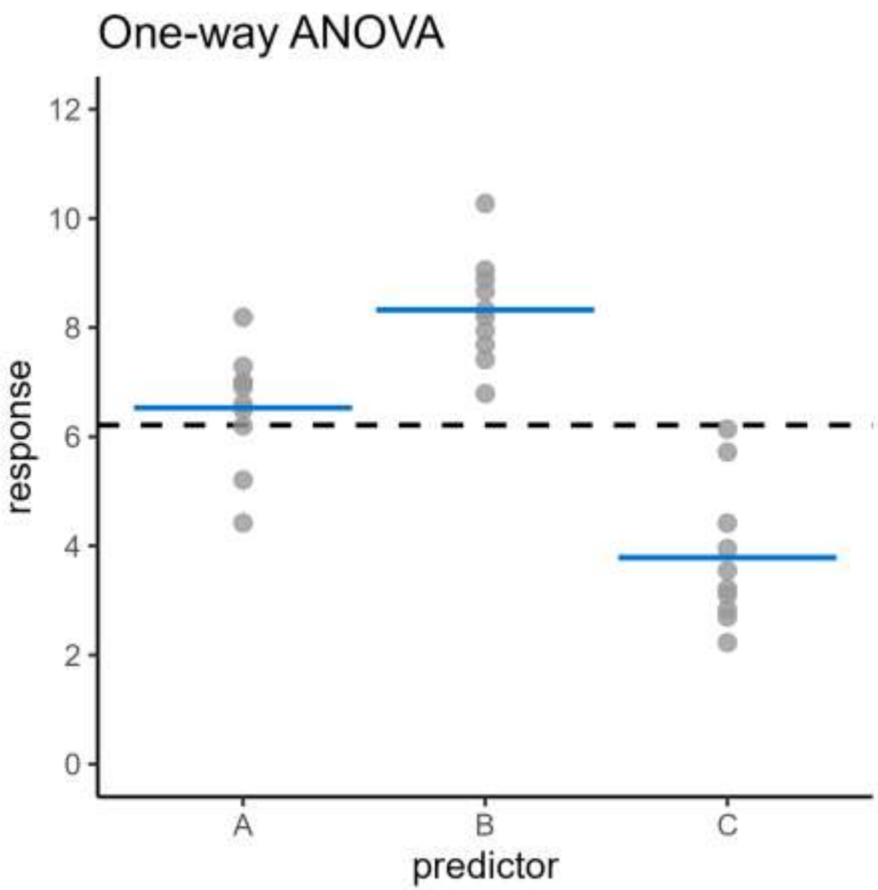
Revisiting one-way ANOVA



$$response = \begin{pmatrix} 6.5 \\ 8.3 \\ 3.8 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \end{pmatrix}$$

$$response = 6.5 + \begin{pmatrix} 0 \\ 1.8 \\ -2.7 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \end{pmatrix}$$

Revisiting one-way ANOVA



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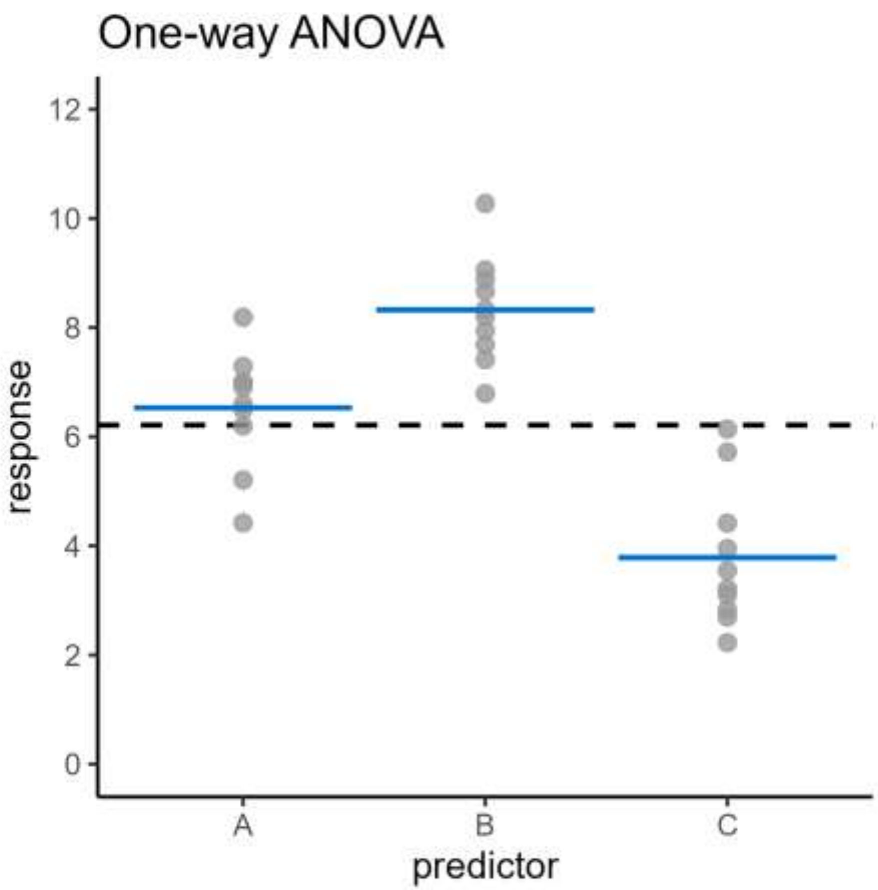
$$y = \beta_0 + \beta_1 x$$

Revisiting one-way ANOVA



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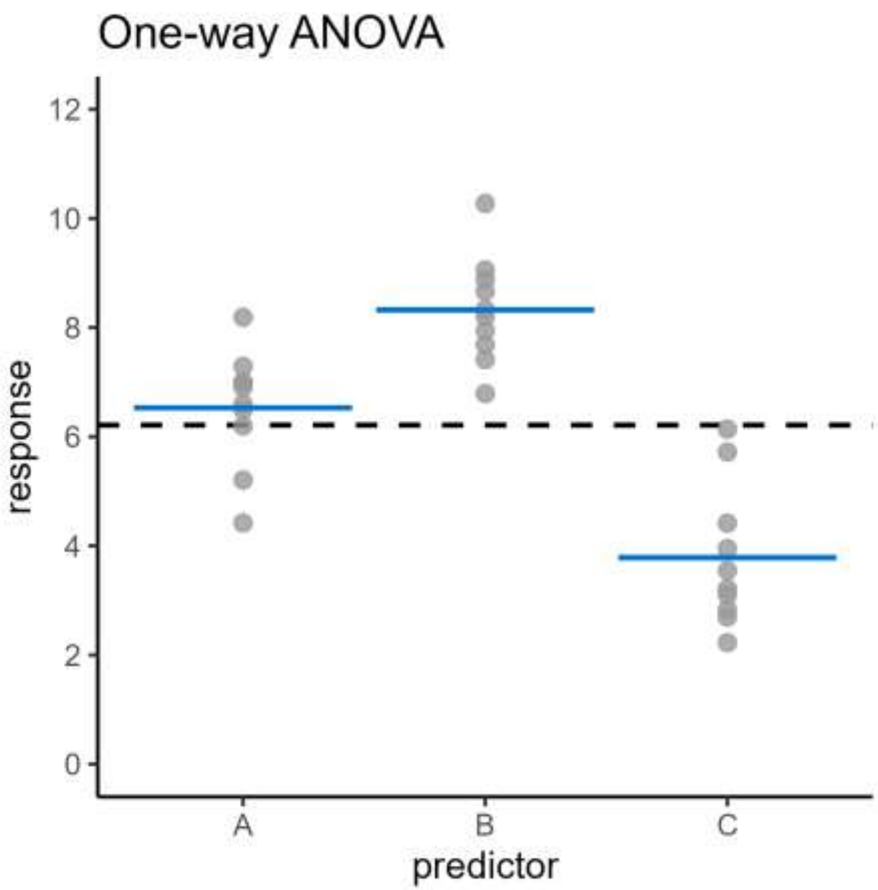
$$response = 6.5 + \begin{pmatrix} 0 \\ 1.8 \\ -2.7 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \end{pmatrix}$$

$$y = \beta_0 + \beta_1 x$$

y = response

x = predictor

Revisiting one-way ANOVA



$$response = \begin{pmatrix} 6.5 \\ 8.3 \\ 3.8 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \end{pmatrix}$$

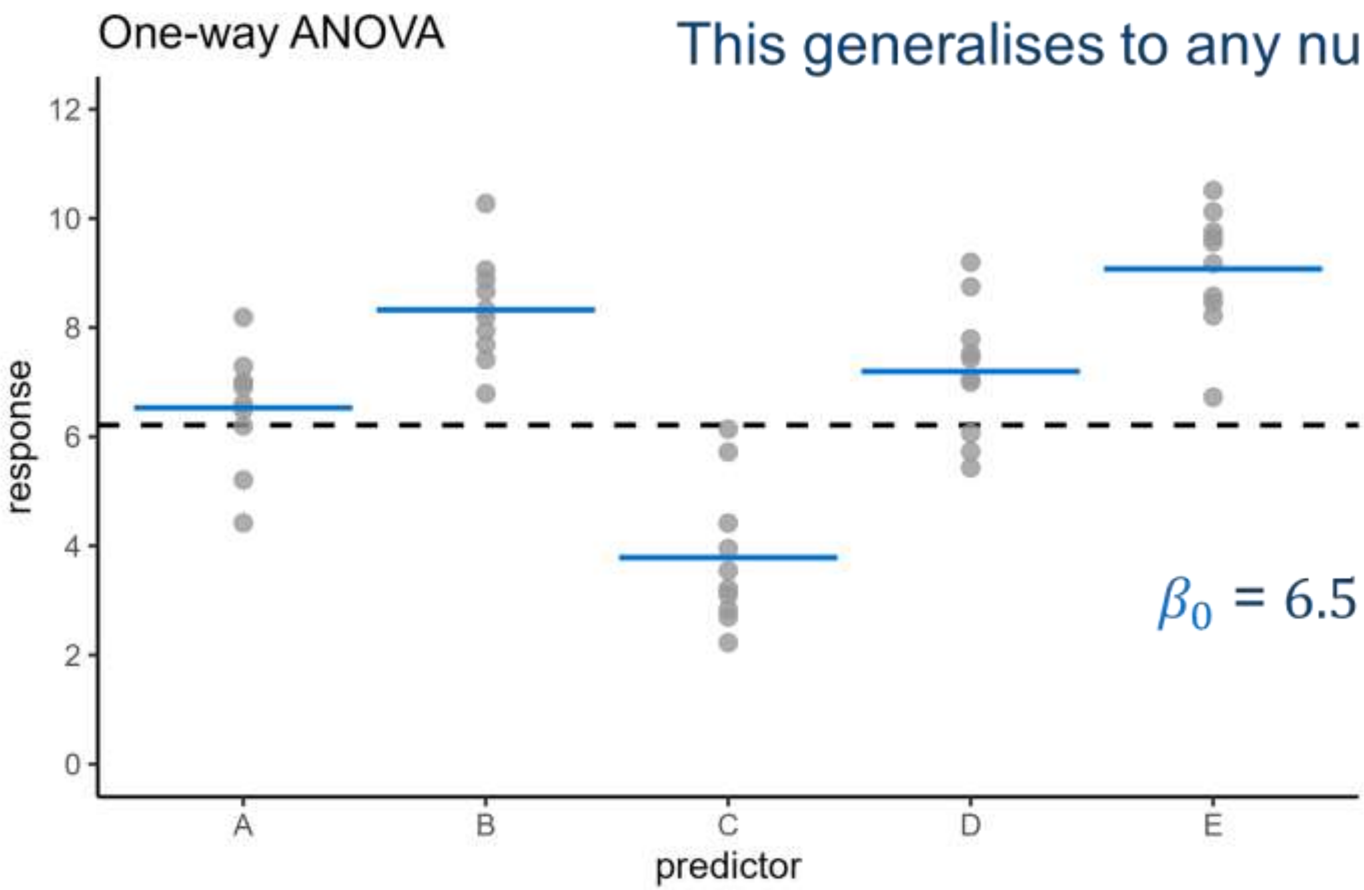
$$response = 6.5 + \begin{pmatrix} 0 \\ 1.8 \\ -2.7 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \end{pmatrix}$$

$$y = \beta_0 + \beta_1 x$$

y = response $\quad \beta_0 = 6.5$

x = predictor $\quad \beta_1 = \begin{pmatrix} 0 \\ 1.8 \\ -2.7 \end{pmatrix}$

Revisiting one-way ANOVA



This generalises to any number of groups

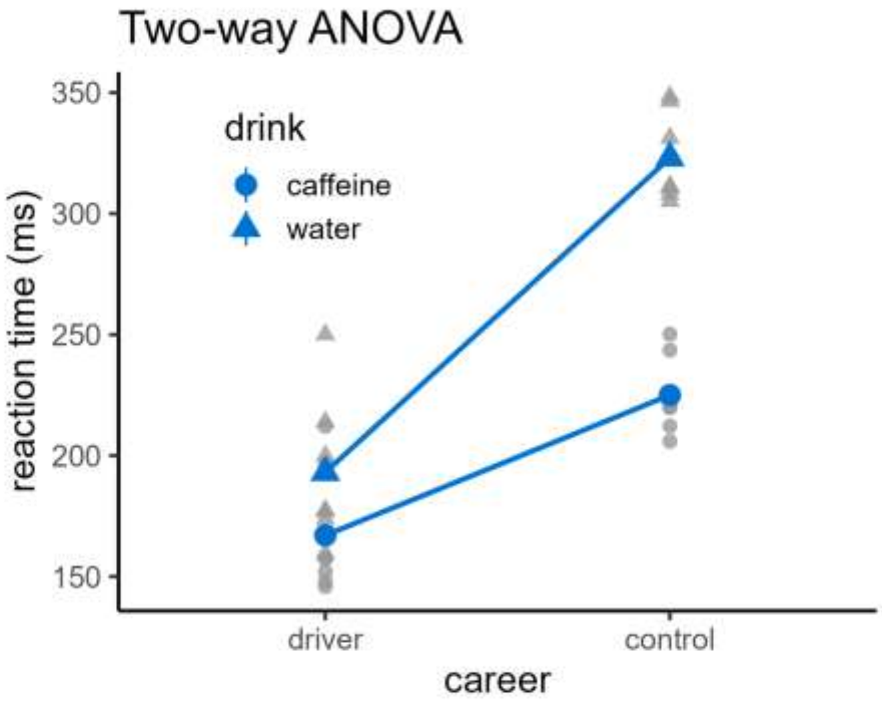
$$\beta_0 = 6.5 \quad \beta_1 = \begin{pmatrix} 0 \\ 1.8 \\ -2.7 \\ 0.7 \\ 2.5 \end{pmatrix}$$

Statistical tests so far

- One-sample, paired & two-sample t-tests
- One-way ANOVA
- Simple linear regression
- Two-way ANOVA
- Grouped linear regression

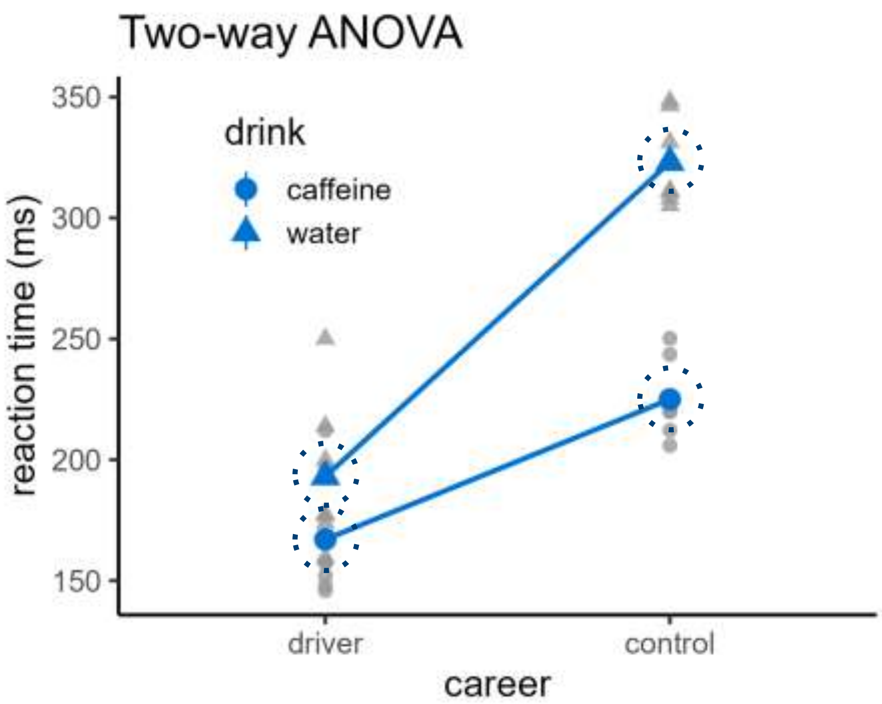
$$y = \beta_0 + \beta_1 x$$

Revisiting two-way ANOVA



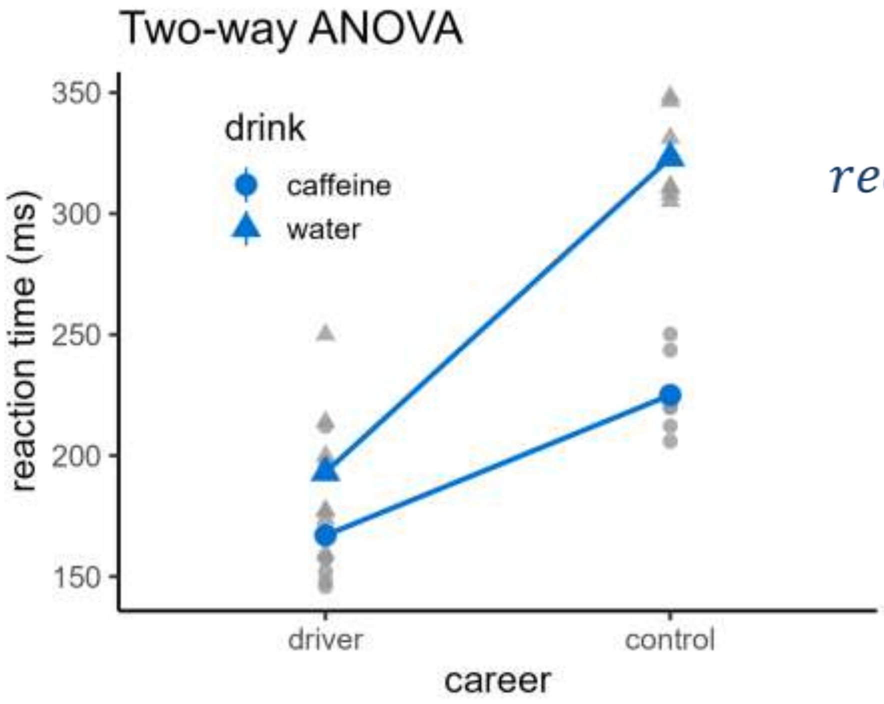
What is the model?

Revisiting two-way ANOVA



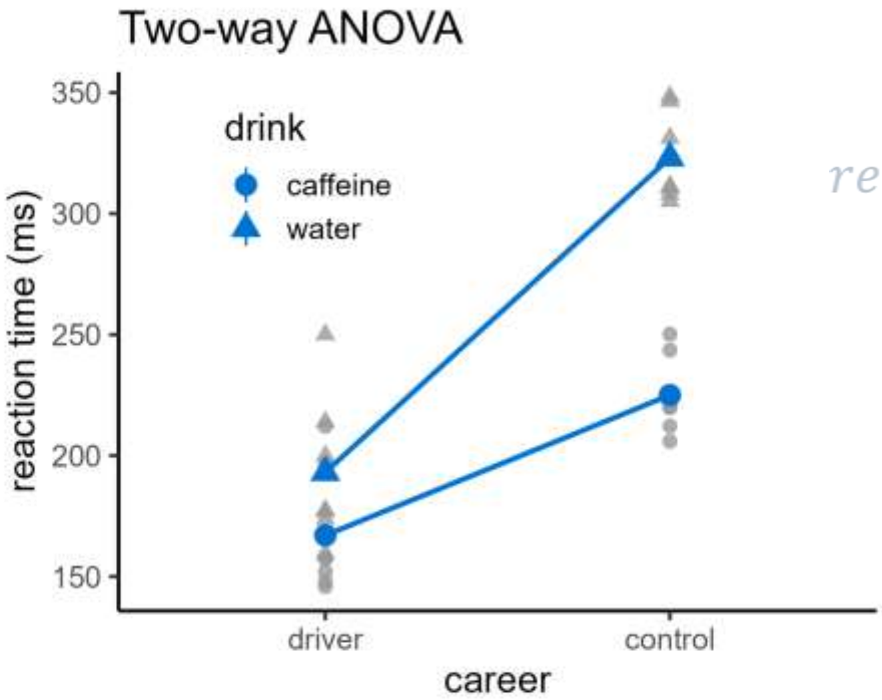
The model is the group means (again)

Revisiting two-way ANOVA



$$reaction = \begin{pmatrix} 167 \\ 193 \\ 225 \\ 323 \end{pmatrix} \begin{pmatrix} driver: caffeine \\ driver: water \\ control: caffeine \\ control: water \end{pmatrix}$$

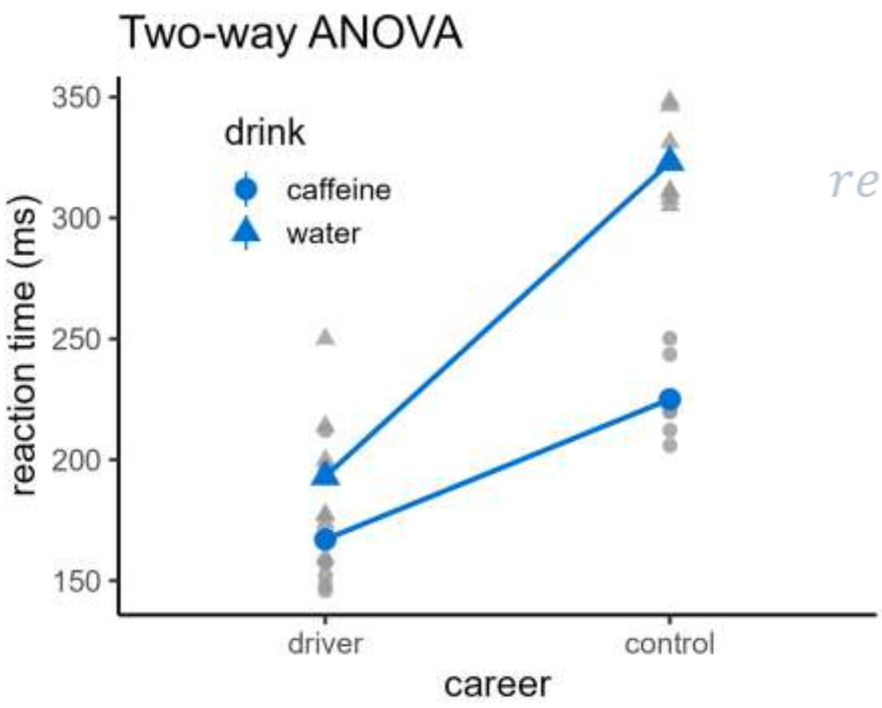
Revisiting two-way ANOVA



$$reaction = \begin{pmatrix} 167 \\ 193 \\ 225 \\ 323 \end{pmatrix} \begin{pmatrix} driver\!:caffeine \\ driver\!:water \\ control\!:caffeine \\ control\!:water \end{pmatrix}$$

A reference group is chosen (again)

Revisiting two-way ANOVA

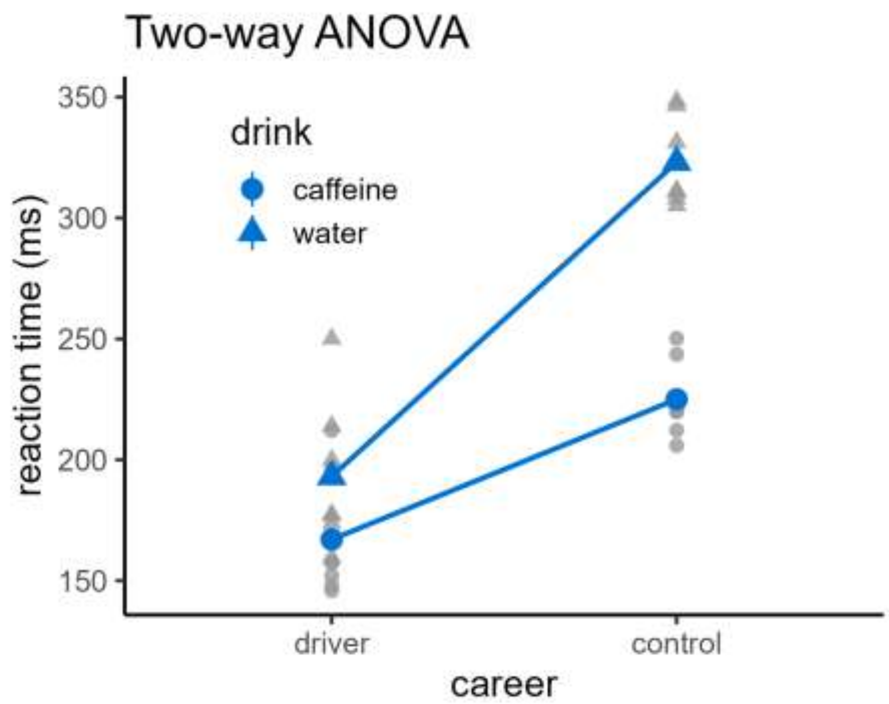


$$reaction = \begin{pmatrix} 167 \\ 193 \\ 225 \\ 323 \end{pmatrix} \begin{pmatrix} driver: caffeine \\ driver: water \\ control: caffeine \\ control: water \end{pmatrix}$$

A reference group is chosen (again)

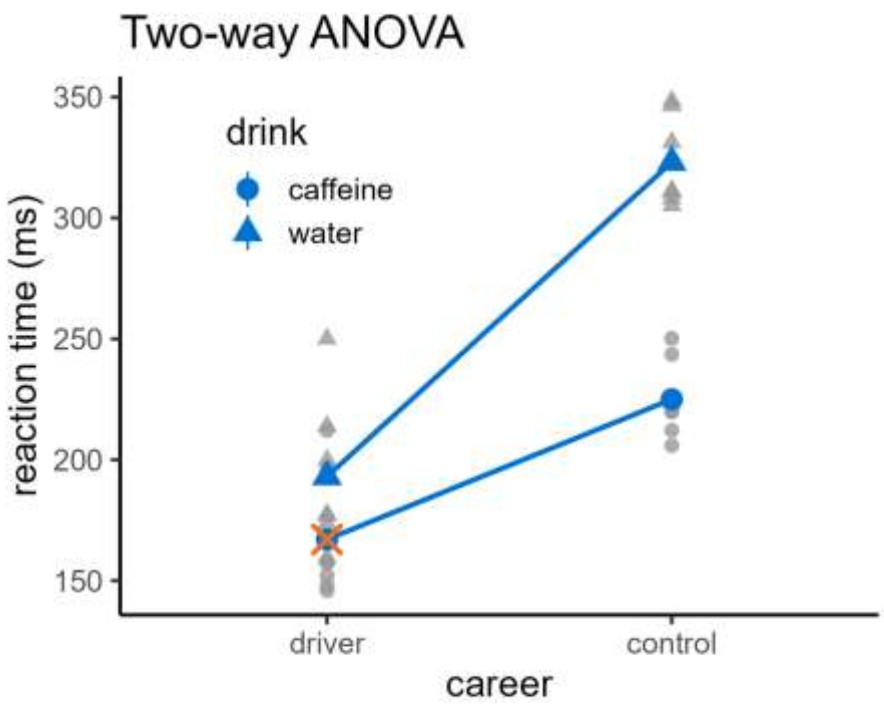
$$reaction = 167 + \begin{pmatrix} 0 \\ 58 \end{pmatrix} \begin{pmatrix} driver \\ control \end{pmatrix} + \begin{pmatrix} 0 \\ 26 \end{pmatrix} \begin{pmatrix} caffeine \\ water \end{pmatrix} + \begin{pmatrix} 0 \\ 72 \end{pmatrix} \begin{pmatrix} other \\ control: water \end{pmatrix}$$

Revisiting two-way ANOVA



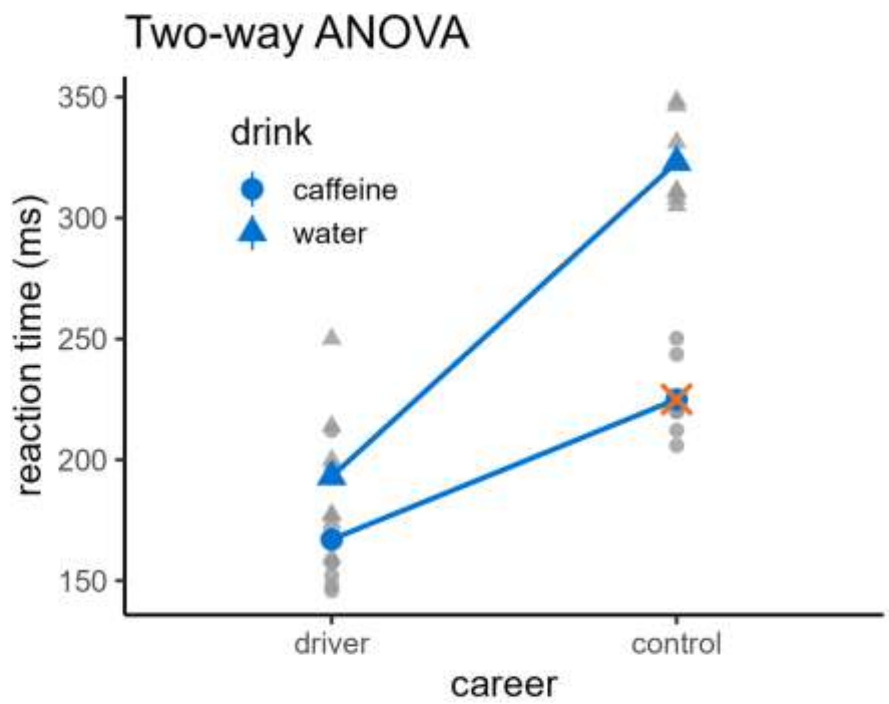
$$reaction = 167 + \begin{pmatrix} 0 \\ 58 \end{pmatrix} \begin{pmatrix} driver \\ control \end{pmatrix} + \begin{pmatrix} 0 \\ 26 \end{pmatrix} \begin{pmatrix} caffeine \\ water \end{pmatrix} + \begin{pmatrix} 0 \\ 72 \end{pmatrix} \begin{pmatrix} other \\ control{:}water \end{pmatrix}$$

Revisiting two-way ANOVA



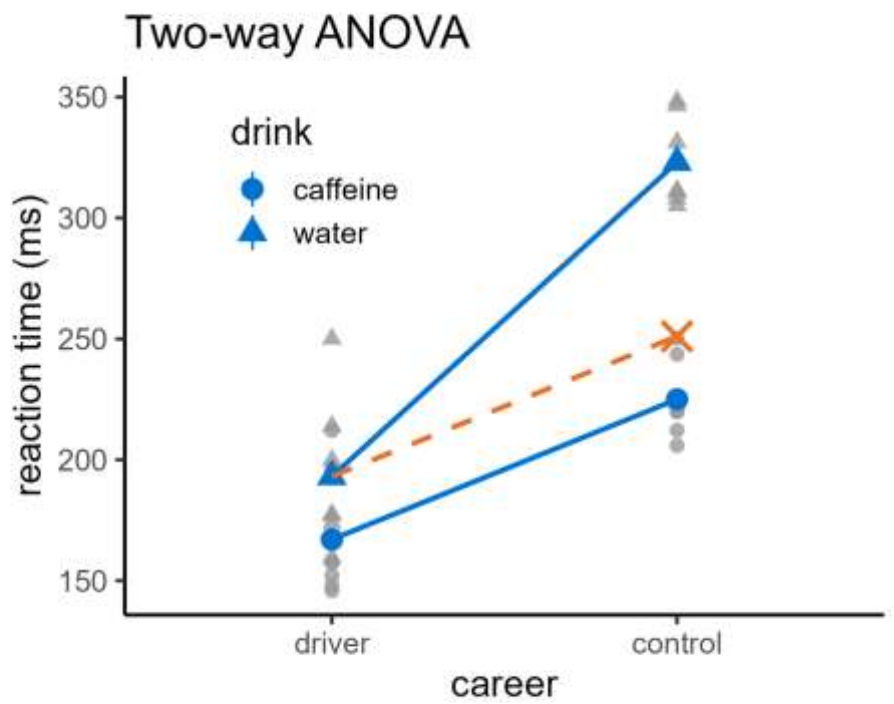
$$reaction = 167 + \binom{0}{58}\binom{driver}{control} + \binom{0}{26}\binom{caffeine}{water} + \binom{0}{72}\binom{other}{control\!:\!water}$$

Revisiting two-way ANOVA



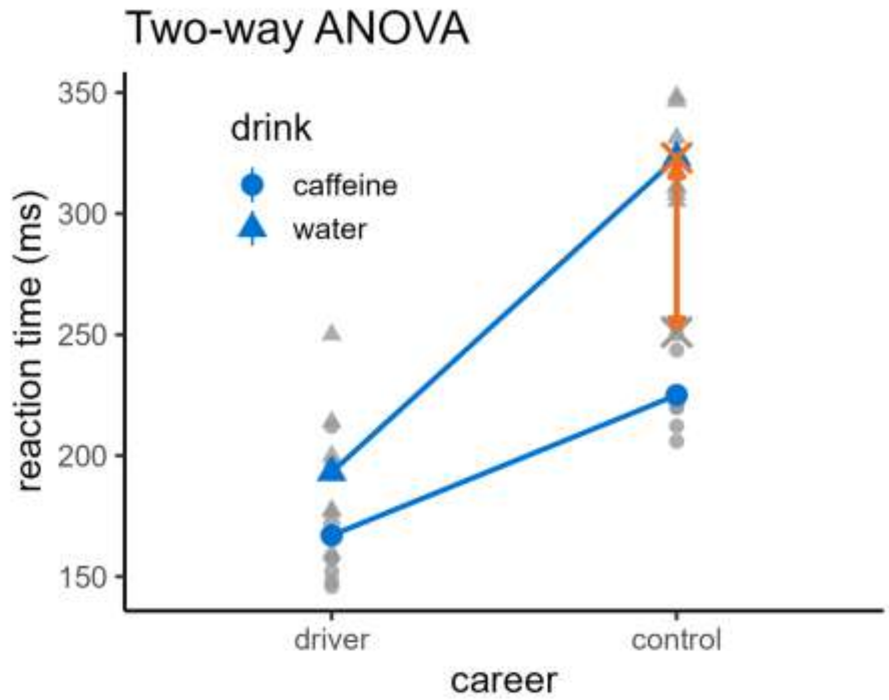
$$reaction = 167 + \binom{0}{58}\binom{driver}{control} + \binom{0}{26}\binom{caffeine}{water} + \binom{0}{72}\binom{other}{control\!:\!water}$$

Revisiting two-way ANOVA



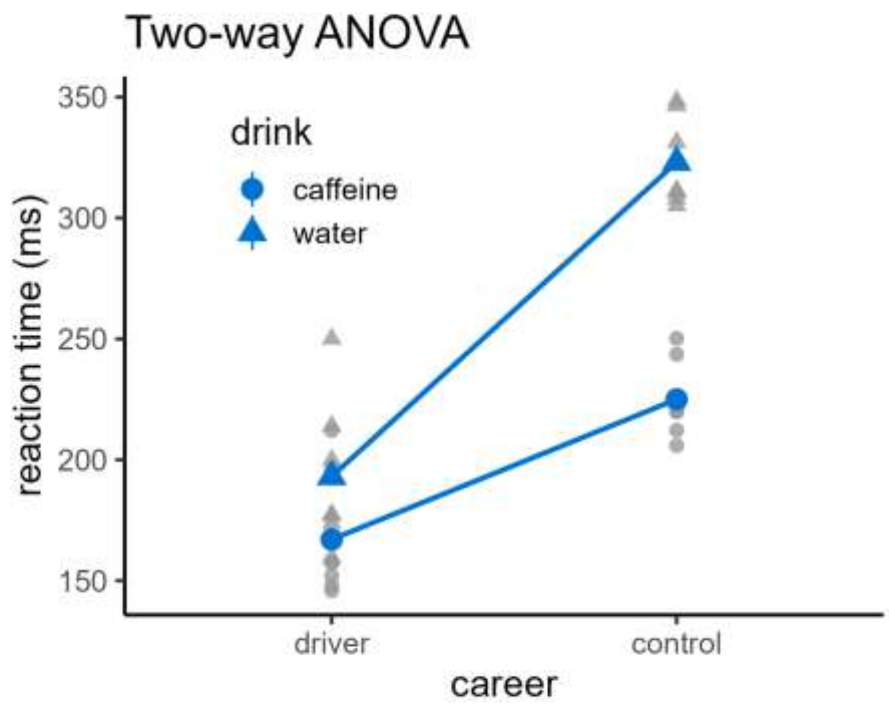
$$reaction = 167 + \begin{pmatrix} 0 \\ 58 \end{pmatrix} \begin{pmatrix} driver \\ control \end{pmatrix} + \begin{pmatrix} 0 \\ 26 \end{pmatrix} \begin{pmatrix} caffeine \\ water \end{pmatrix} + \begin{pmatrix} 0 \\ 72 \end{pmatrix} \begin{pmatrix} other \\ control{:}water \end{pmatrix}$$

Revisiting two-way ANOVA



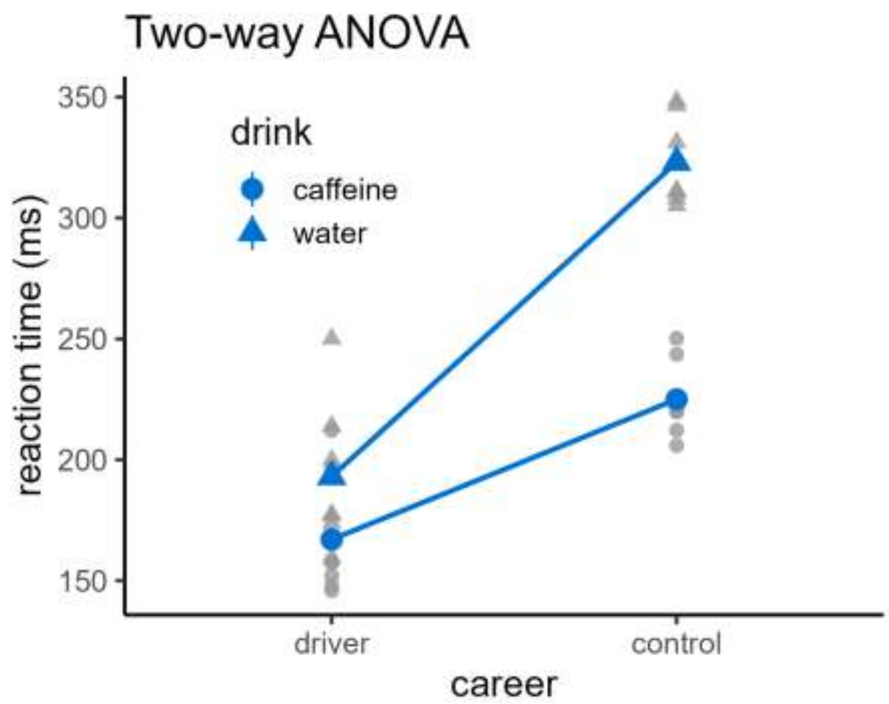
$$reaction = 167 + \binom{0}{58}\binom{driver}{control} + \binom{0}{26}\binom{caffeine}{water} + \binom{0}{72}\binom{other}{control{:}water}$$

Revisiting two-way ANOVA



$$reaction = \beta_0 + \beta_1 career + \beta_2 drink + \beta_3 career\colon drink$$

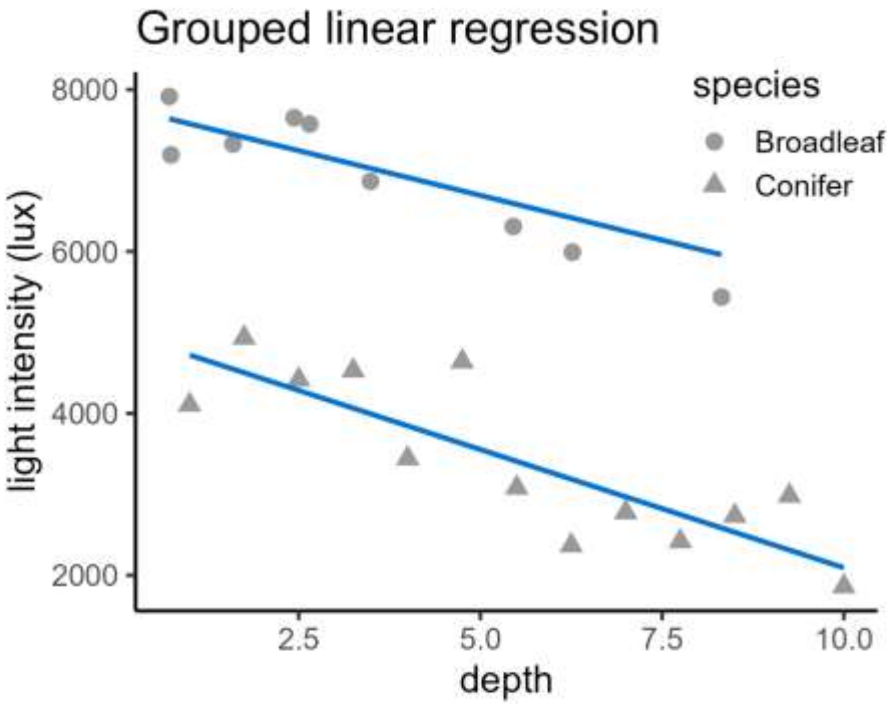
Revisiting two-way ANOVA



$$reaction = \beta_0 + \beta_1 career + \beta_2 drink + \beta_3 career\!:drink$$

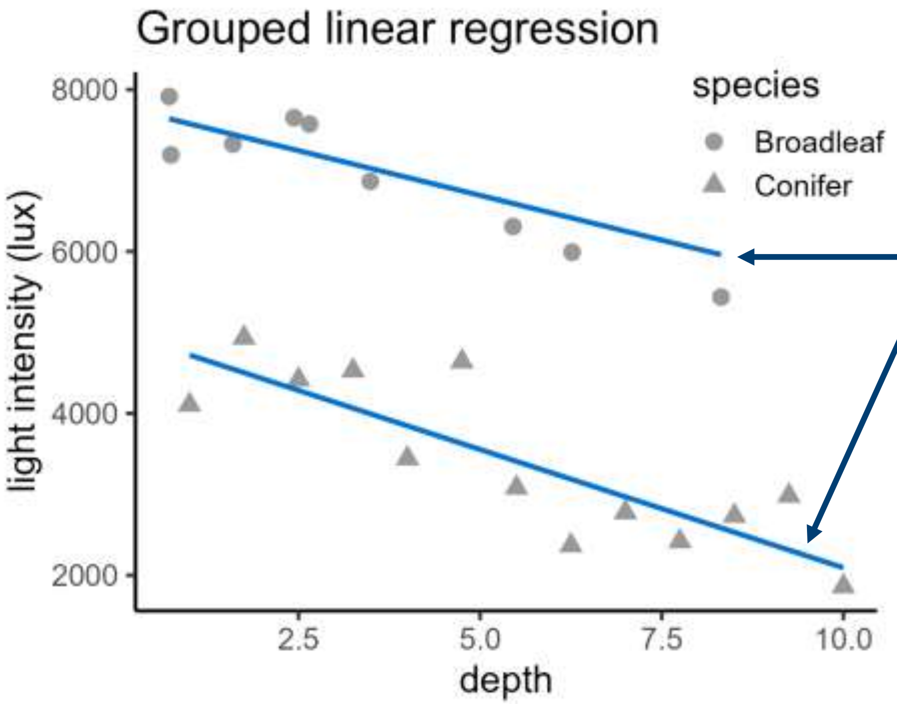
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$$

Revisiting grouped regression



What is the model?

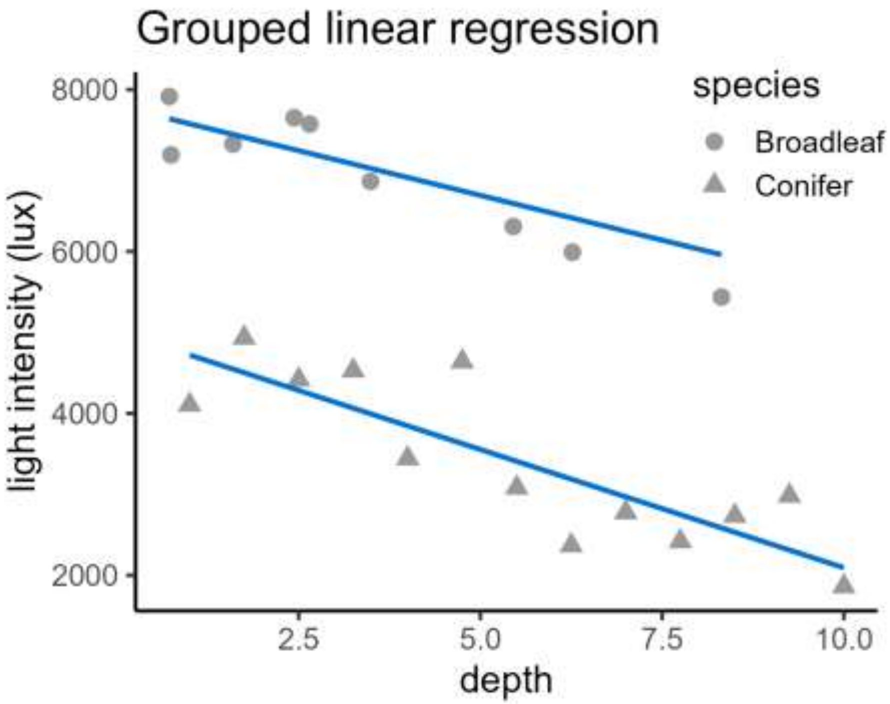
Revisiting grouped regression



What is the model?

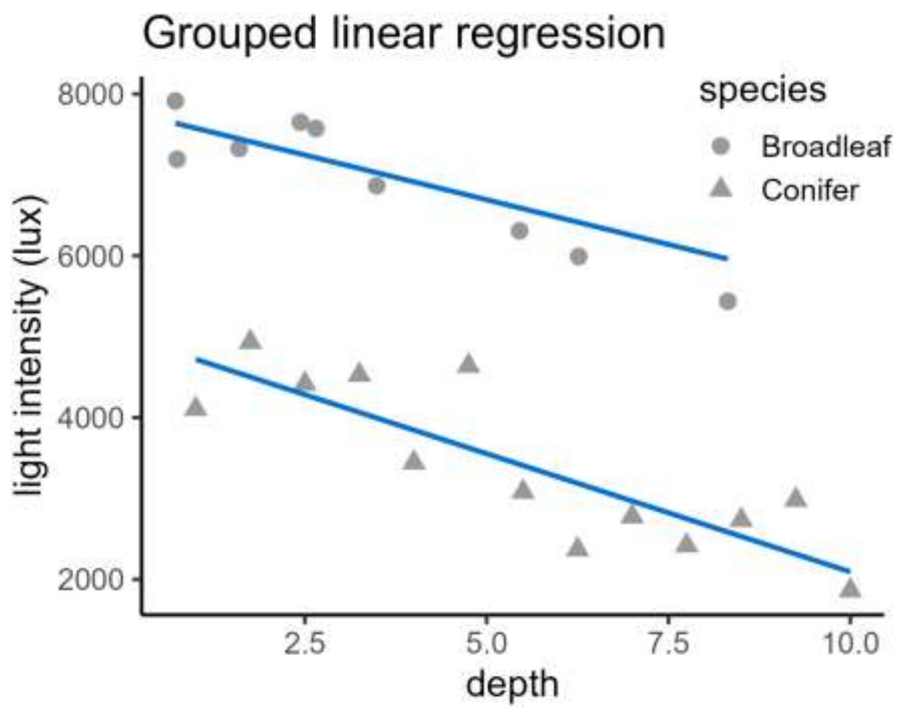
Lines of best fit

Revisiting grouped regression



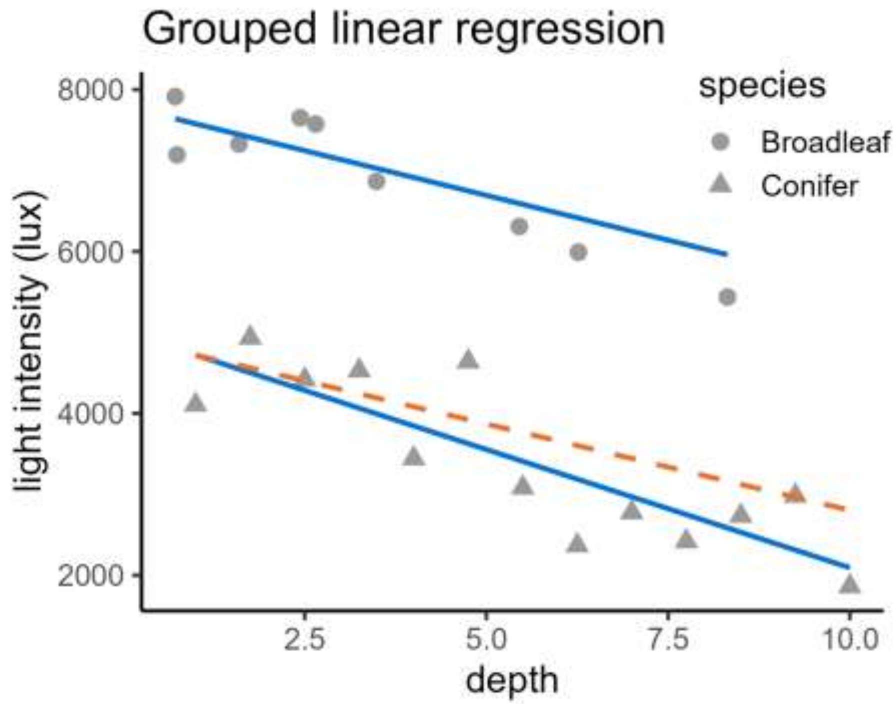
$$light = \begin{pmatrix} 7799 - 221 \times Depth \\ 5014 - 292 \times Depth \end{pmatrix}$$

Revisiting grouped regression



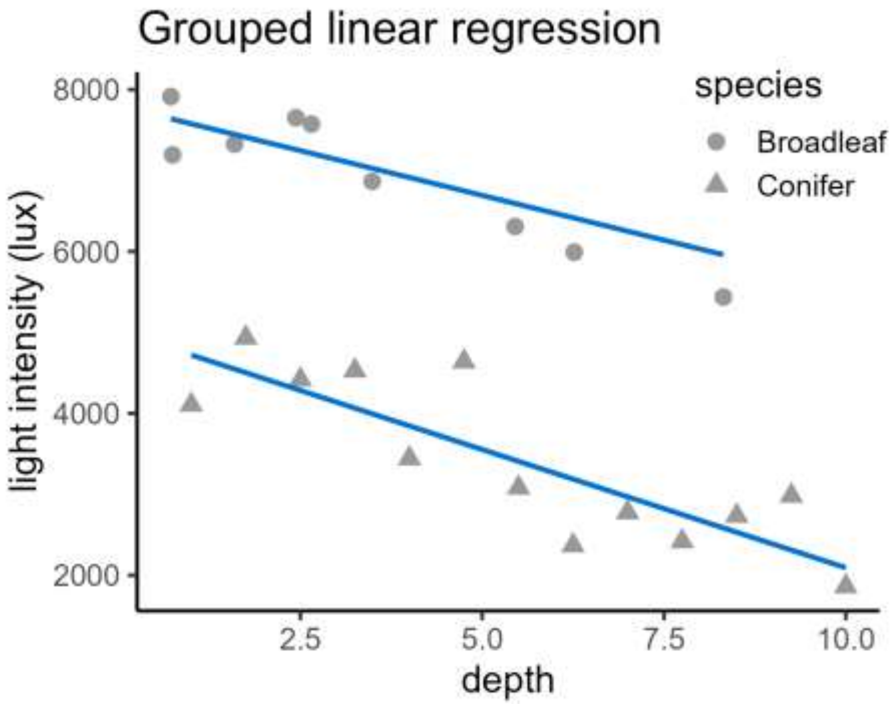
$$light = 7799 - 221 \times depth + \begin{pmatrix} 0 \\ -2785 \end{pmatrix} \begin{pmatrix} Broadleaf \\ Conifer \end{pmatrix} + \begin{pmatrix} 0 \times depth \\ -71 \times depth \end{pmatrix} \begin{pmatrix} Broadleaf \\ Conifer \end{pmatrix}$$

Revisiting grouped regression



$$light = 7799 - 221 \times depth + \begin{pmatrix} 0 \\ -2785 \end{pmatrix} \begin{pmatrix} Broadleaf \\ Conifer \end{pmatrix} + \begin{pmatrix} 0 \times depth \\ -71 \times depth \end{pmatrix} \begin{pmatrix} Broadleaf \\ Conifer \end{pmatrix}$$

Revisiting grouped regression



$$light = \beta_0 + \beta_1 depth + \beta_2 species + \beta_3 depth\!:\!species$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$$

Statistical tests so far

- One-sample, paired & two-sample t-tests
- One-way ANOVA
- Simple linear regression

$$y = \beta_0 + \beta_1 x$$

- Two-way ANOVA
- Grouped linear regression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$$

2. Linearity

What makes these models linear?

What is a linear model?

A linear model describes the relationship between:

- A **response variable**
  e.g., height
- And any number of **predictor variables**
  e.g., age, diet, gender, mother’s height, time of day, country of birth, star sign, type of pet they own, number of Instagram followers

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 \ldots$$

What is a linear model?

Variables can be:

- Continuous — age
- Categorical — country of birth
- Interactions — age:country
- Transformed variables — age^2

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 \ldots$$

What is a linear model?

Variables can be:

- Continuous — age
- Categorical — country of birth
- Interactions — age:country
- Transformed variables — age^2

Linearity refers to the β coefficients

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 \ldots$$

What is a linear model?

$$y = \beta_0 \ln(\beta_1 x_1 + \beta_2 x_2)$$

$$y = \beta_0 e^{\beta_1 x}$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2^2$$

$$y = \frac{\beta_0}{\beta_1 + \beta_2 \mathrm{e}^{\beta_3 x}}$$

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Summary so far…

- T-tests
- ANOVA
- Linear regression

All “special” cases of linear models

Summary so far…

- T-tests
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All "special" cases of linear models

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 \ \ldots$$

Summary so far…

- T-tests
- ANOVA
- Linear regression

All "special" cases of linear models

Explaining a response variable

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 \ldots$$

Summary so far…

- T-tests
- ANOVA
- Linear regression

All “special” cases of linear models

Explaining a response variable in terms of a number of predictor variables

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 \ldots$$

Summary so far…

- T-tests
- ANOVA
- Linear regression

All "special" cases of linear models

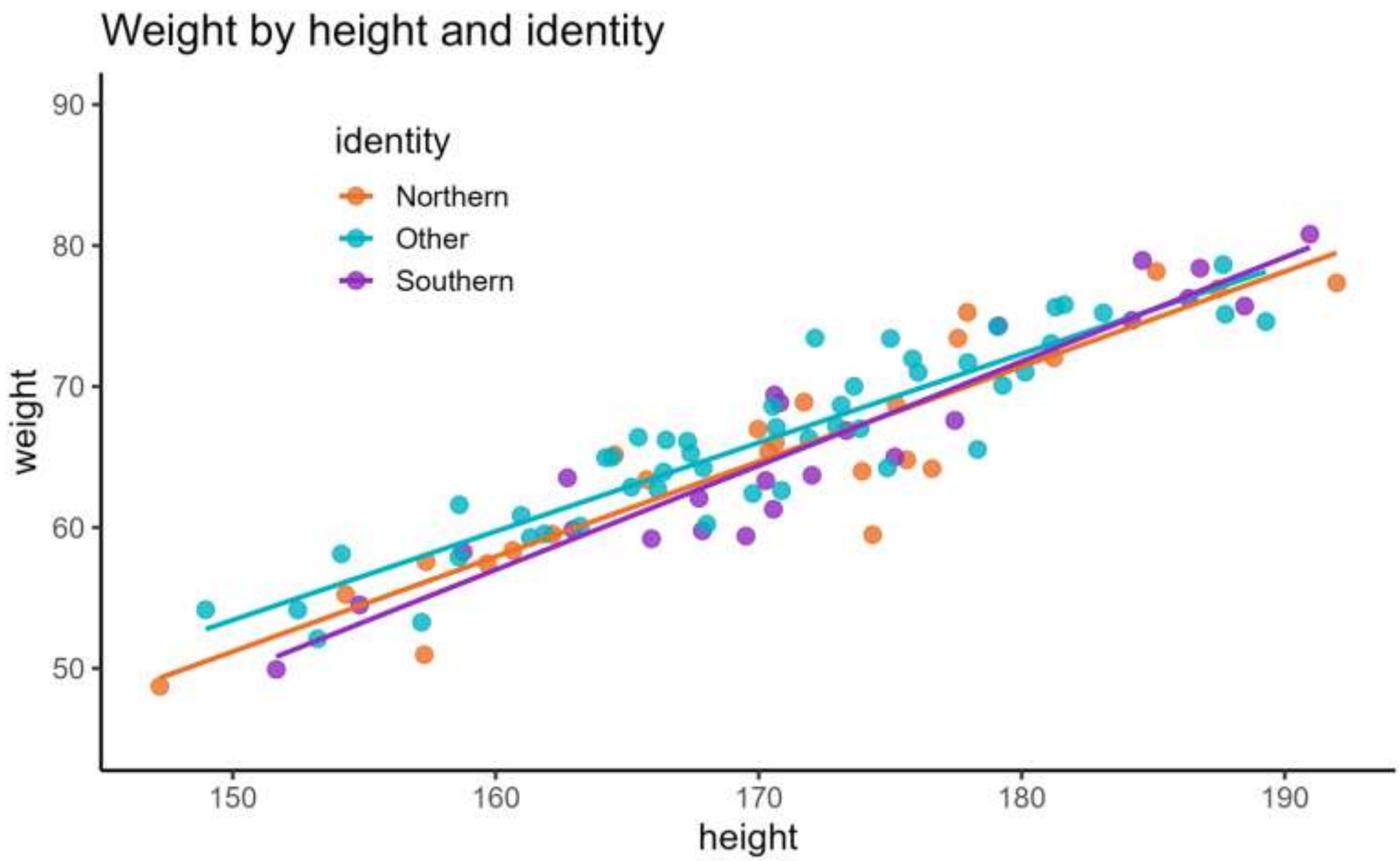
Explaining a response variable in terms of a number of predictor variables where all β coefficients are linear

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 \ldots$$

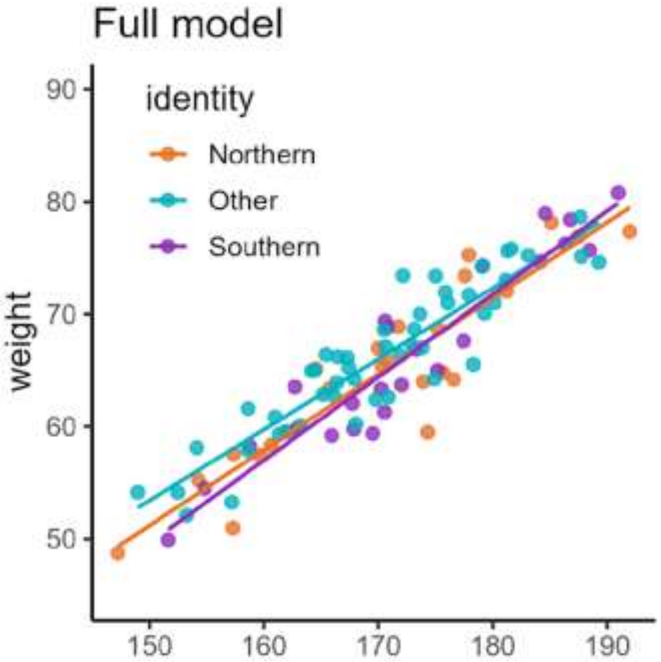
3. Model comparison

Choosing which variables to include in model

Choosing predictors

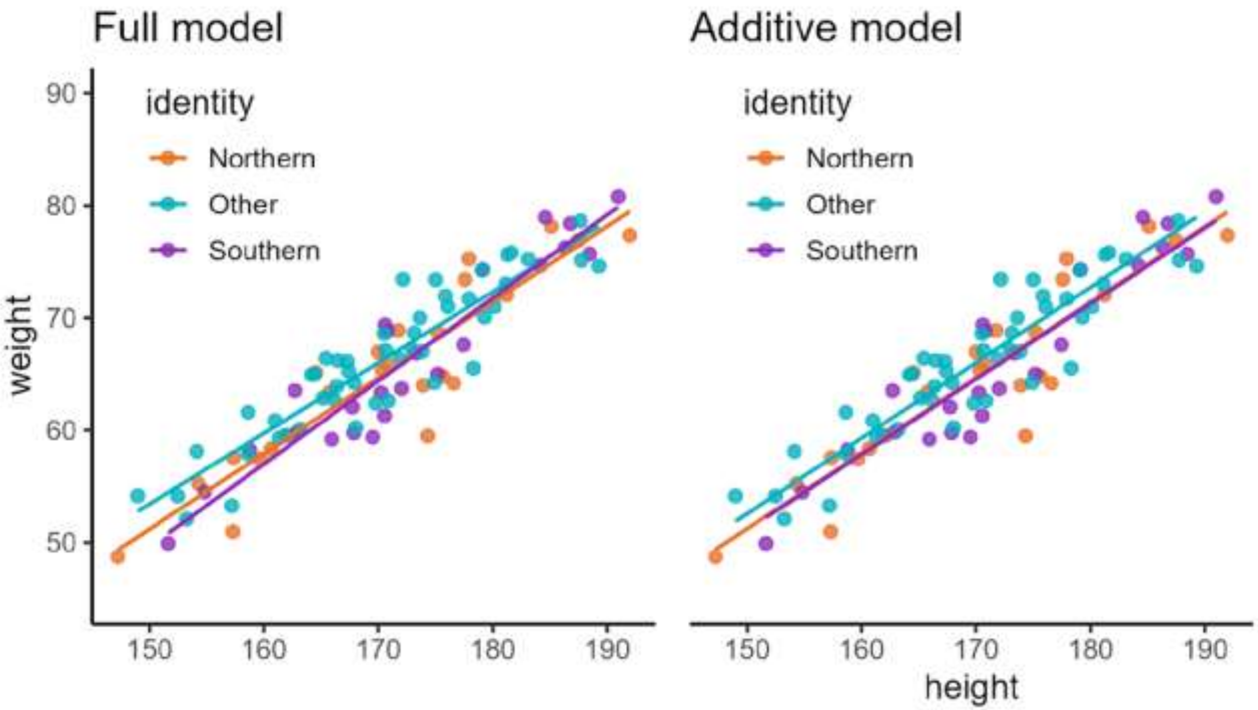


Choosing predictors



weight ~ height + identity
+ height:identity

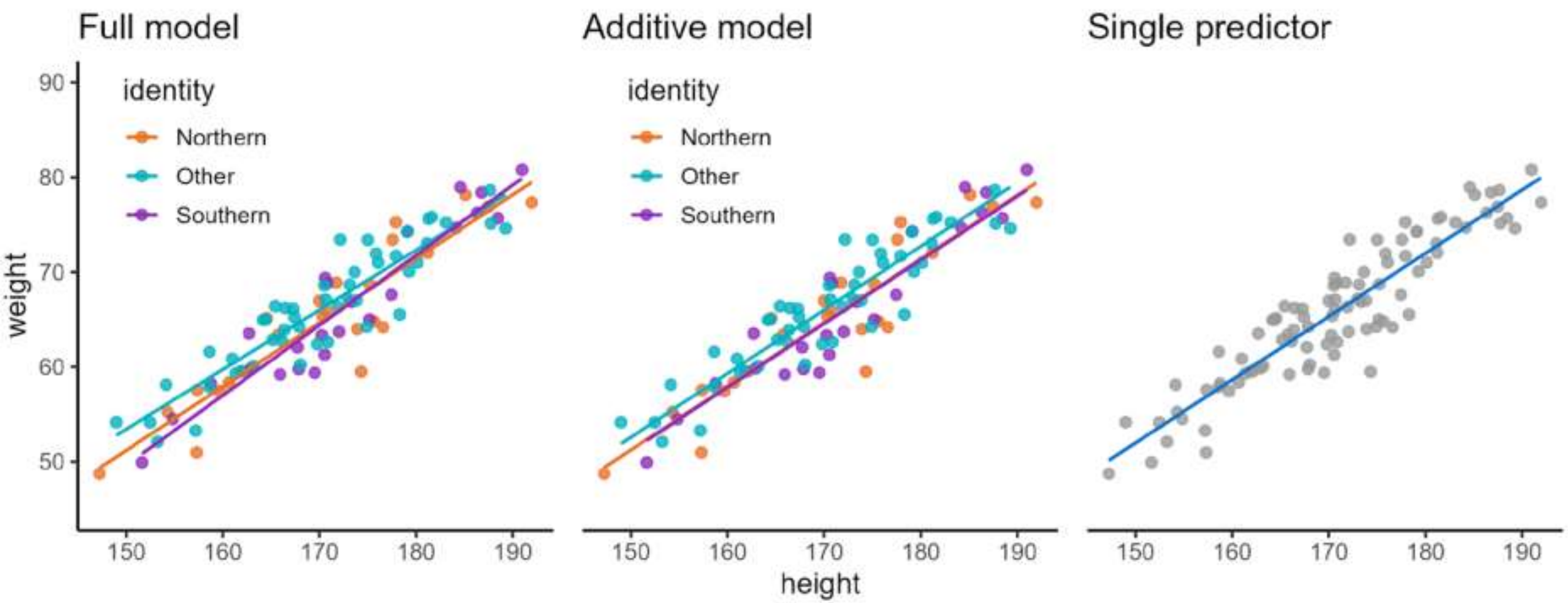
Choosing predictors



weight ~ height + identity + height:identity

weight ~ height + identity

Choosing predictors



weight ~ height + identity + height:identity

weight ~ height + identity

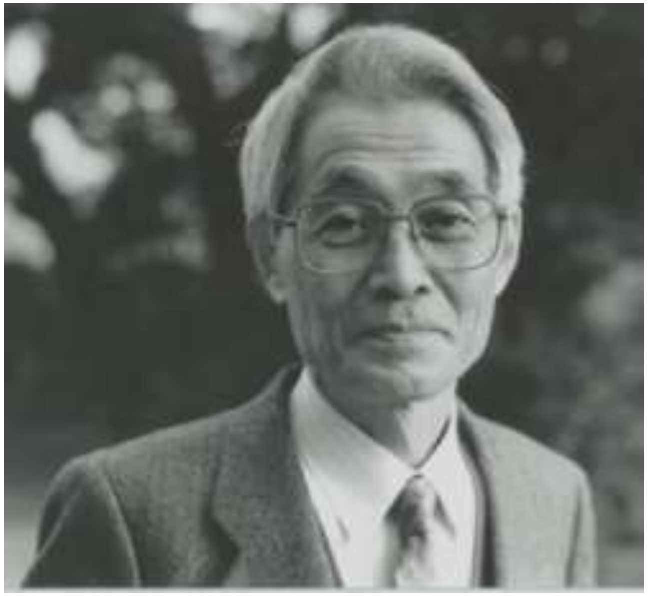
weight ~ height

Model comparison

Akaike Information Criterion (AIC)

Measures trade-off between:

- **Goodness-of-fit**
  (amount of error)
- **Complexity**
  (number of predictors)



Hirotugu Akaike

Model comparison

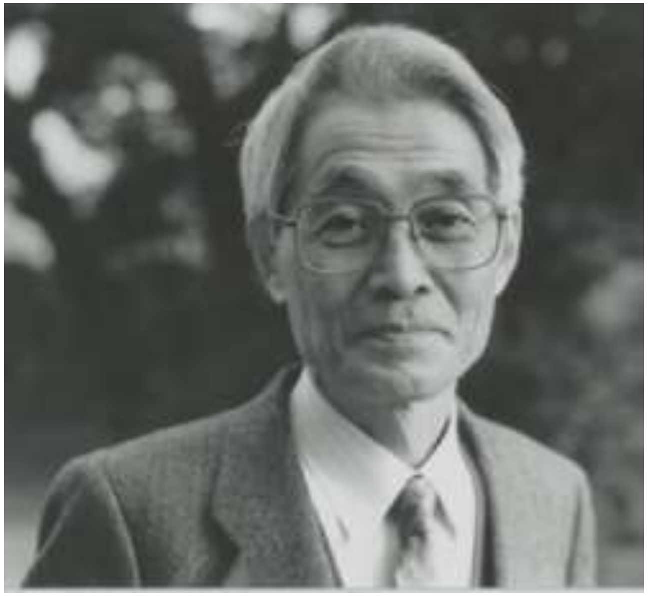
Akaike Information Criterion (AIC)

Measures trade-off between:

- **Goodness-of-fit**
  (amount of error)
- **Complexity**
  (number of predictors)

Used for comparing models:

- Lower AIC means higher quality
- Relative (within dataset) only
- Looking for a difference ≥ 2



Hirotugu Akaike

Model comparison

Bayesian Information Criterion (BIC)

- Alternative to AIC
- Can be used in the same way/in addition to AIC
- Also based on goodness-of-fit vs complexity
- Lower BIC means higher quality

Hirotugu Akaike

Gideon Schwarz

Model comparison

Procedures like backwards stepwise elimination use AIC (or similar) to select a model

Backwards stepwise elimination

Consider a full model with three predictors

$$y \sim \underbrace{A \;+\; B \;+\; C}_{\text{Main effects}} + \underbrace{A{:}B \;+\; A{:}C \;+\; B{:}C}_{\text{Two-way interactions}} + \underbrace{A{:}B{:}C}_{\text{Three-way interaction}}$$

Backwards stepwise elimination

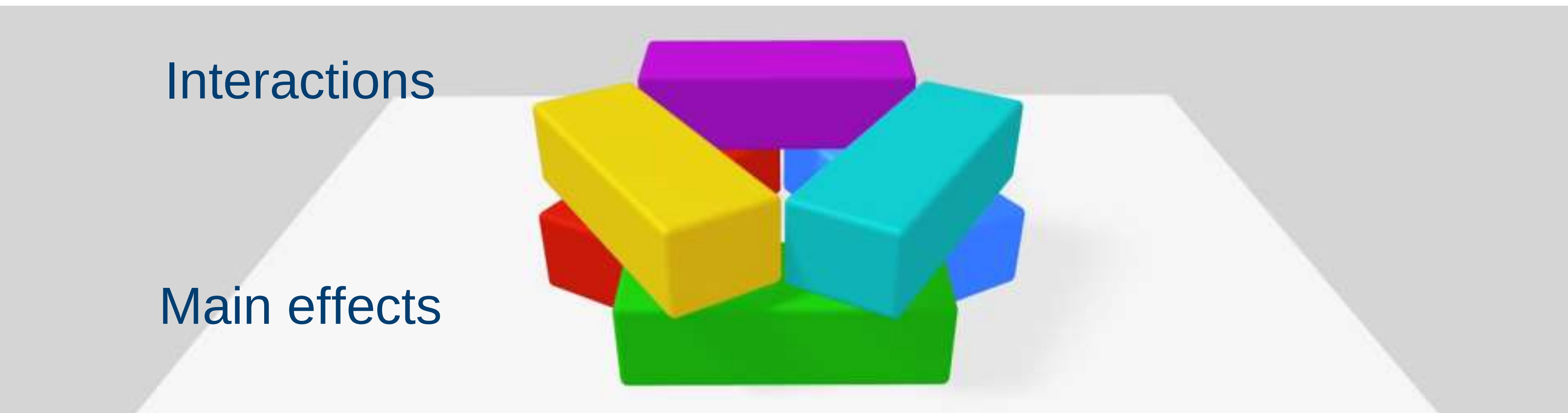
Consider a full model with three predictors

$$y \sim \underbrace{A + B + C}_{\text{Main effects}} + \underbrace{A{:}B + A{:}C + B{:}C}_{\text{Two-way interactions}}$$

Backwards stepwise elimination

Consider a full model with three predictors

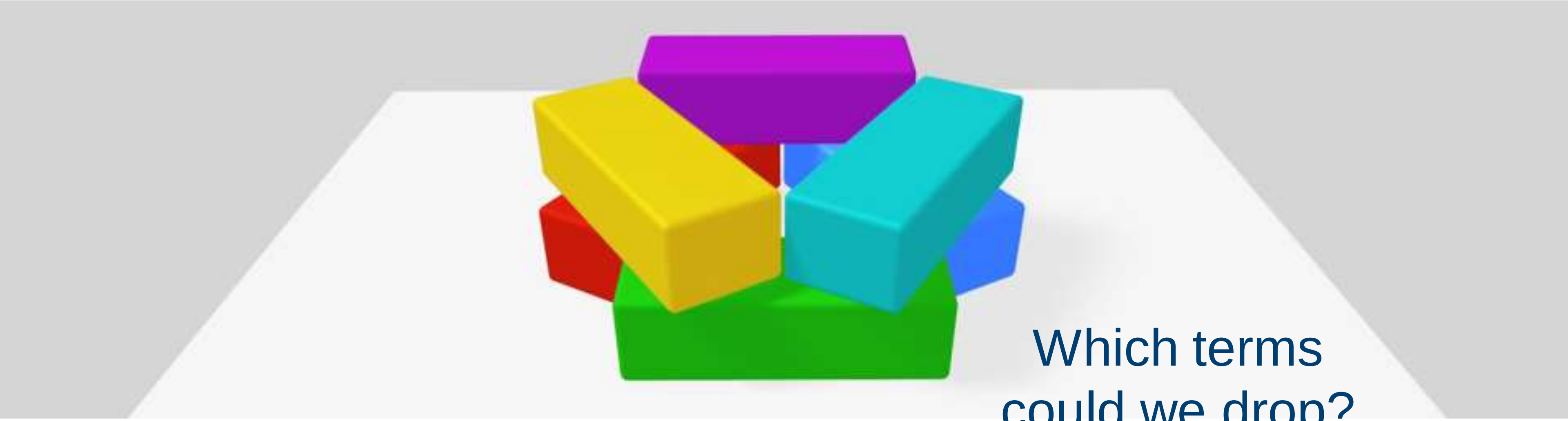
$$y \sim A + B + C + A{:}B + A{:}C + B{:}C$$



Backwards stepwise elimination

Consider a full model with three predictors

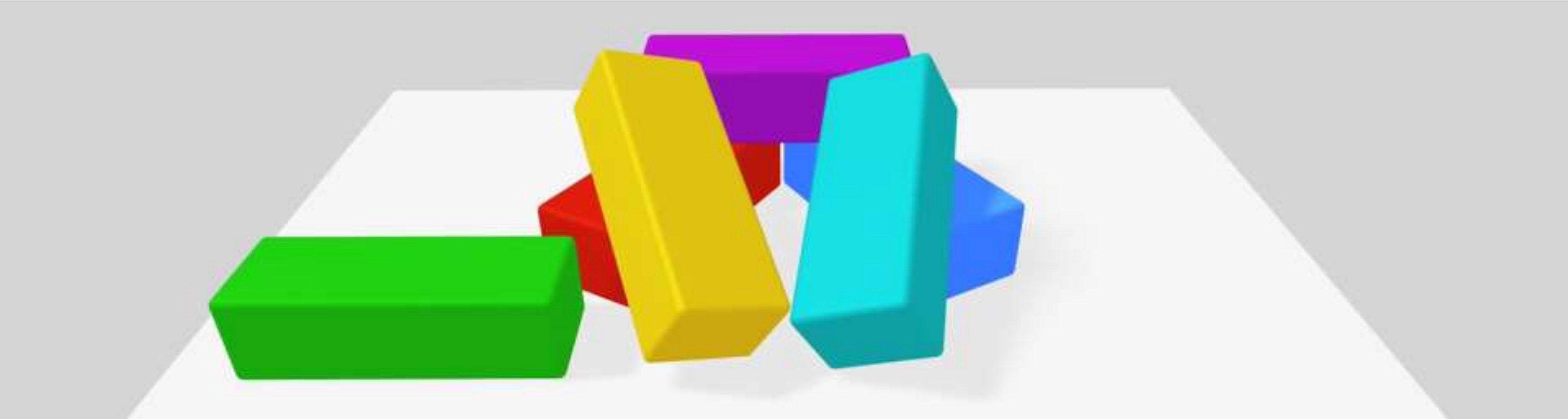
$$y \sim A + B + C + A{:}B + A{:}C + B{:}C$$



Which terms could we drop?

Backwards stepwise elimination

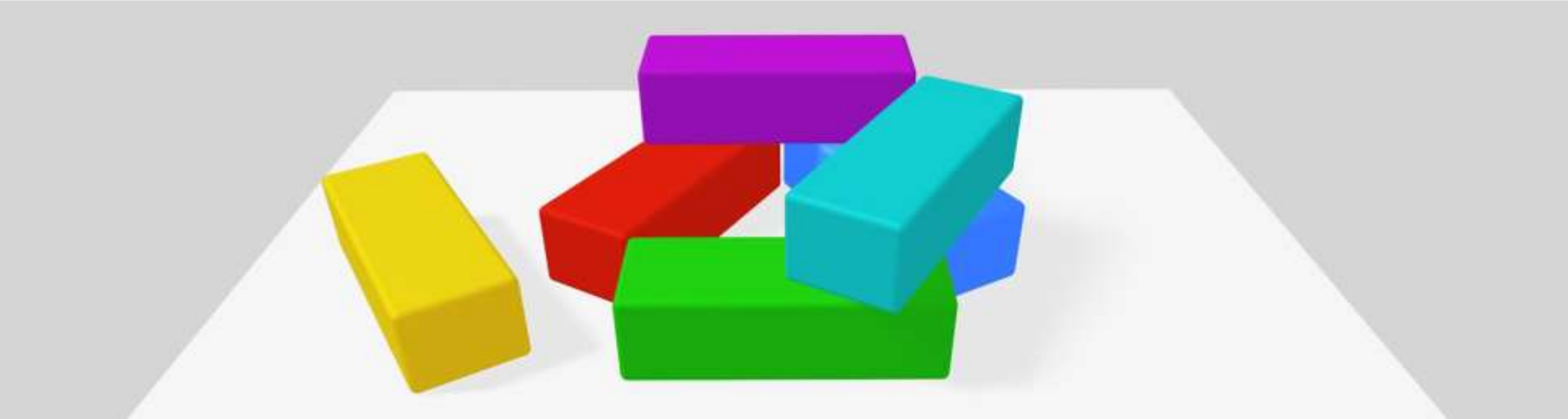
Cannot drop main effects that are part of an existing interaction



Backwards stepwise elimination

Cannot drop main effects that are part of an existing interaction

But could try dropping the interactions



Backwards stepwise elimination

Full model: $y \sim A + B + C + A{:}B + A{:}C + B{:}C$

Create three new candidate models:

Backwards stepwise elimination

Full model: $y \sim A + B + C + A{:}B + A{:}C + B{:}C$

Create three new candidate models:

drop.lm1:	$y \sim A + B + C + A{:}C + B{:}C$	drop A:B
drop.lm2:	$y \sim A + B + C + A{:}B + B{:}C$	drop A:C
drop.lm3:	$y \sim A + B + C + A{:}B + A{:}C$	drop B:C

Backwards stepwise elimination

Full model:	$y \sim A + B + C + A\!:\!B + A\!:\!C + B\!:\!C$	123

Create three new candidate models:

		AIC
drop.lm1:	$y \sim A + B + C + A\!:\!C + B\!:\!C$	121
drop.lm2:	$y \sim A + B + C + A\!:\!B + B\!:\!C$	118
drop.lm3:	$y \sim A + B + C + A\!:\!B + A\!:\!C$	131

Backwards stepwise elimination

Full model:	$y \sim A + B + C + A{:}B + A{:}C + B{:}C$	123

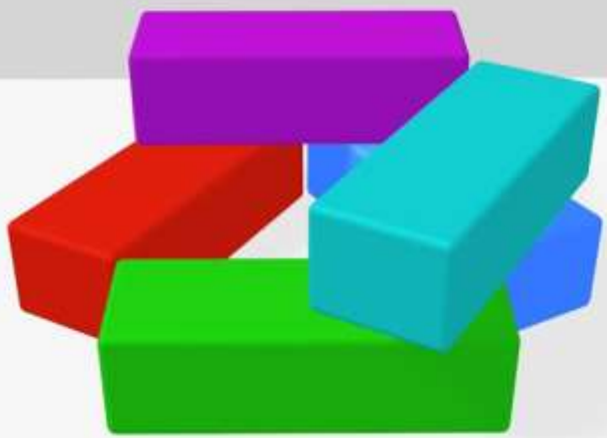
Create three new candidate models:

		AIC
drop.lm1:	$y \sim A + B + C + A{:}C + B{:}C$	121
drop.lm2:	$y \sim A + B + C + A{:}B + B{:}C$	118
drop.lm3:	$y \sim A + B + C + A{:}B + A{:}C$	131

Choose model with lowest AIC as working model: drop.lm2

Backwards stepwise elimination

Working model: $y \sim A + B + C + A{:}B + B{:}C$ 118



Which terms could we drop?

Backwards stepwise elimination

Working model: $y \sim A + B + C + A{:}B + B{:}C$ 118

Create two new candidate models:

drop.lm4: $y \sim A + B + C + A{:}B$ drop B:C

drop.lm5: $y \sim A + B + C + B{:}C$ drop A:B

Backwards stepwise elimination

Working model: $y \sim A + B + C + A{:}B + B{:}C$ 118

Create two new candidate models:

		AIC
drop.lm4:	$y \sim A + B + C + A{:}B$	124
drop.lm5:	$y \sim A + B + C + B{:}C$	114

Backwards stepwise elimination

Working model: $y \sim A + B + C + A{:}B + B{:}C$ 118

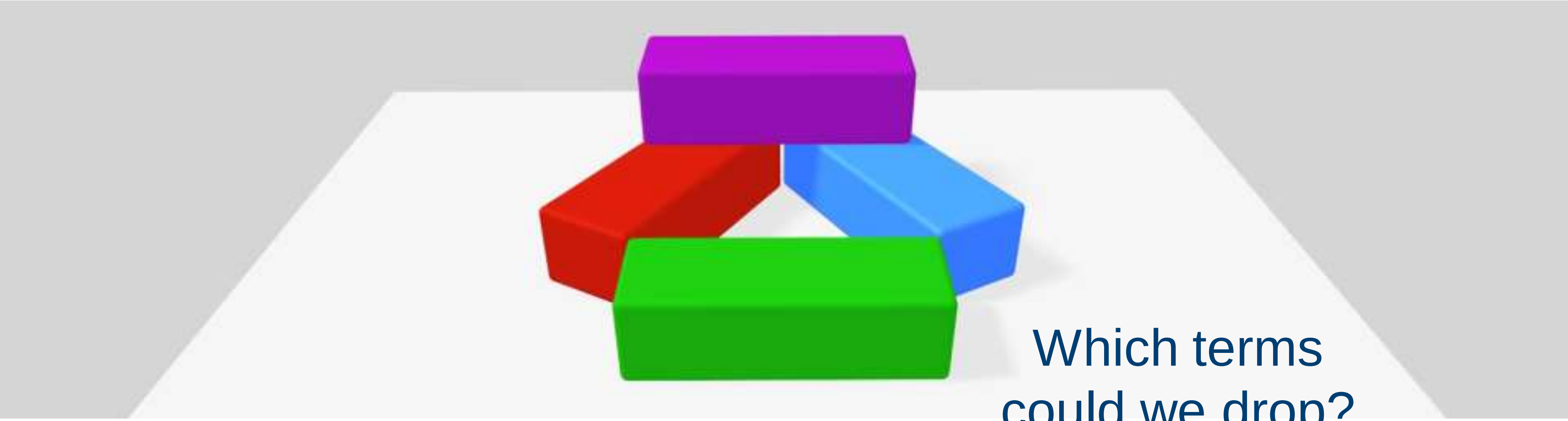
Create two new candidate models:

		AIC
drop.lm4:	$y \sim A + B + C + A{:}B$	124
drop.lm5:	$y \sim A + B + C + B{:}C$	114

Choose model with lowest AIC as working model: drop.lm5

Backwards stepwise elimination

Working model: $y \sim A + B + C + B{:}C$ 114



Backwards stepwise elimination

Working model: $y \sim A + B + C + B{:}C$ 114

Create two new candidate models:

drop.lm6: $y \sim B + C + B{:}C$ drop A
drop.lm7: $y \sim A + B + C$ drop B:C

Backwards stepwise elimination

Working model: $y \sim A + B + C + B{:}C$ 114

Create two new candidate models:

		AIC
drop.lm6:	$y \sim B + C + B{:}C$	130
drop.lm7:	$y \sim A + B + C$	127

Backwards stepwise elimination

Working model: $y \sim A + B + C + B{:}C$ 114

Create two new candidate models:

		AIC
drop.lm6:	$y \sim B + C + B{:}C$	130
drop.lm7:	$y \sim A + B + C$	127

No further decrease in AIC – keep current working model

Session 5: Multiple predictors

Lecture summary:

- All parametric tests so far can be thought of as special cases of a linear model
- Linear model = response ~ predictors, where the beta coefficients are linear
- Information criteria can be used to compare models with different predictors (e.g., Akaike information criterion)

Session 5: Multiple predictors

Course materials:

Fitting more complex linear models

- Datasets with 3+ predictors
- Testing assumptions

Practising backwards stepwise elimination using AIC values

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Core statistics

Session 5: Multiple predictors

Cambridge Centre for Research Informatics Training

UNIVERSITY OF CAMBRIDGE

Core statistics

Session 6: Uncertainty

Cambridge Centre for Research Informatics Training

Plan for the session

Session 6: **Uncertainty**

1. Errors & power
2. Effect sizes
3. Multiple comparisons

1. Errors & power

How likely is it that your p-value is “correct”?

Uncertainty in significance testing

Hypothesis tests are a decision: **Is this result significant or not?**

This decision can be wrong in two ways:

Uncertainty in significance testing

Hypothesis tests are a decision: **Is this result significant or not?**

This decision can be wrong in two ways:

False positive (type I) error

- Fail to retain the null hypothesis even though it's true
- i.e., you see something that isn't there

Uncertainty in significance testing

Hypothesis tests are a decision: **Is this result significant or not?**

This decision can be wrong in two ways:

False positive (type I) error

- Fail to retain the null hypothesis even though it's true
- i.e., you see something that isn't there

False negative (type II) error

- Fail to reject the null hypothesis even though it's false
- i.e., you miss something

Uncertainty in significance testing

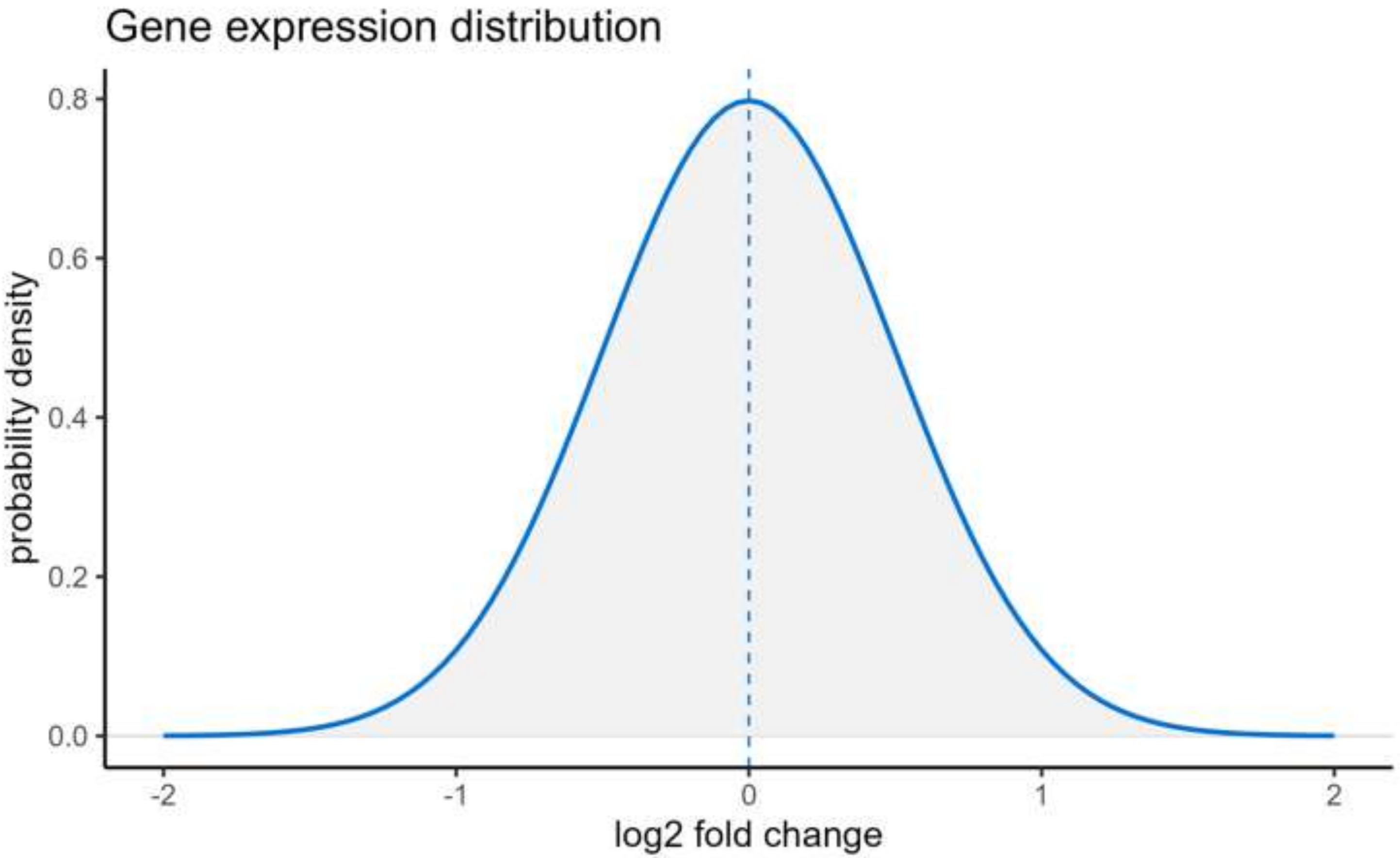
Hypothesis tests are a decision: **Is this result significant or not?**

This decision can be wrong in two ways:

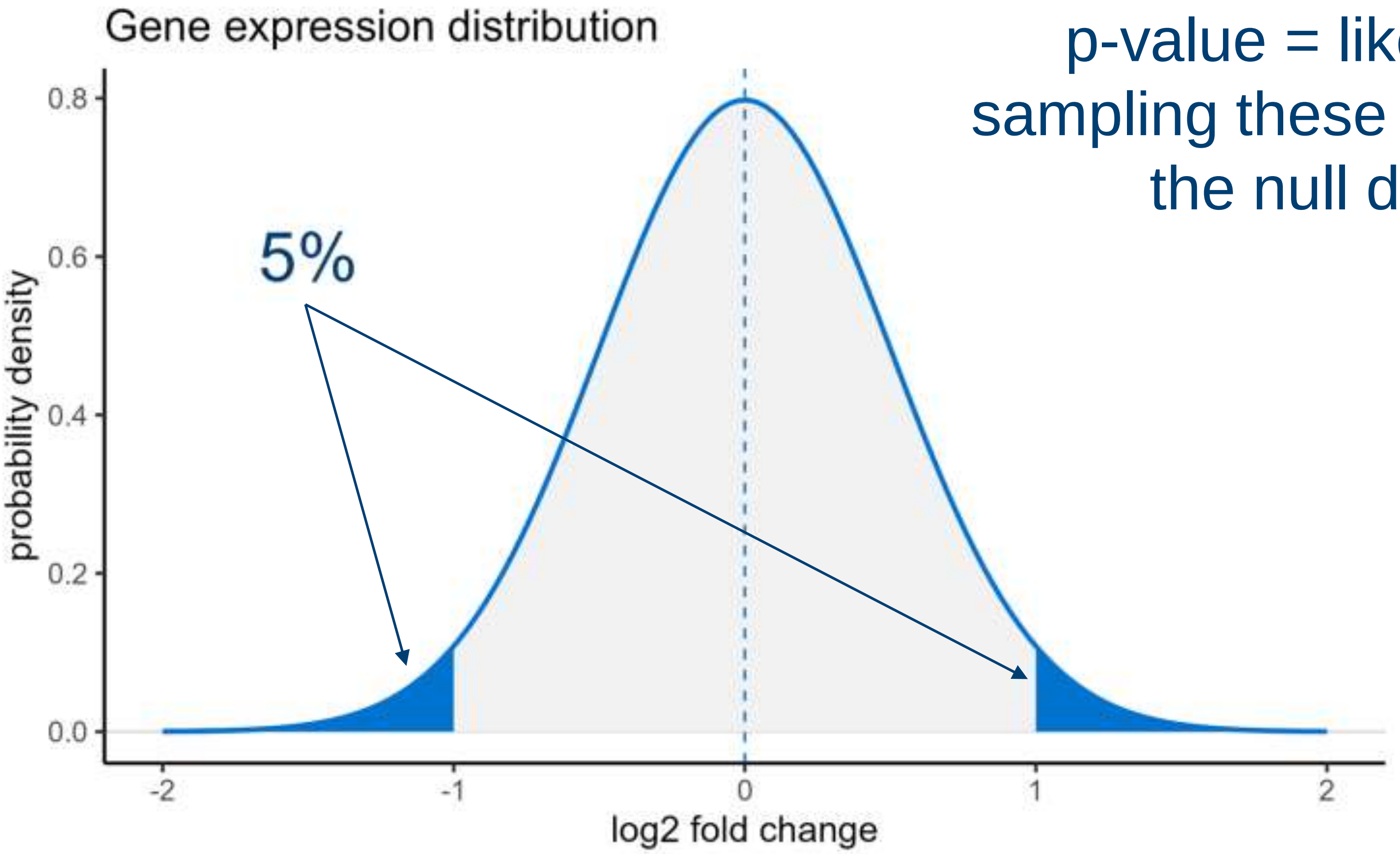
False positive (type I) error

False negative (type II) error

False positive (type I) error

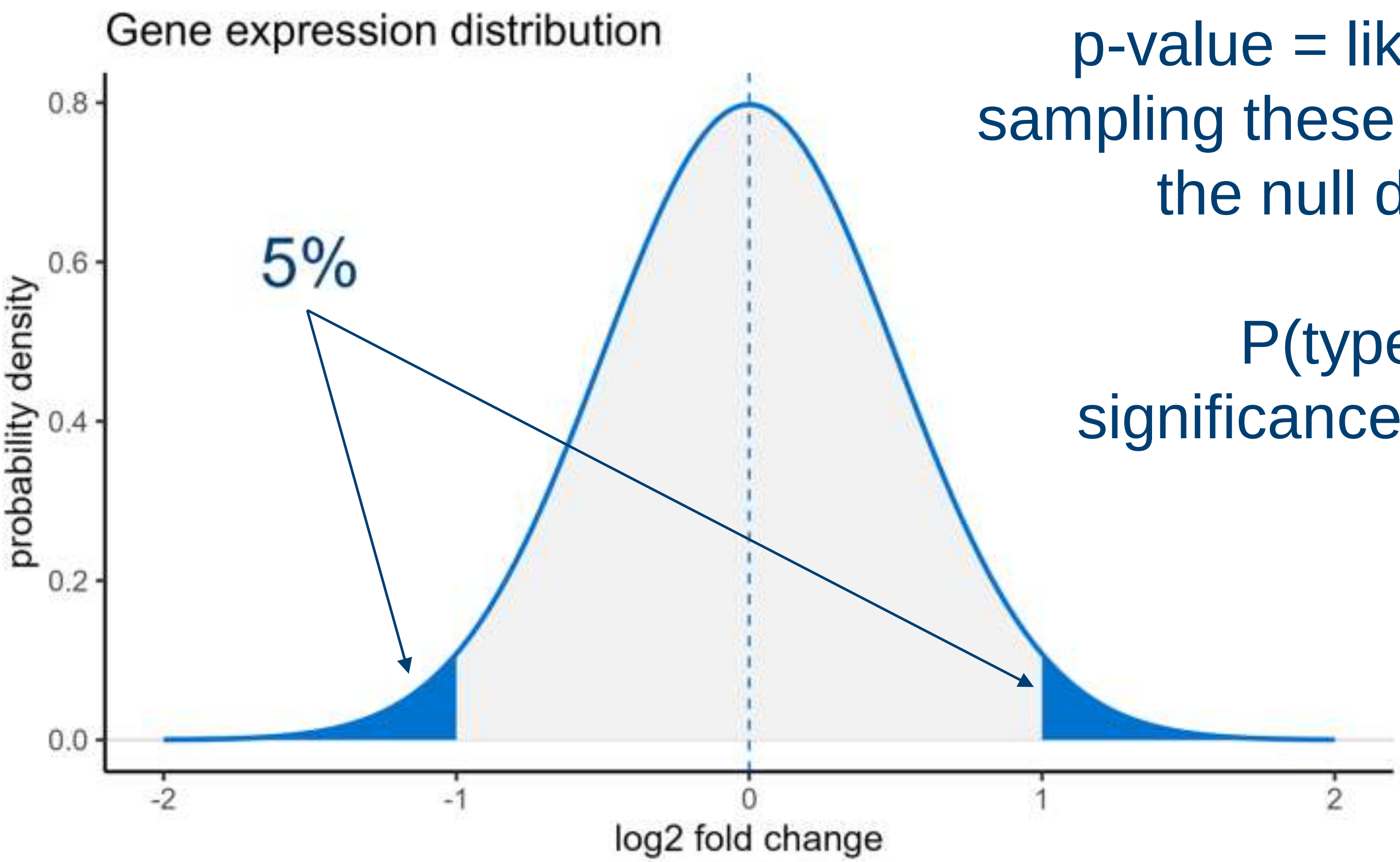


False positive (type I) error



p-value = likelihood of sampling these data from the null distribution

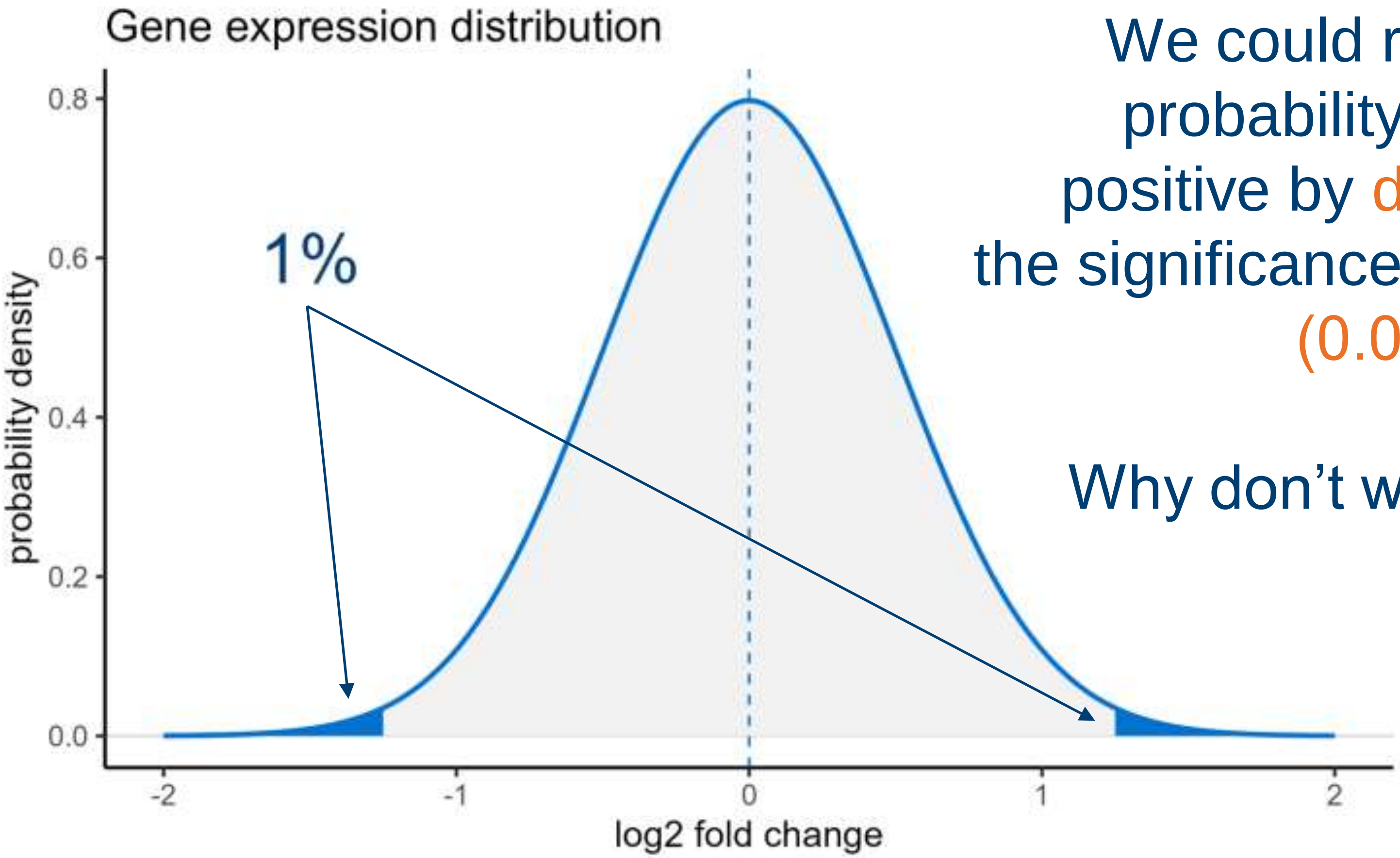
False positive (type I) error



p-value = likelihood of sampling these data from the null distribution

P(type I error) = significance threshold

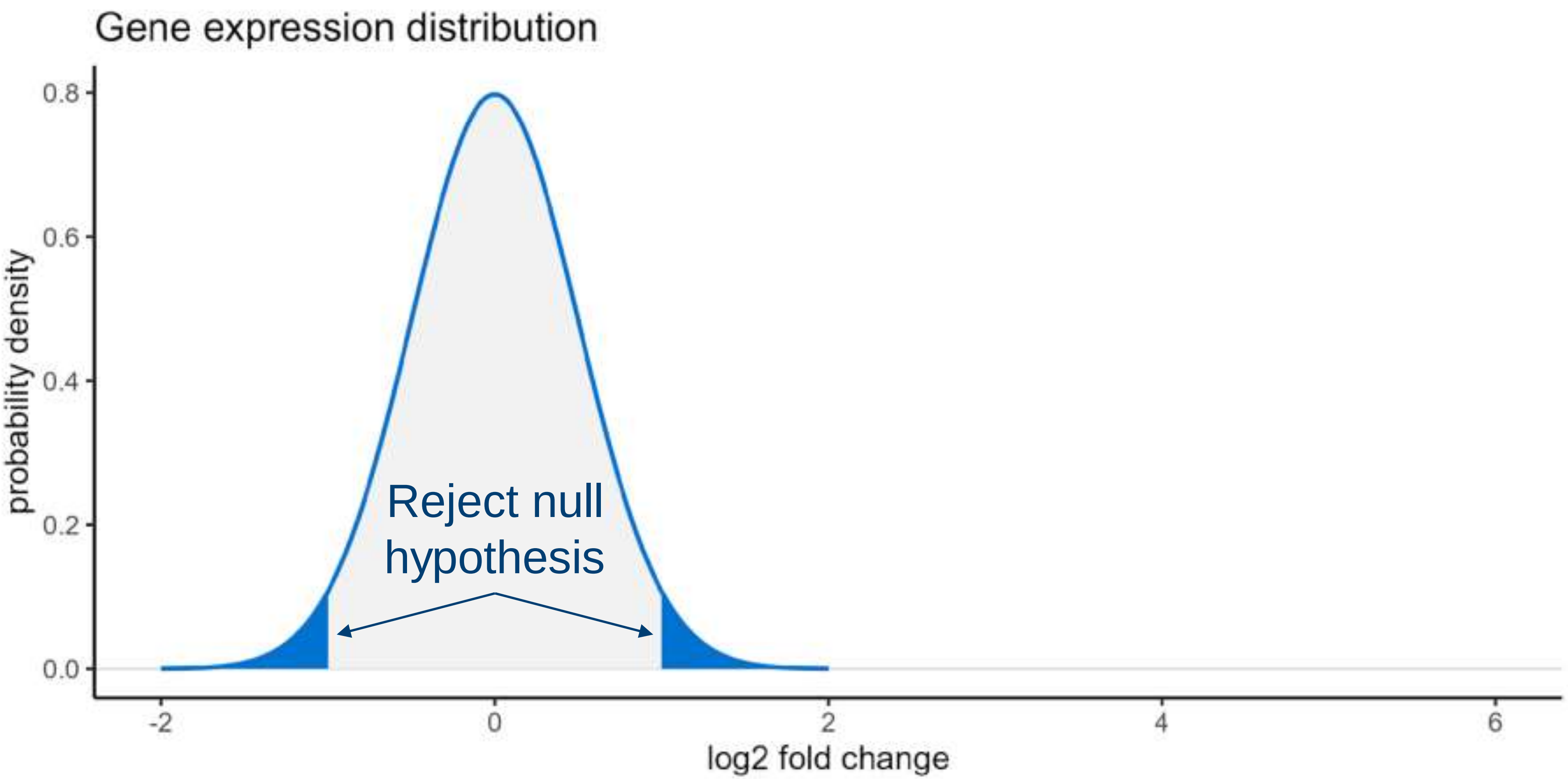
False positive (type I) error



We could reduce the probability of a false positive by decreasing the significance threshold (0.05 → 0.01)

Why don’t we do this?

False negative (type II) error



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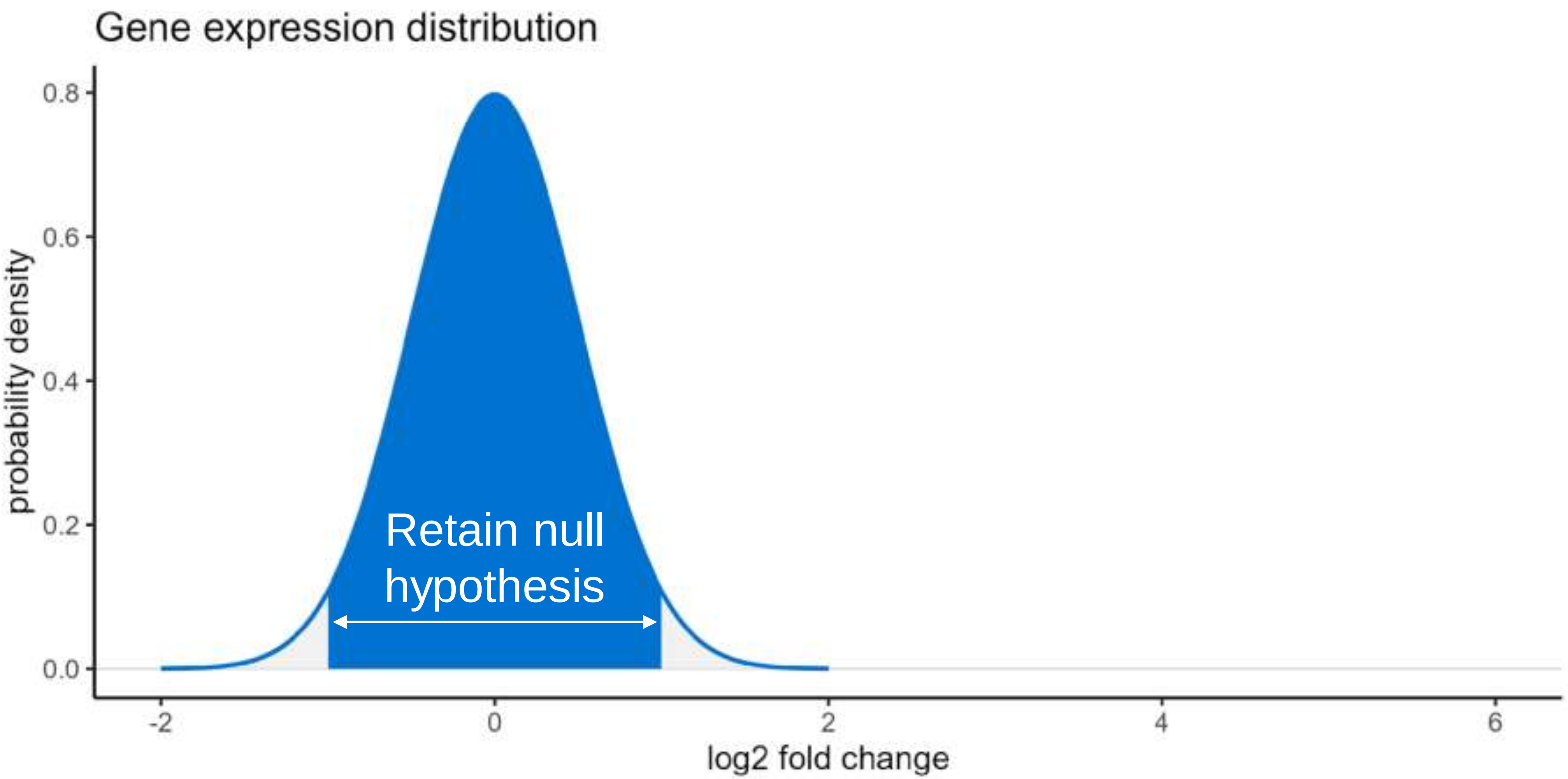
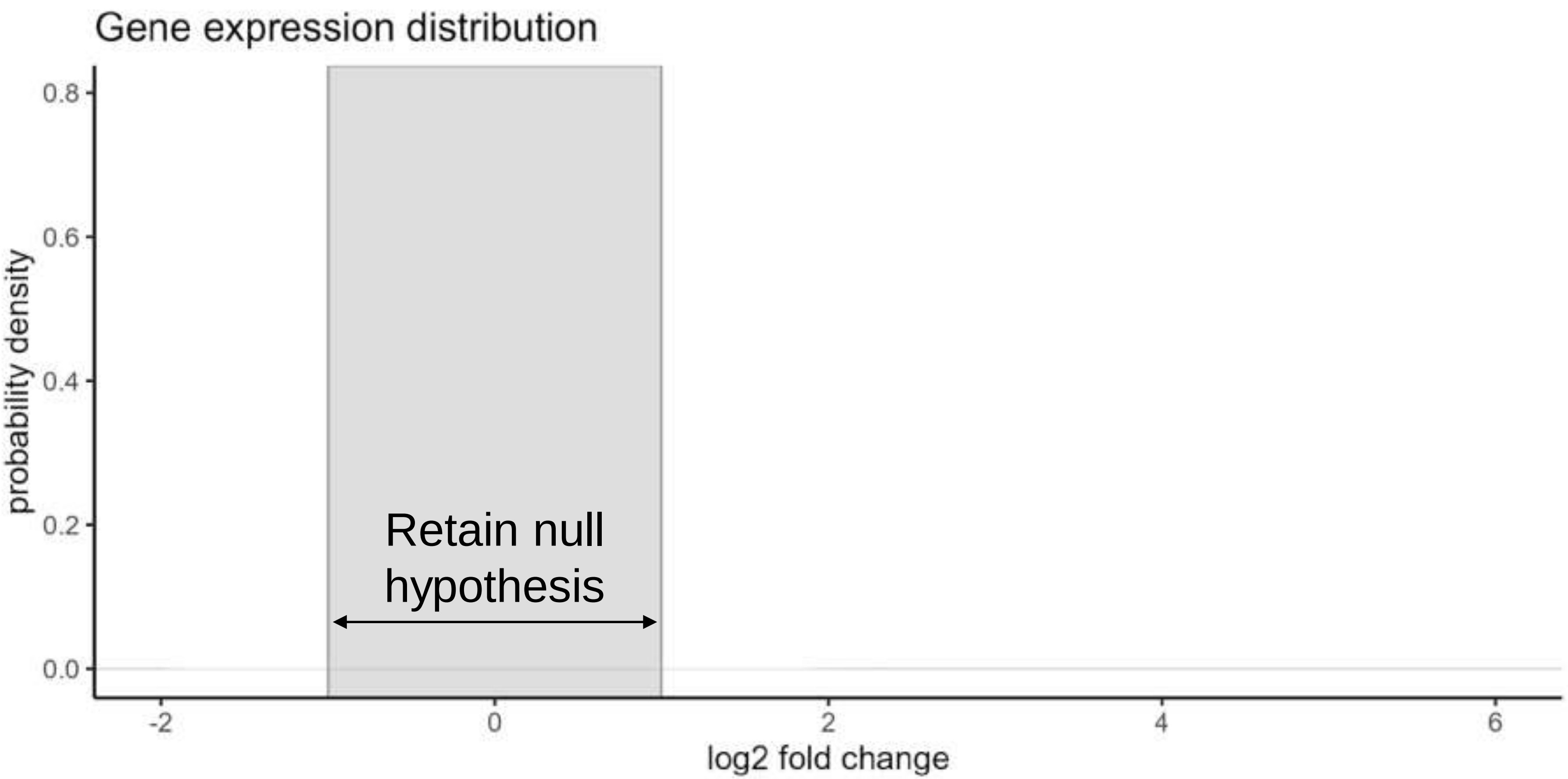
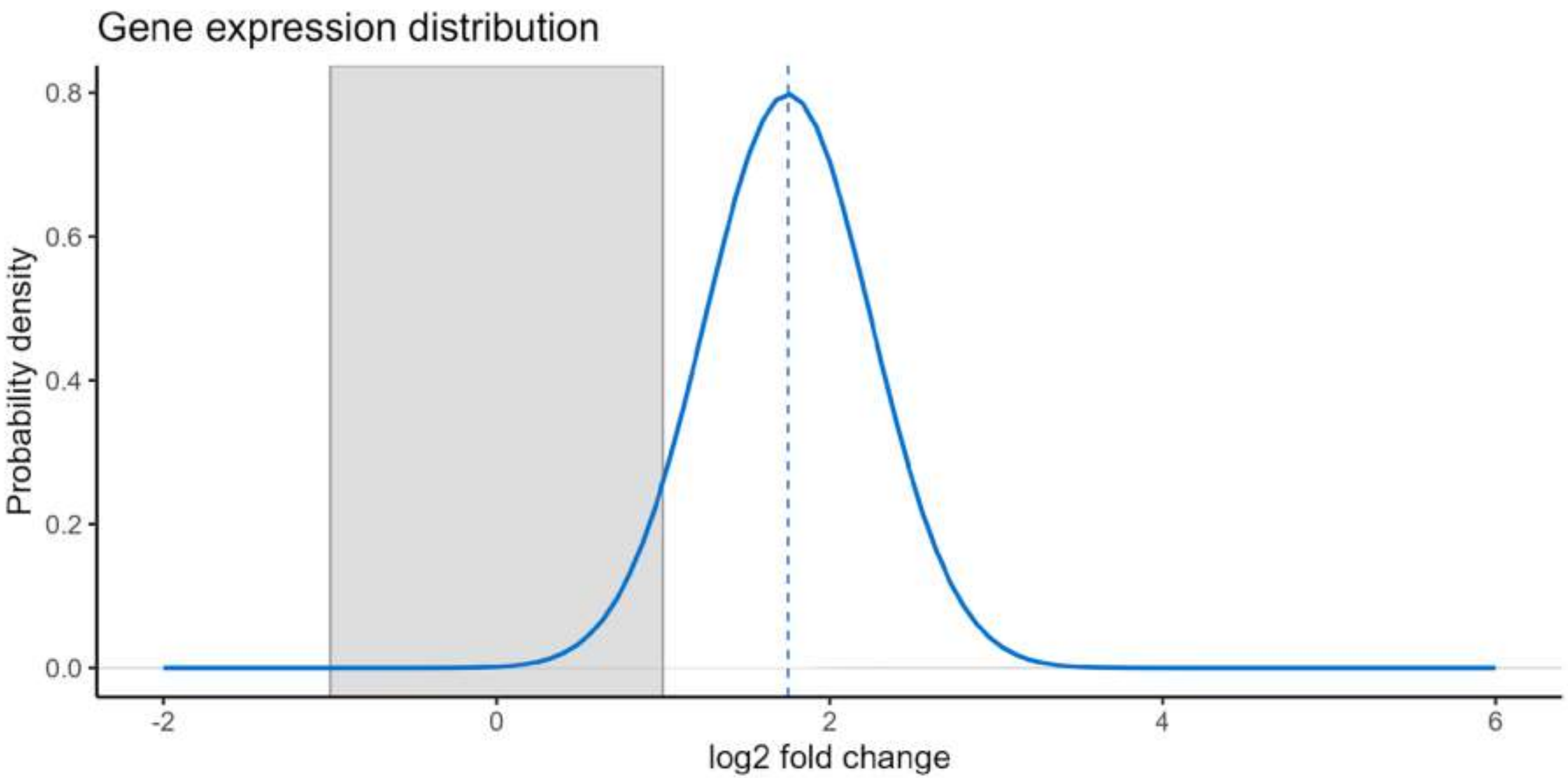
False negative (type II) error



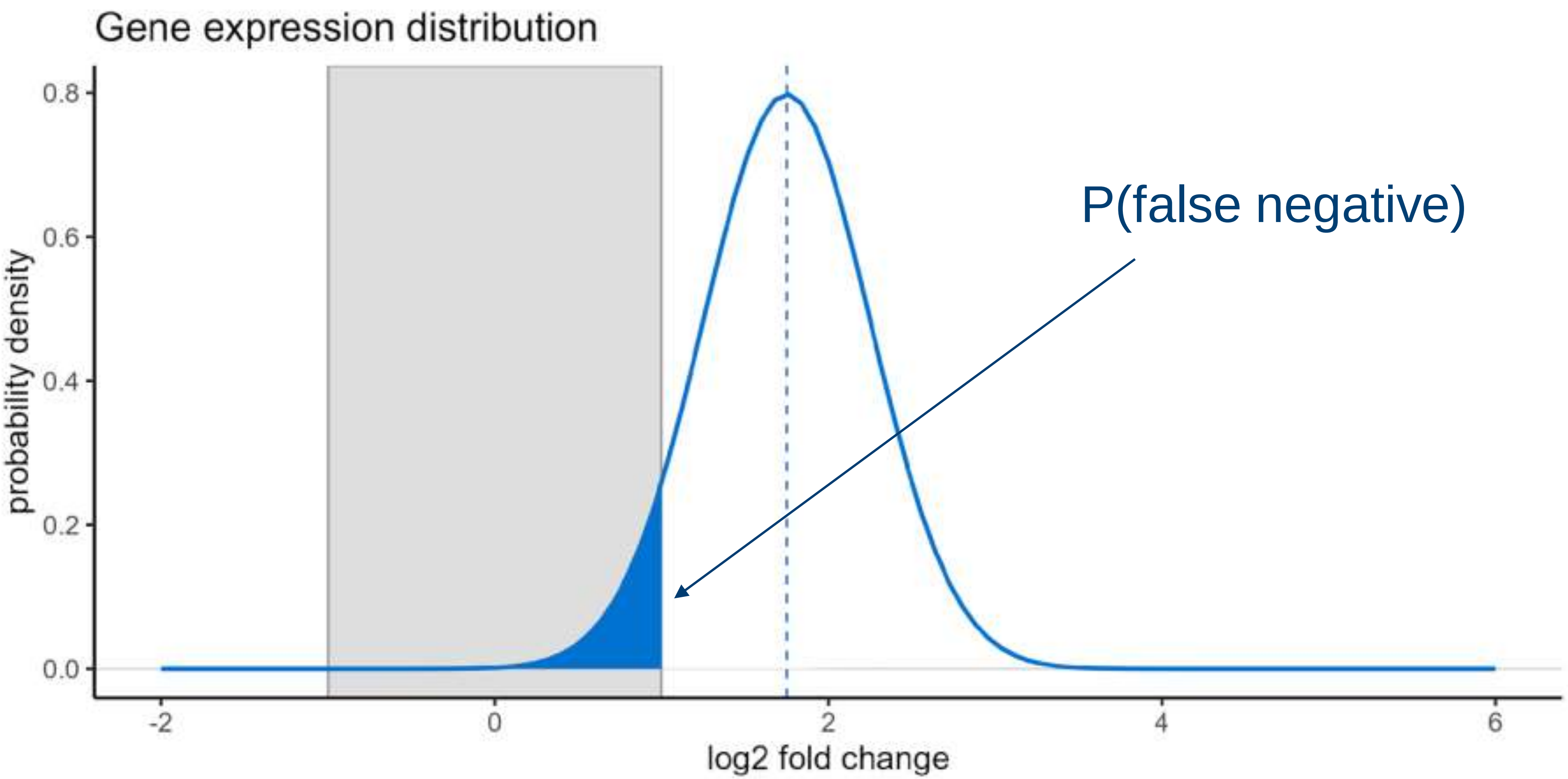
False negative (type II) error



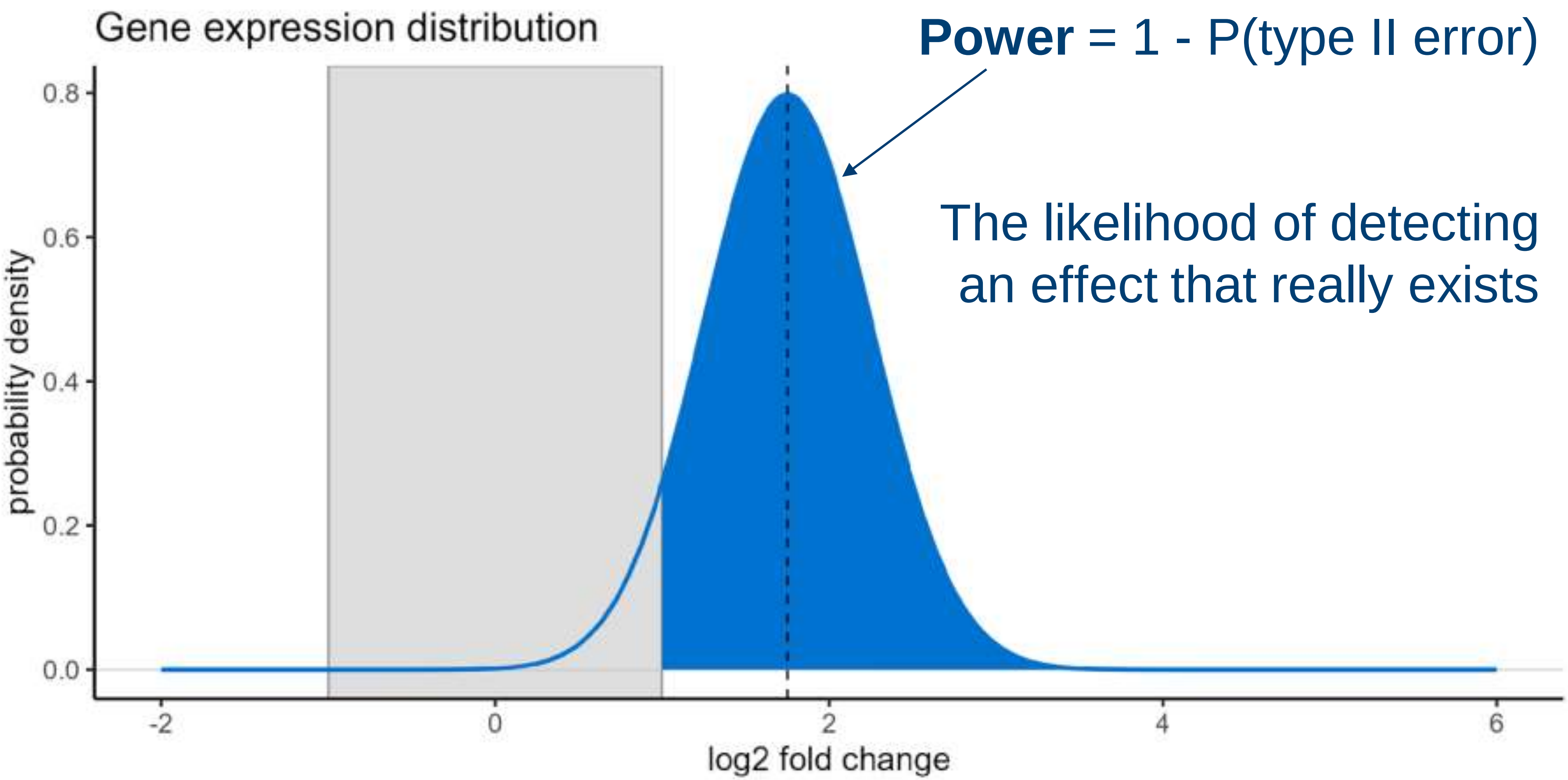
False negative (type II) error



False negative (type II) error



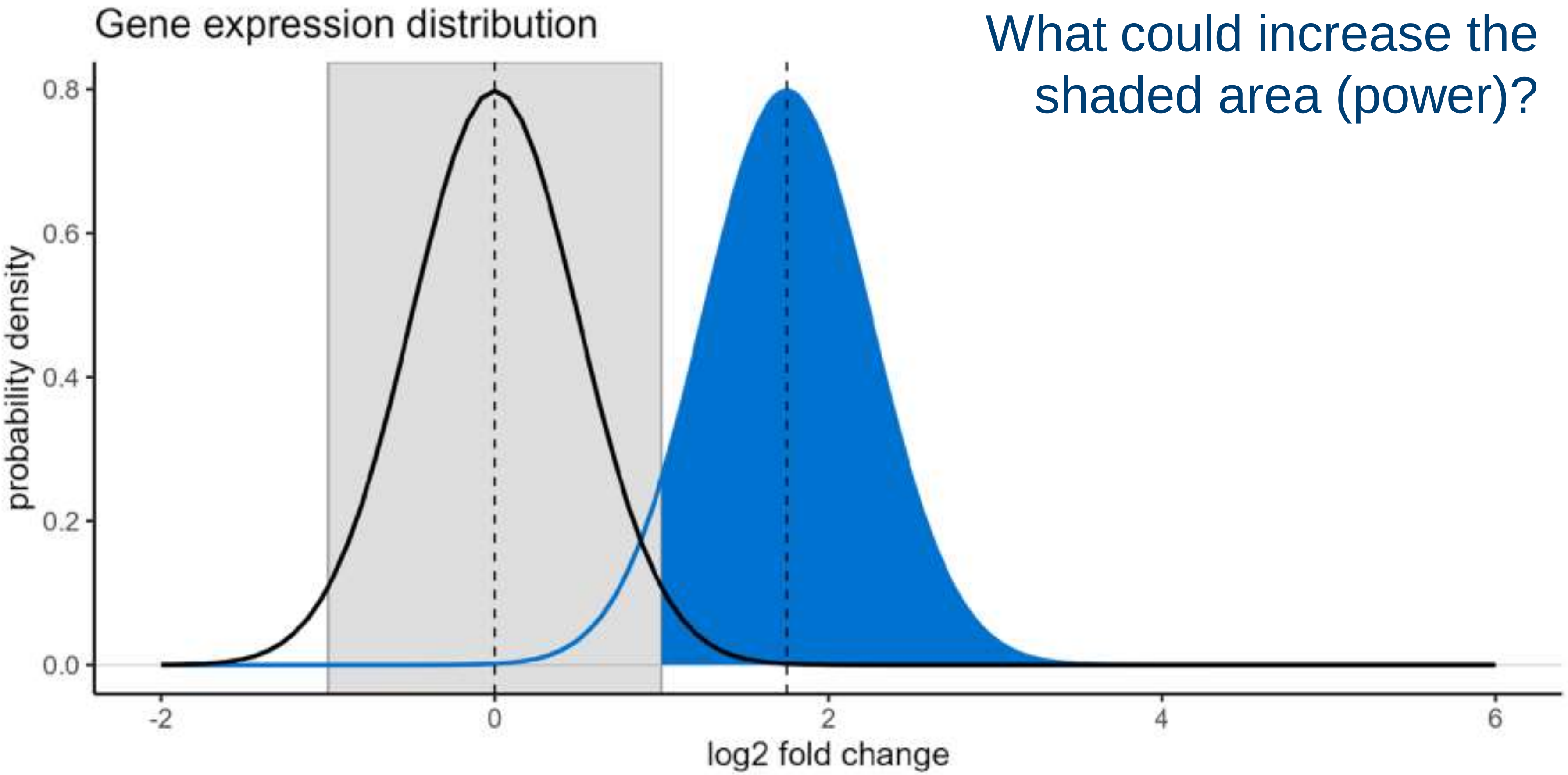
Power



Power = 1 - P(type II error)

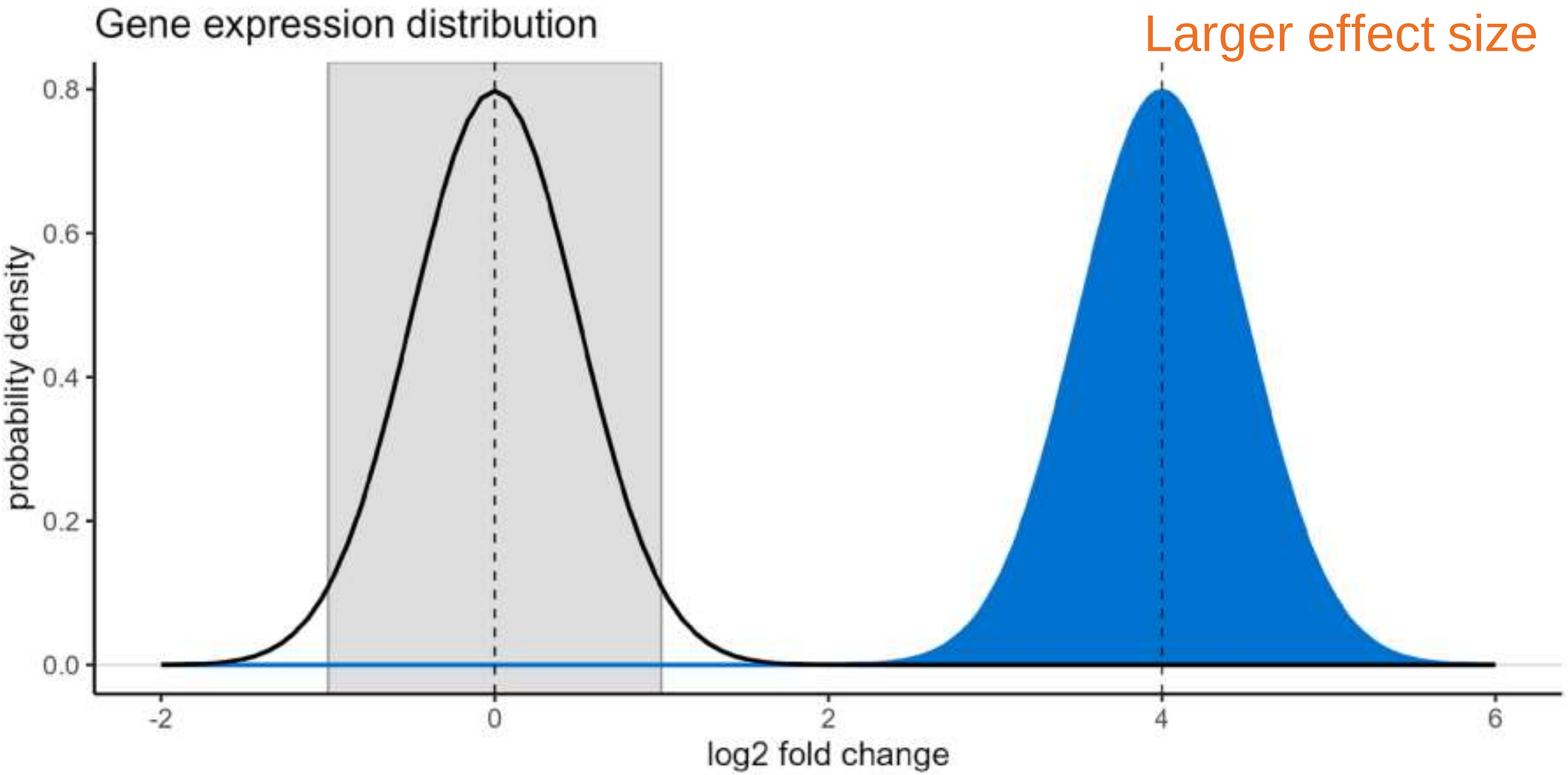
The likelihood of detecting an effect that really exists

Power



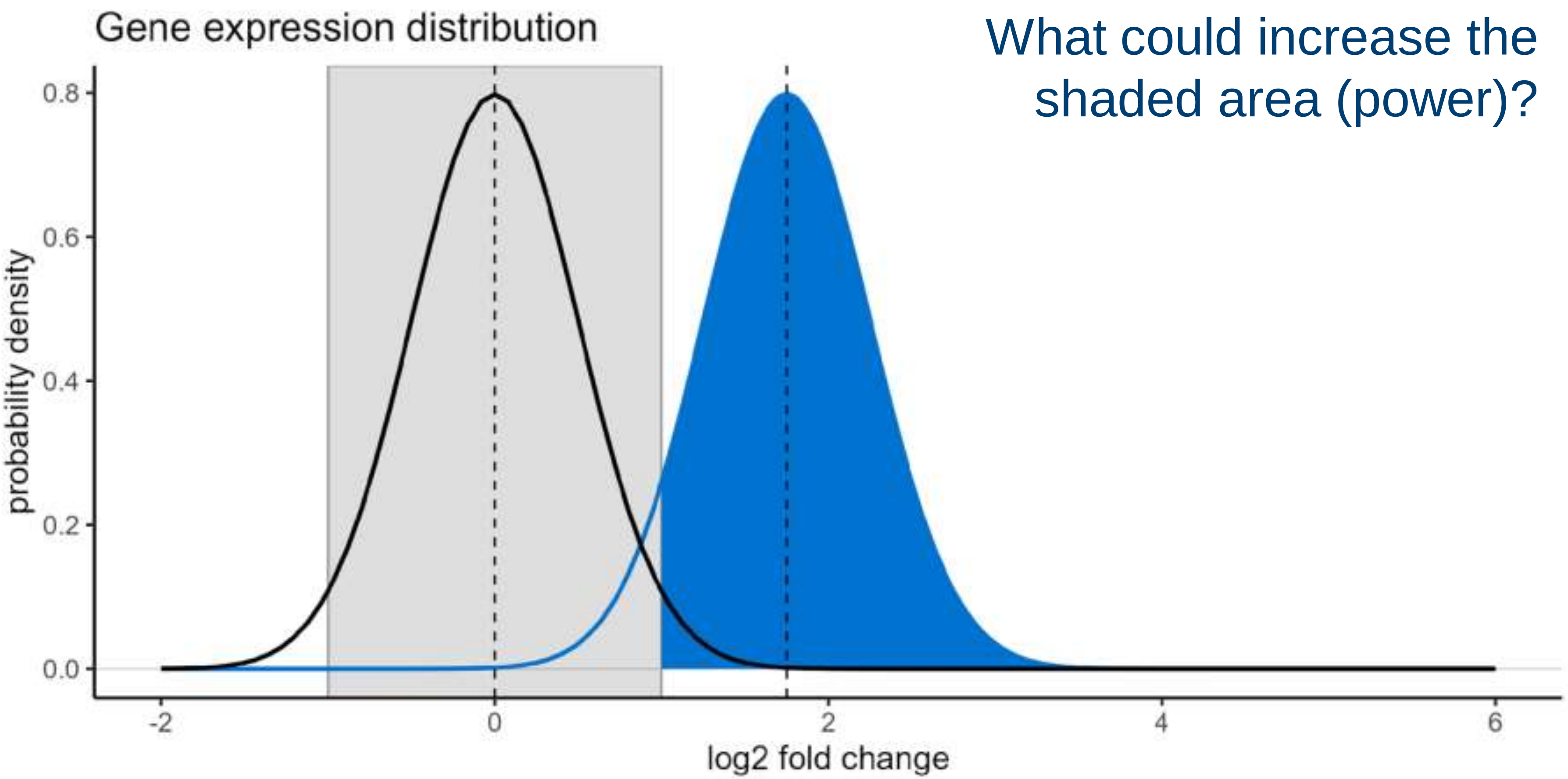
What could increase the shaded area (power)?

Power



Gene expression distribution
Larger effect size
probability density
0.0
0.2
0.4
0.6
0.8
-2
0
2
4
6
log2 fold change

Power



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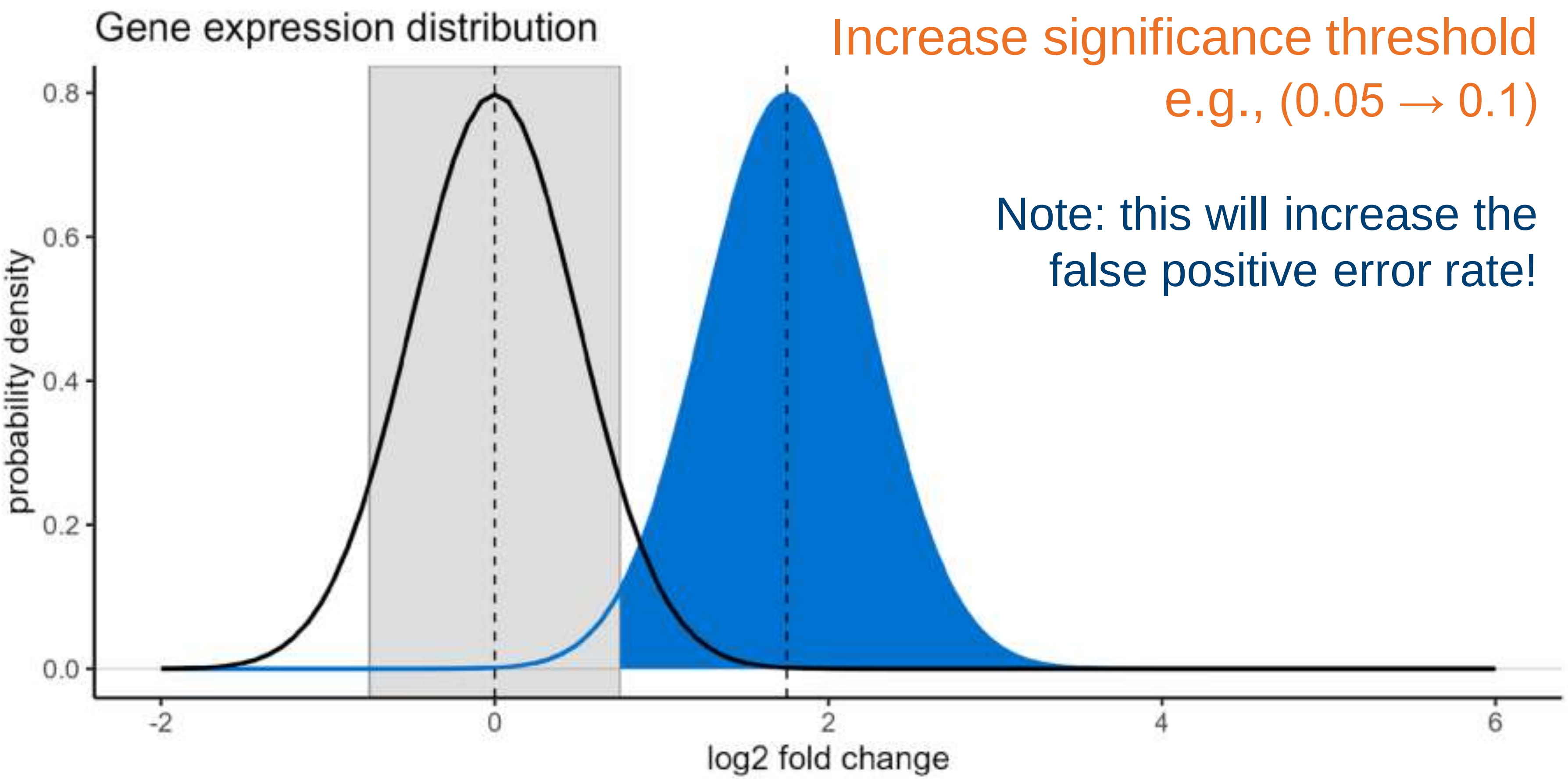
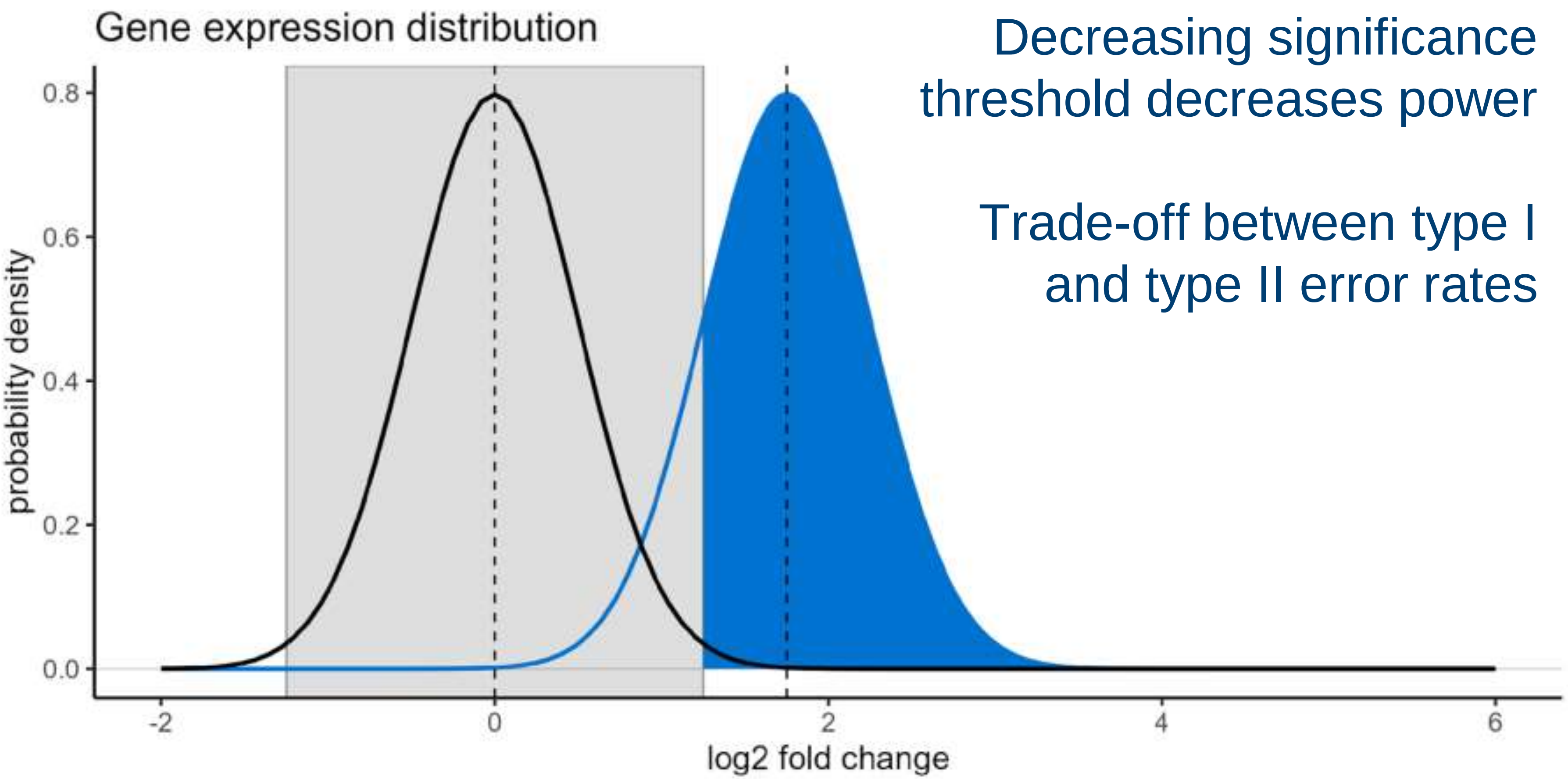
Power



Increase significance threshold
e.g., (0.05 → 0.1)

Note: this will increase the false positive error rate!

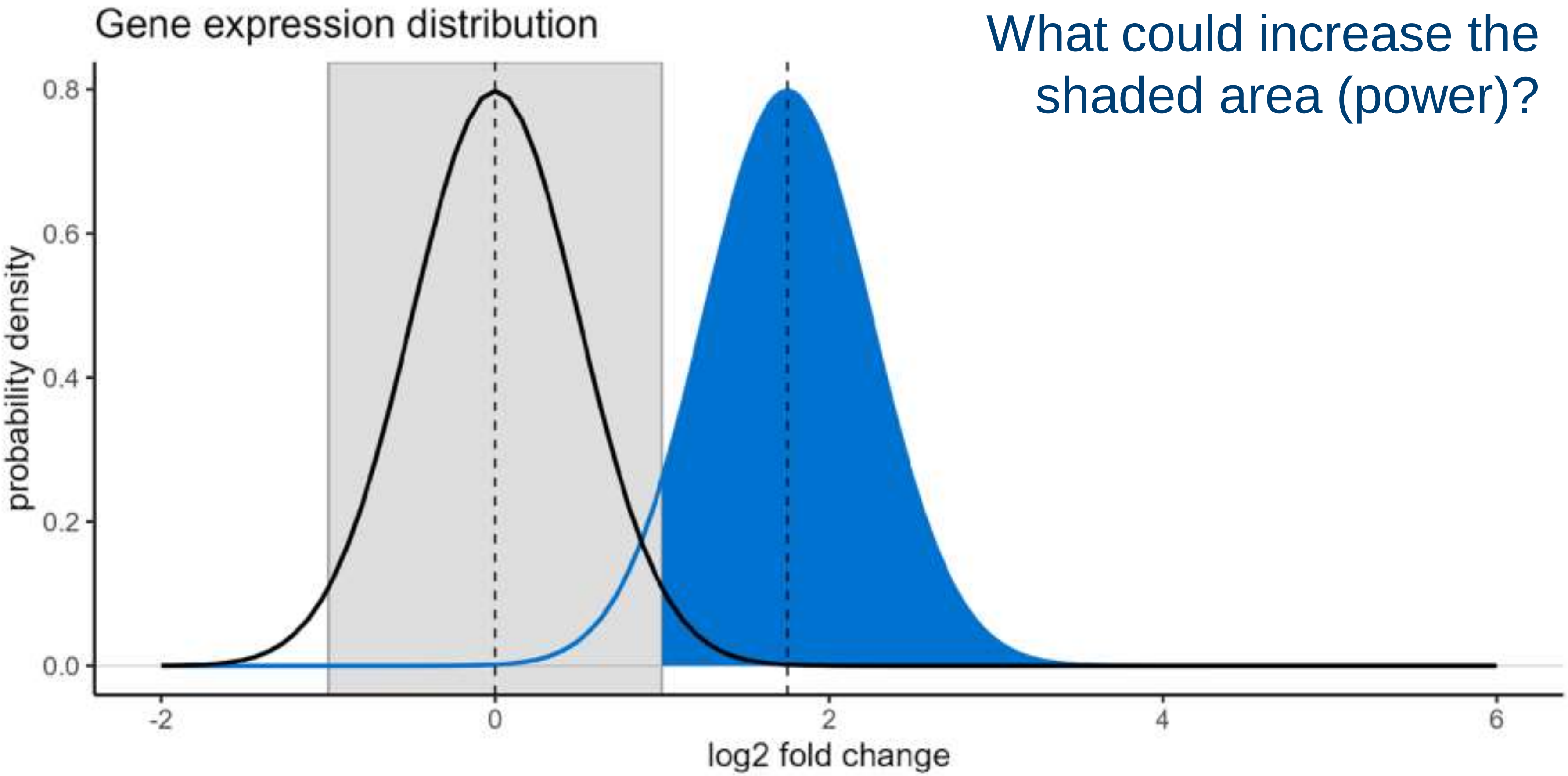
Power



Decreasing significance threshold decreases power

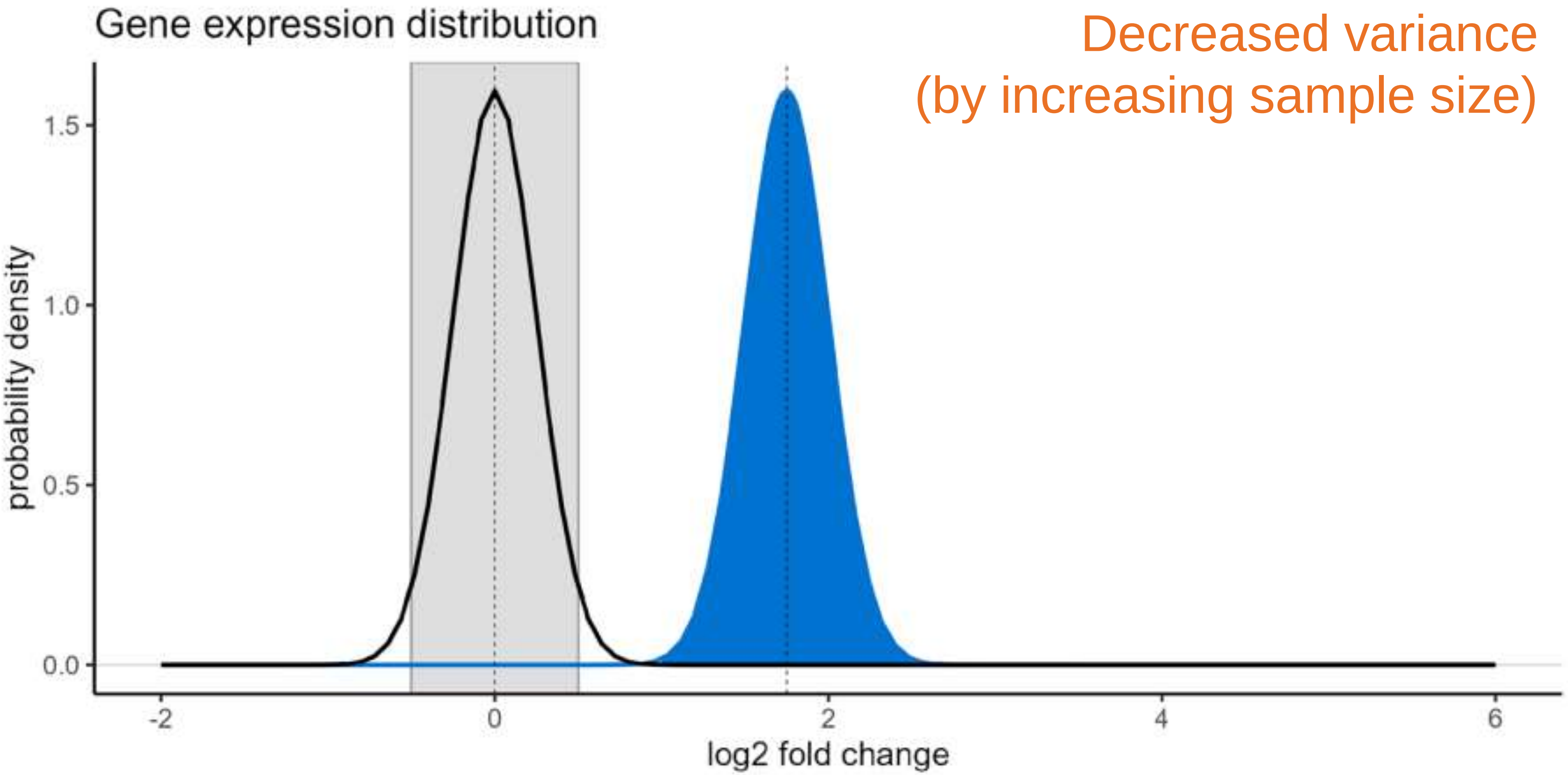
Trade-off between type I and type II error rates

Power



What could increase the shaded area (power)?

Power



Decreased variance
(by increasing sample size)

Power

Increased statistical power due to:

- Larger effect size
- Increased significance threshold (increased false positives)
- Decreased variance (increased sample size)

Power

Increased statistical power due to:

- Larger effect size
- Increased significance threshold (increased false positives)
- Decreased variance (increased sample size)

Power

Increased statistical power due to:

- Larger effect size
- Increased significance threshold (increased false positives)
- Decreased variance (increased sample size)

Power analysis

Effect size	Sample size
Significance threshold	Power

Four values are linked

If you know three, you can calculate the fourth

Power analysis

Effect size

Sample size

Significance threshold

Power

***A priori* power analysis**
Work out the required sample size, given:

- Significance threshold
- Desired power (80+%)
- Estimated effect size

Power analysis

Effect size

Sample size

Significance threshold

Power

***A priori* power analysis**
Work out the required sample size, given:

- Significance threshold
- Desired power (80+%)
- Estimated effect size

2. Effect sizes

Going beyond significance/p-values

Effect size

Effect size = the magnitude of the relationship between response & predictor variables

Effect size

Effect size = the magnitude of the relationship between response & predictor variables

Typically standardised (unitless)

Not based on hypothesis testing or p-values

Another form of statistical inference
(sample effect size → population effect size)

Effect size

Effect size = the magnitude of the relationship between response & predictor variables

Types of effect size:

- Difference between group means
- Amount of variance explained
- Relative risk/odds ratios (categorical variables)

Effect size

Effect size = the magnitude of the relationship between response & predictor variables

Types of effect size:

- Difference between group means
- Amount of variance explained
- Relative risk/odds ratios (categorical variables)

Cohen’s d

Cohen’s d is a measure of **difference in group means**

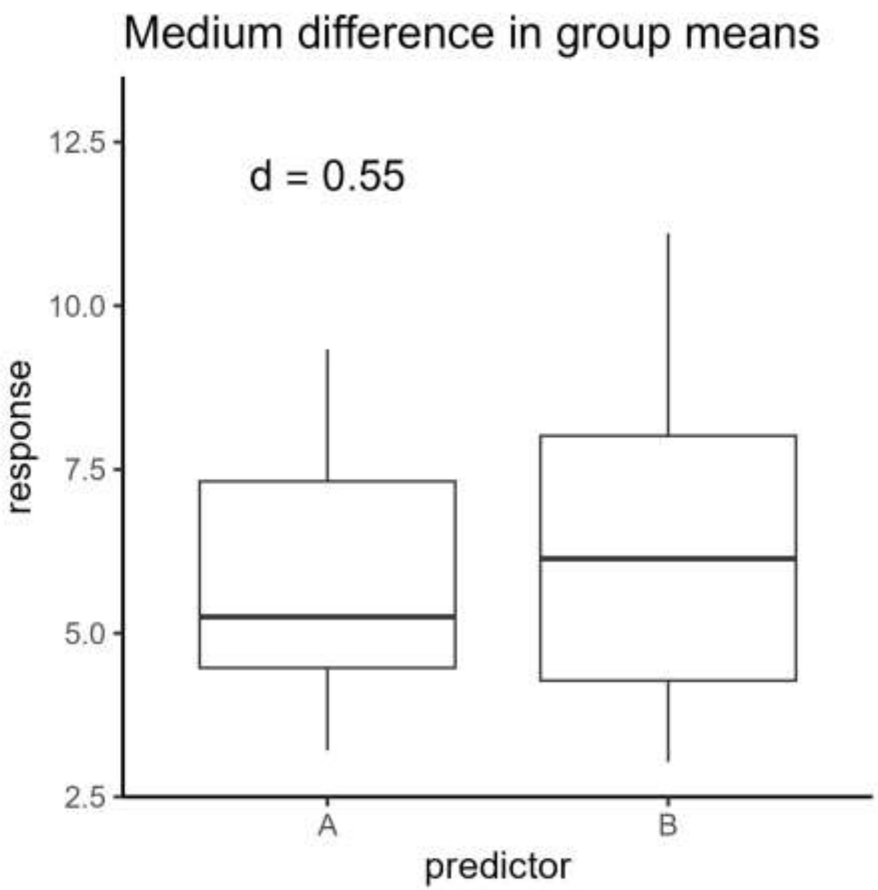
Calculated using:

- Difference in group means
- Standard deviation of data

On average:

- Small ≤ 0.2
- Medium = 0.5
- Large ≥ 0.8

Cohen's d

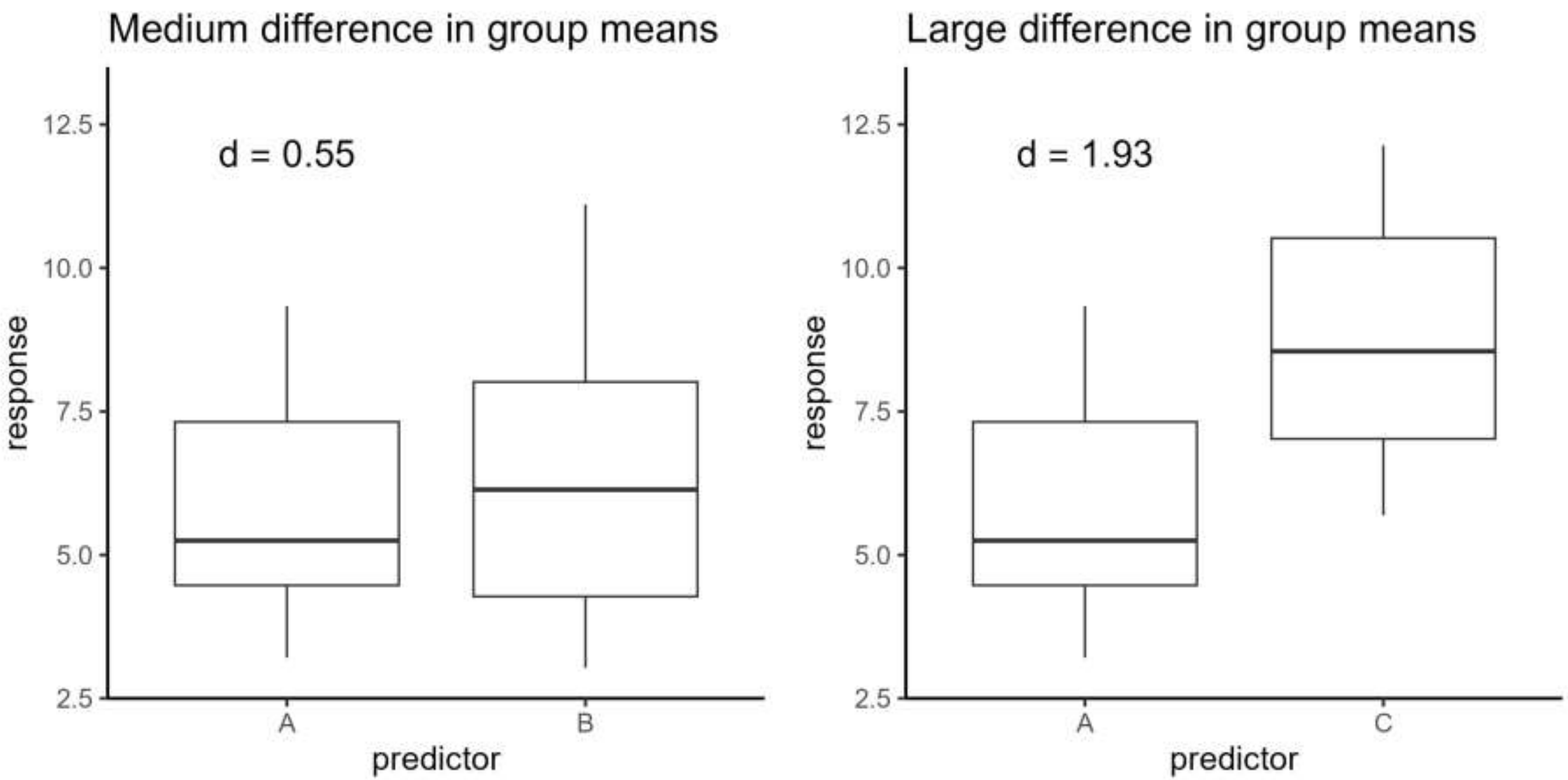


Calculated using:

- Difference in group means
- Standard deviation of data

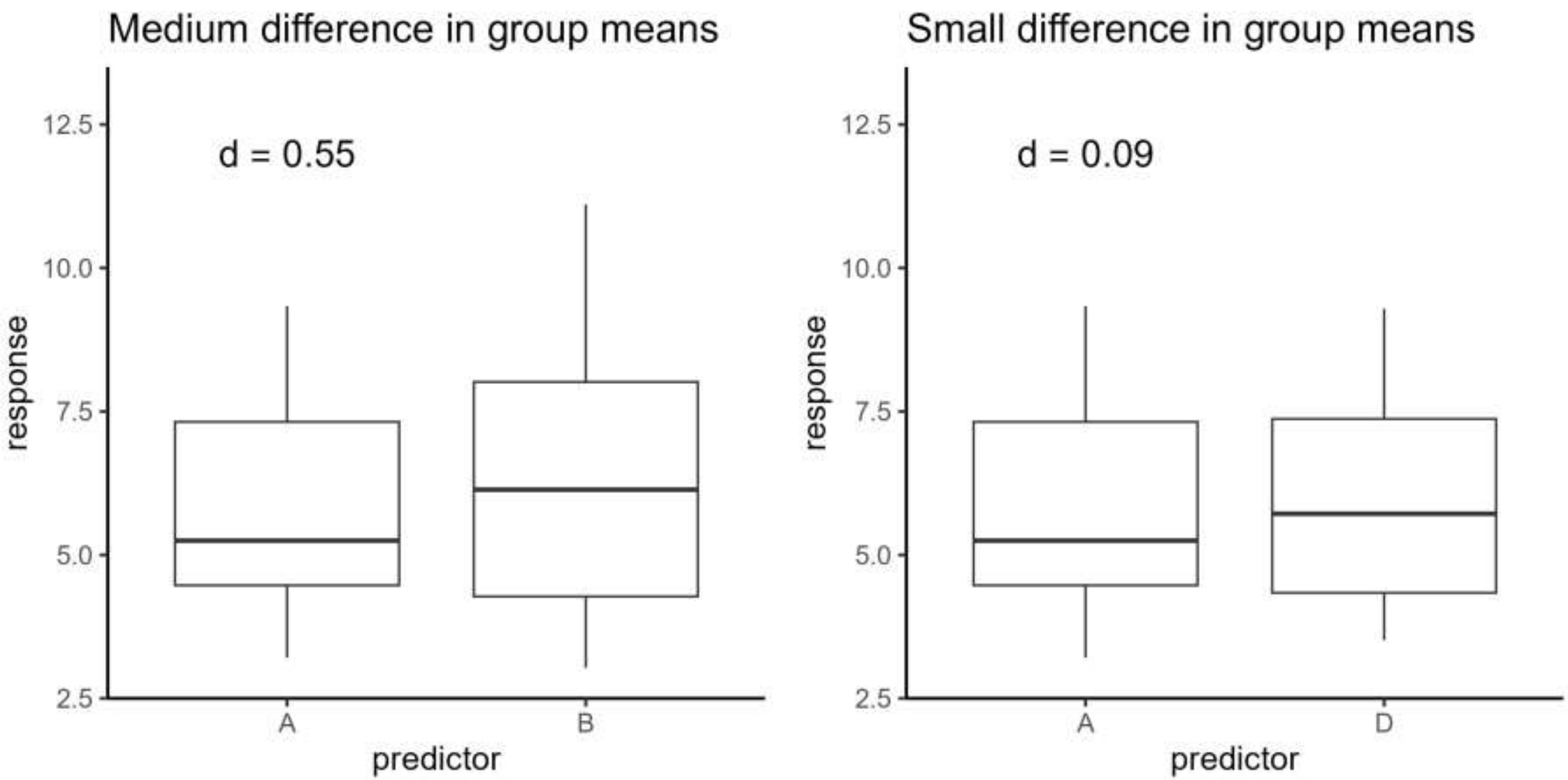
Small ≤ 0.2
Medium = 0.5
Large ≥ 0.8

Cohen's d



Medium difference in group means
d = 0.55
response
12.5
10.0
7.5
5.0
2.5
A
B
predictor
Large difference in group means
d = 1.93
response
12.5
10.0
7.5
5.0
2.5
A
C
predictor

Cohen's d



Medium difference in group means
d = 0.55
12.5
10.0
7.5
5.0
2.5
response
A
B
predictor
Small difference in group means
d = 0.09
12.5
10.0
7.5
5.0
2.5
response
A
D
predictor

R^2 and f^2

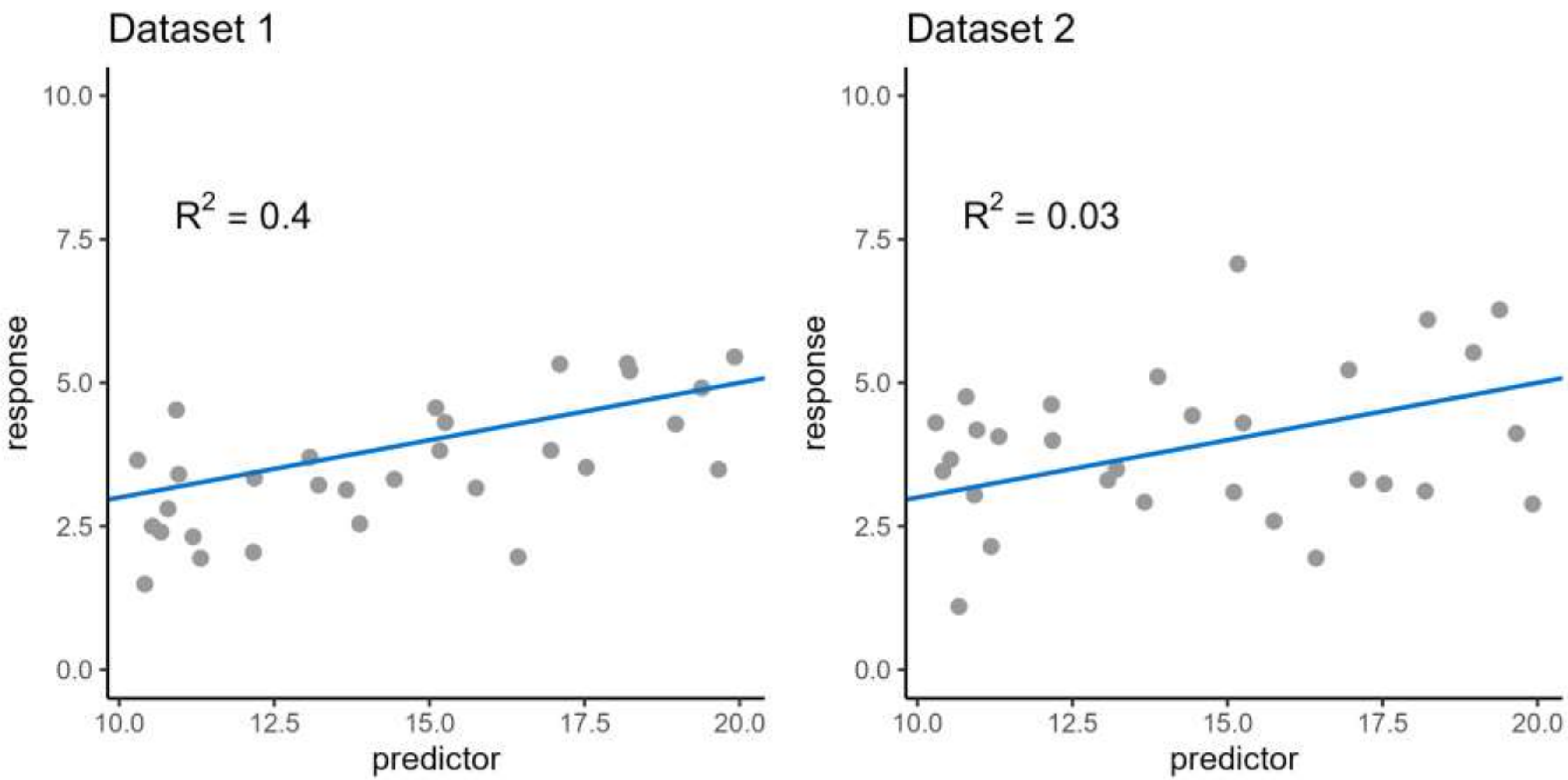
R^2 is a measure of **amount of variance explained**

Calculated using:

- Amount of variance in response variable
- Amount of variance in residuals (sum of squares)

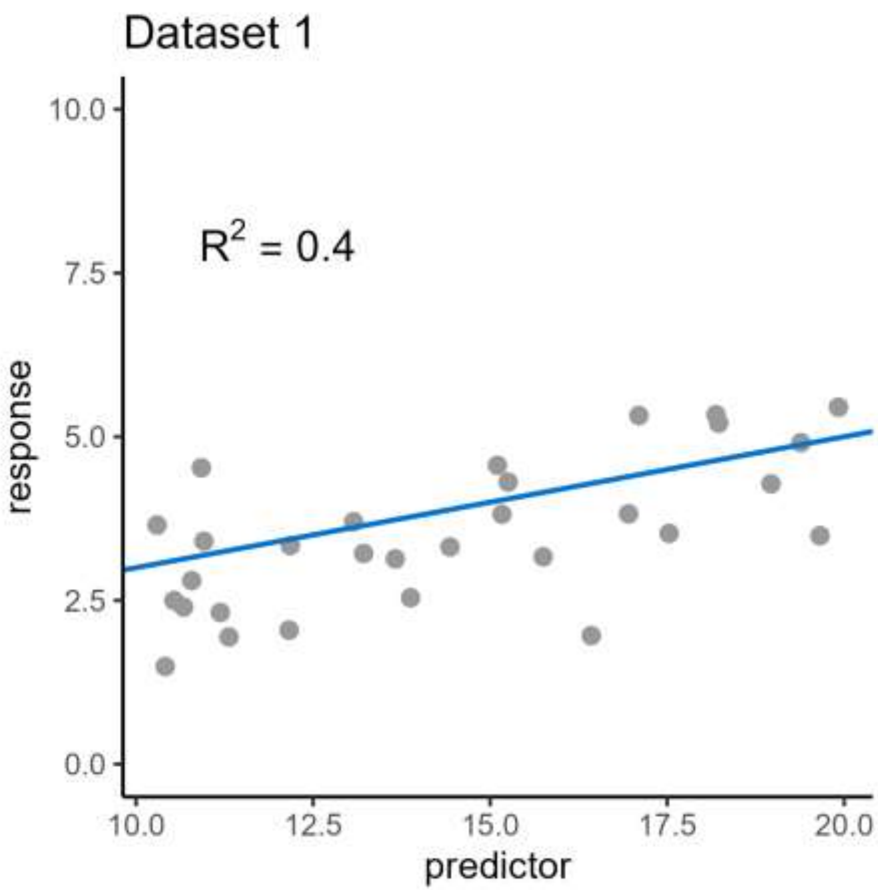
A related effect size is **Cohen's f^2**

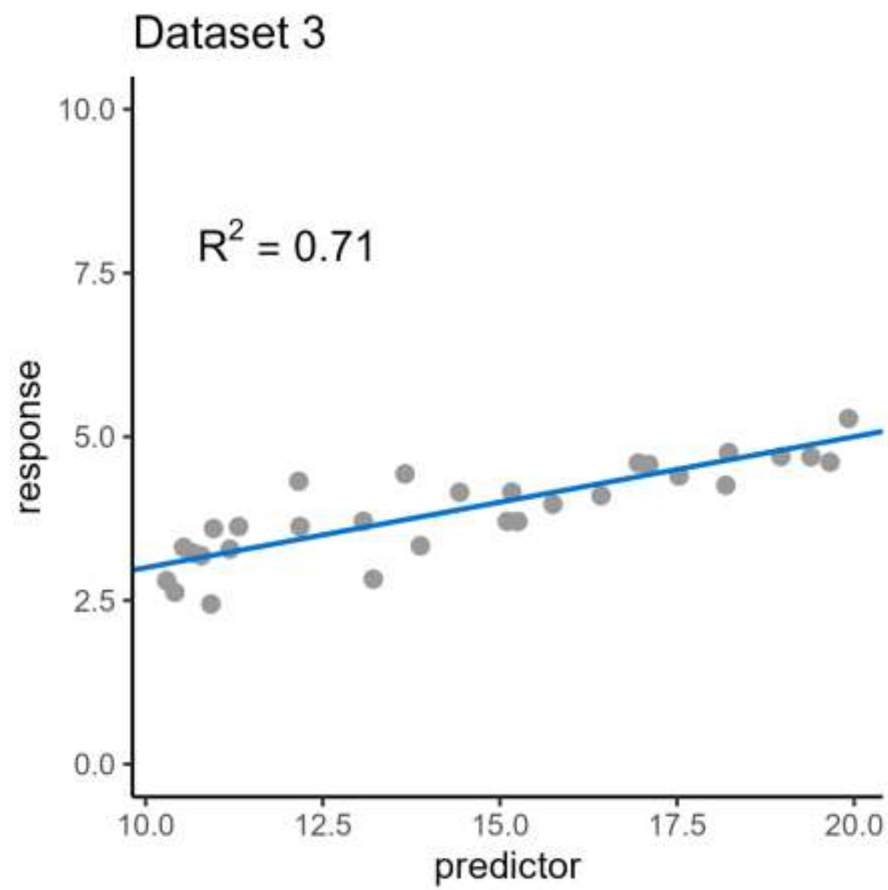
R^2



Dataset 1
R² = 0.4
response
predictor
10.0
7.5
5.0
2.5
0.0
10.0
12.5
15.0
17.5
20.0
Dataset 2
R² = 0.03
response
predictor
10.0
7.5
5.0
2.5
0.0
10.0
12.5
15.0
17.5
20.0

R^2


Dataset 1
R^2 = 0.4
response
predictor
10.0
7.5
5.0
2.5
0.0
10.0
12.5
15.0
17.5
20.0


Dataset 3
R^2 = 0.71
response
predictor
10.0
7.5
5.0
2.5
0.0
10.0
12.5
15.0
17.5
20.0

Uses of effect size

Why do we care about effect size?

1. **Statistical inference**
   Report alongside significance/p-value
   (sample effect size → population effect size)

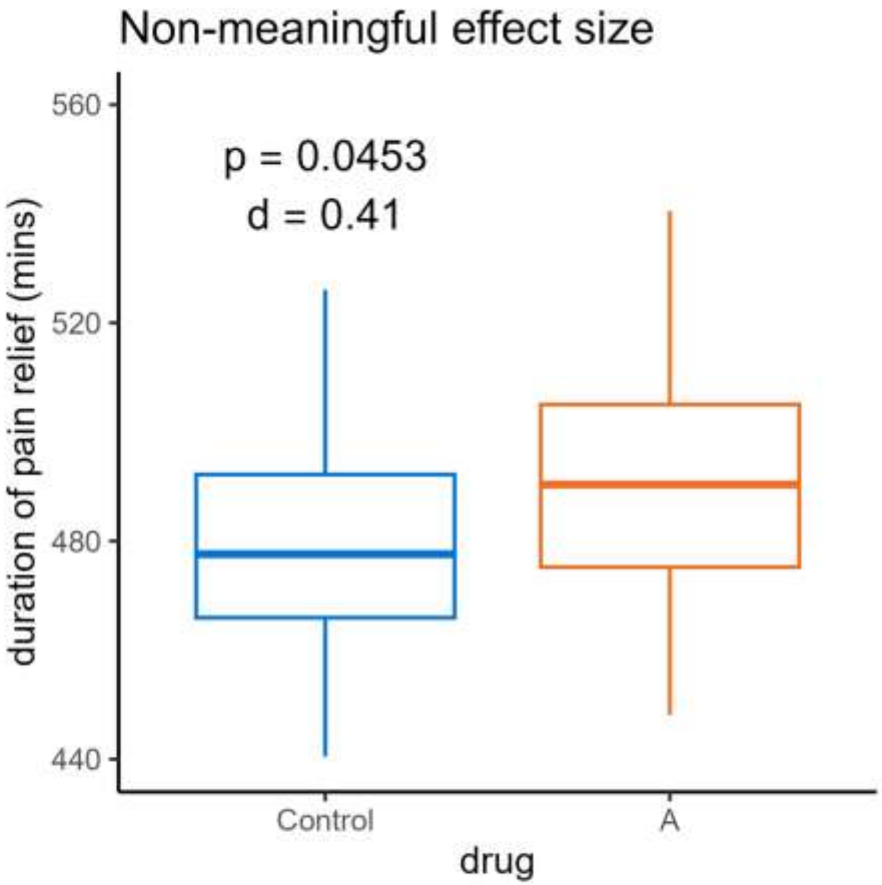
1. **Power analysis**
   Estimated effect size for *a priori* power analysis

Estimating an effect size

Where to get an estimated effect size?

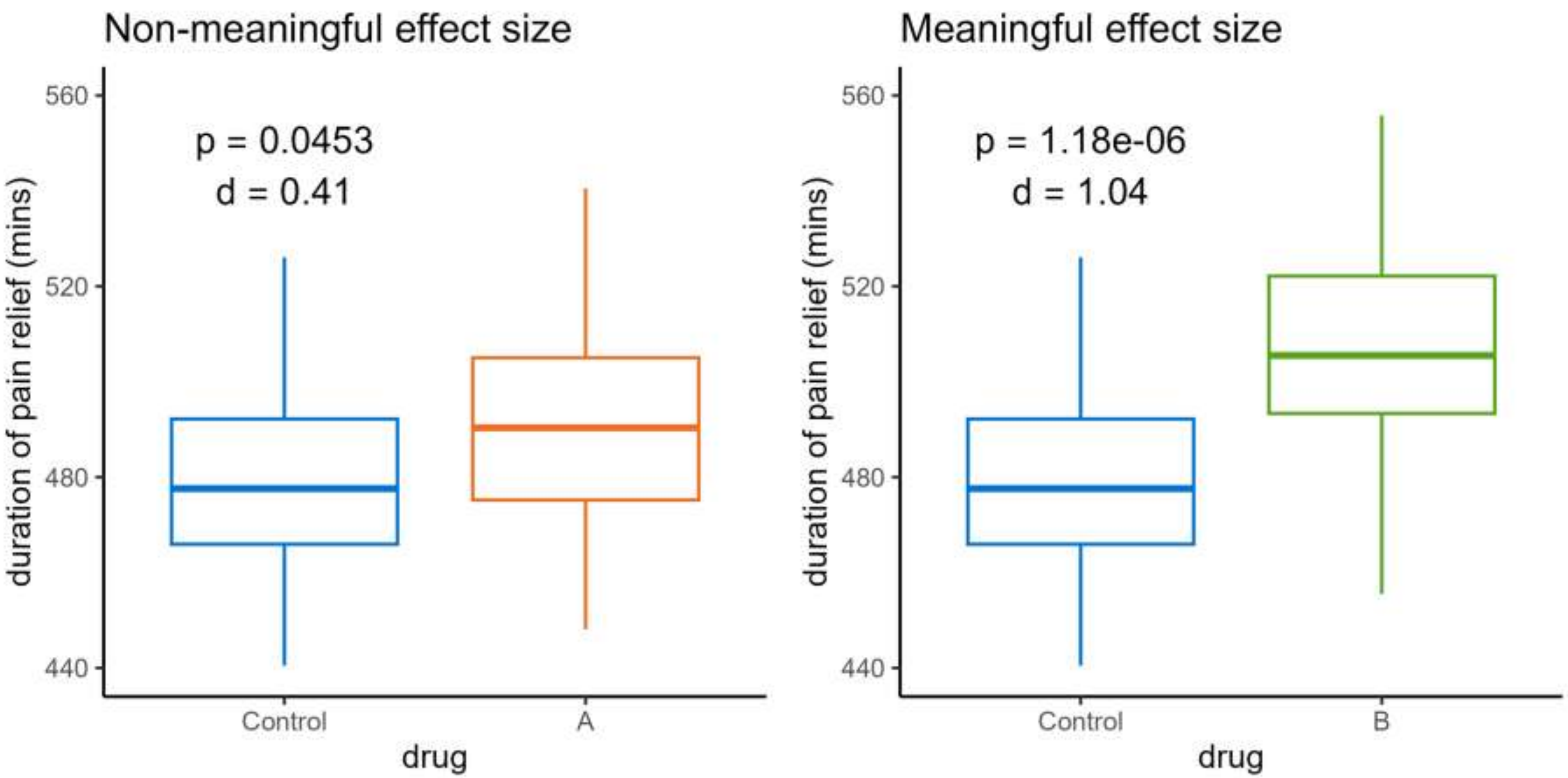
- Pilot study
- Papers/theory
- Desired “meaningful” effect size

Meaningful effect size



When is an effect size **meaningful**?

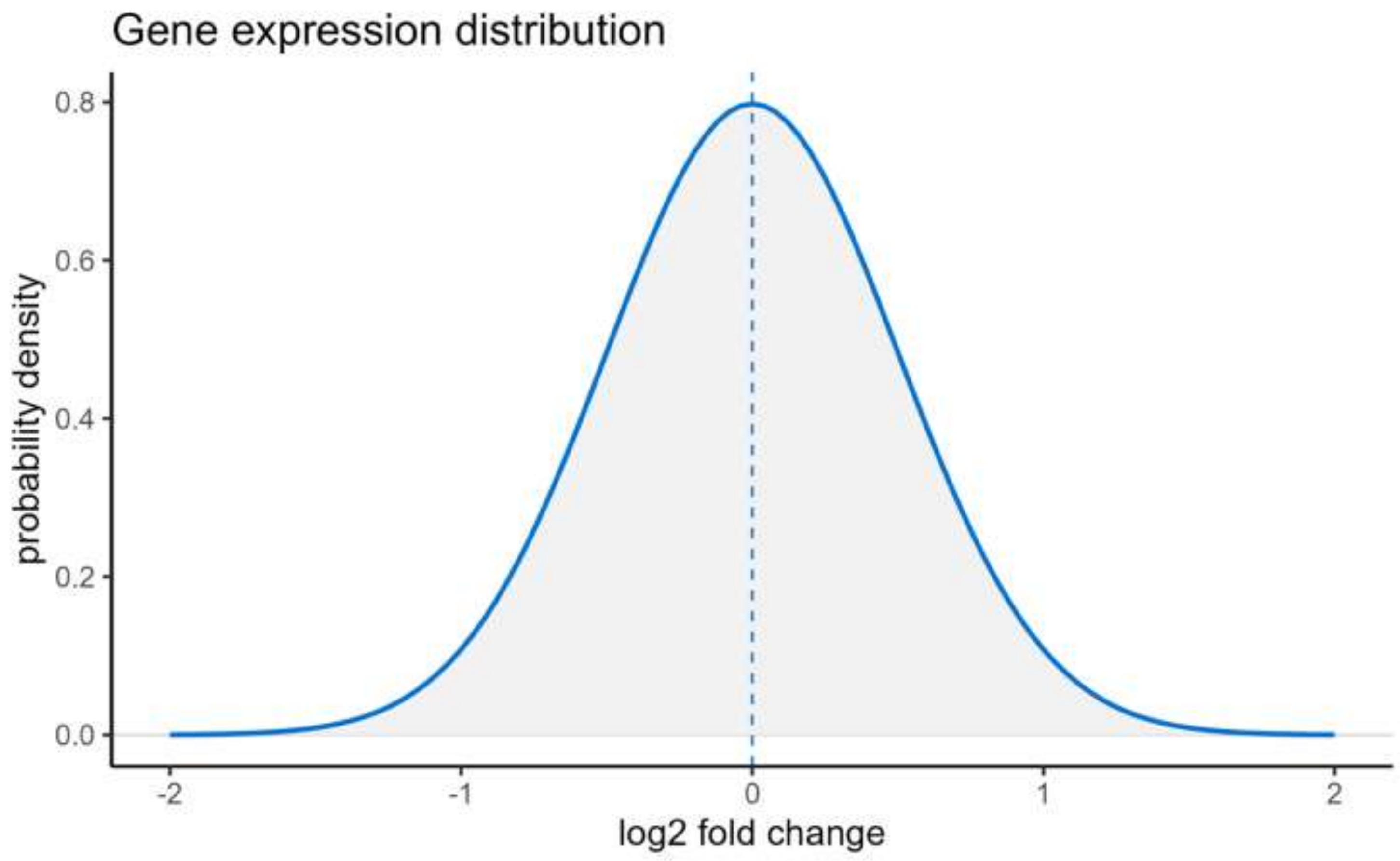
Meaningful effect size



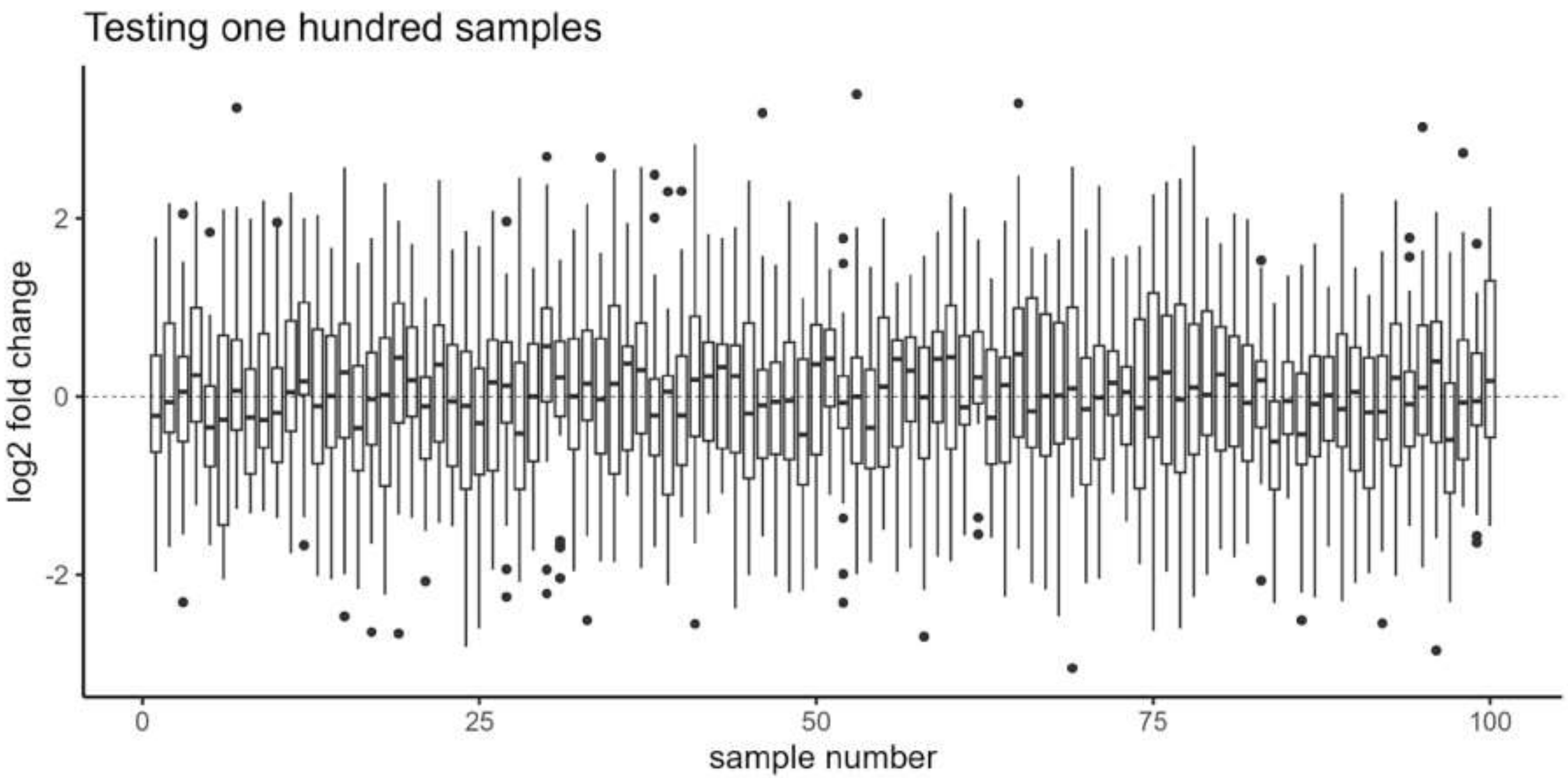
3. Multiple comparisons

Correcting p-values to avoid false positives

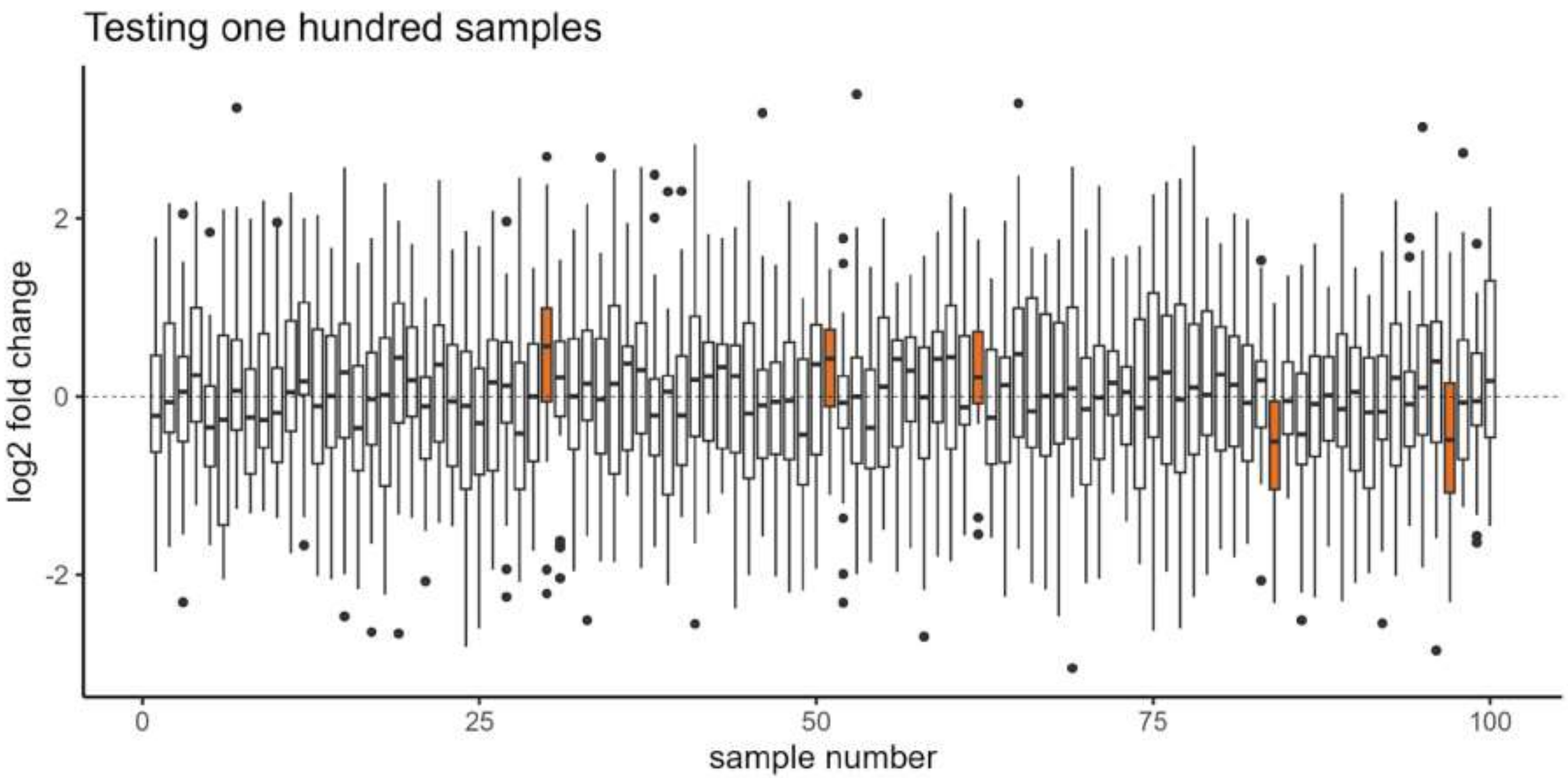
Why are we worried about multiple comparisons?



Why are we worried about multiple comparisons?



Why are we worried about multiple comparisons?



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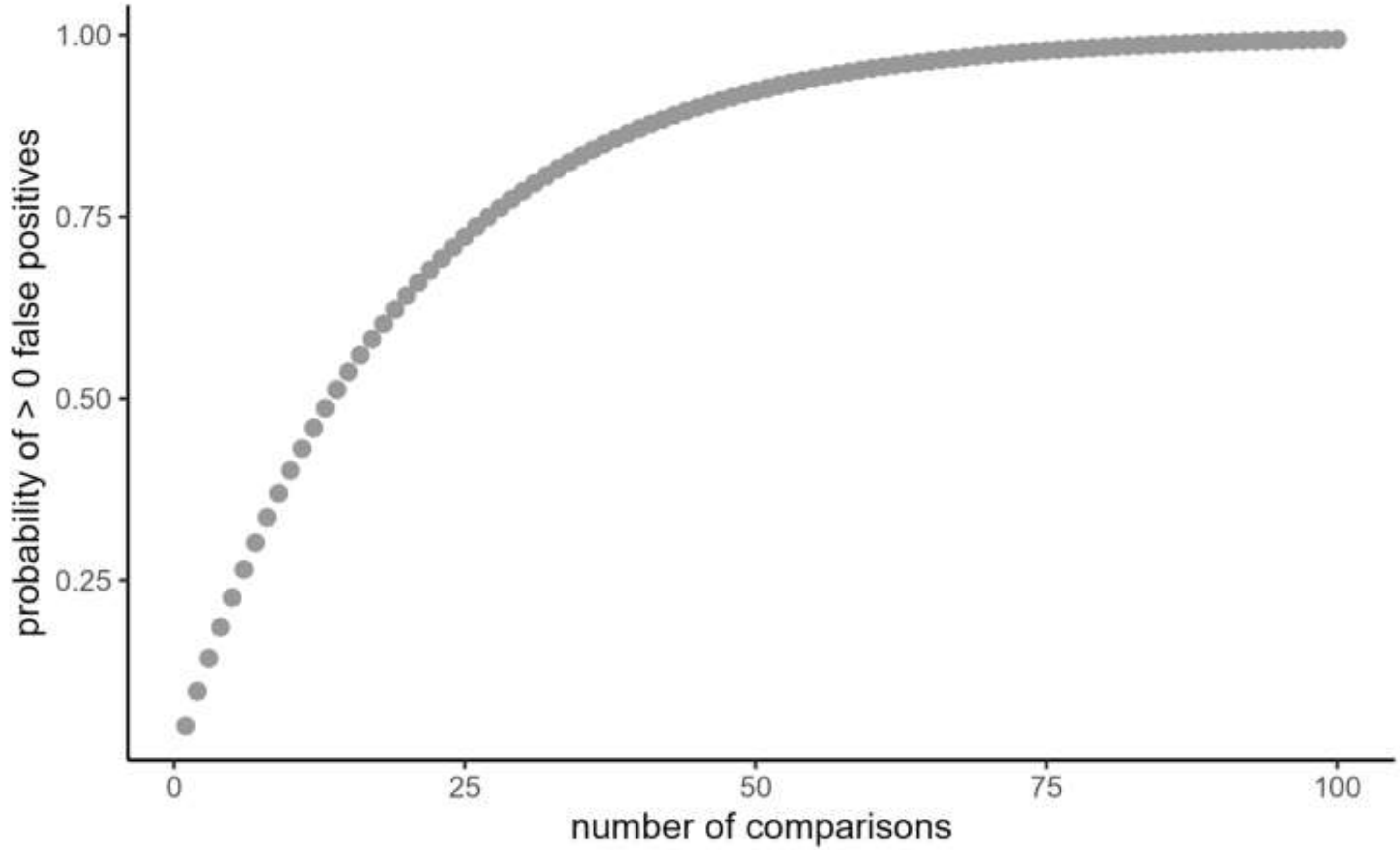
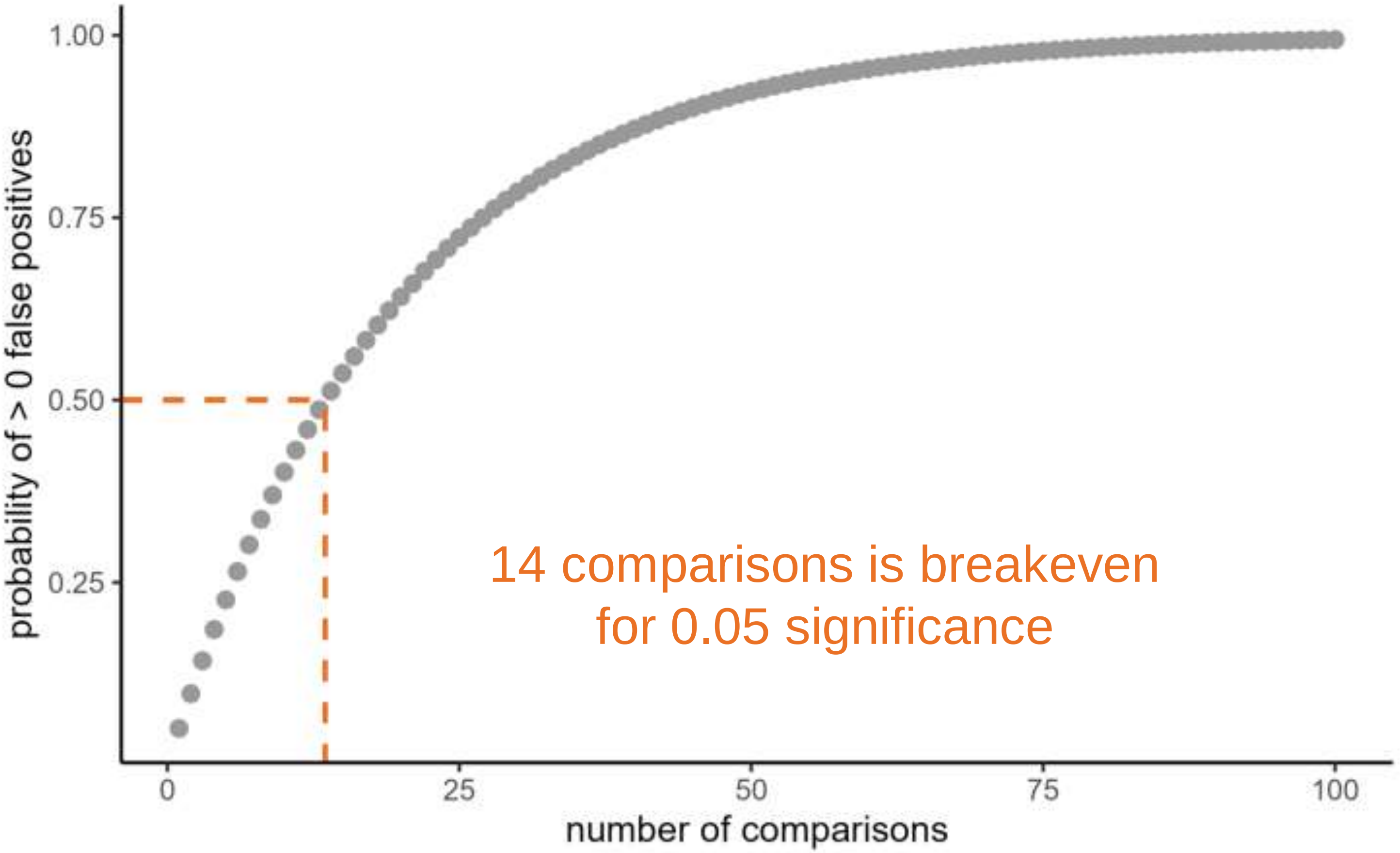
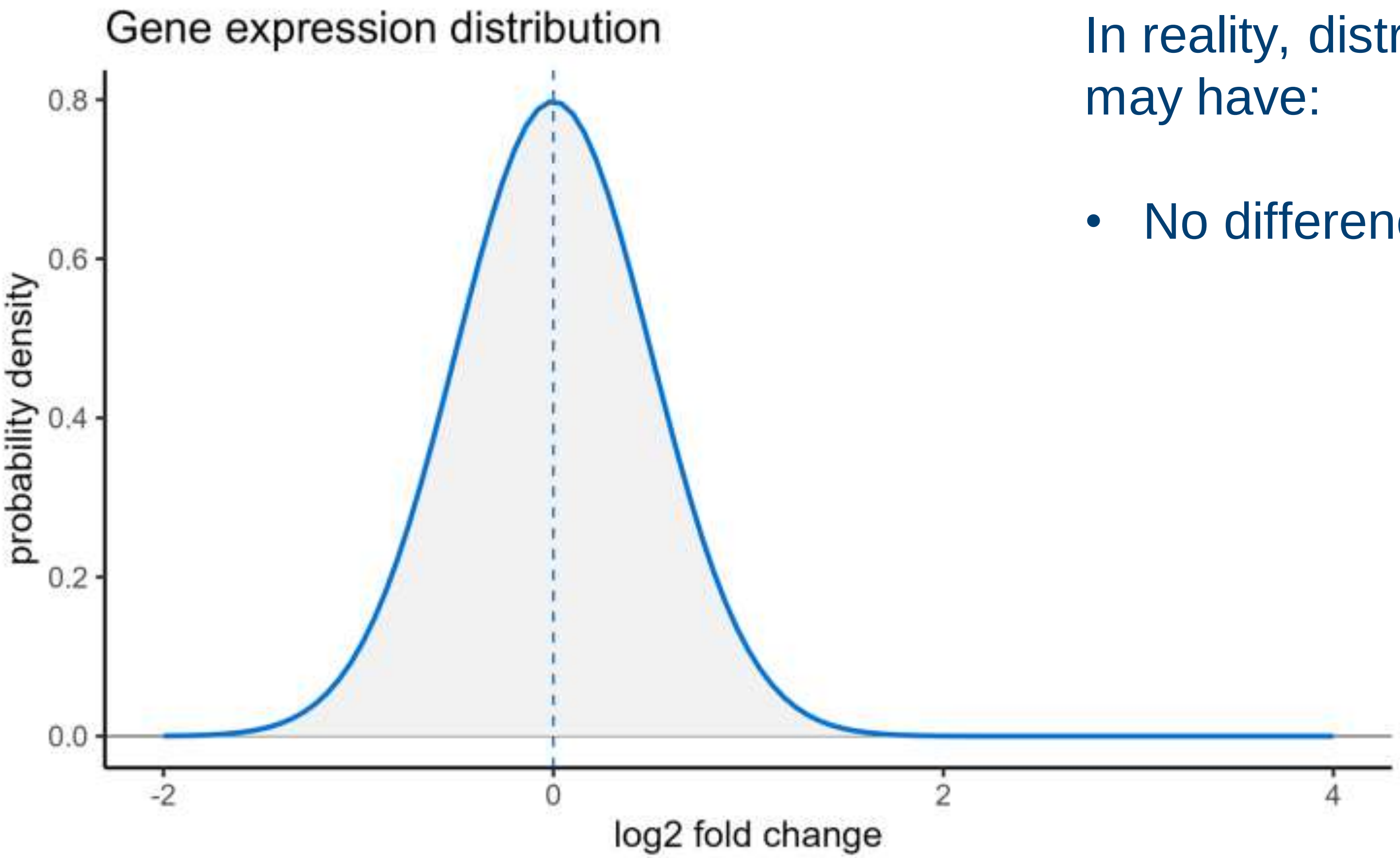
Why are we worried about multiple comparisons?



Why are we worried about multiple comparisons?



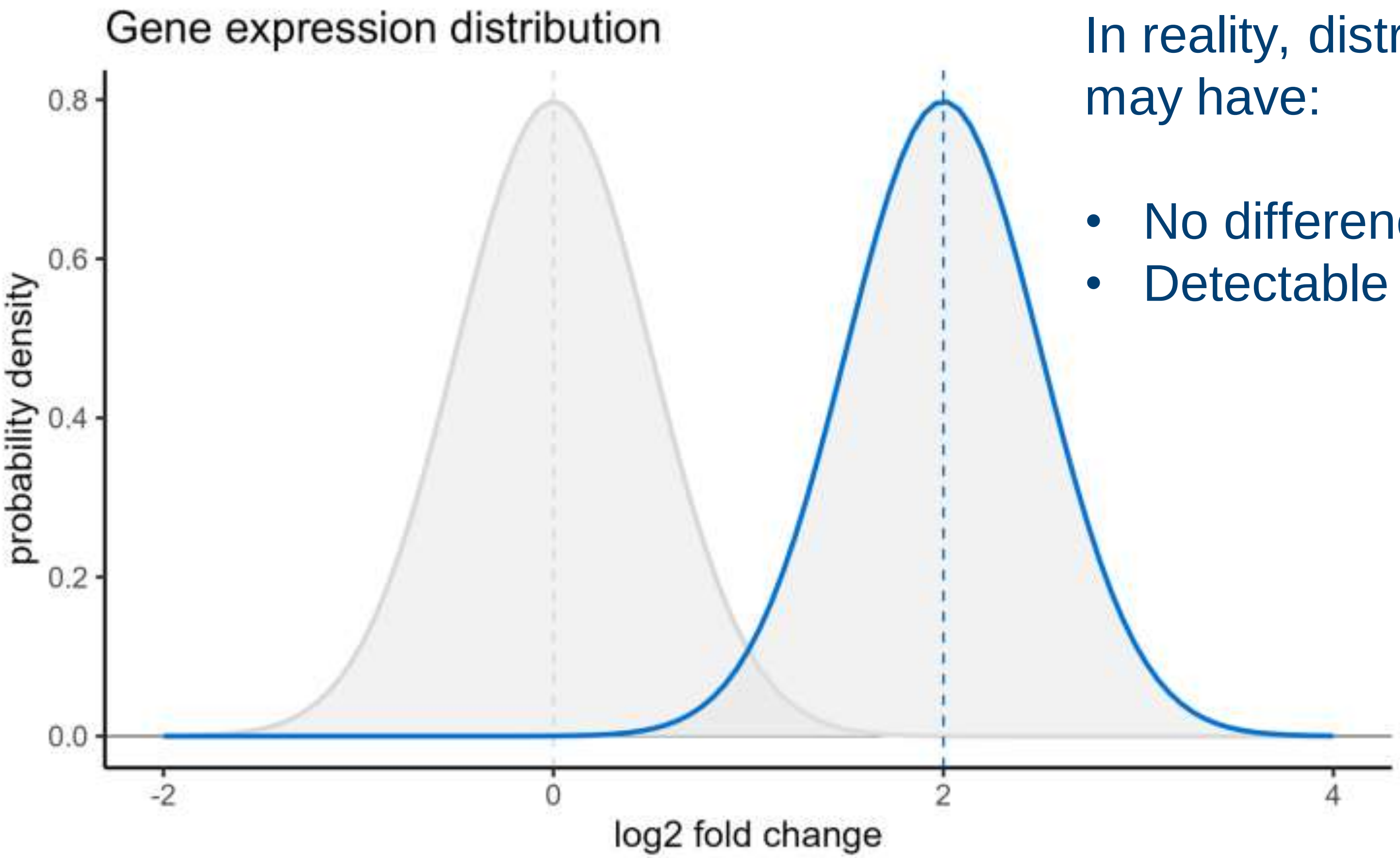
Multiple comparisons in practice



In reality, distributions may have:

- No difference

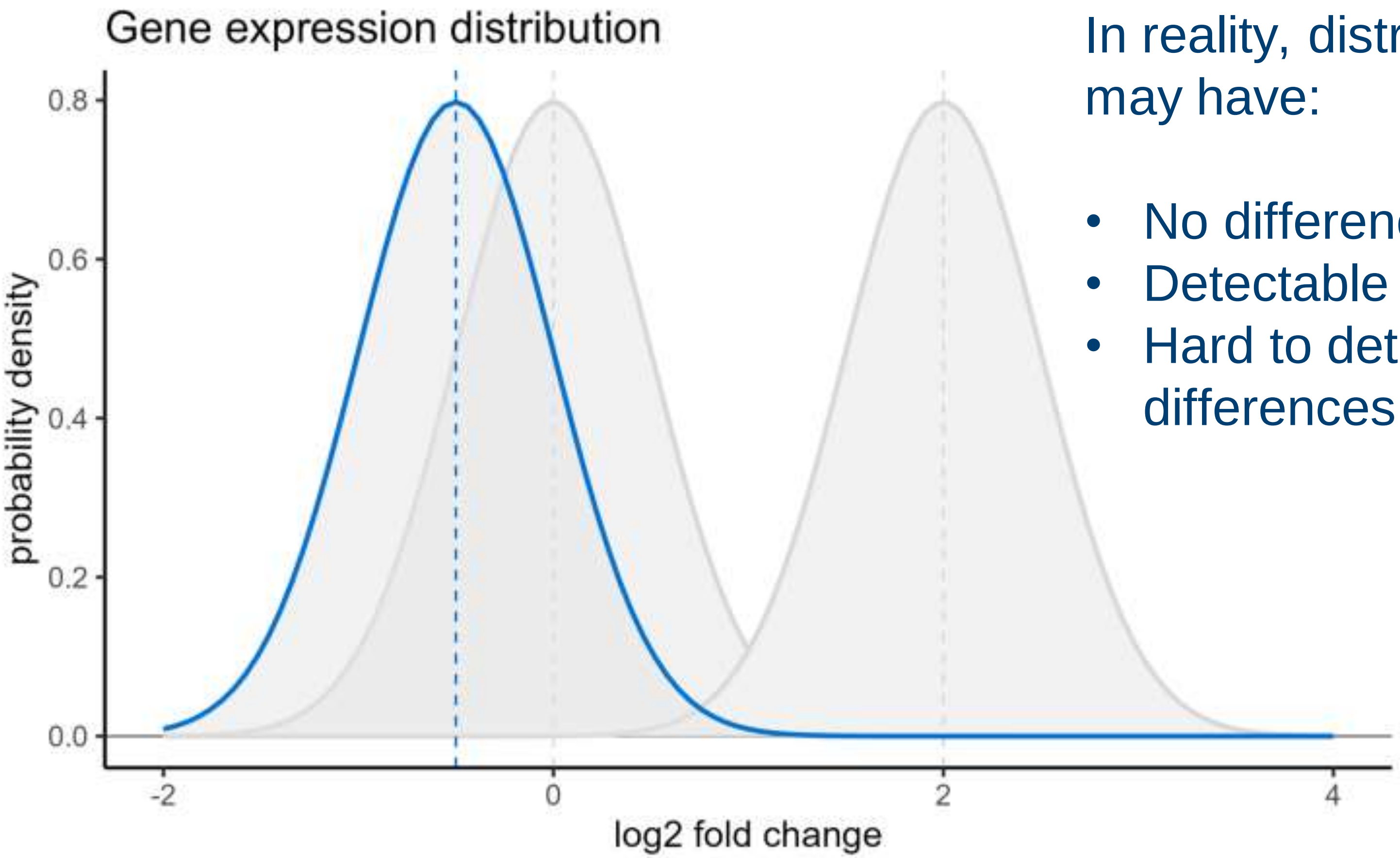
Multiple comparisons in practice



In reality, distributions may have:

- No difference
- Detectable difference

Multiple comparisons in practice



In reality, distributions may have:

- No difference
- Detectable difference
- Hard to detect differences (low power)

Multiple comparisons in practice

		Reality	
		boring (H_0 true)	interesting (H_0 false)
Test	insignificant (don't reject H_0)		
	significant (reject H_0)		

Multiple comparisons in practice

		Reality		
		boring (H_0 true)	interesting (H_0 false)	
Test	insignificant (don't reject H_0)			
	significant (reject H_0)			
				1000

1000 comparisons

Multiple comparisons in practice

		Reality: boring (H_0 true)	Reality: interesting (H_0 false)	
Test	insignificant (don't reject H_0)			
Test	significant (reject H_0)			
				1000

1000 comparisons, looking for a rare event (2%)

Multiple comparisons in practice

Reality

		boring (H_0 true)	interesting (H_0 false)	
Test	insignificant (don't reject H_0)			
	significant (reject H_0)			
		980	20	1000

1000 comparisons, looking for a rare event (2%)

Multiple comparisons in practice

		Reality		
		boring (H_0 true)	interesting (H_0 false)	
Test	insignificant (don't reject H_0)			
	significant (reject H_0)			
		980	20	1000

1000 comparisons, looking for a rare event (2%)
Standard significance threshold (0.05)

Multiple comparisons in practice

		Reality		
		boring (H_0 true)	interesting (H_0 false)	
Test	insignificant (don't reject H_0)	931		
	significant (reject H_0)	49		
		980	20	1000

1000 comparisons, looking for a rare event (2%)
Standard significance threshold (0.05)

Multiple comparisons in practice

		Reality		
		boring (H_0 true)	interesting (H_0 false)	
Test	insignificant (don't reject H_0)	931		
	significant (reject H_0)	49		
		980	20	1000

1000 comparisons, looking for a rare event (2%)
Standard significance threshold (0.05) with 80% power

Multiple comparisons in practice

		Reality		
		boring (H_0 true)	interesting (H_0 false)	
Test	insignificant (don't reject H_0)	931	4	
	significant (reject H_0)	49	16	
		980	20	1000

1000 comparisons, looking for a rare event (2%)
Standard significance threshold (0.05) with 80% power

Multiple comparisons in practice

		Reality		
		boring (H_0 true)	interesting (H_0 false)	
Test	insignificant (don't reject H_0)	931	4	
	significant (reject H_0)	49	16	
		980	20	1000

The problem: we don't know **which** of our results are correct!

Multiple comparisons in practice

		Reality: boring (H_0 true)	Reality: interesting (H_0 false)	
Test	insignificant (don't reject H_0)	931	4	935
Test	significant (reject H_0)	49	16	65
		980	20	1000

The problem: we don't know **which** of our results are correct!

Multiple comparisons in practice

		Reality		
		boring (H_0 true)	interesting (H_0 false)	
Test	insignificant (don't reject H_0)	931	4	935
	significant (reject H_0)	49	16	65
		980	20	1000

So what **can** we measure?

Multiple comparisons in practice

		Reality		
		boring (H_0 true)	interesting (H_0 false)	
Test	insignificant (don't reject H_0)	931	4	935
	significant (reject H_0)	49	16	65
		980	20	1000

Family-wise error rate (FWER):
Probability of finding any false positives $1 - (1 - 0.05)^{980} \approx 1$

Multiple comparisons in practice

		Reality: boring (H_0 true)	Reality: interesting (H_0 false)	
Test	insignificant (don't reject H_0)	931	4	935
	significant (reject H_0)	49	16	65
		980	20	1000

False discovery rate (FDR):
Expected proportion of positives that are false $\quad {}^{49}/_{65} \approx 0.75$

Correcting p-values

We can correct p-values by adjusting these rates

Correcting p-values

We can correct p-values by adjusting these rates

Family-wise error rate

Change significance level

- Reduces probability of getting **any** false positives
- Also has a big impact on power/false negative rate

Correcting p-values

We can correct p-values by adjusting these rates

Family-wise error rate

Change significance level

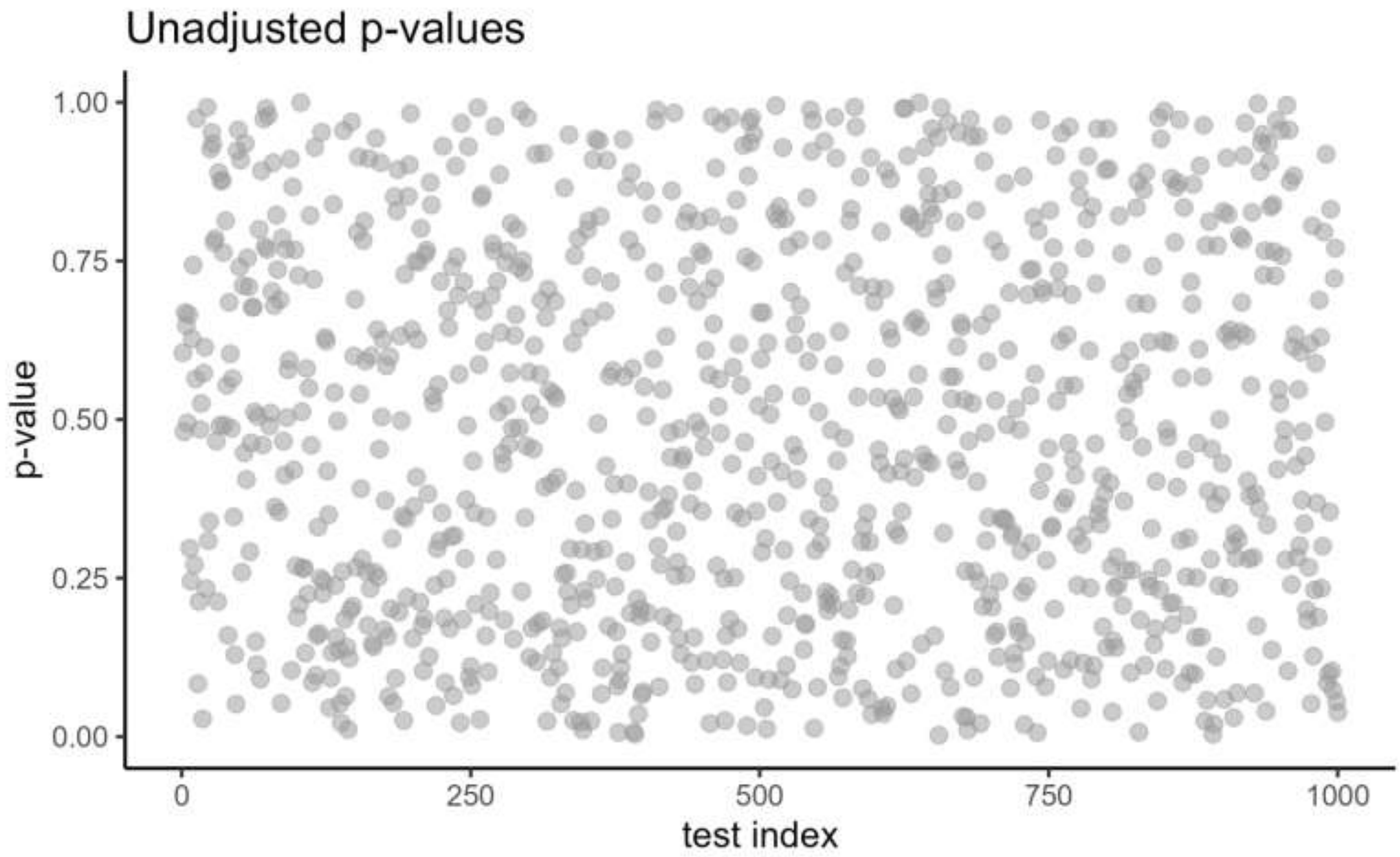
- Reduces probability of getting **any** false positives
- Also has a big impact on power/false negative rate

False discovery rate

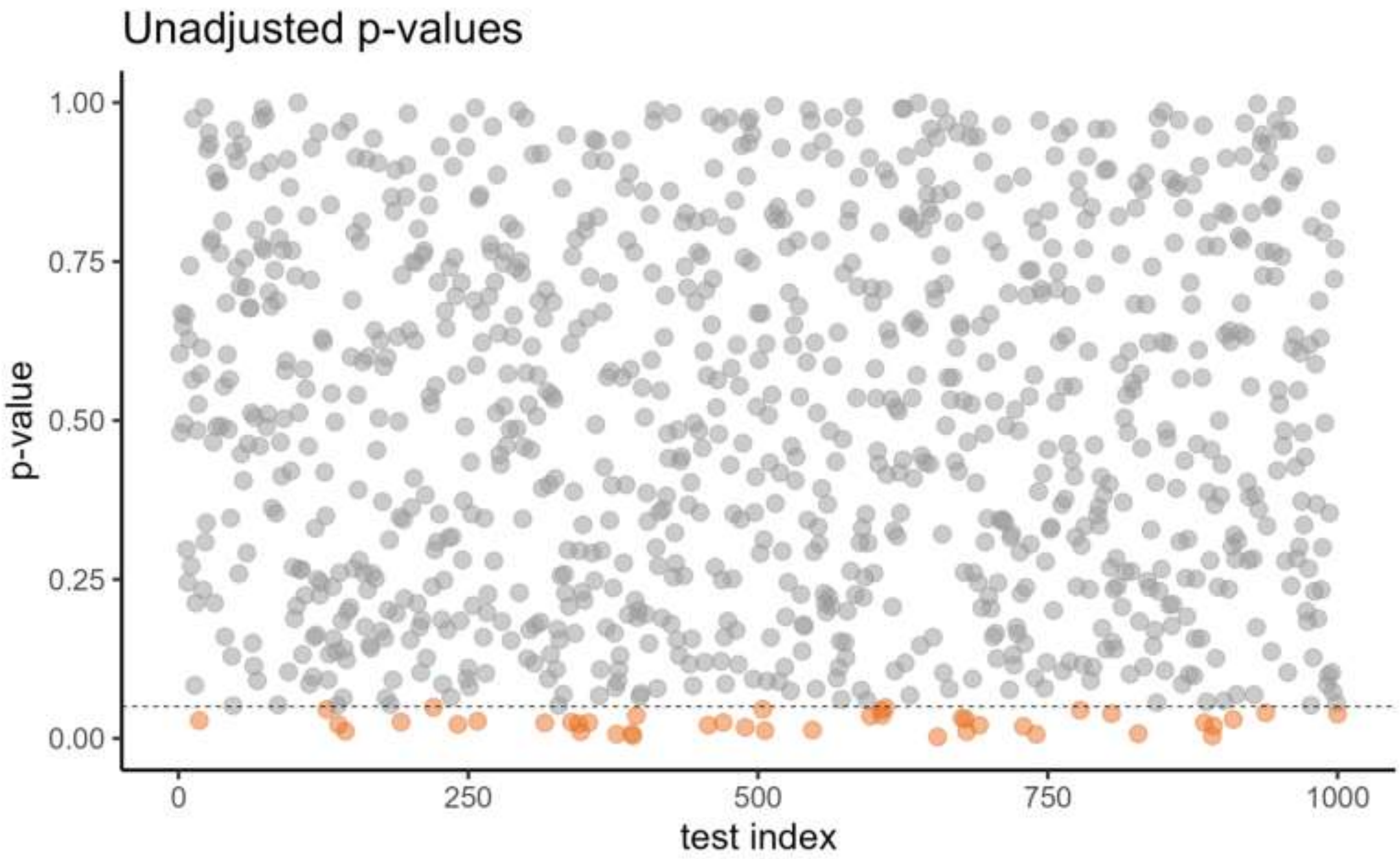
Set a fixed "acceptable" false discovery rate

- Accept some false positives
- Retain (more) power

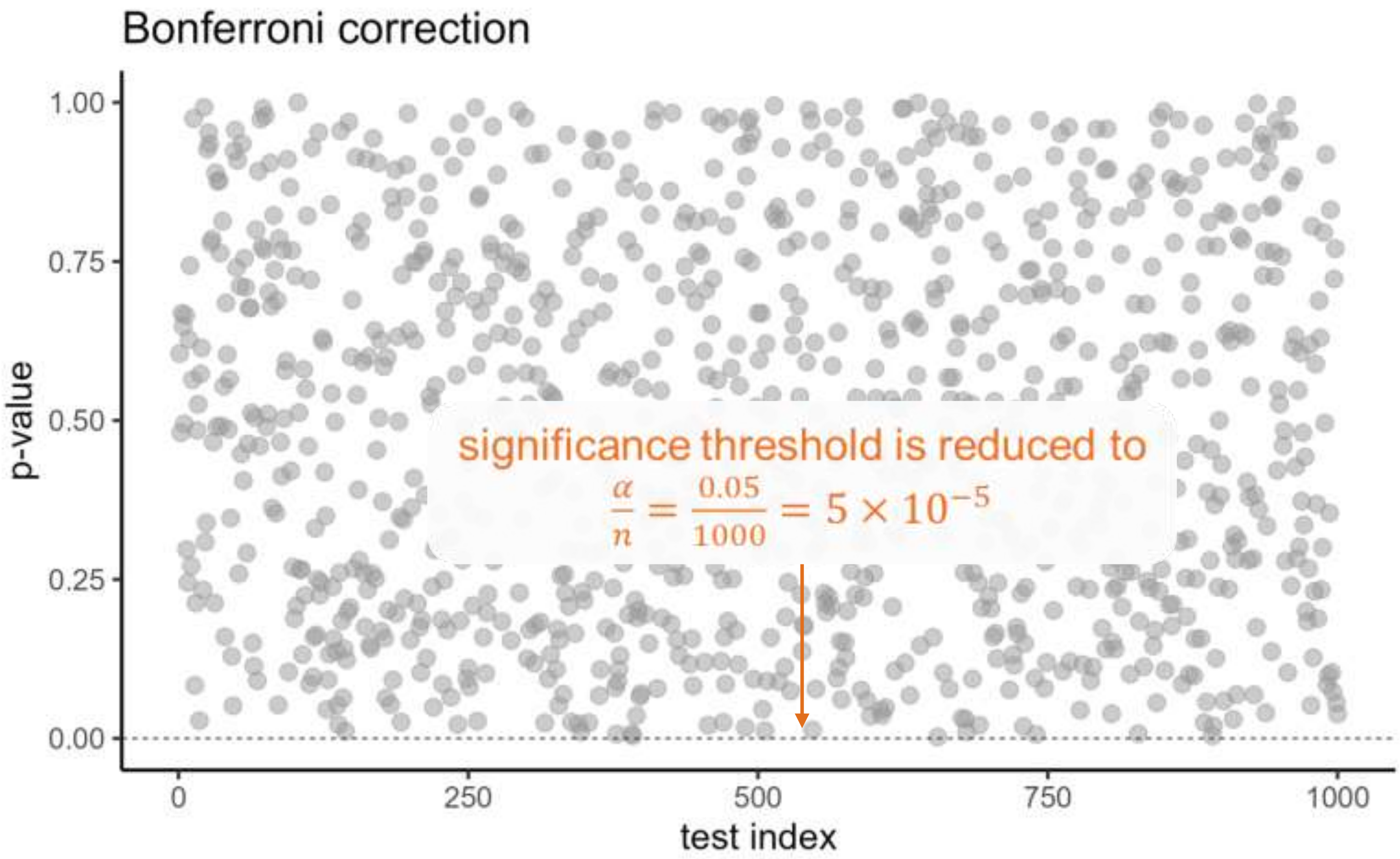
Correcting p-values



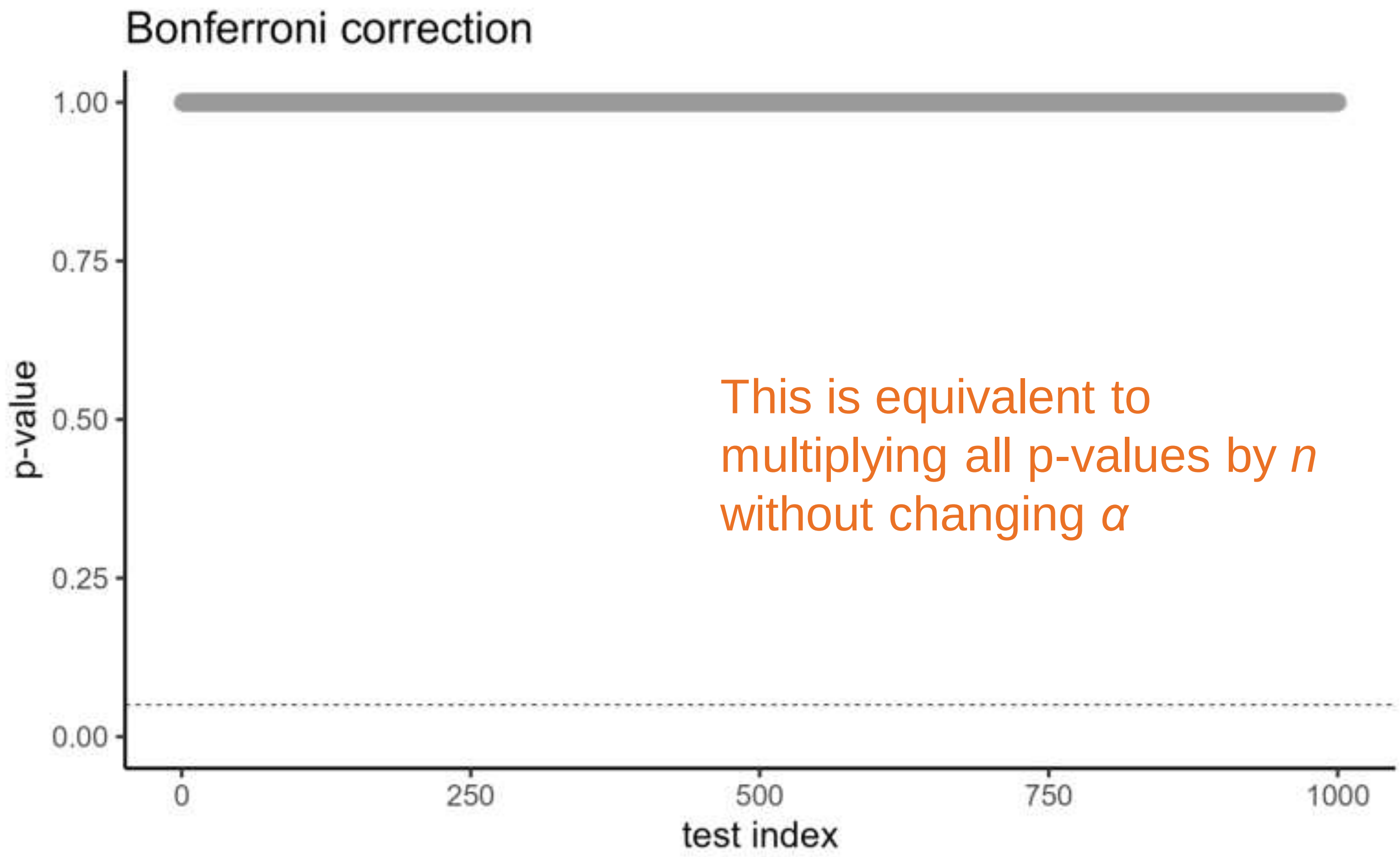
Correcting p-values



Correcting p-values

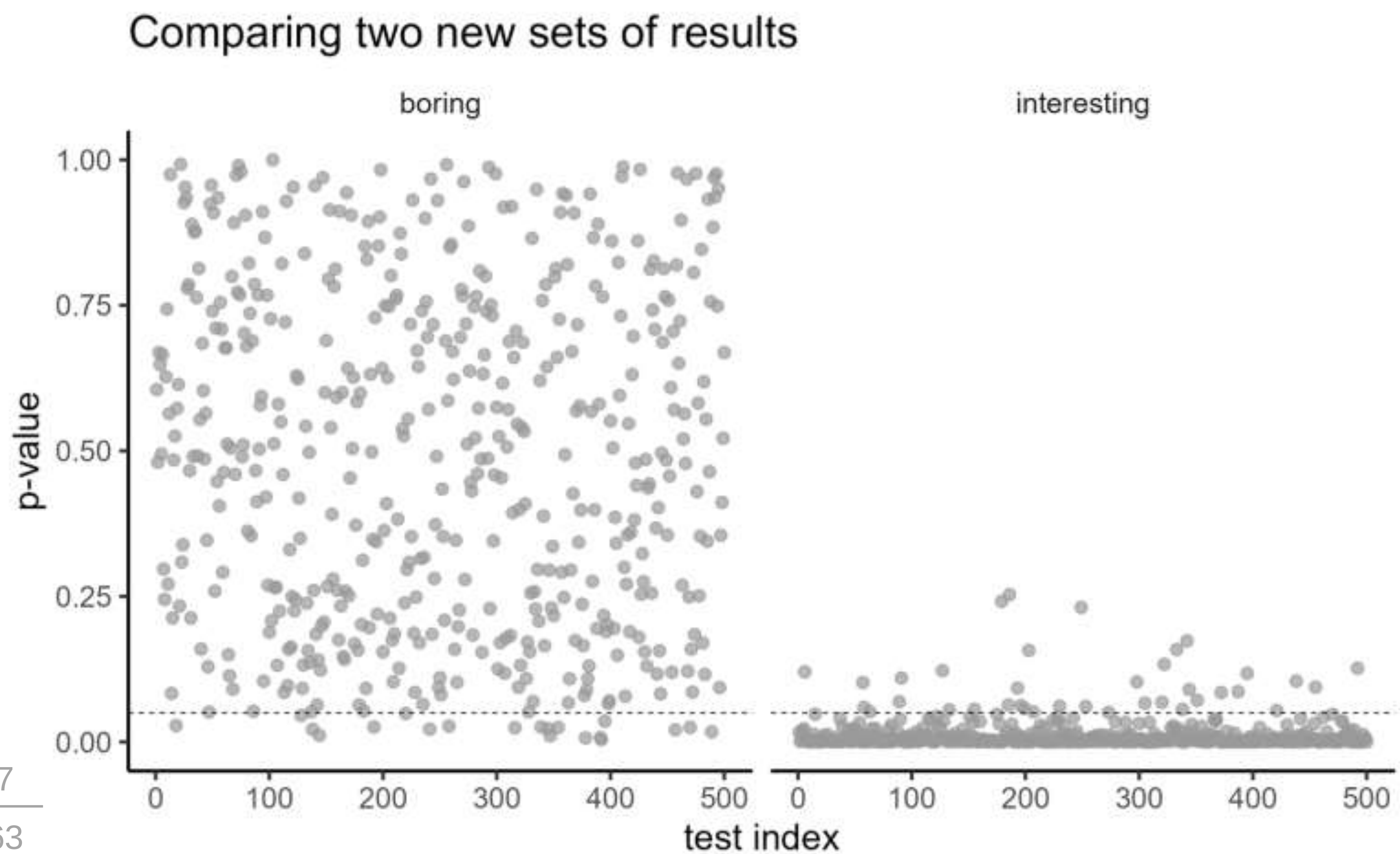


Correcting p-values



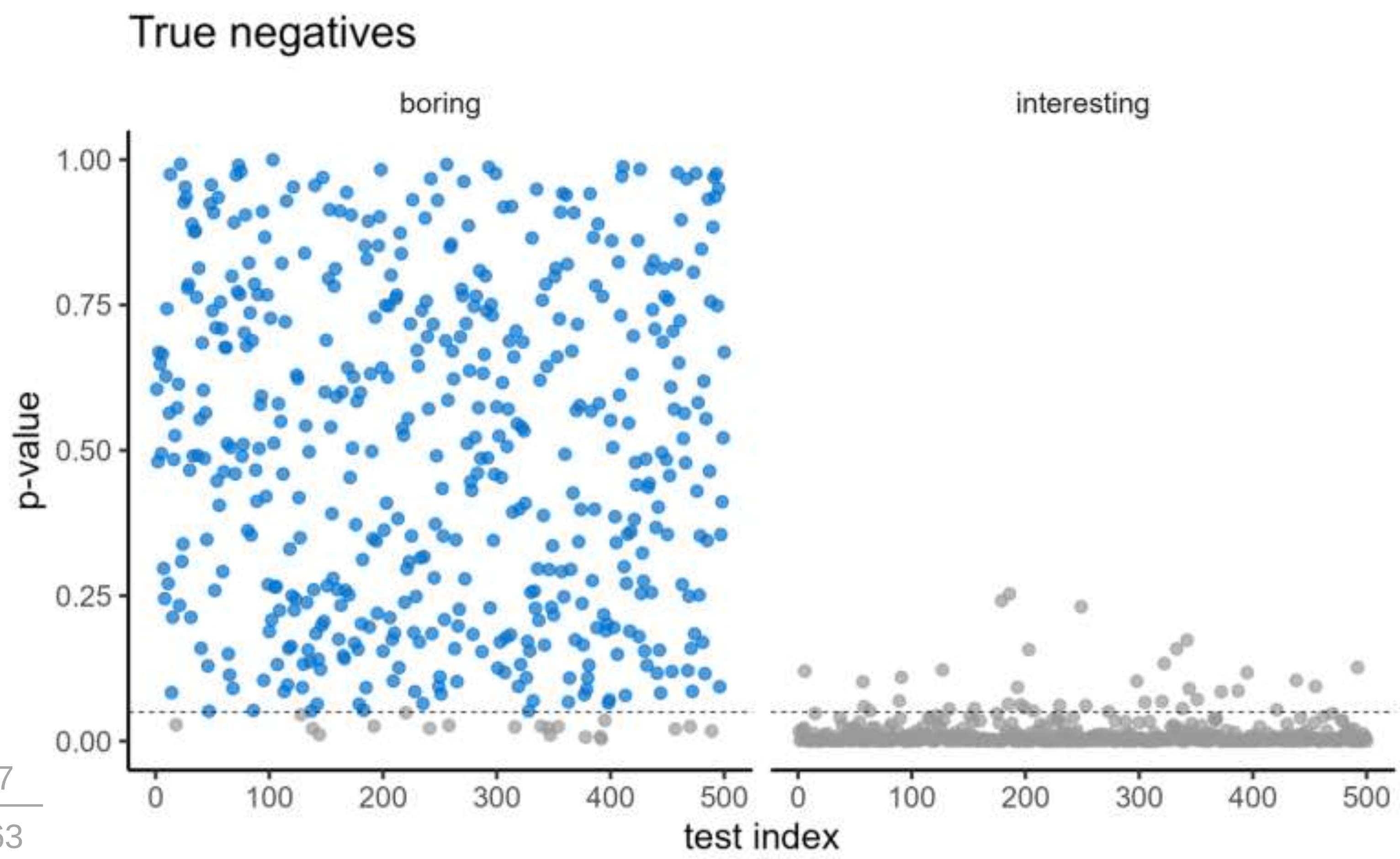
This is equivalent to multiplying all p-values by *n* without changing *α*

Correcting p-values



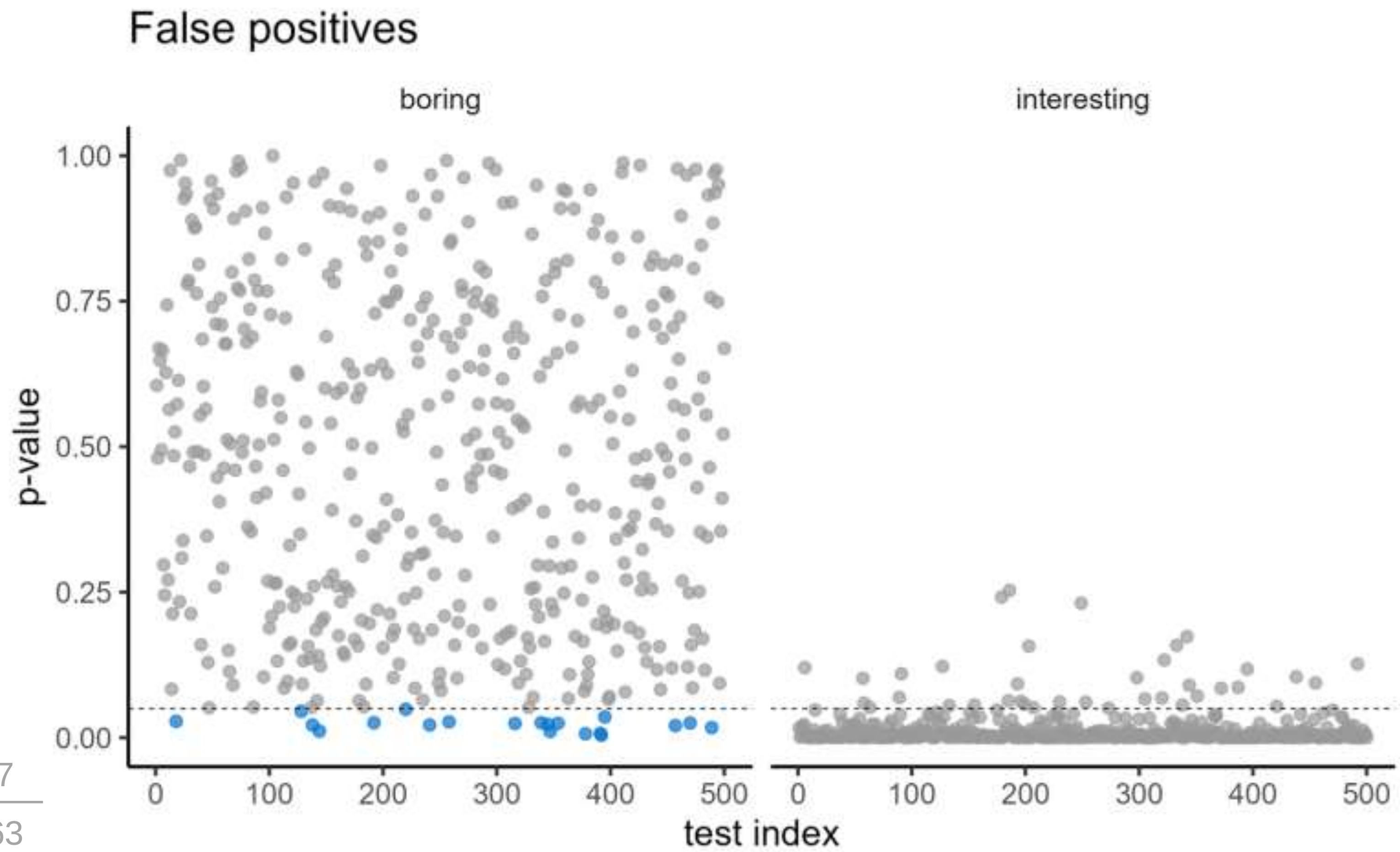
480	37
20	463

Correcting p-values



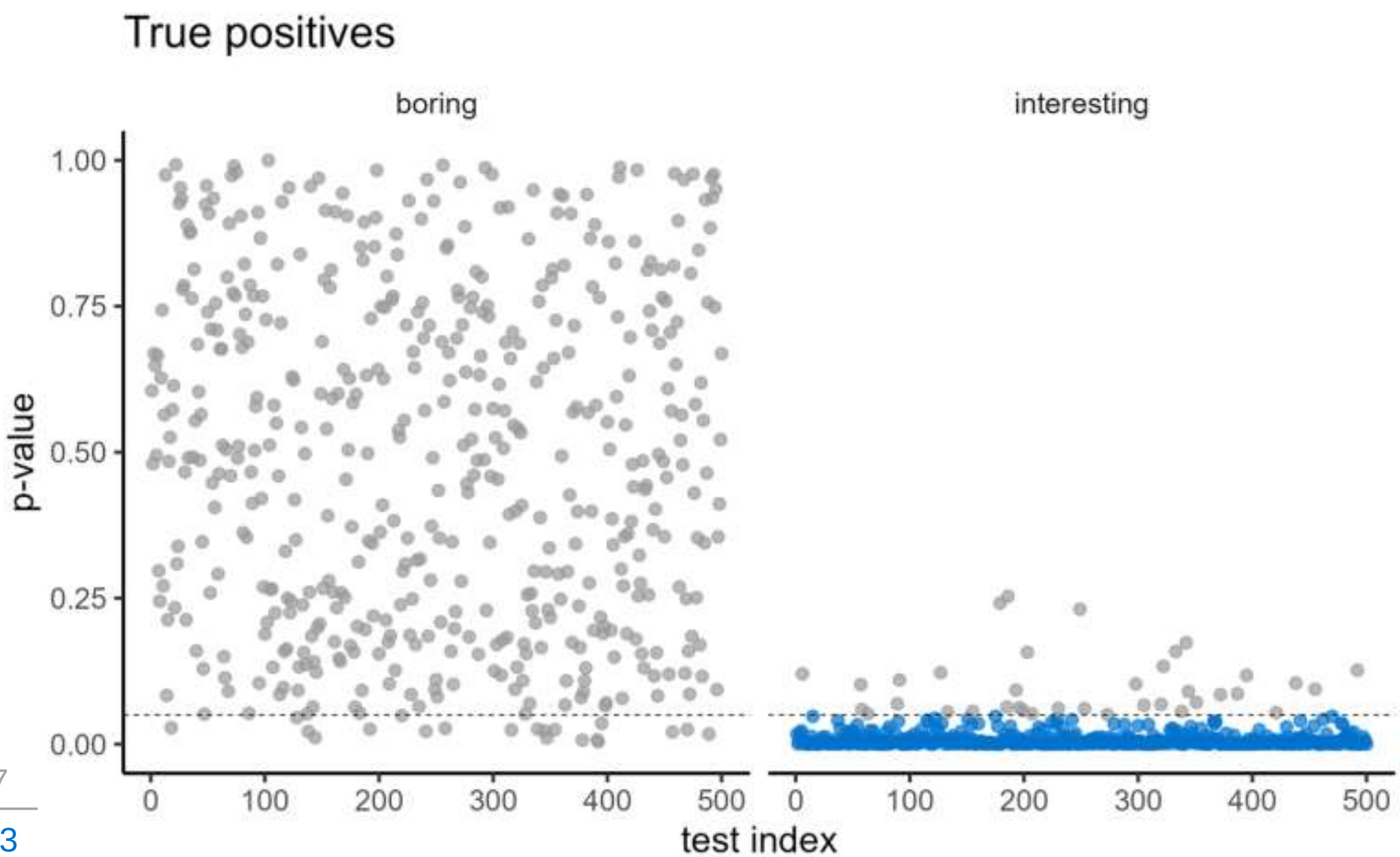
480	37
20	463

Correcting p-values



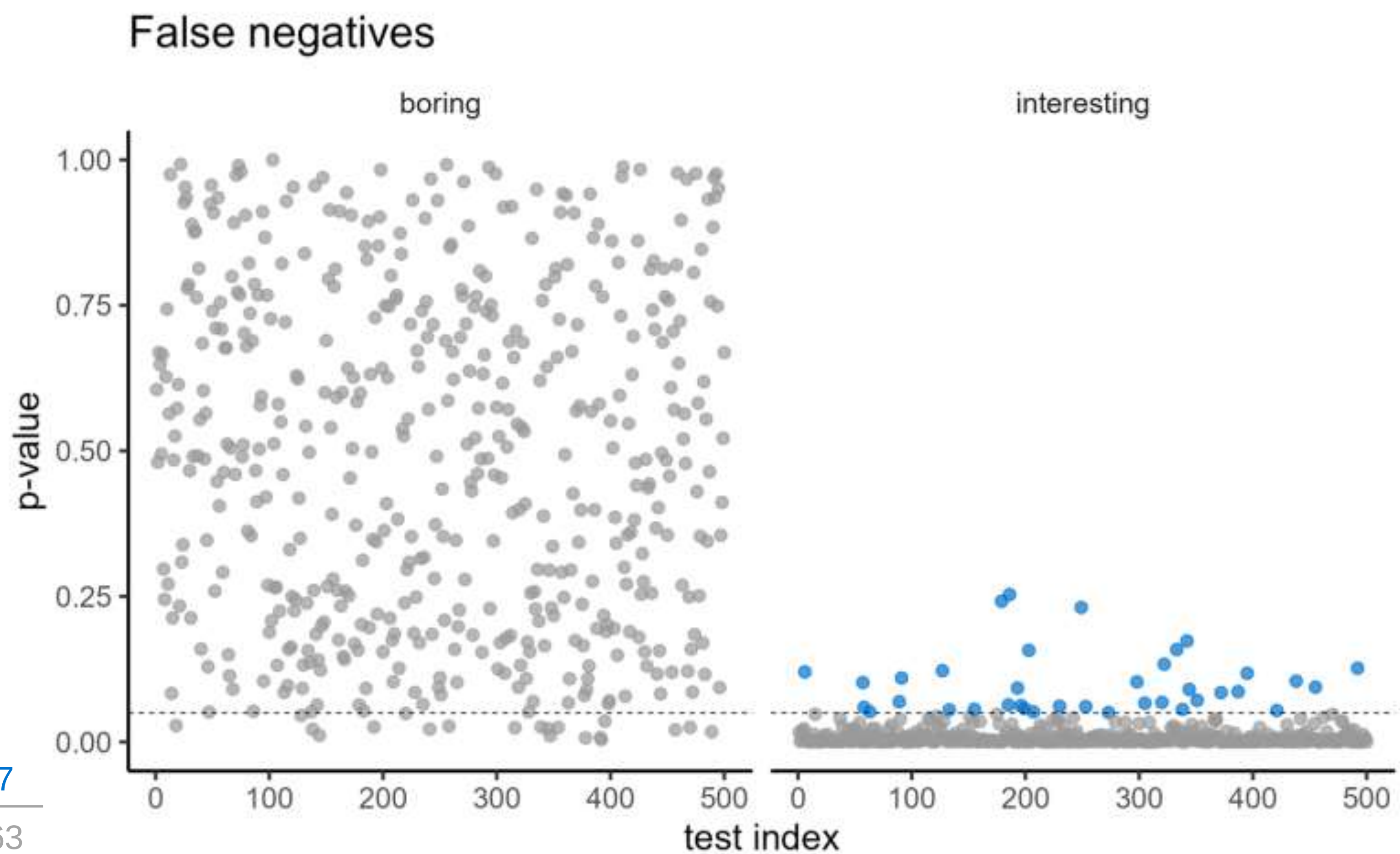
480	37
20	463

Correcting p-values



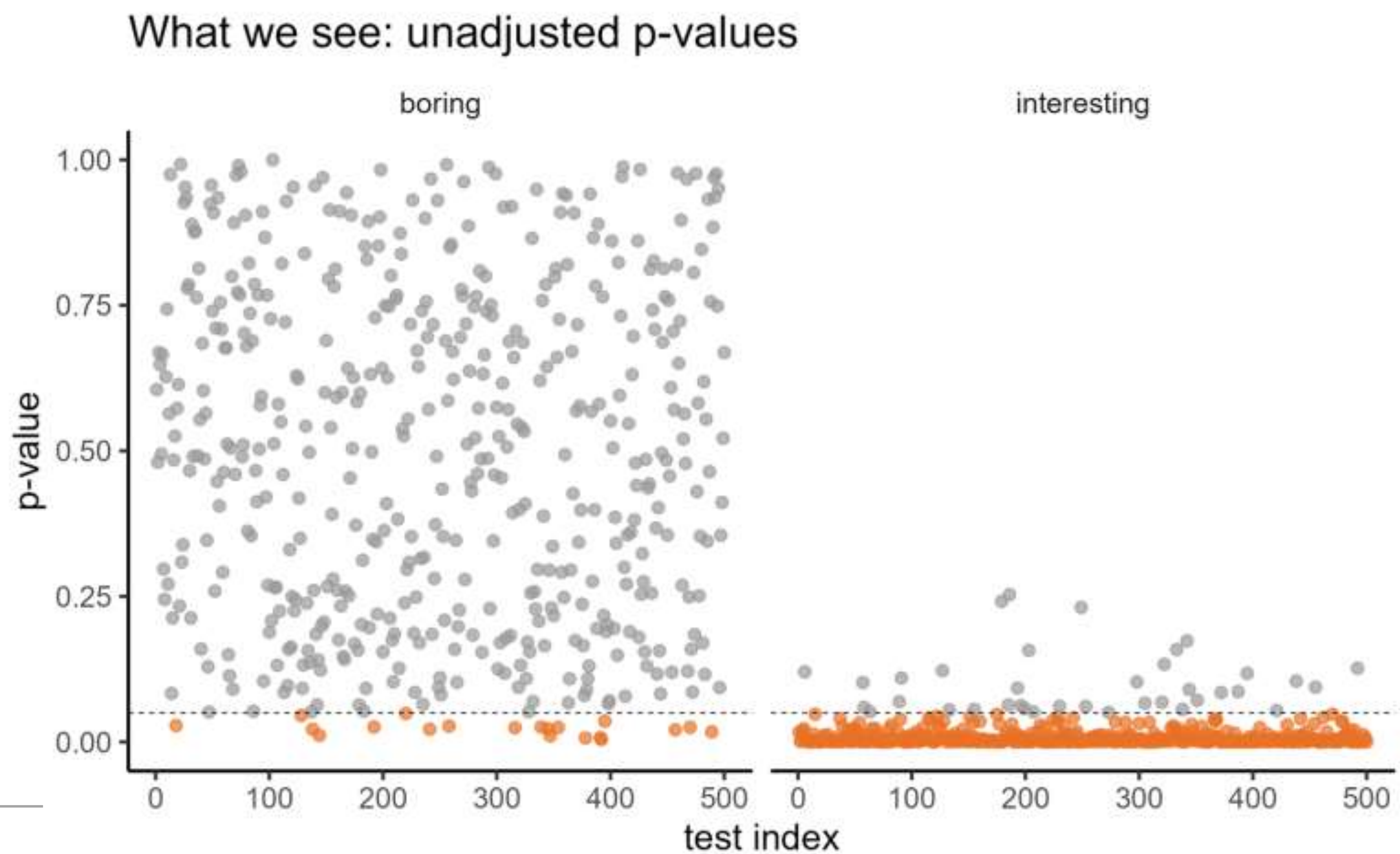
480	37
20	463

Correcting p-values



480	37
20	463

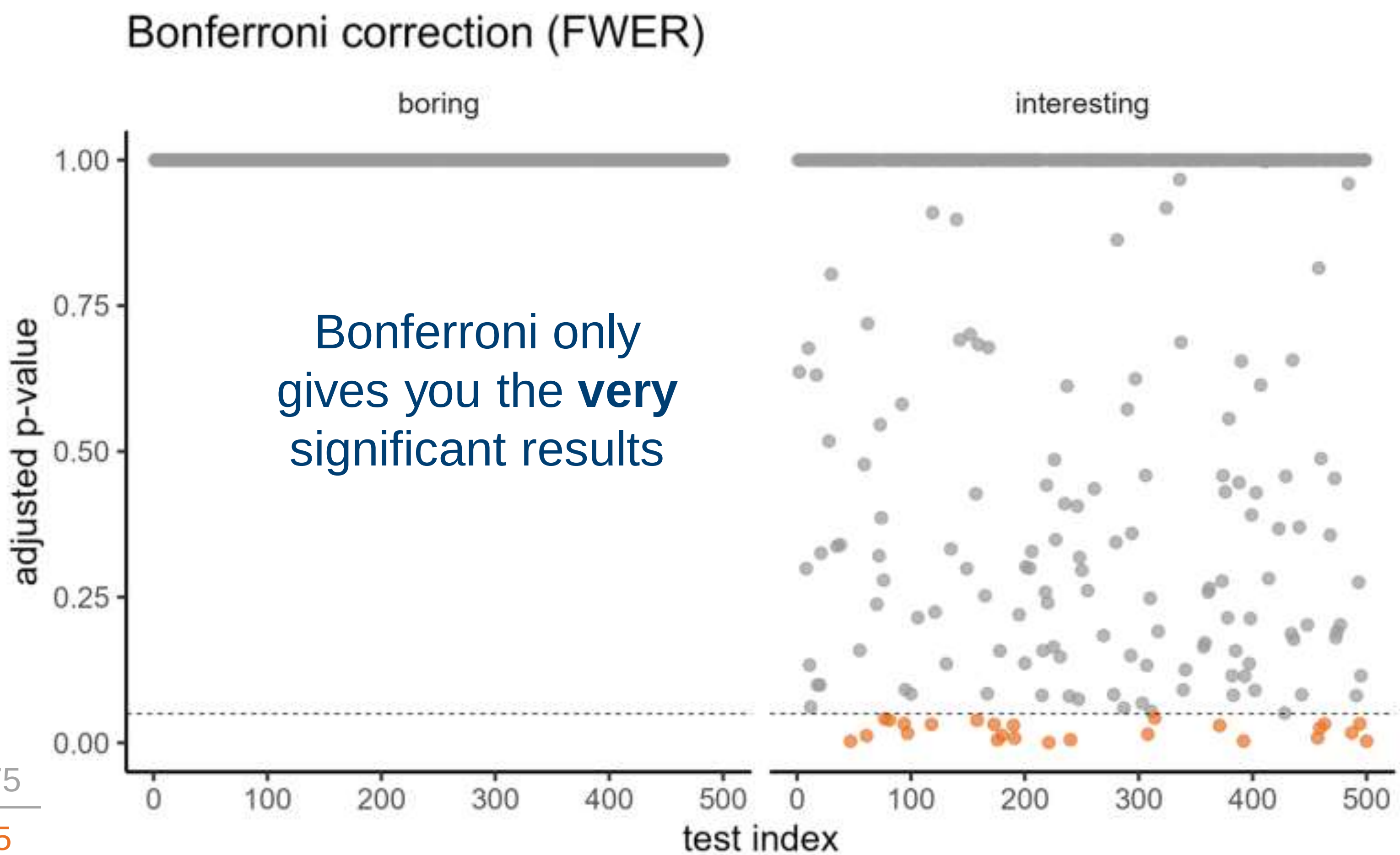
Correcting p-values



517

483

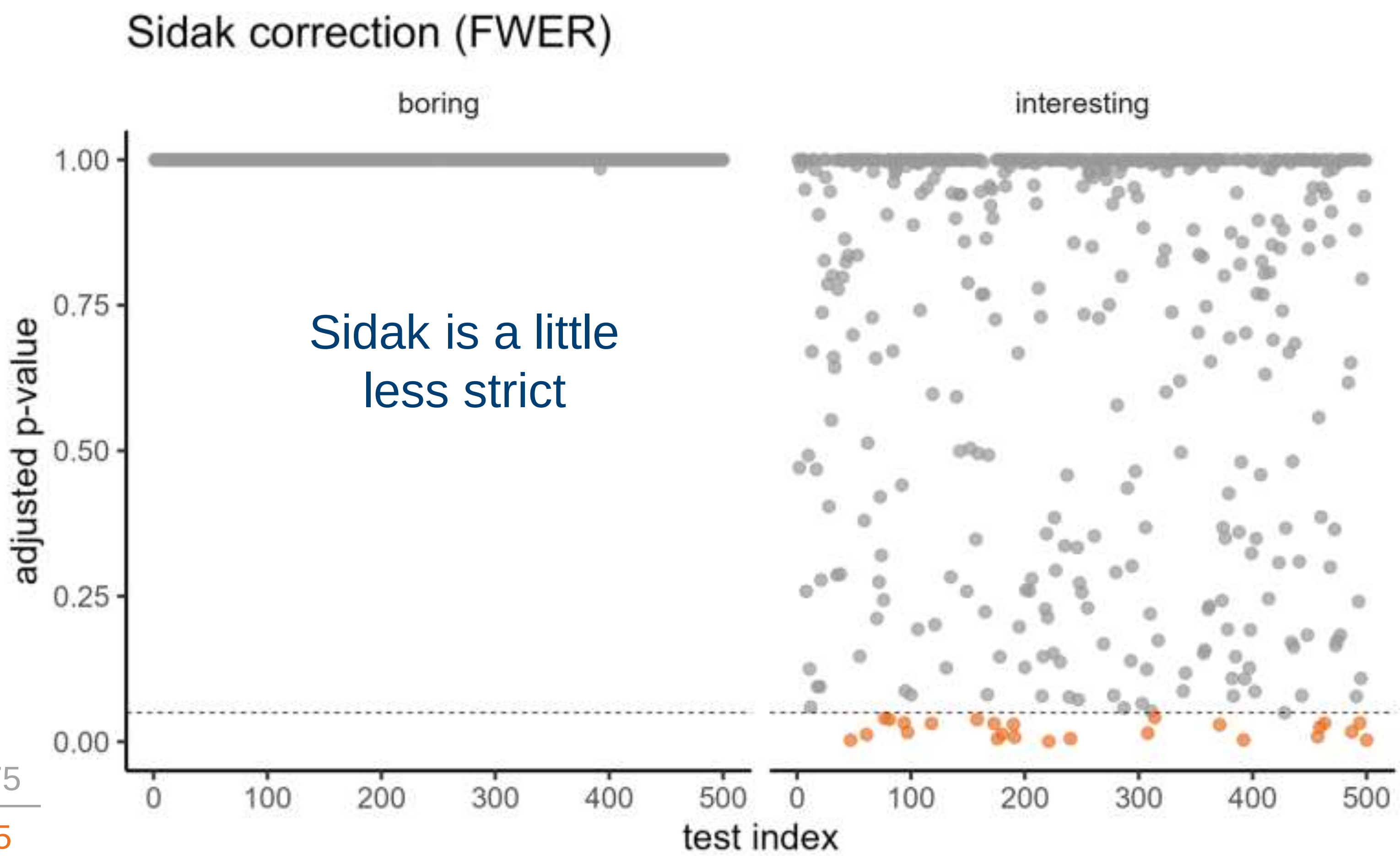
Correcting p-values



Bonferroni only gives you the **very** significant results

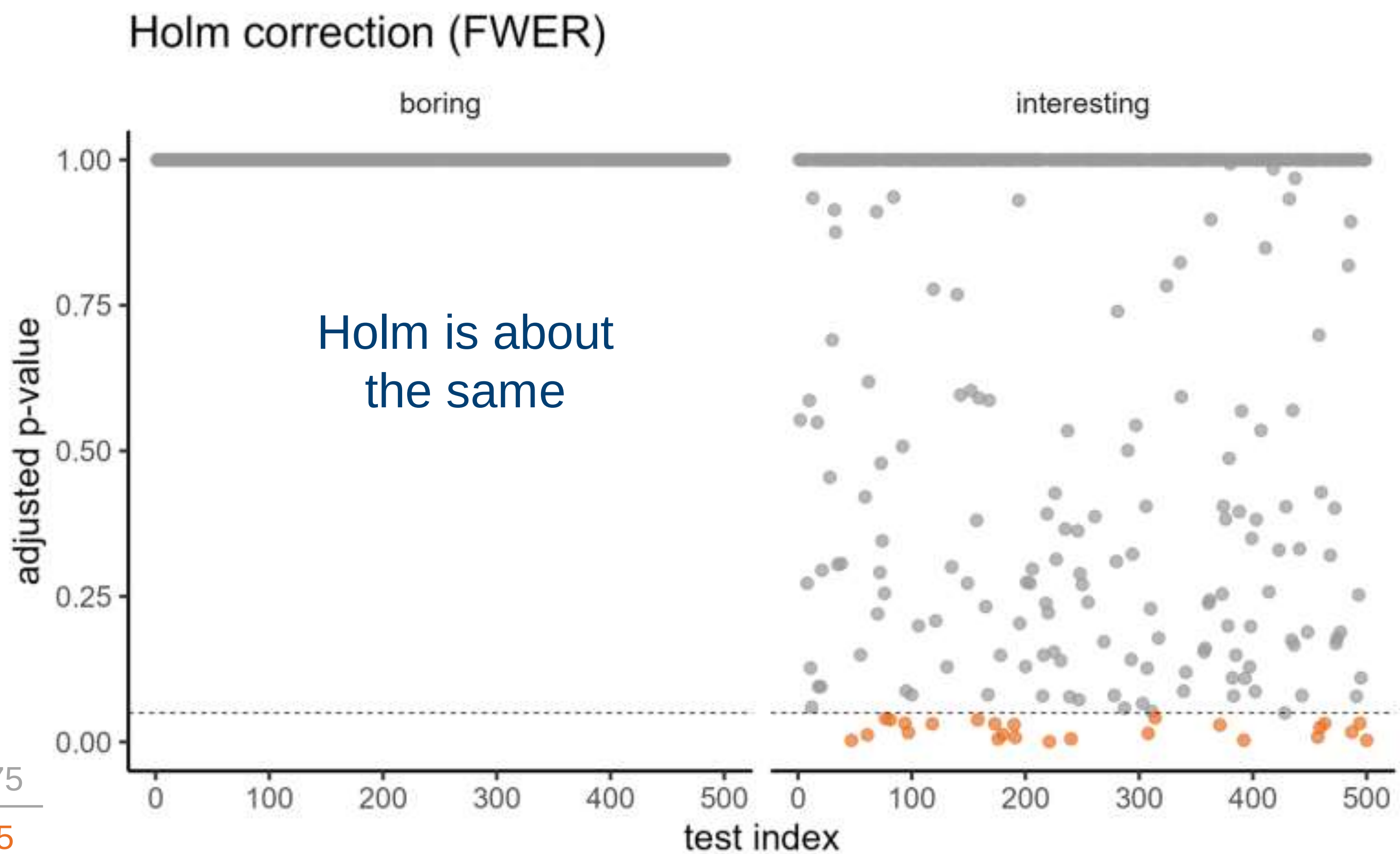
500	475
0	25

Correcting p-values



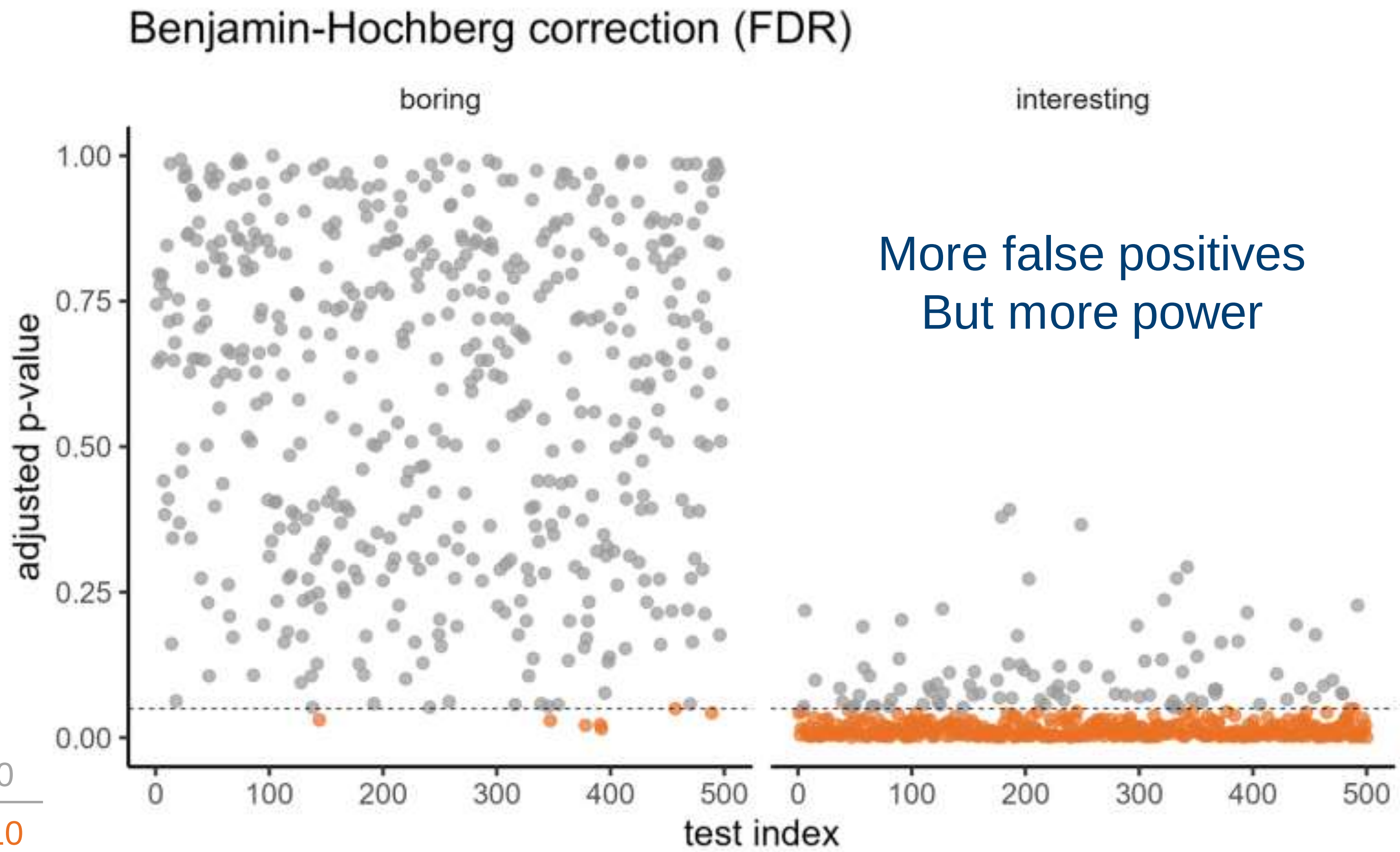
500	475
0	25

Correcting p-values



500	475
0	25

Correcting p-values



More false positives
But more power

493	90
7	410

Correcting p-values

What type of correction should you choose?

Correcting p-values

FWER methods

E.g., clinical trials

- Bonferroni
- Holm
- Hochberg
- Hommel
- Sidak

For when you need to be sure/confident about a set of analyses

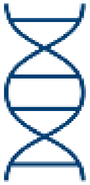
Correcting p-values

FWER methods

E.g., clinical trials

FDR methods

E.g., genome-wide



- Benjamini-Hochberg
- Benjamini-Hochberg-Yekutieli

Typically used when running a very large number of tests

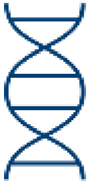
Correcting p-values

FWER methods

E.g., clinical trials

FDR methods

E.g., genome-wide



No correction

E.g., COVID tests

If you’re trying to maximise power, and
false positives are less of a problem

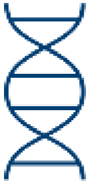
Correcting p-values

FWER methods

Avoid type I error

FDR methods

Control false leads



No correction

Avoid type II error

Power AND false positives increasing

Session 6: Uncertainty

Lecture summary:

- Hypothesis tests can be wrong in two ways: false positive (type I) and false negative (type II) errors

- Statistical power is the probability of a "true positive"
- We can test and plan our experiment's power using power analysis

- Effect sizes measure the magnitude of relationship between variables, and are required for power analysis

- To control error rates, we can apply p-value correction

Session 6: Uncertainty

Course materials:

Calculate effect sizes from data

- Cohen's d, f^2

Calculate required sample sizes (*a priori* power analysis)

- T-tests
- Linear models

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Core statistics

Session 6: Uncertainty

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Final notes

- Course materials & slides will remain online
- Any scripts will be shared via the course site

- Feedback survey accessible via the course site

Related courses:

- Generalised linear models
- Linear mixed effects models

Core statistics

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